

Plant Breeding Tools

USER'S MANUAL



Plant Breeding Tools

Version 1.3
May 2014

A NOTE TO THE READER:
An electronic copy of this manual is included in the PBTools installer. The PBTools User's Manual may be printed/copied and distributed to any number of users. PBTools is a freeware developed for non-profit use. Hence, selling of either the software or the manual is prohibited.

Biometrics and Breeding Informatics
Plant Breeding, Genetics and Biotechnology Division
INTERNATIONAL RICE RESEARCH INSTITUTE

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1. Introduction

Plant Breeding Tools (PBTools) is a software that has been developed to assist plant breeders in the design of experiments and analysis of data. It has an easy-to-navigate graphical user interface that does not require users to have programming skills to perform data manipulation and analysis. It uses R functions that were specifically written for the development of this software and those that are available in the following R packages: ade4, agricolae, dae, DiGGer, doBy, fields, fitdstrplus, gdata, gplots, gtools, Hmisc, leaps, lme4, lmerTest, MASS, multcomp, mvtnorm, numDeriv, pastecs, permute, plotrix, plyr, qtl, R.methodsS3, R.oo, rJava, spam, stringr and vegan.

Featured Modules

PBTools provides the following modules:

- Data management in spreadsheet view
- Randomization for commonly used experimental designs in plant breeding
- Single-environment analysis
- Multiple-environment analysis
- QTL analysis
- Selection index
- Commonly used mating designs
- Generation mean analysis

New Features in PBTools 1.3

The following are the revisions made for the latest version of PBTools:

- Randomization for Latinized Alpha Lattice design is now an option under the randomization for Alpha Lattice design
- Randomization for Latinized Row-Column design is now an option under the randomization for Row-Column design
- Genotype×Environment (G×E) Analysis module which consists of Stability Models and Multiplicative Models is added under multi-environment analysis.
- Stability analyses, which are options under multi-environment analysis in the previous versions, are transferred to Stability Models module.
- AMMI and GGE analyses, which are options under multi-environment analysis in the previous versions, are transferred to Multiplicative Models module.
- GGE biplots are presented in symmetric, environment and genotype views.

About the Manual

This manual provides step-by-step instructions on how to perform certain tasks that are of interest to users. Tasks are discussed using sample datasets that are included when PBTools is installed. In most tasks, dialog boxes are displayed and the user has to specify required fields and additional options as desired. Descriptions and limitations of these fields and options are also provided. Screen images are also included for additional references.

This manual is written using the following format:

- Menu items, names of dialogs and form controls appear in **bold** letters
- Filenames, folder names, variables names and directories are *italicized*
- Menu items appear in the form **Project | New Project**, which means "choose **New Project** from the **Project** menu"

Citing the Software

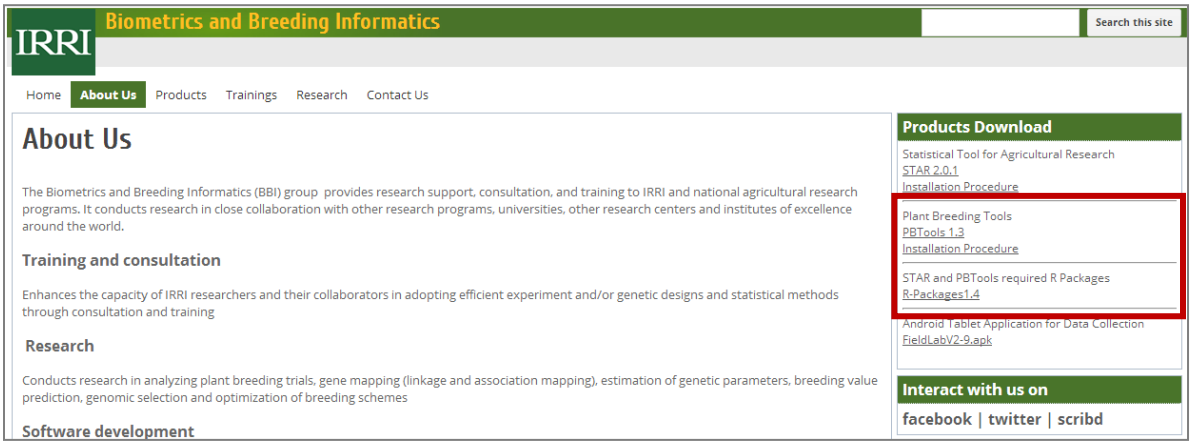
The suggested citation of PBTools in publications is as follows:

PBTools, version 1.3. 2014. Biometrics and Breeding Informatics, PBGB Division, International Rice Research Institute, Los Baños, Laguna.

2. Getting Started

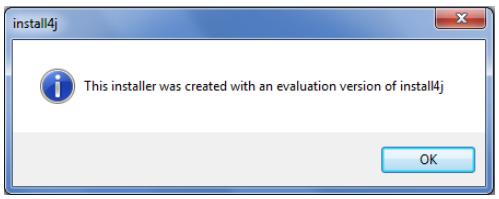
Installation

The installer of PBTools 1.3 and its required R packages can be downloaded from *bbi.irri.org*. To download the installers, click the links labeled as *PBTools 1.3* and *R-Packages1.4* at the right-hand side of the page.

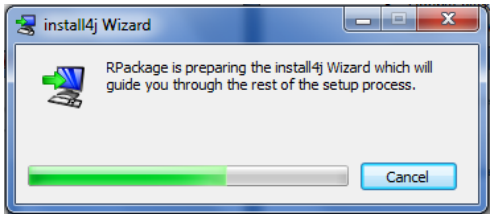


The steps to install the required R packages are as follows:

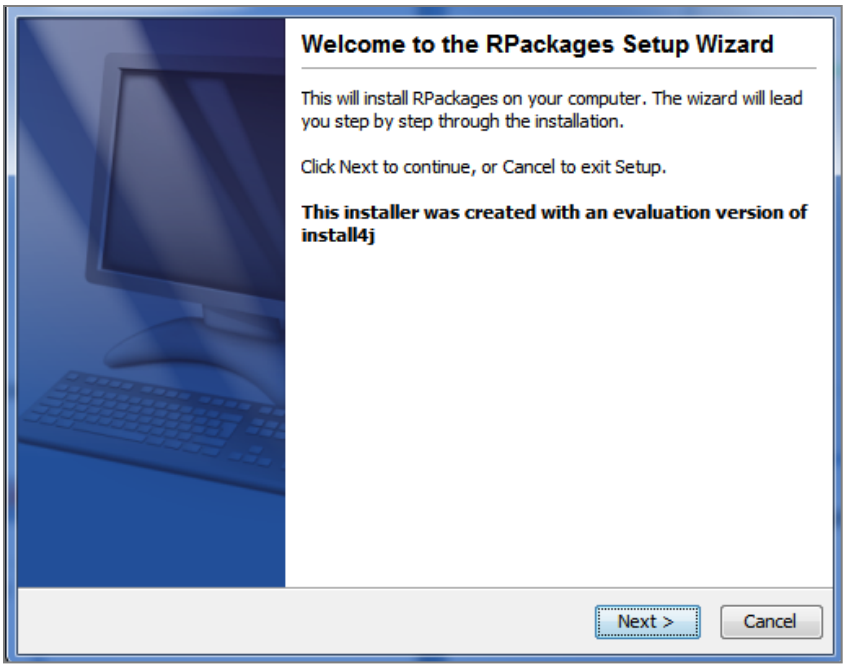
- Double-click the installer icon RPackages-1.4-windows-setup-32bit to launch the setup. A dialog as shown below is displayed. Click **OK**.



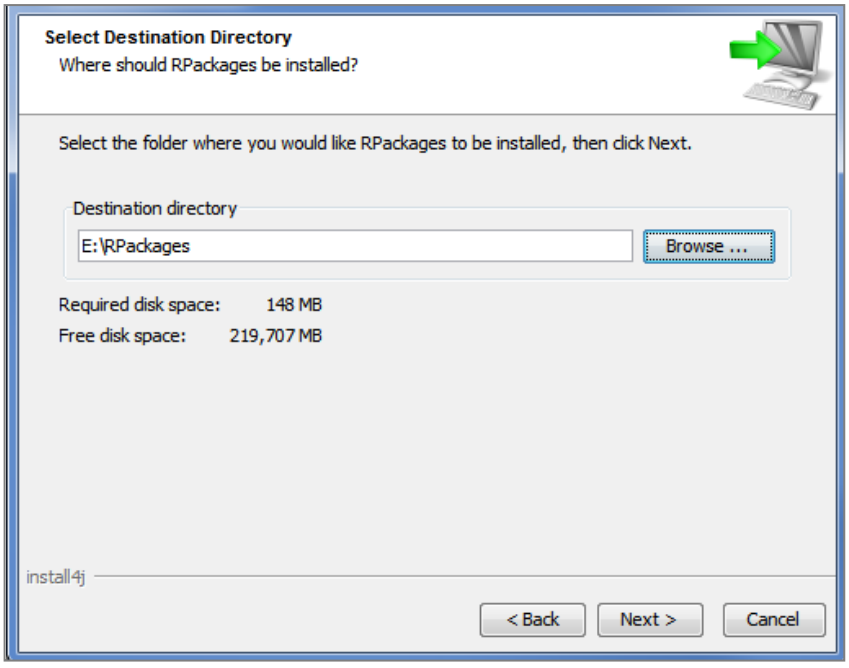
- The next step is the preparation of the wizard. A dialog with progress bar is displayed. Wait until the preparation is completed.



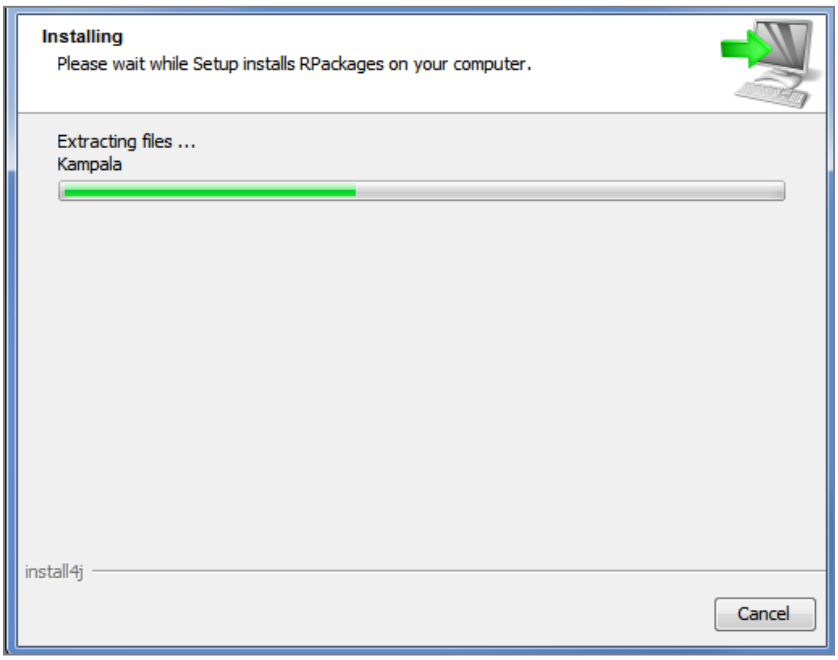
- When installing R Packages for the first time, the **Welcome** dialog as shown below is displayed. Click **Next**.



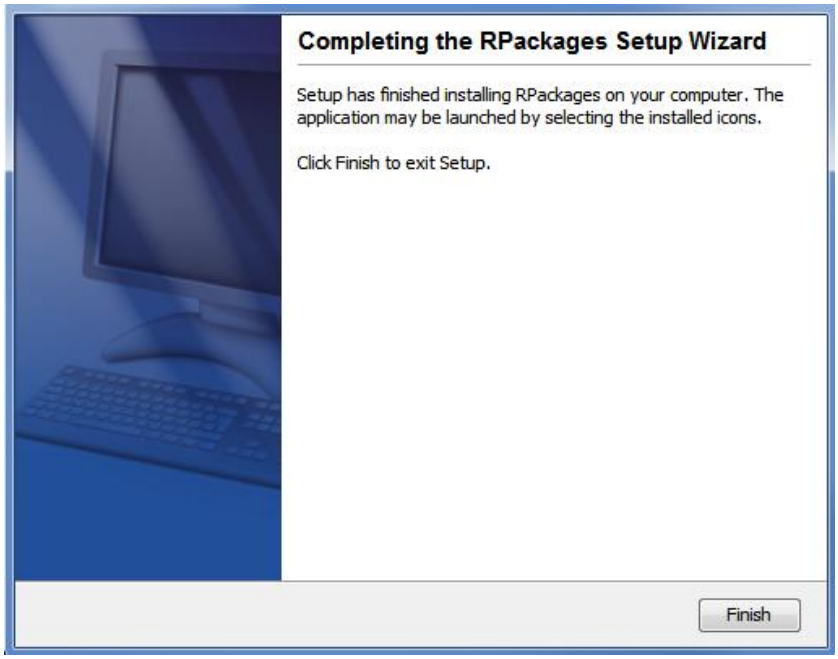
- Select the destination folder, preferably **not in C:\Program Files** as problems were encountered when RPackages is installed in this location. Click **Next**.



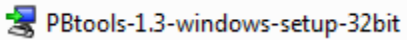
- Wait until the installation is completed.



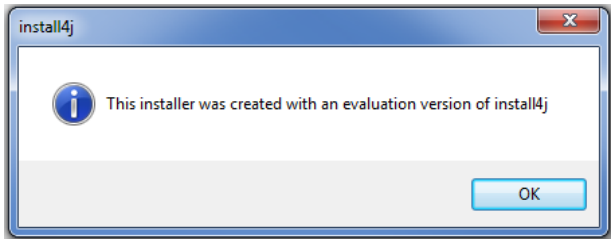
- Click **Finish** to exit Setup.



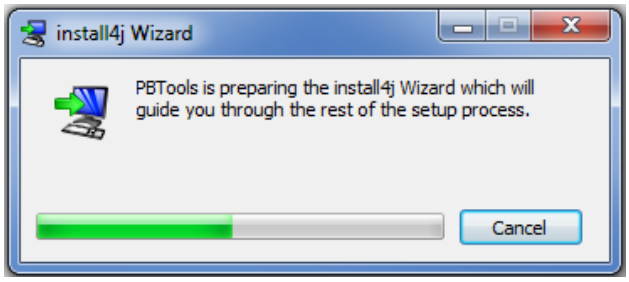
The steps to install PBTools are as follows:

- Double-click the installer icon 

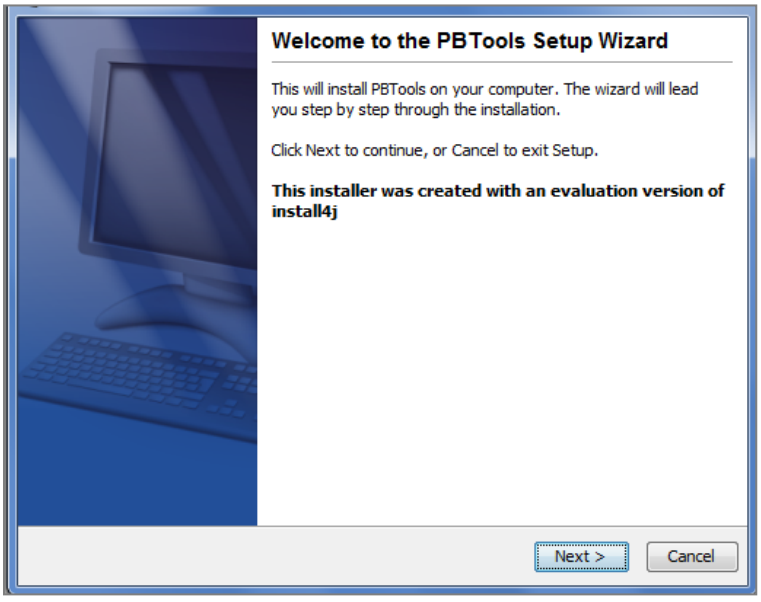
to launch the setup. A dialog as shown below is displayed. Click **OK**.



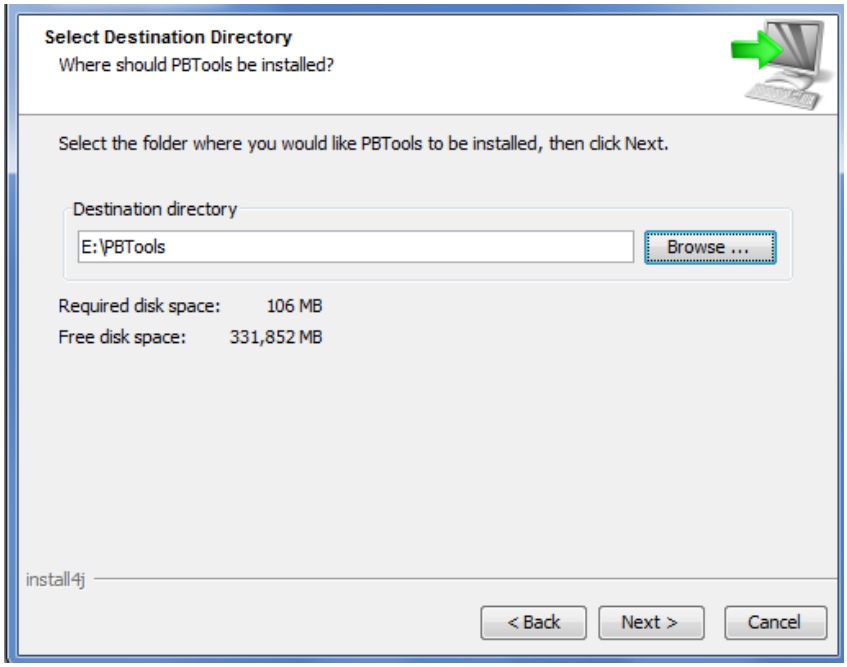
- The next step is the preparation of the wizard. A dialog with progress bar is displayed. Wait until the preparation is completed.



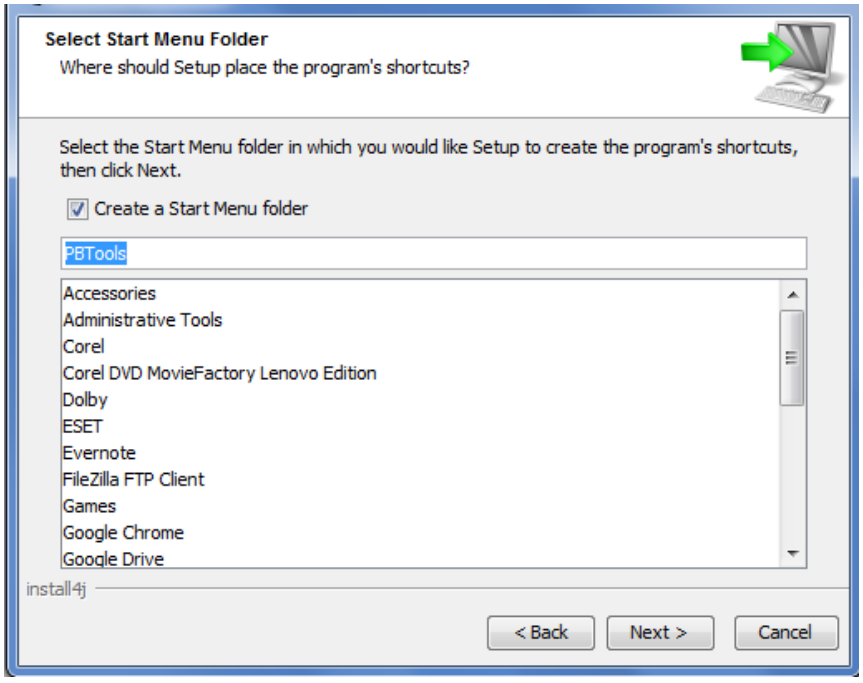
- When installing PBTools for the first time, the **Welcome** dialog as shown below is displayed. Click **Next**.



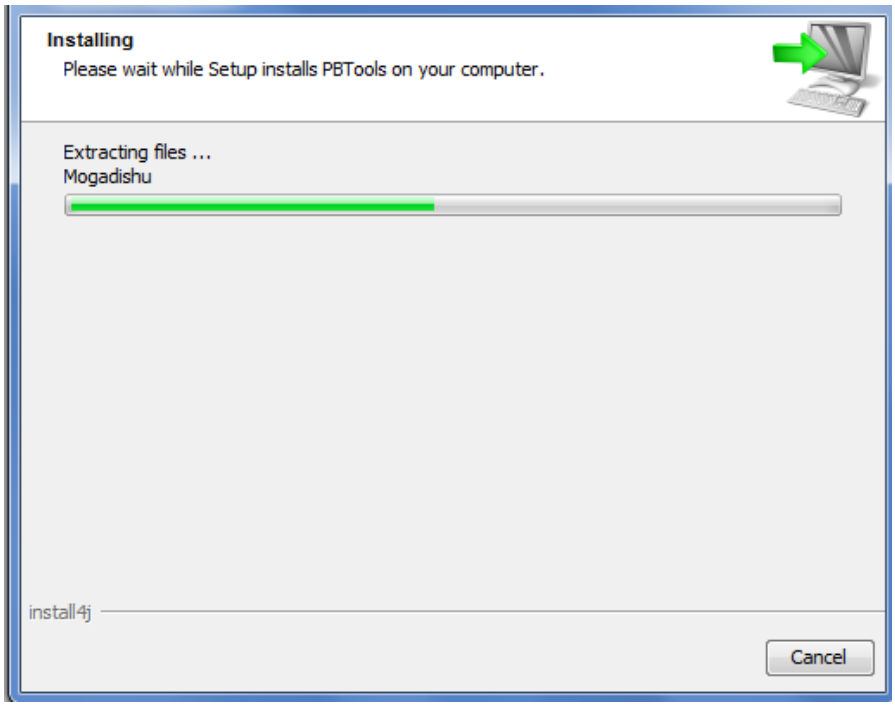
- Select the destination folder, preferably **not in C:\Program Files** as problems were encountered when PBTools is installed in this location. Click **Next**.



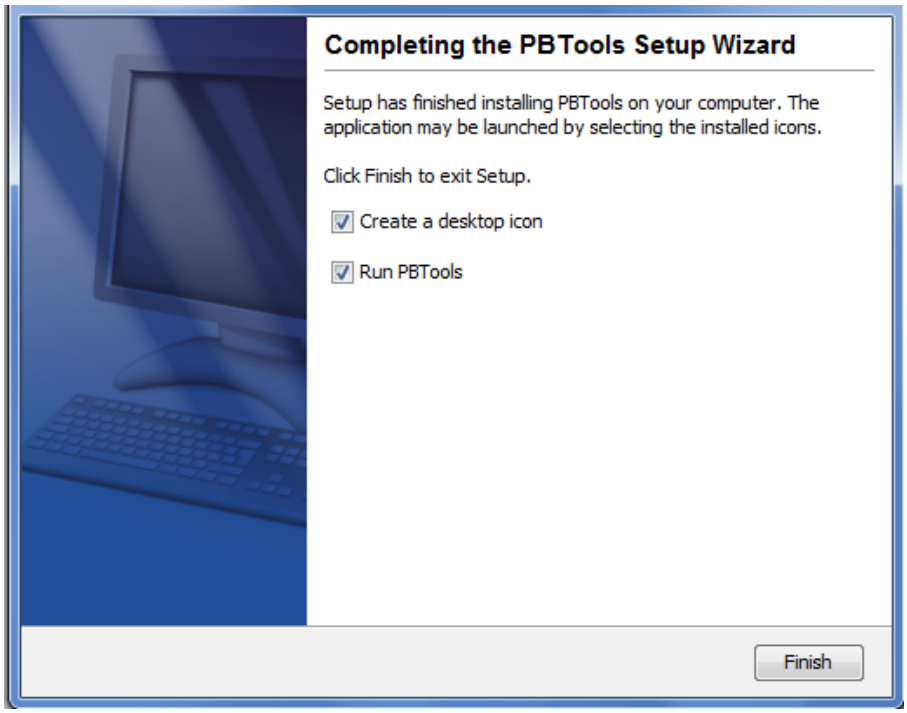
- Select which start menu folder you want to place the PBTools icons in and click **Next**.



- Wait until the installation is completed.



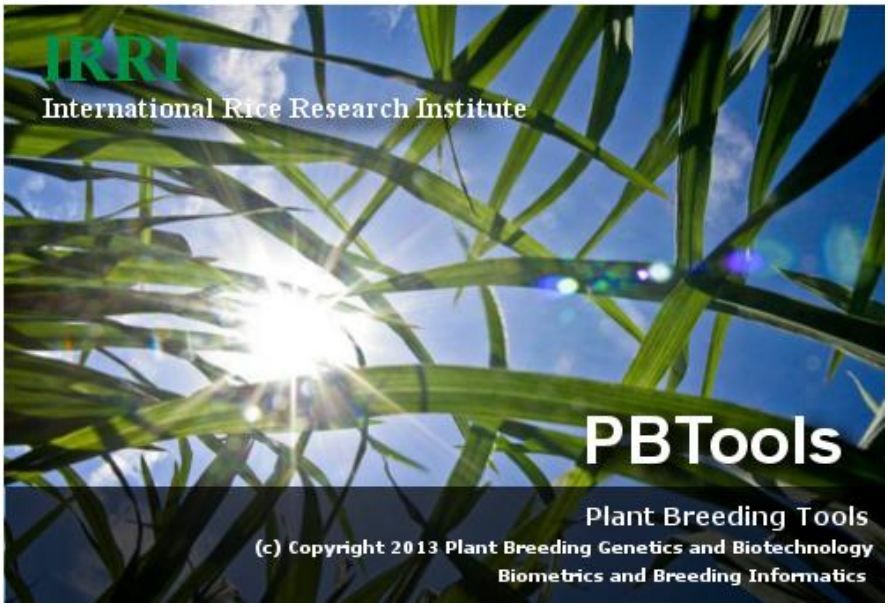
- Finally, specify if you want to create a desktop icon and run PBTools. Click **Finish**.



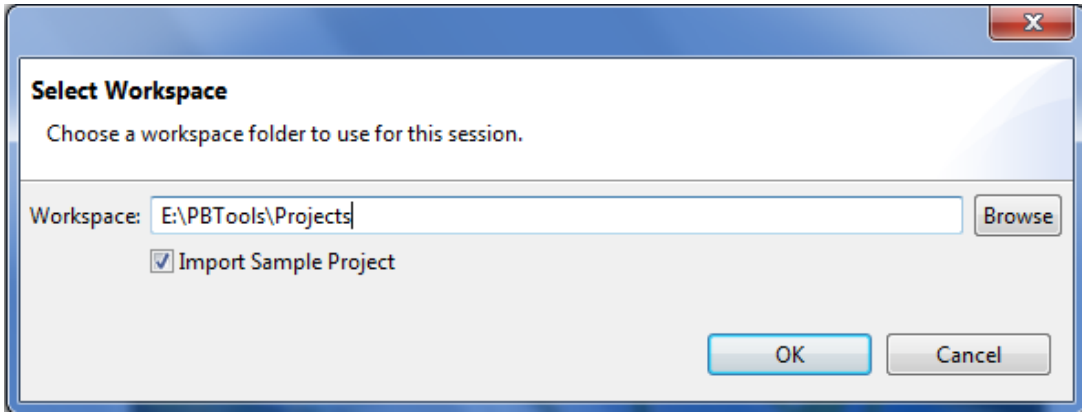
Launching PBTools

To launch PBTools, the user can either double-click the PBTools shortcut icon on the desktop (if there is one) or click Start on the Windows task bar, choose All Programs, then click on the *PBTools* folder and click the PBTools icon.

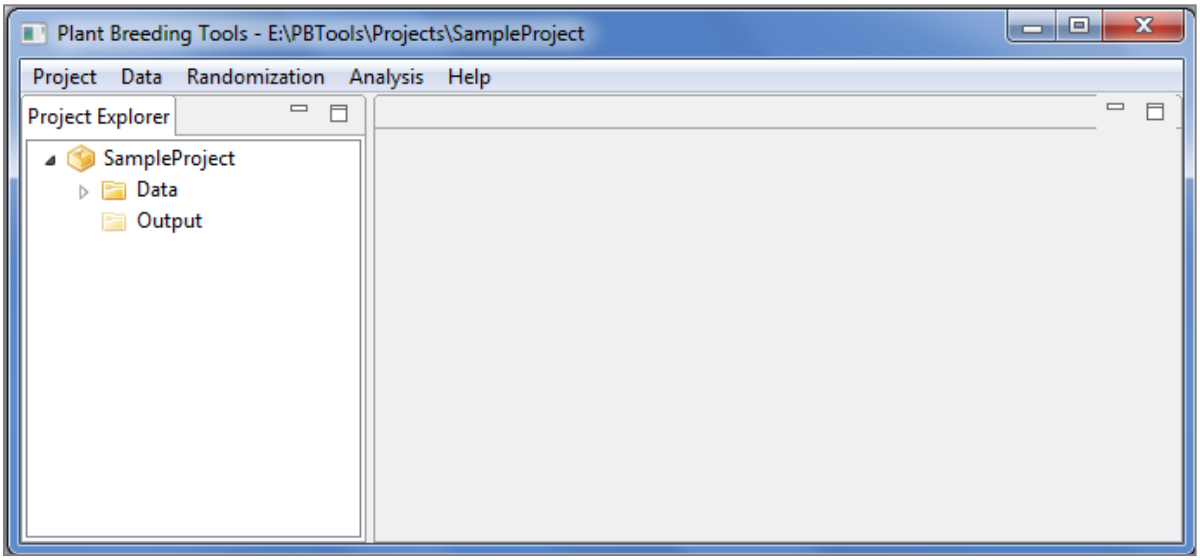
Upon launching, the splash image as shown below will appear.



When PBTools is launched for the first time, the **Select Workspace** dialog is displayed. A default workspace folder is indicated in the **Workspace** entry box**.** To indicate a different workspace folder, click the **Browse** button and select the new workspace folder. By default, the checkbox **Import Sample Project** is selected. Then click the **OK** button.

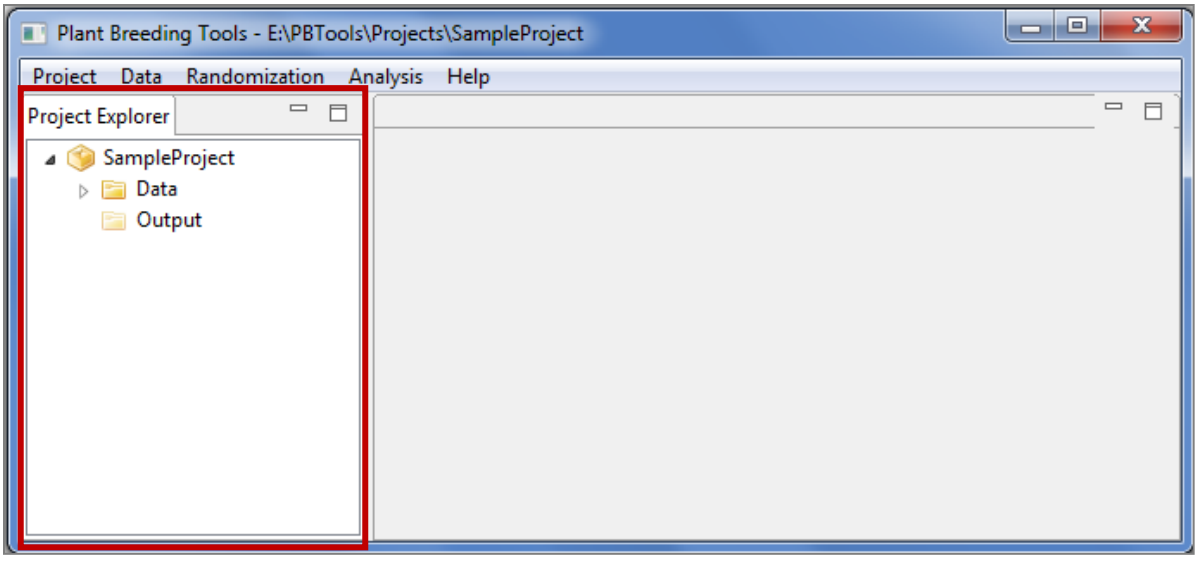


The main window will appear.



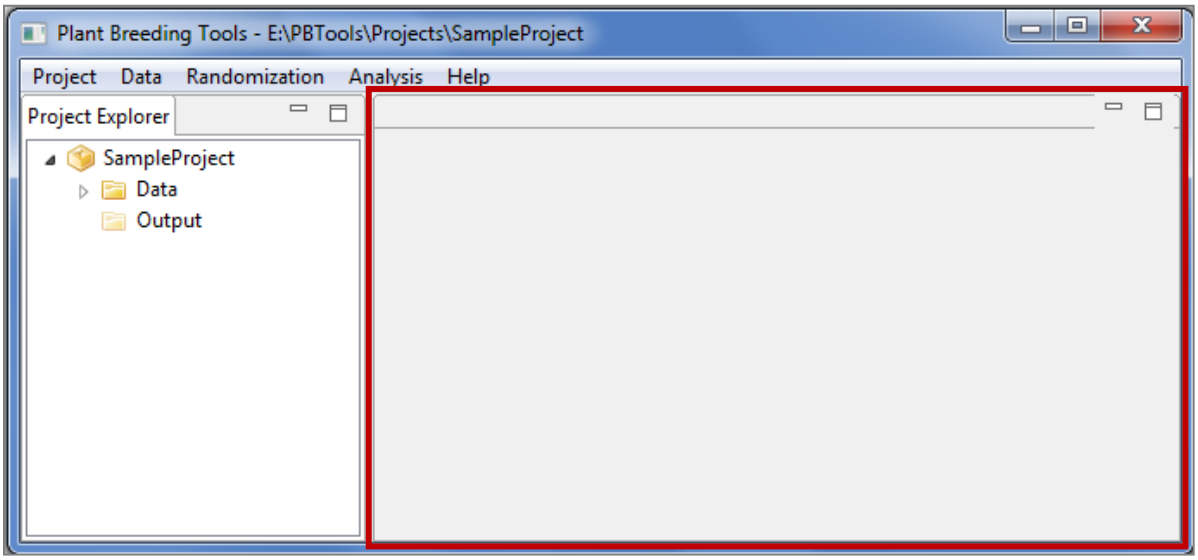
Project Explorer Panel

Project Explorer can be seen at the left-hand side of the main window. It serves as a file manager of the active project, where data and results of the analysis are displayed in a tree. It displays the last opened project from the previous PBTools session. When the 'Import Sample Project' option is selected in the **Select Workspace** dialog, a default project named *SampleProject* is displayed with *Data* folder and *Output* folder inside it. The *Data* folder contains sample datasets that will be used in this manual. The *Output* folder, on the other hand, will contain all output files that will be created when an analysis is performed.



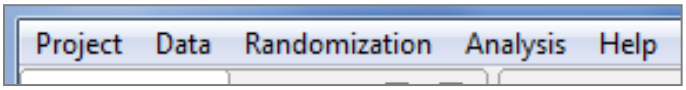
Editor Panel

Editor panel is located at the right-hand side of the main window. It consists of **Data Viewer** tab and the **Result** tab. By default, the editor panel is empty.

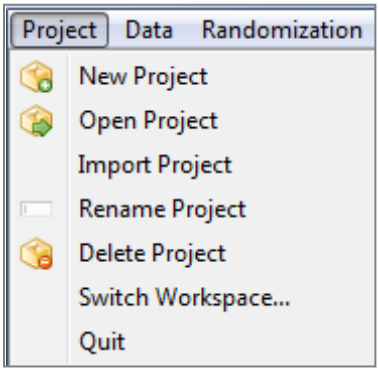


Menu Bar

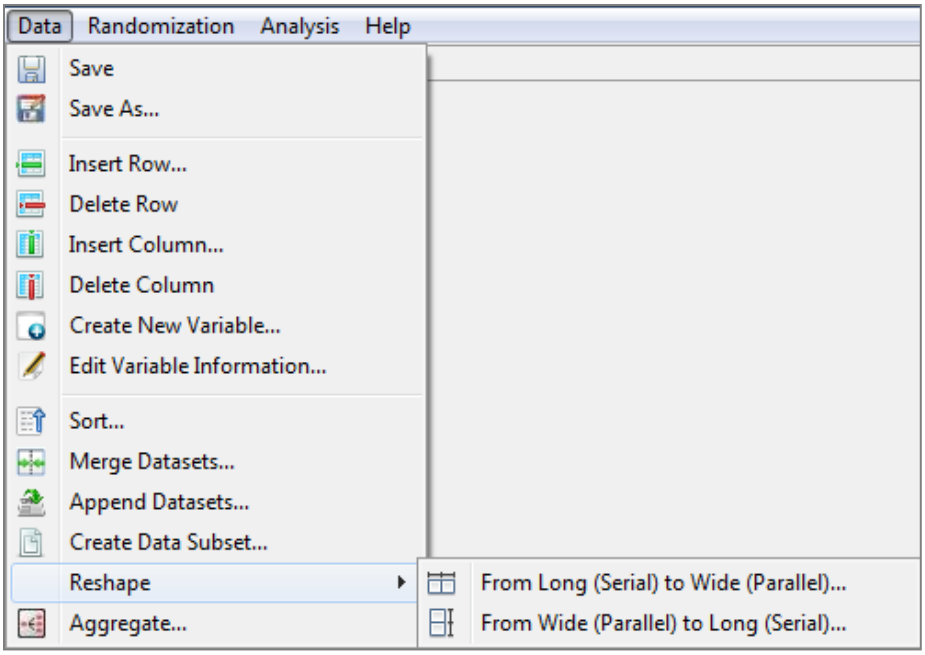
PBTools has five items in the menu bar as shown below:



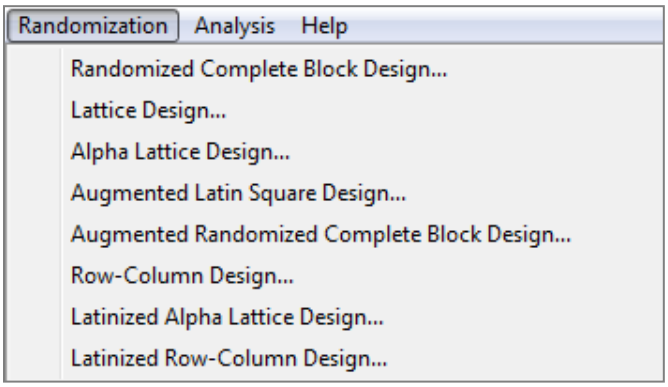
Project Menu contains functions for creating and managing projects. This menu can also be used to quit or terminate PBTools session. It contains the following submenu items:



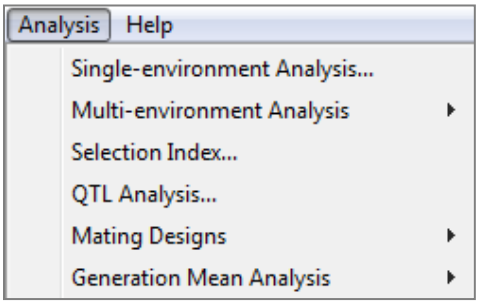
Data Menu contains functions for reading, managing and manipulating datasets. Items under this menu can be used only when a data in CSV format is displayed in the active tab of the editor panel. It contains the following submenu items:



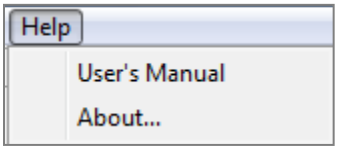
Randomization Menu contains functions for generating random assignment of factor levels for commonly used experimental designs in plant breeding. It contains the following submenu items:



Analysis Menu contains functions to perform statistical analysis. Items under this menu can be used only when a data in CSV format is displayed in the active tab of the editor panel. It contains the following submenu items:

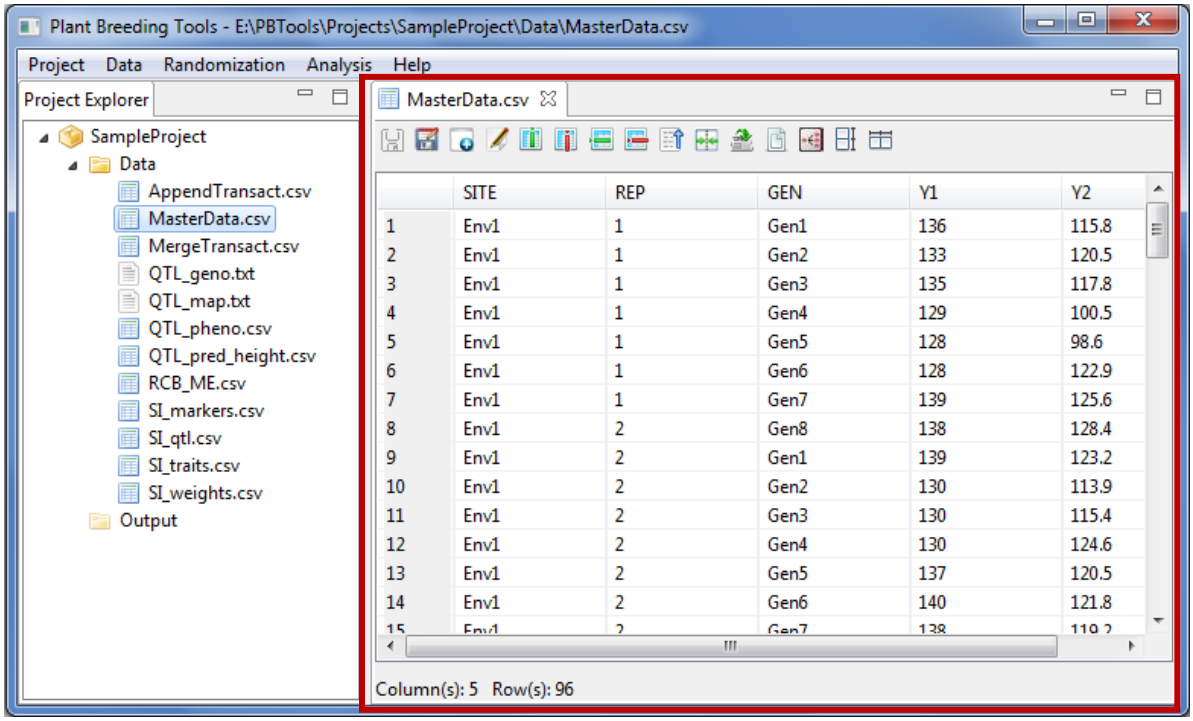


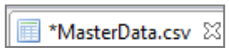
Help Menu is used to access PBTools user's manual and some information about the software.



Data Viewer

Data Viewer is displayed in the editor panel when a CSV file is double-clicked from the Project Explorer tree. It is used for viewing data file in spreadsheet format, editing data values and performing data manipulation. Several Data Viewers can be opened simultaneously inside the Editor Panel but only one is considered to be the active tab. Each Data Viewer contains toolbars which can be used for managing the data. The toolbars have the same functionalities as the options under the **Data** menu.

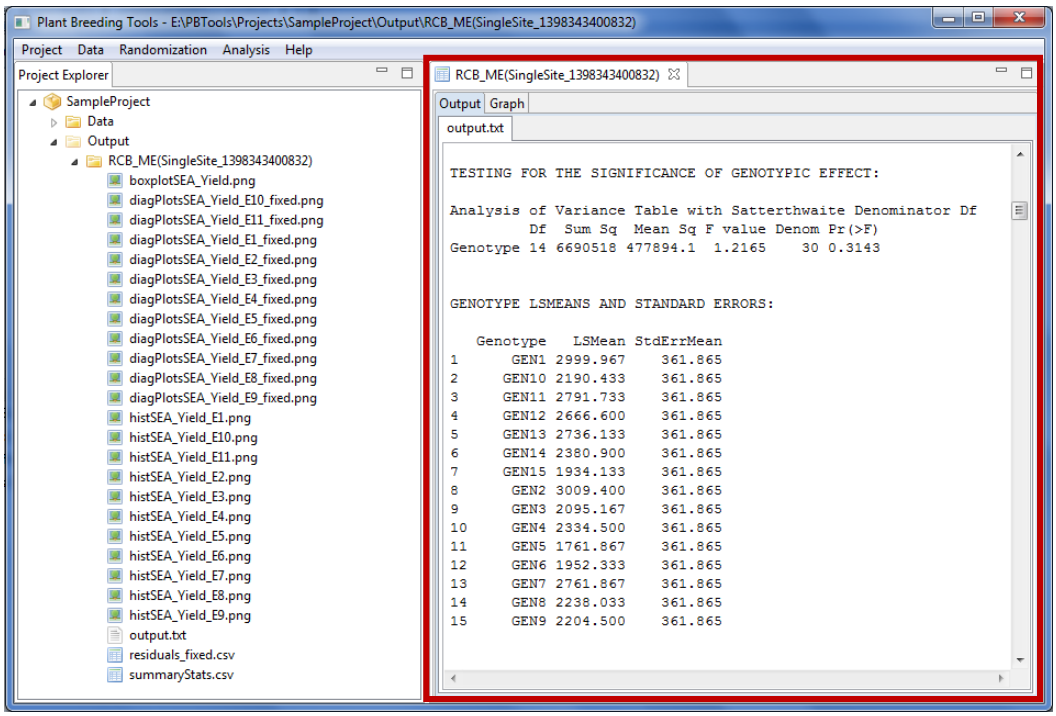


The name of the tab indicates the name of the dataset. When an asterisk appears in the tab name 

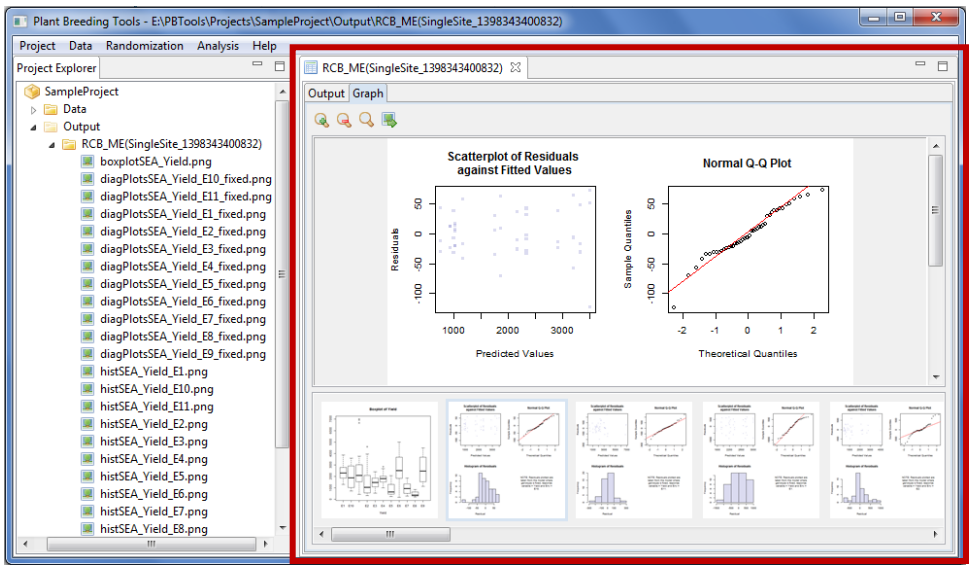
, it indicates that there has been modification made in the data. If data manipulation or analysis is performed on the data on the active tab, PBTools captures all information in the active tab (even unsaved modifications) and submit it for manipulation or analysis. If a Data Viewer with unsaved changes is closed, the user will be prompted to save the changes.

Results Viewer

The Results Viewer is displayed in the editor panel when an analysis is performed or by double-clicking a results folder inside the *Output* folder. Depending on the contents of the results folder, the Results Viewer may have an *Output* page and/or *Graph* page.



All outputs in *txt* format are displayed in the *Output* page while all graphical outputs in *png* format are displayed in the *Graph* page. The *Graph* page has toolbars that can be used to minimize, maximize, and view the graph to normal size and to export graph to external sources. To view other graphs in the page, the left and right arrow keys can be used.

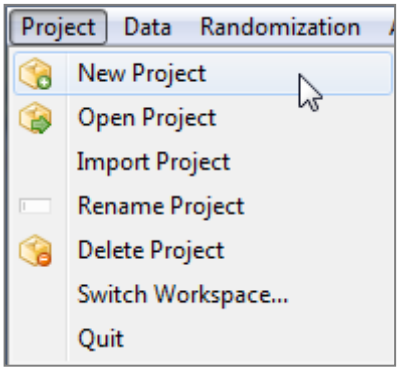


3. Project and File Management

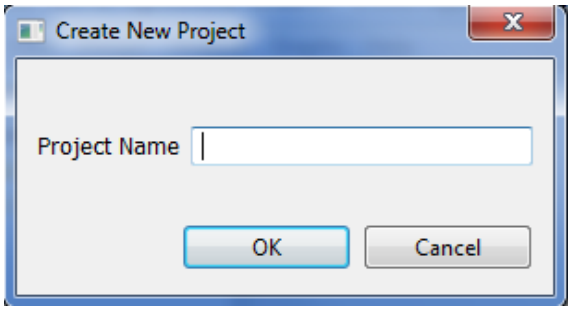
Creating New Project

The user can create several projects in PBTools. However, only one project can be considered as the active project and is visible in the Project Explorer. The steps for creating new project are listed below.

- Select **Project | New Project** from the main menu.



- The **Create New Project** dialog box will appear. In the **Project Name** field, specify a name of the new project.

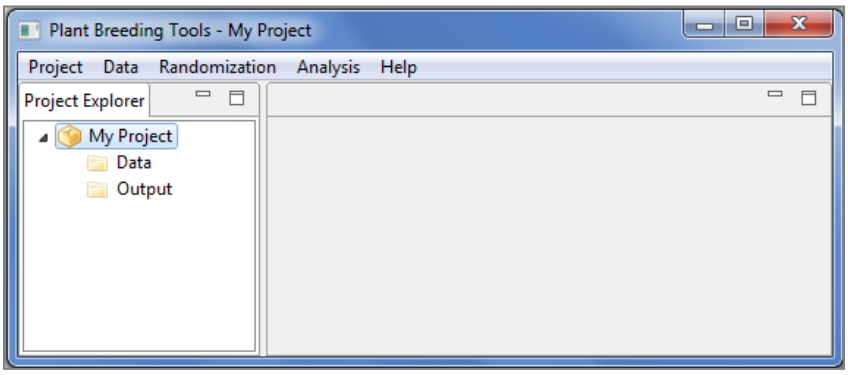


In naming a project, the following rules apply:

- The name must start with a letter. The remaining characters can be a letter, digit, period, underscore, blank or dash.
- The last character of the name should not be a period, underscore, blank or dash.
- The length of the name should not be less than 4 characters.
- The name must be different from the existing project names.
- The name is not case sensitive.

- For the example, type *My Project*. Click **OK**.

- The new project named *My Project*, will now be displayed in the **Project Explorer** panel. By default, two empty folders under the project name will be automatically created, the *Data* and *Output* folders.



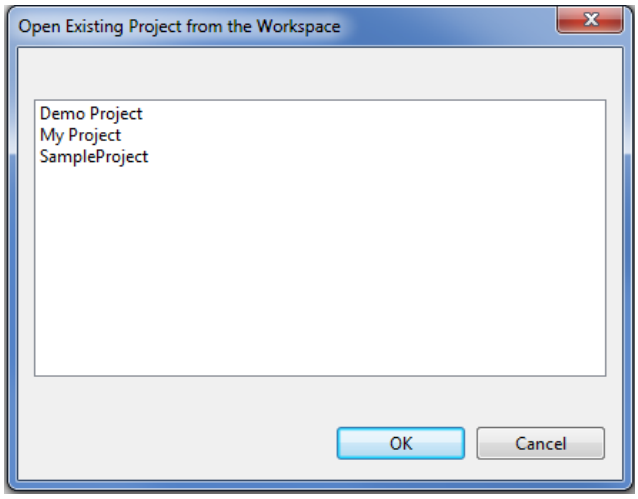
- NOTE: As part of the example, create another project named *Demo Project*.

Opening Existing Project

The steps for opening existing project are listed below.

- Select **Project | Open Project** from the main menu.

- The **Open Existing Project from the Workspace** dialog box will appear. The list of all existing projects in the workspace can be seen.

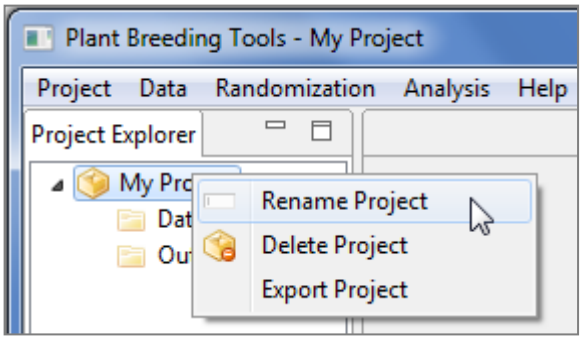


- For the example, select the project named *My Project*. Click **OK**. *My Project* should now be the active project in the Project Explorer.

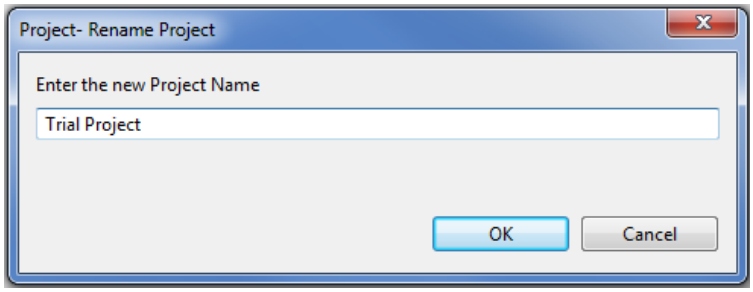
Renaming Project

The steps for renaming the active project are listed below.

- Right-click on the active project, say *My Project* then select **Rename Project** or select **Project | Rename Project** from the main menu.



- Enter a new project name, say *Trial Project*. Click **OK**.

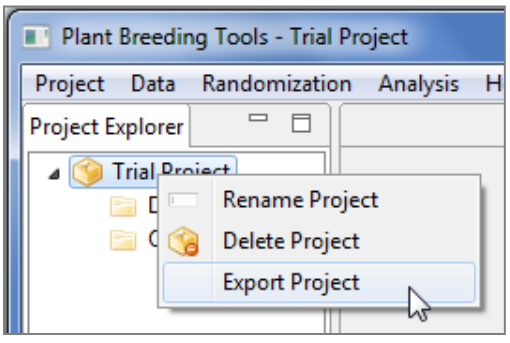


- The project should be displayed in the Project Explorer with the new name.

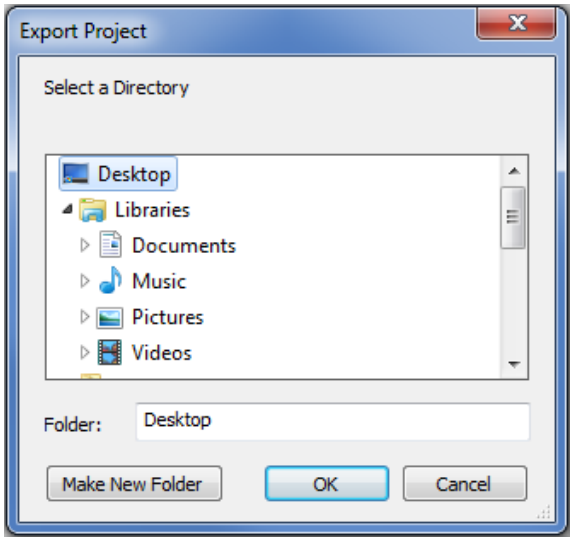
Exporting Project

If the user wants to save the active project to a different directory, the export project feature can be used. The steps are listed below.

- Right click on the active project, say *Trial Project* then select **Export Project**.



- Select the directory where the active project will be saved. For the example, choose *Desktop*. Click **OK**.

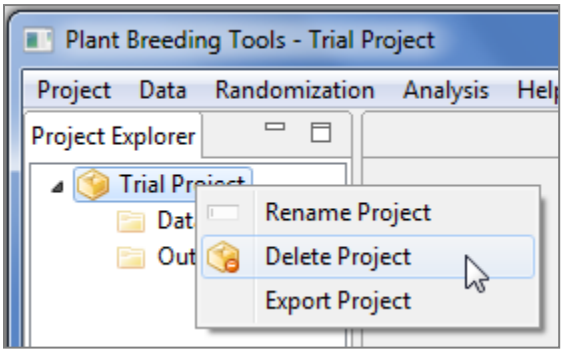


- A *Trial Project* folder is now on the desktop.

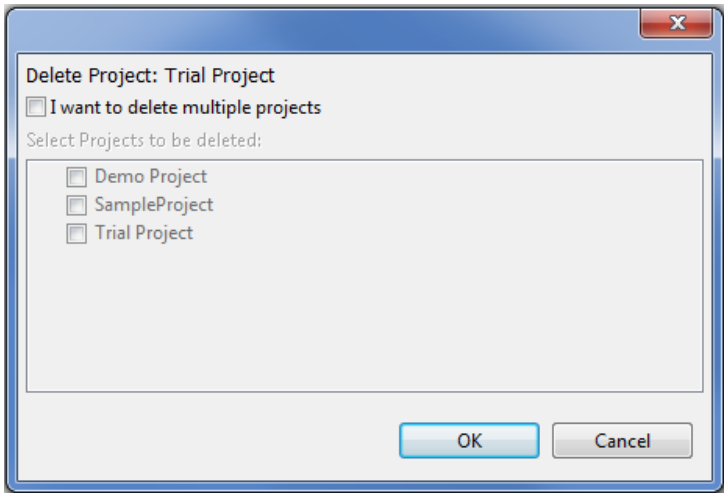
Deleting Project

The steps for deleting an existing project inside the PBTools workspace are listed below.

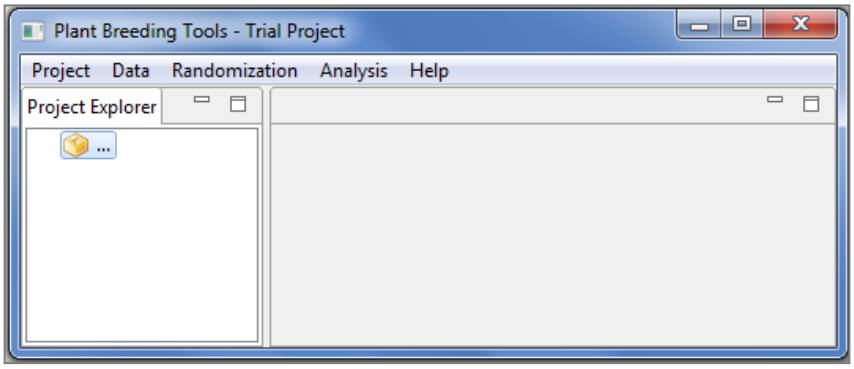
- Right-click on the active project, say *Trial Project* then select **Delete Project** or select **Project | Delete Project** from the main menu.



- Another dialog will be displayed. If the user wants to delete multiple projects, click on the check box labeled 'I want to delete multiple projects' then select from the list the projects to be deleted. For the example, delete only *Trial Project* by clicking **OK**.



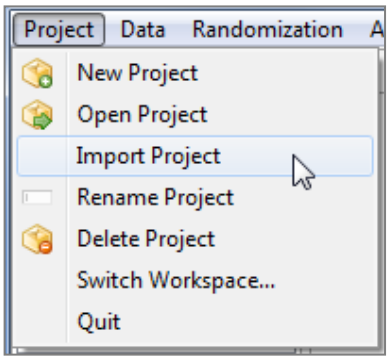
- Since *Trial Project* is now deleted from the PBTools workspace, there is no active project displayed in the Project Explorer as shown below.



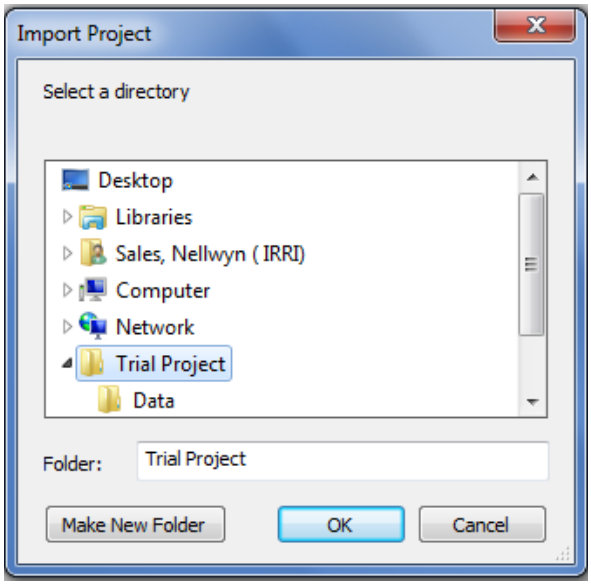
Importing Project

Projects, as long as it is a valid PBTools project, can be saved inside the PBTools workspace by using the import project feature. The steps are listed below.

- Click **Project | Import Project** from the main menu.



- Select the project folder that will be imported. For the example, select *Trial Project* folder on the desktop. Click **OK**.



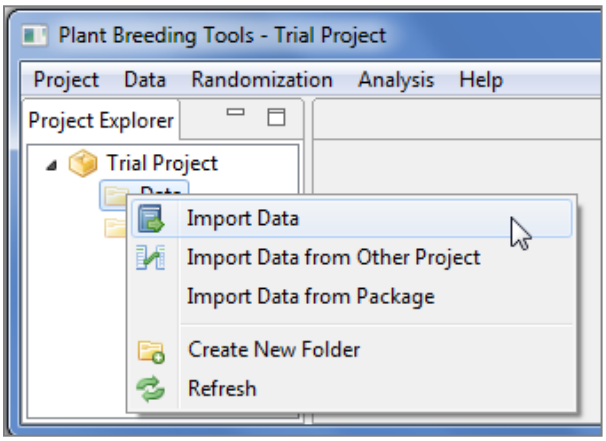
- *Trial Project* is now the active project in the Project Explorer.

Importing Data from External Source

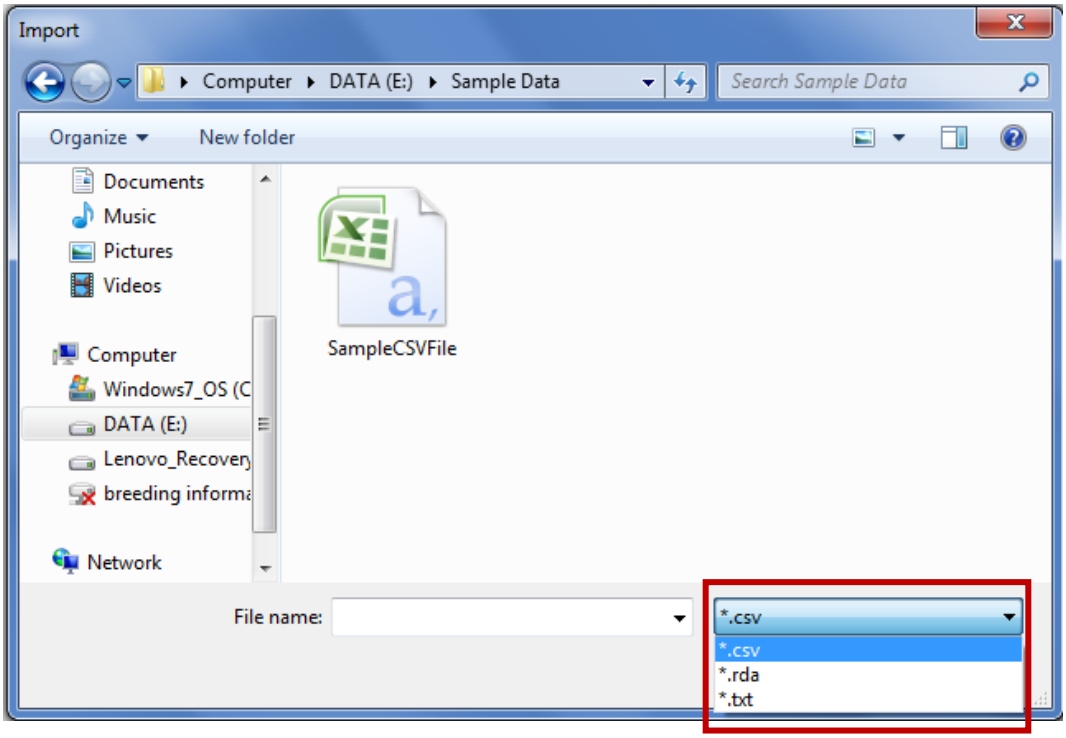
There are three file formats that PBTools accepts to import from external source namely: R datasets; text files with tab, space, comma and semi-colon delimiter; and comma separated value files.

The general steps for importing data from external source are listed below:

- On the **Project Explorer** panel, right-click the *Data* folder of the project named *Trial Project*. Choose **Import Data**.

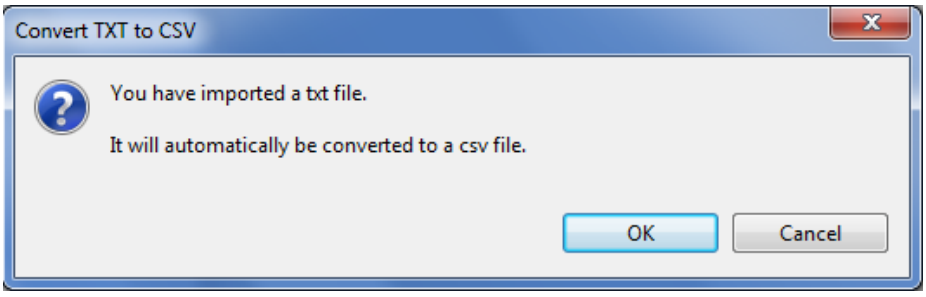


- The **Import** dialog box will appear. Choose the appropriate file type to be imported in the File of Types drop-down box. The default file type is ***.csv**. Go to the directory where the file to be imported is located. Select the file or type the file name on the **File name** text box.

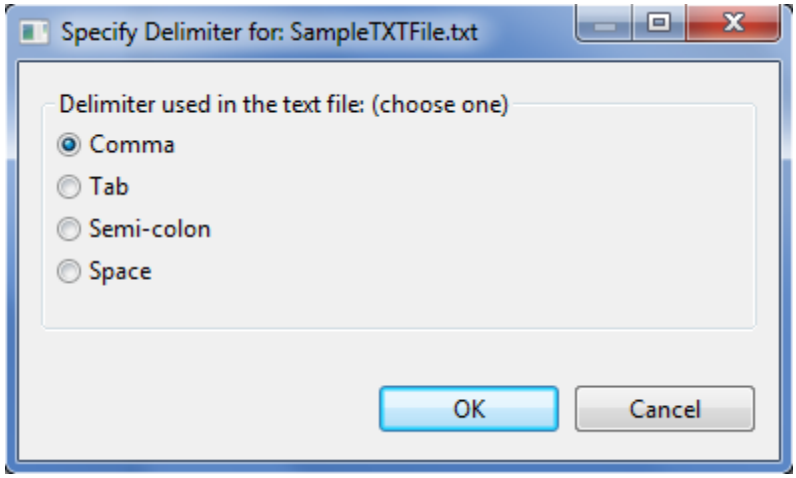


- Click the **Open** button.

If the ***.txt** or the ***.rda** file type is chosen, a message dialog box will appear that will prompt the user that the text or r data files will be automatically converted to a csv file. Click **OK**.



Further, if the ***.txt** file type is chosen, the **Specify Delimiter** dialog box will appear. Choose the appropriate delimiter of the text file to be open then click **OK**.

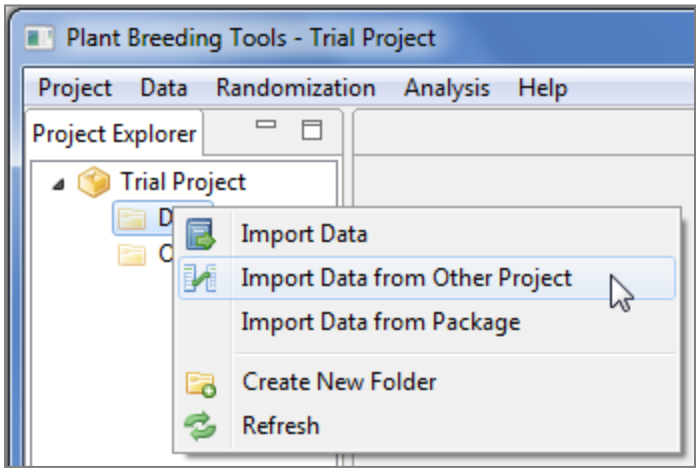


- The newly imported data will now be displayed inside the *Data* folder of the active project in the Project Explorer. To view the data in a spreadsheet format, double-click the icon of the data file.

Importing Data from Other Project

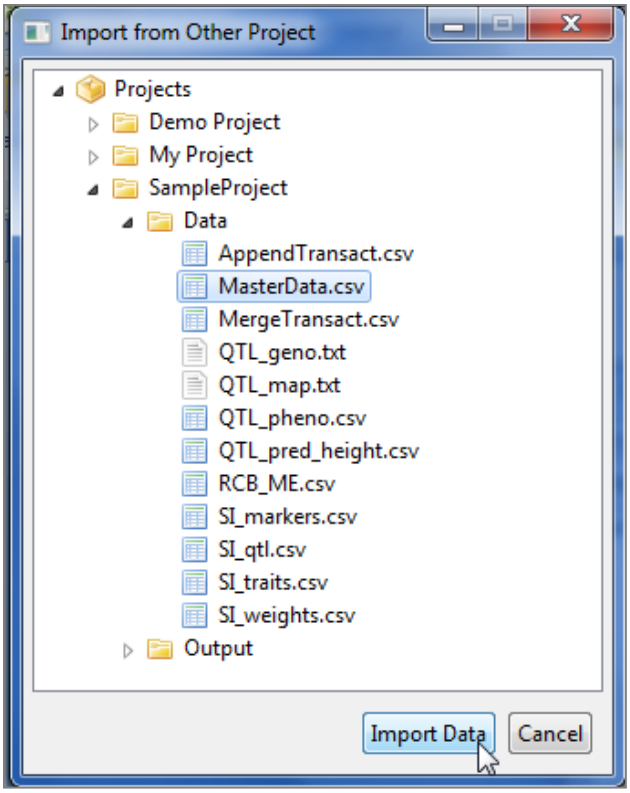
The steps for importing data from other project are listed below:

- On the **Project Explorer** panel, right-click the *Data* folder of the project named *Trial Project*. Choose **Import Data from Other Project**.



- The **Import from Other Project** dialog box will appear. This dialog box contains all existing PBTools projects in the workspace. The user can select one or several project folders, one or several folders within projects or one or several data files. Choose the data files to be imported.

- For the example, choose MasterData.csv from Projects | SampleProject | Data.

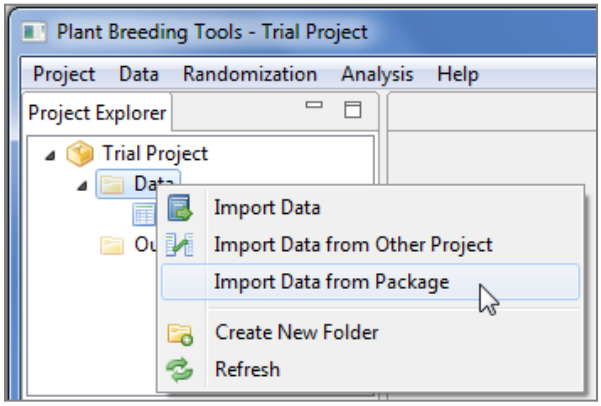


- Click the **Import Data** button. The newly imported data will now be displayed inside the *Data* folder of the active project in the Project Explorer. To view the data in a spreadsheet format, double-click the icon of the data file.

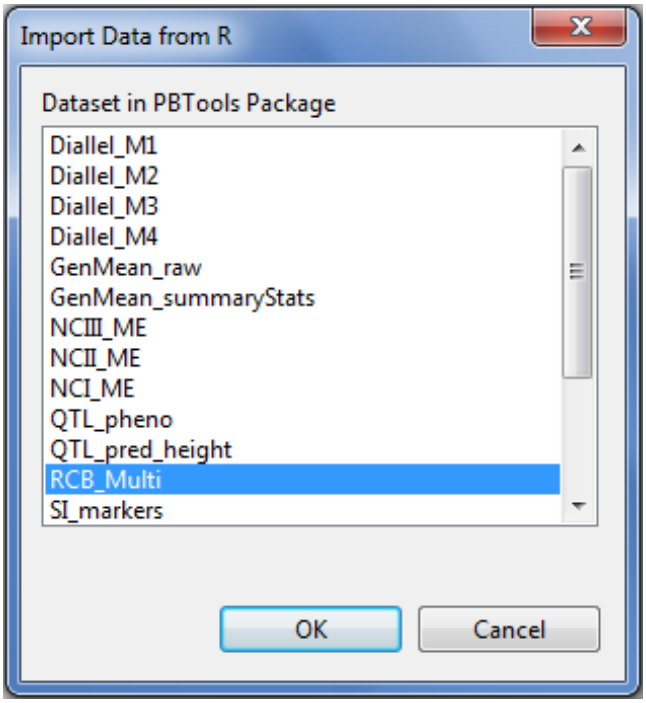
Importing Data from Package

Datasets in the R package named *PBTools* can be imported. The steps are listed below:

- On the **Project Explorer** panel, right-click the *Data* folder of the project named *Trial Project*. Choose **Import Data from Package**.



- The **Import Data from R** dialog box will appear. Choose one dataset to be imported. Only one data set can be imported at a time. For the example, choose *RCB_Multi*.



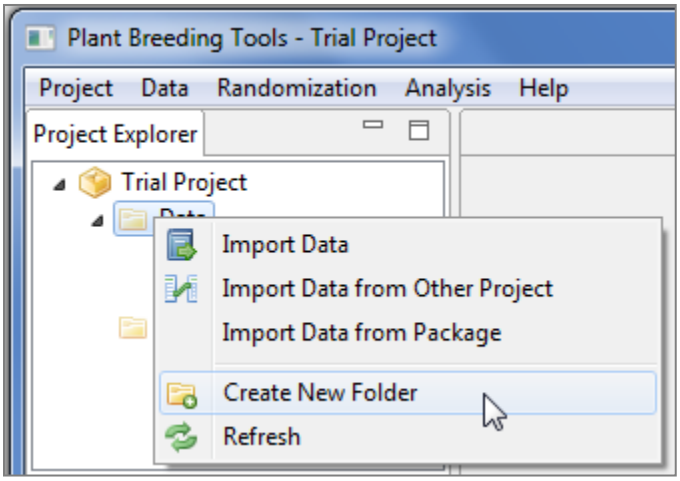
- Click **OK**. A dialog will appear confirming the data is successfully imported. The newly imported data will now be displayed inside the *Data* folder of the active project in the Project Explorer. To view the data in a spreadsheet format, double- click the icon of the data file.

Creating New Folder

To organize the data files imported in a project or results of an analysis, user may want to create a sub-folder inside the *Data* folder or inside the *Output* folder.

The steps in creating a sub folder are listed below:

- On the **Project Explorer** panel, right-click the folder where you want to create the sub-folder. For the example, right-click the *Data* folder of the project named *Trial Project*. Choose **Create New Folder**.

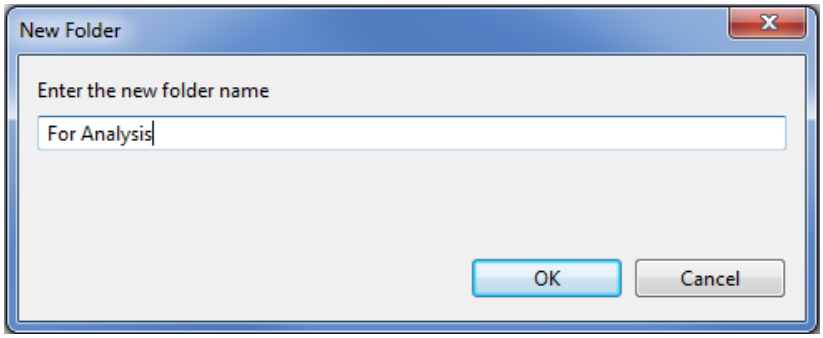


- The **New Folder** dialog box will appear**.** In the **Enter the new folder name** field, user can specify the name for the new folder. If the sub folder will be created within the *Data* folder, the default folder name is *Data*. If the sub folder will be created within the *Output* folder, the default folder name is *Output*.

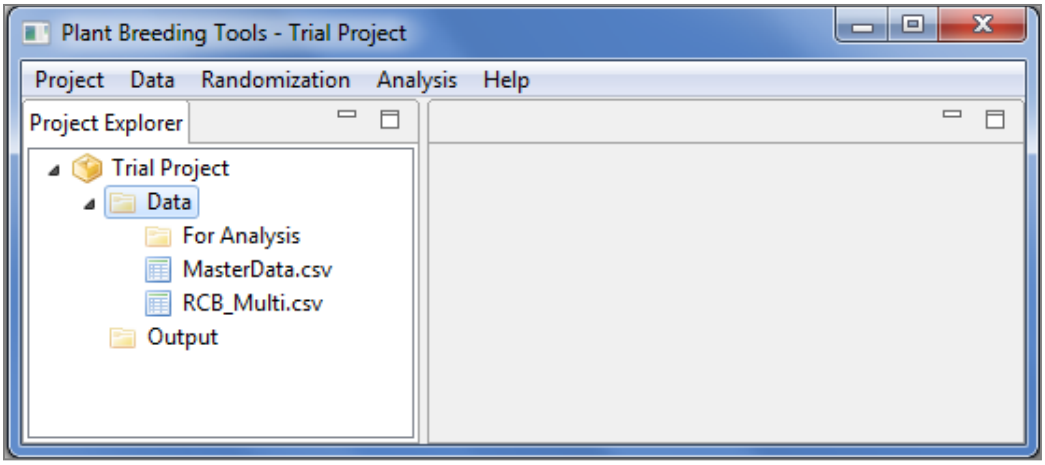
In naming a folder, the following rules apply:

- The name must start with a letter. The remaining characters can be any letter, any digit, a period, underscore, blank or dash.
- The last character of the name should not be a period, underscore, blank or dash.
- The length of the name should not be less than 4 characters.
- The name must be different from the existing folder names inside a parent folder.
- The name is not case sensitive.

- For the example, type *For Analysis* as the new folder name.



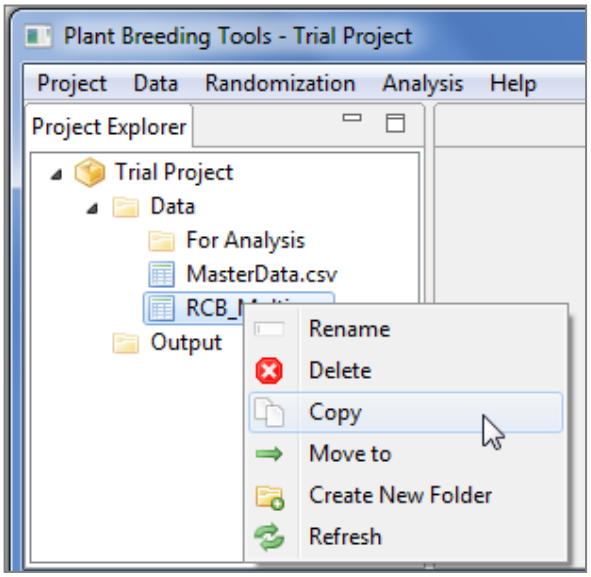
- A new sub-folder inside *Data* folder is now created and can be seen in the Project Explorer.



Copying File / Folder

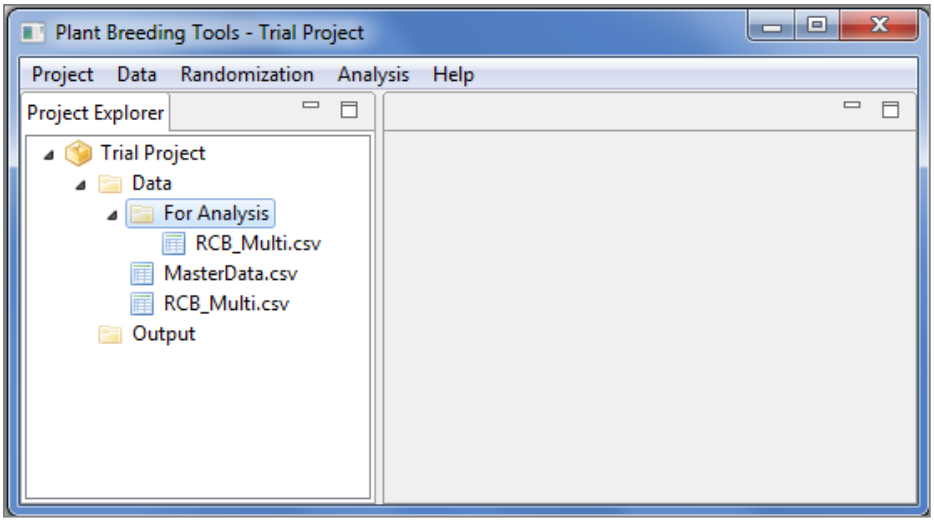
The steps to copy data files to other location are listed below:

- On the **Project Explorer** panel, right-click the file you want to copy. For the example, right-click *RCB_Multi.csv* inside the *Data* folder of the project named *Trial Project*. Choose **Copy**.



- Go to the desired destination folder, which can be within or outside PBTools workspace, then right-click then choose **Paste.** For the example, right-click on the *For Analysis* folder then choose **Paste.**

- The file is now copied inside the destination folder.

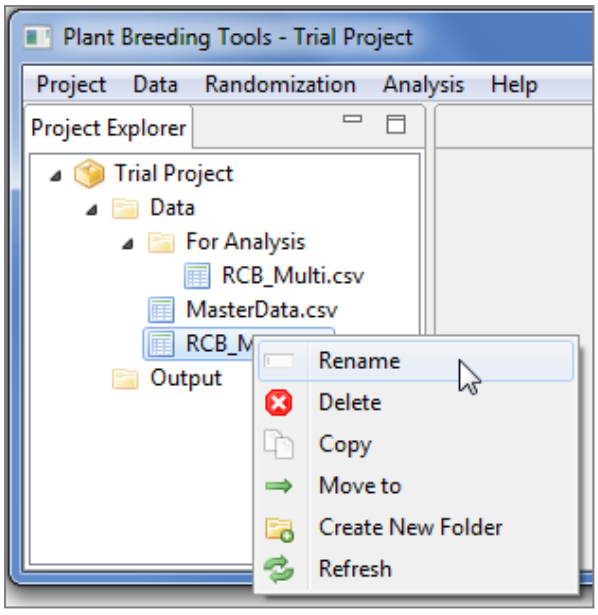


These steps also apply when copying sub-folders within the *Data* and *Output* folders.

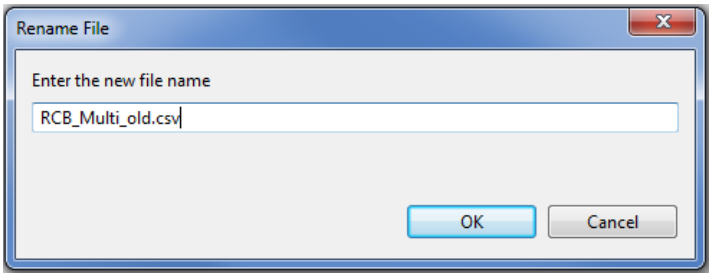
Renaming File / Folder

The steps to rename files are listed below:

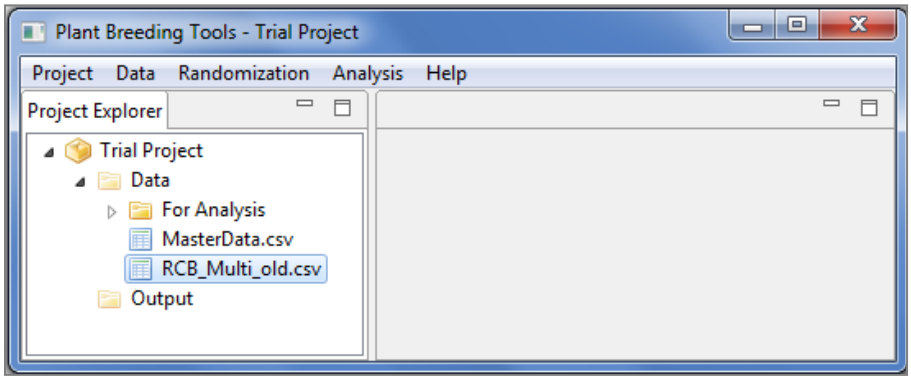
- On the **Project Explorer** panel, right-click the file you want to rename. For the example, right-click *RCB_Multi.csv* inside the *Data* folder of the project named *Trial Project*. Choose **Rename**.



- The **Rename File** dialog box will appear. In the **Enter the new file name** field, user can specify the new filename. For the example, set the new filename to *RCB_Multi_old.csv*.



- Click **OK**. The renamed data file should appear in the Project Explorer panel.

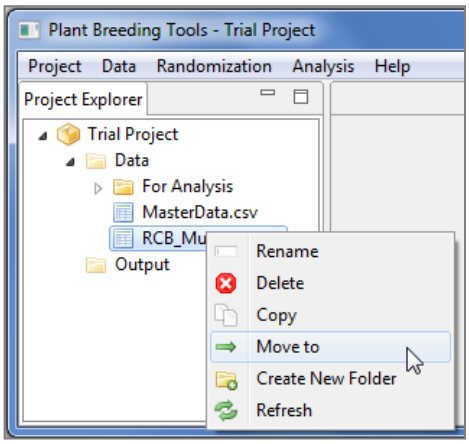


These steps also apply when renaming sub-folders within the *Data* and *Output* folders. Note that the *Data* and *Output* folders are required folders for a PBTools project so it cannot be renamed.

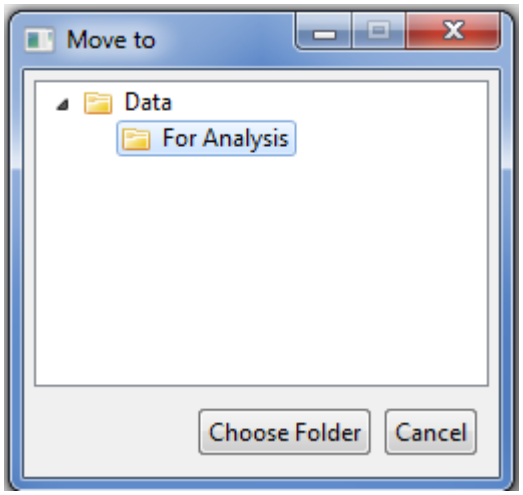
Moving File / Folder

The steps to move data files to other location in the PBTools workspace are listed below:

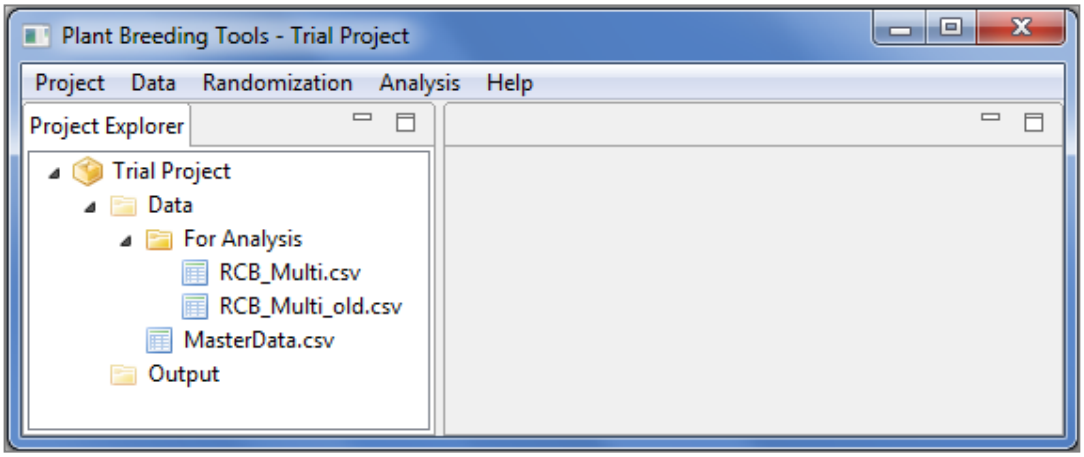
- On the **Project Explorer** panel, right-click the file you want to rename. For the example, right-click *RCB_Multi_old.csv* inside the *Data* folder of the project named *Trial Project*. Choose **Move to**.



- The **Move to** dialog box will appear. Select the destination folder. For the example, choose the *For Analysis* folder inside the *Data* folder.



- Click **Choose Folder**. The file is now moved to the destination folder.

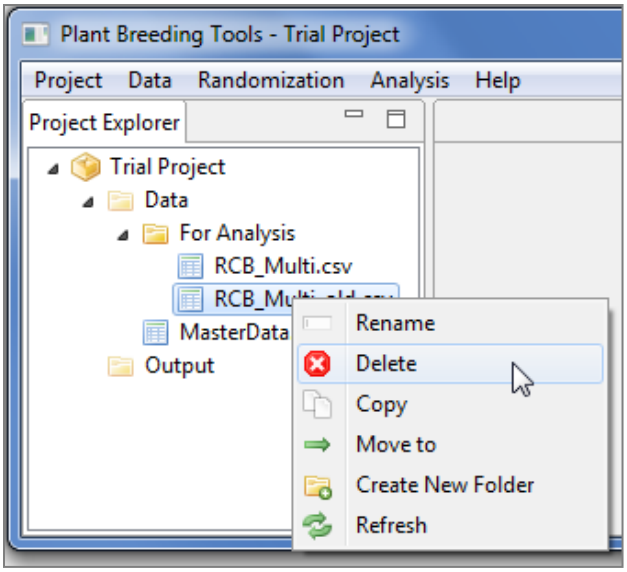


These steps also apply when moving sub-folders within the *Data* and *Output* folders.

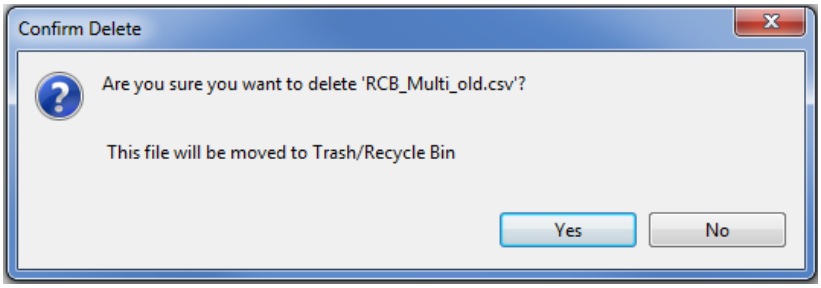
Deleting File / Folder

The steps to delete data files are listed below:

- On the **Project Explorer** panel, right-click the file you want to delete. For the example, right-click *RCB_Multi_old.csv* inside *For Analysis* folder in the *Data* folder of the project named *Trial Project*. Choose **Delete**.



- The **Confirm Delete** dialog box will appear.



- Click **Yes** to delete.

These steps also apply when deleting sub-folders inside the *Data* and *Output* folders.

4. Data Handling and Manipulation

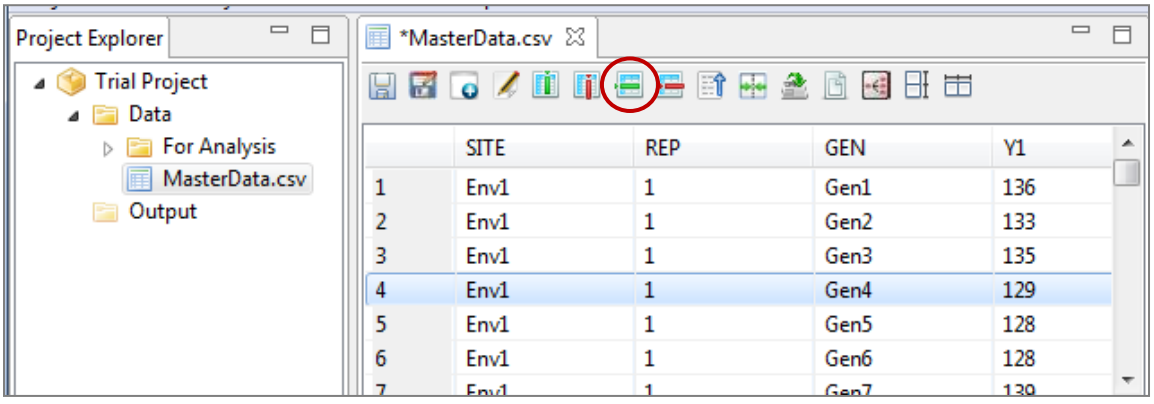
Representing Missing Value(s)

In PBTools, missing values can be represented by using any of the following: *NA*, space, period or blank. Using other symbol, say asterisk (*), to represent missing values will affect how PBTools identifies the attributes of the columns of the dataset. A column with numeric values but contains asterisks will be classified as a factor column. This misclassification may cause problems later when the data is subjected to manipulation or analysis.

Inserting Row(s)

The steps to insert row(s) are listed below:

- On the Project Explorer, locate the *MasterData.csv* from the *Data* folder of the project named *Trial Project*. Double-click the file to view it in the Data Viewer.
- Select any row(s) or any cell in the row(s) where the user wants to insert new row(s) above it. To select several cells/rows, click any cell/row, then hold the Ctrl key and click on another cell/row. The number of selected rows is the number of row that will be inserted.
- For the example, select the 4th row.
- Click **Data** | **Insert Row…** from the menu bar or click the Insert Row icon  in the **Data Viewer** toolbar.



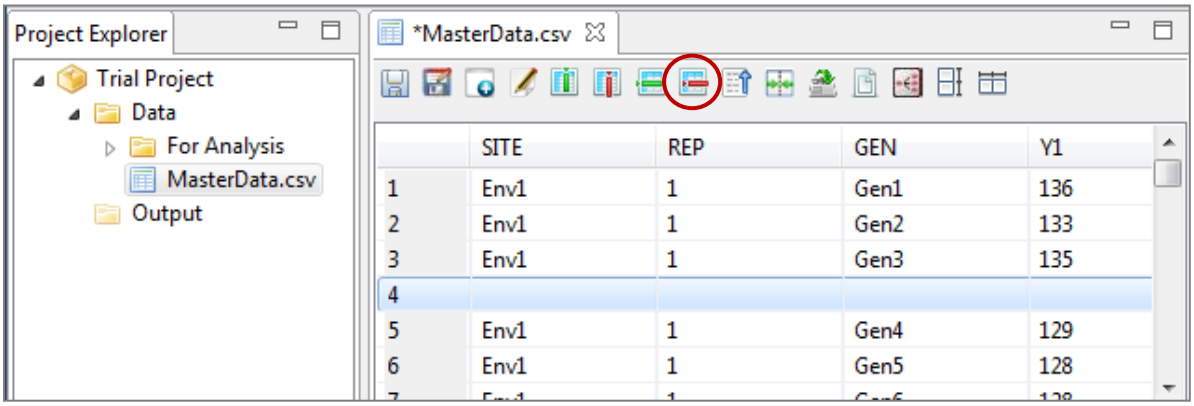
- The data now contains the newly inserted row.

If there is no row selected before executing the Insert Row function, the user will be prompted if the user wants to insert a row after the last row of the data or not.

Deleting Row(s)

The steps to delete row(s) are listed below:

- For the example, use the modified and unsaved *MasterData.csv* from the previous section that is displayed in the Data Viewer.
- Select any row(s) or any cell in the row(s) that the user wants to delete. To select several cells/rows, click any cell/row, then hold the Ctrl key and click on another cell/row. The number of selected rows is the number of row that will be deleted.
- For the example, select the 4th row.
- Click **Data | Delete Row** from the menu bar or click the Delete Row icon  in the Data Viewer tool bar.



- The Delete Row dialog box will appear. Confirm that row deletion is desired by clicking **Yes**.
- The changes made should be reflected in the Data Viewer.
- For the example, close the modified *MasterData.csv* and do not save the changes.

Inserting Column

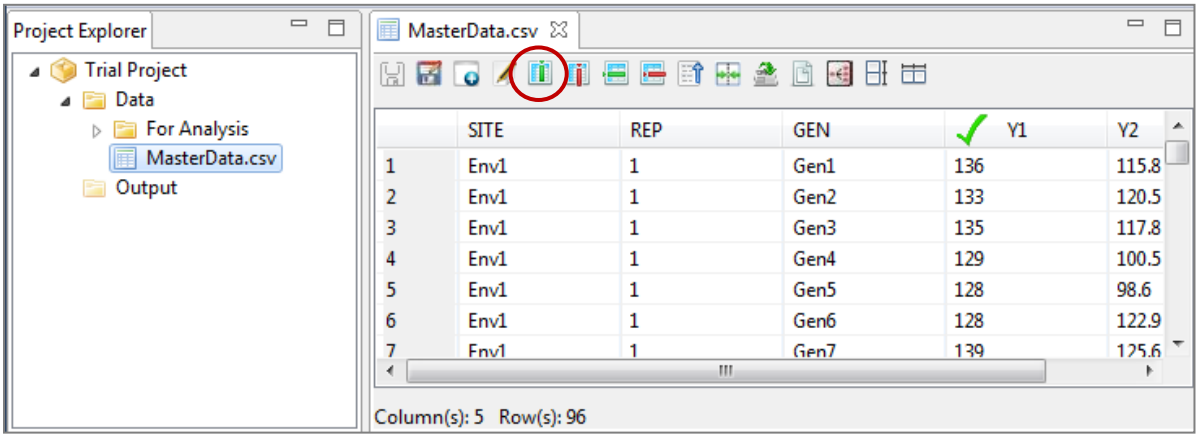
The steps to insert column are listed below:

- On the Project Explorer, locate the *MasterData.csv* from the *Data* folder of the project named *Trial Project*. Double-click the file to view it in the Data Viewer.

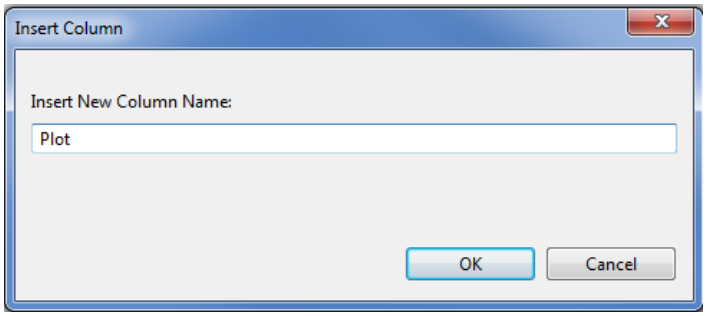
- Select a column where the user wants a new column to be inserted before it by clicking on the column name. A check icon will appear on the column header.

- For the example, select the column named *Y1*.

- Click **Data | Insert Column…** from the menu bar or click the Insert Column icon  in the Data Viewer tool bar.



- The **Insert Column** dialog box will appear. In the **Insert New Column Name** field, user can specify the name of the new column. The default name is *NewColumn*. For the example, type *Plot*.



In naming a column, the following rules apply:

- The name must start with a letter while the remaining characters can be any letter, any digit, a period or underscore.
- The name must be different from the existing column names in the data.
- The name is case sensitive.

- Click **OK**.

- The data now contains the newly inserted column named *Plot*.

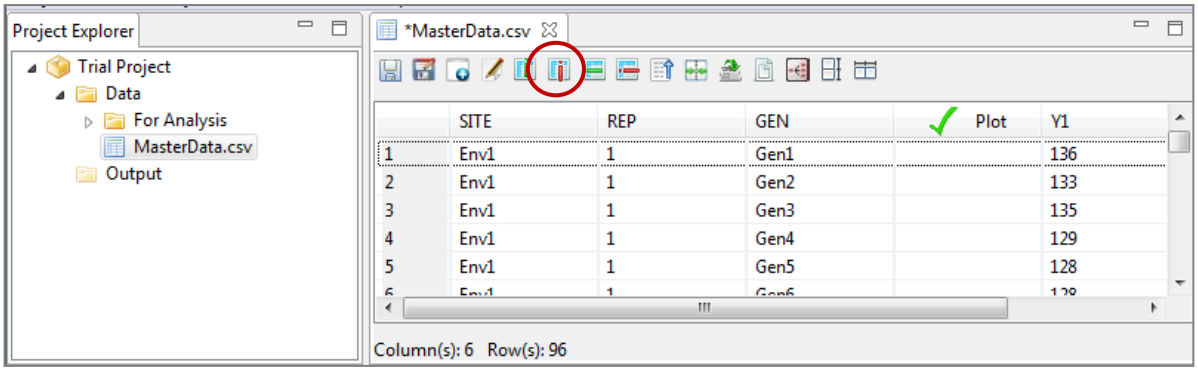
Deleting Column(s)

The steps to delete column(s) are listed below:

- For the example, use the modified and unsaved *MasterData.csv* from the previous section that is displayed in the Data Viewer.

- Select the column or one of the columns to be deleted by clicking on the column name. A check icon will appear on the column header. For the example, select the column named *Plot*.

- Click **Data | Delete Column** from the menu bar or click on the Insert Column icon  in the Data Viewer tool bar.



- A dialog box will appear. To confirm the deletion of the column is desired, click **Yes**.

If the user wants to delete other column(s), select the check box labeled "I want to delete multiple columns" then select the column(s) on the text box that will appear.

- The changes made should be reflected in the Data Viewer. For the example, close the modified *MasterData.csv* and do not save the changes.

Creating a New Variable

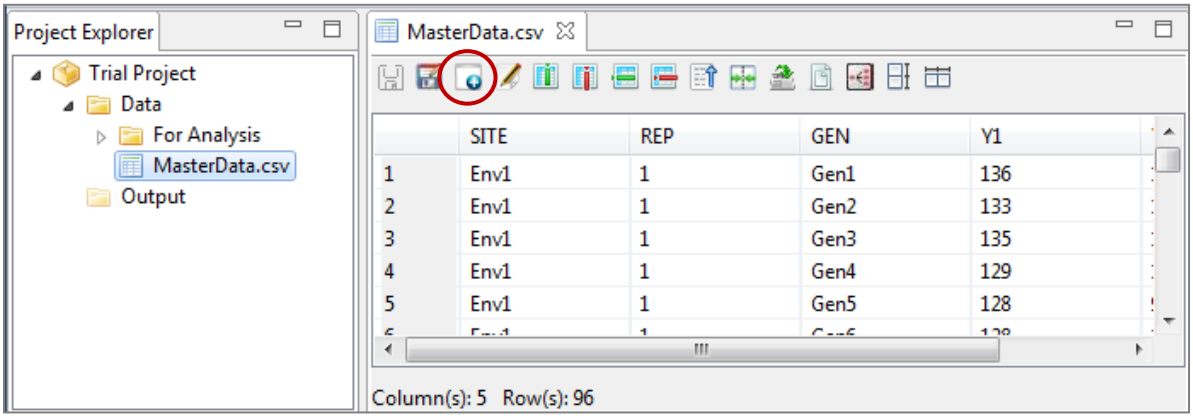
The user can create new variables in the active data set by transforming existing variables or collapsing the values of two or more existing variables.

For transformations, there are six available functions, namely: logarithm, natural logarithm, square root, power, exponential and standardized. For logarithm, natural logarithm, square root and power transformation, the variables to be transformed should not contain negative values. For logarithm and natural logarithm transformation, values of the variable to be transformed will be incremented by 1 if the variable has values equal to 0. For square root and power transformation, values of the variable will be incremented by 0.5 if the variable has values equal to 0.

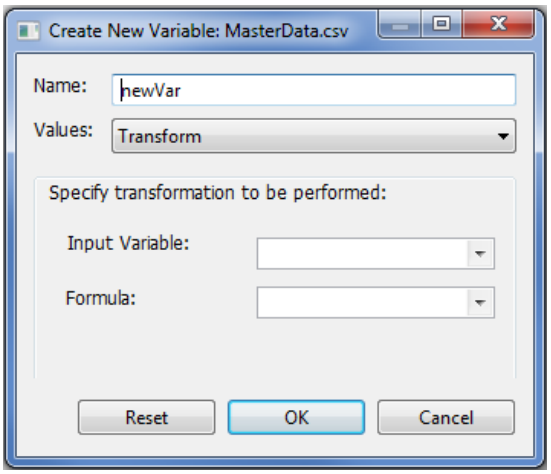
The steps to create new variables are listed below:

- On the Project Explorer, locate the *MasterData.csv* from the *Data* folder of the project named *Trial Project*. Double-click the file to view it in the Data Viewer.

- Click **Data | Create New Variable…** from the menu bar or click on the Create New Variable icon  in the Data Viewer toolbar.



- The **Create New Variable** dialog box will appear.



- Specify the required fields and appropriate options.

Name

This is the name of the new variable to be created. The default variable name is *newVar*. In naming the column, the following rules apply:

- The name must start with a letter while the remaining characters can be any letter, any digit, a period or underscore.
- The name must be different from the existing column names in the data.
- The name is case sensitive.

Values

There are two options to determine the values of the new variable. The user can either transform any existing numeric variables or concatenate the values of any variables in the data set. The default is *Transform*.

If transformation option is selected, the user needs to specify the following:

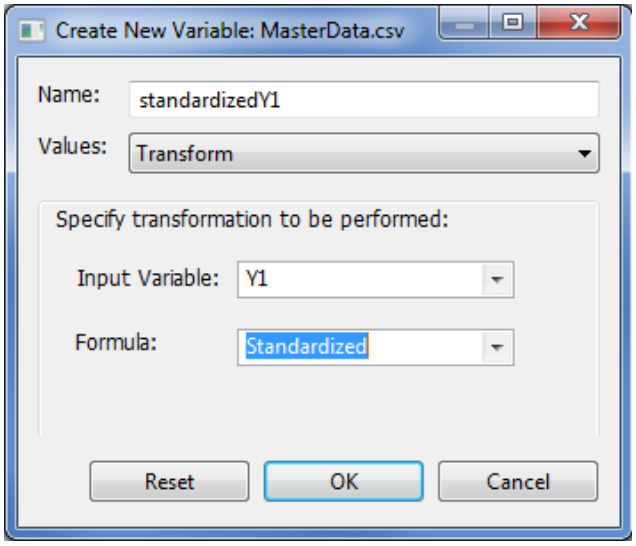
Input Variable

This pertains to the variable where the transformation will be performed. All numeric variables of the active data will be displayed in the drop-down box. Select one variable where the transformation will be performed.

Formula

This pertains to the formula that will be used in the transformation. The available options are logarithm, natural logarithm, square root, power, exponential and standardized.

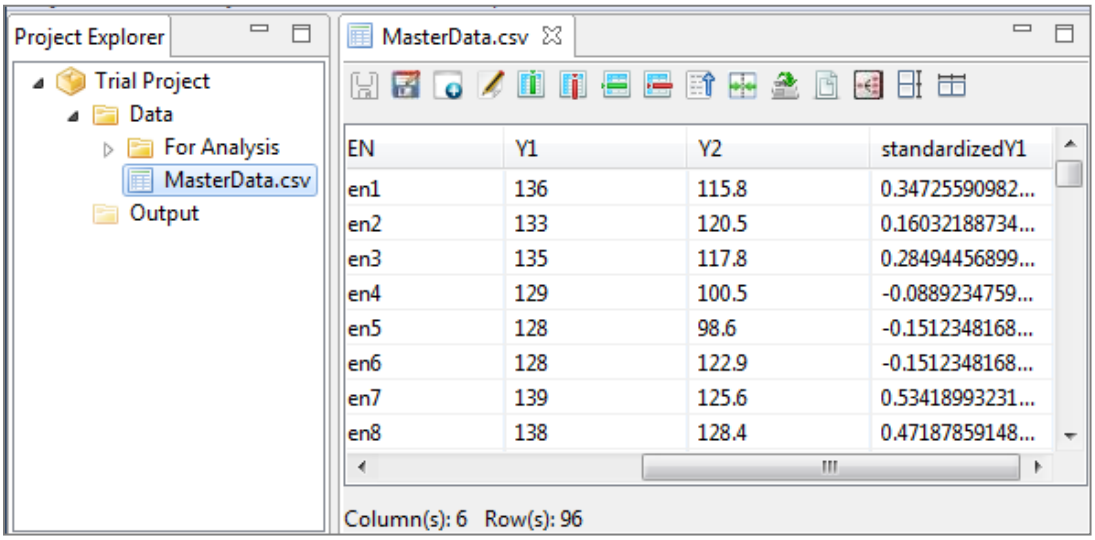
For the example, suppose we want to create a new variable named *standardizedY1* by standardizing the values of *Y1*. The completed dialog box should appear as shown below:



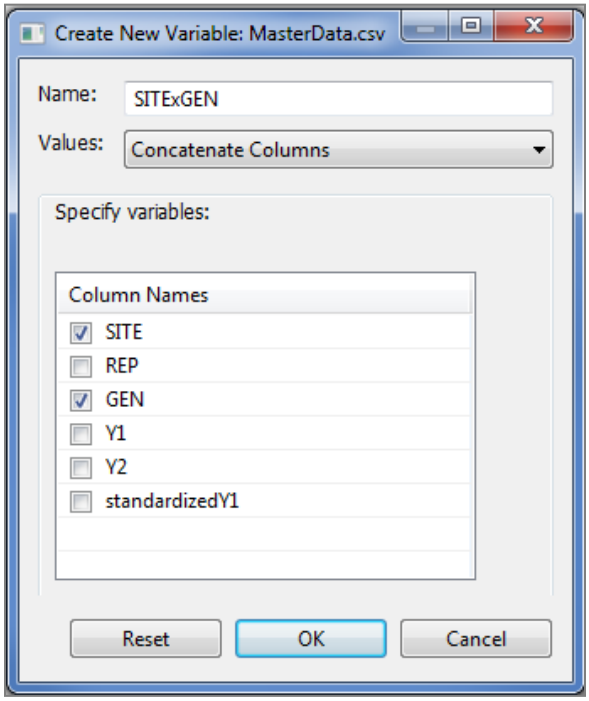
If concatenation of columns is selected, a table containing all the columns of the active data will appear at the lower part of the dialog box. The user needs to specify the columns to concatenate by ticking at least two column names.

- Click **OK**.

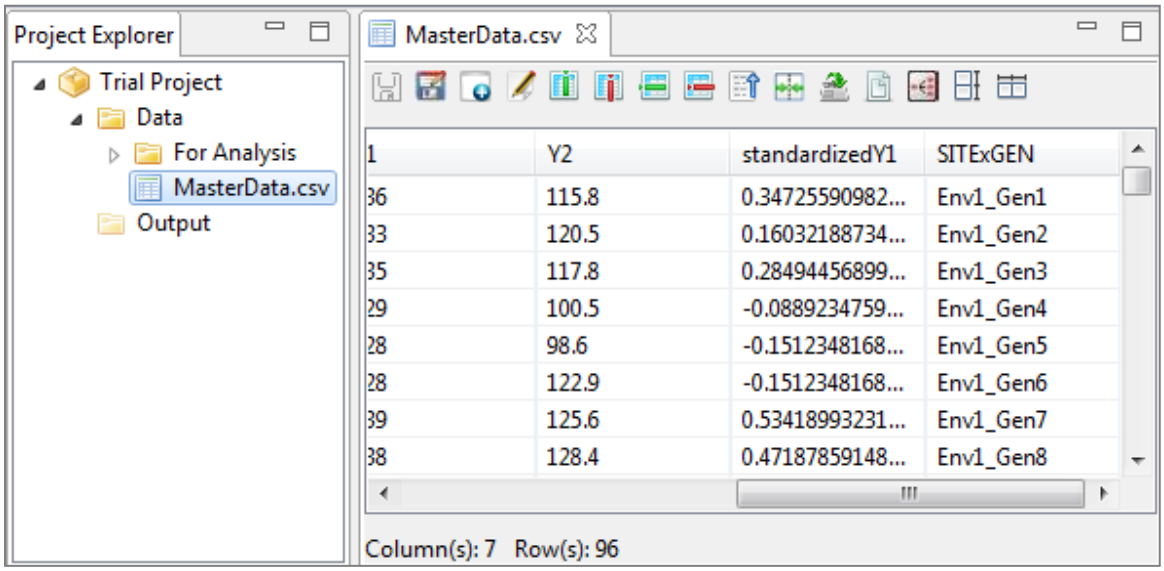
- The changes are saved to the original dataset and the file is displayed in the Data Viewer.



Suppose the user also wants to create a new variable named *SITExGEN* by concatenating the values of *SITE* and *GEN*, click on the Create New Variable icon. The completed dialog box should appear as shown below:



- Click the **OK**. The changes are saved to the original dataset and the file is displayed in the Data Viewer.

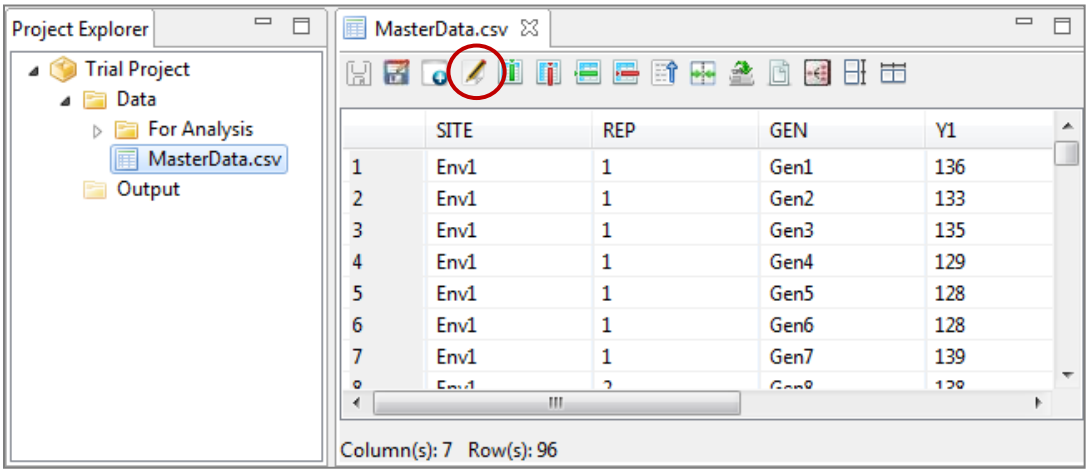


Editing Variable Information

The steps to edit the variable information are listed below:

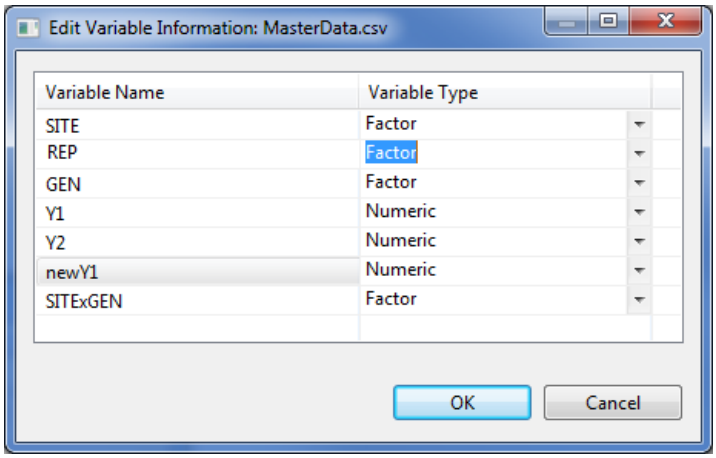
- On the Project Explorer, locate the *MasterData.csv* from the *Data* folder of the project named *Trial Project*. Double-click the file to view it in the Data Viewer.

- Click **Data | Edit Variable Information** from the menu bar or click the Edit Variable Information icon in the Data Viewer toolbar.



- The **Edit Variable Information** dialog box will appear. The variables in the active data set are listed and classified as either a factor or numeric variable. The user can modify the name and type of any variable in the dataset.

- For the example, change the variable type of *REP* from Numeric to Factor and rename *standardizedY1* to *newY1*. The complete dialog box should appear as shown below:



- Click **OK**.

- A message dialog box will appear prompting that changes have been saved. Click **OK**.

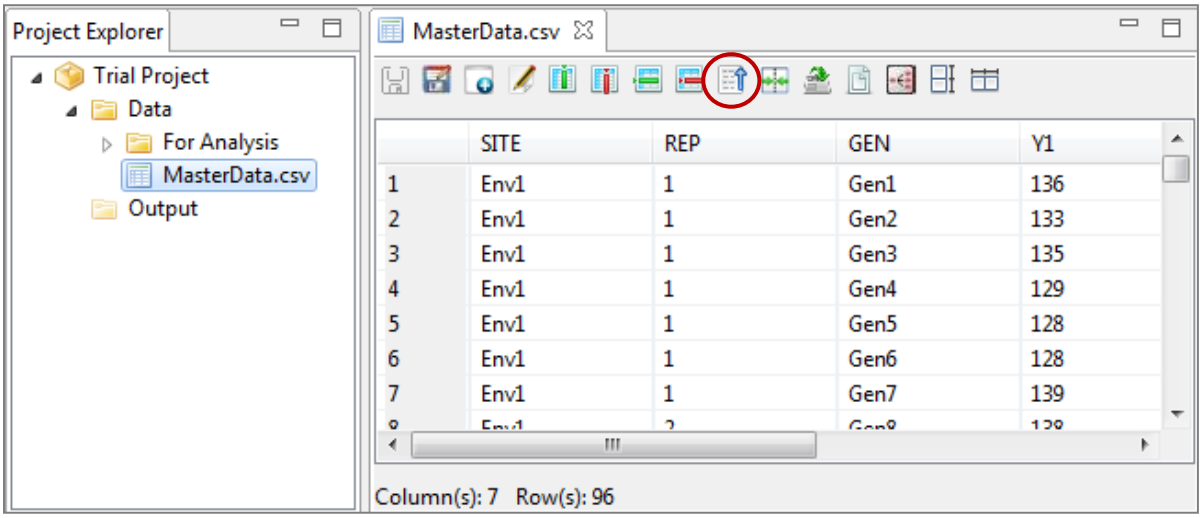
- The changes made should be reflected in the Data Viewer. For the example, close the modified *MasterData.csv* and do not save the changes.

Sorting data

Re-arranging or sorting the rows or cases of the data file is often useful and sometimes necessary for certain types of analysis. The steps to re-arrange the rows based on the value of one or more sorting variables are listed below:

- On the Project Explorer, locate the *MasterData.csv* from the *Data* folder of the project named *Trial Project*. Double-click the file to view it in the Data Viewer. The data file is arranged by *SITE*, *REP* then by *GEN*.

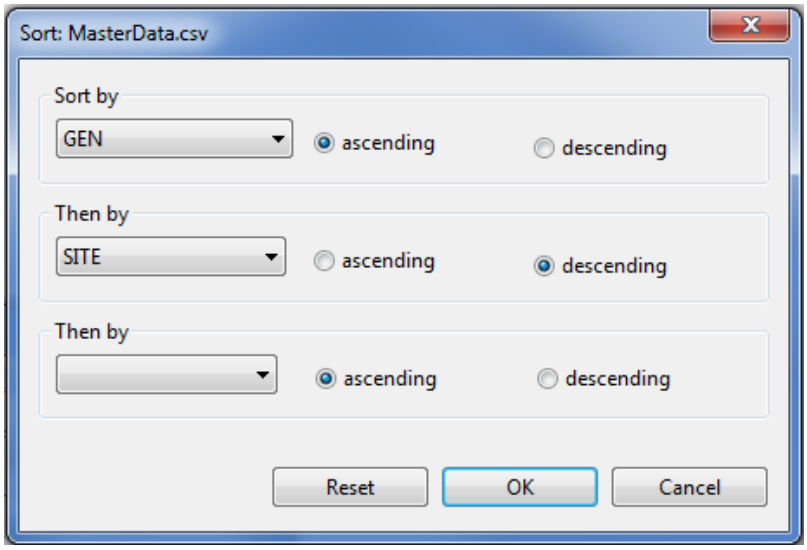
- Choose **Data** | **Sort…** from the menu bar or click the Sort icon  in the Data Viewer tool bar.



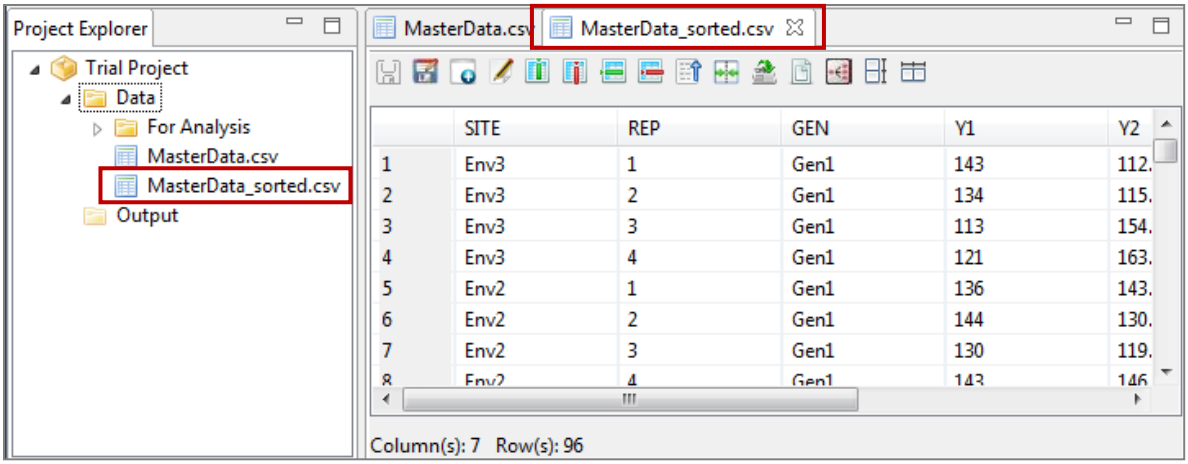
- The **Sort** dialog box will appear.

- The user can specify up to three variables as the basis for sorting and the order of sorting. Click the drop-down list box to identify the sorting variables. The rows can be re-arranged or sorted in ascending or descending order, with ascending order as the default option. If two or three variables are selected, rows are sorted for each variable within categories of the preceding variables. For character variables, uppercase letters precede their lowercase counterparts.

- For the example, suppose we want to re-arrange the rows of the data based on ascending order of the variable *GEN* then by descending order of the variable *SITE*. The completed dialog box should appear as illustrated below:



- Click the **OK**. The sorted data is created and saved in the parent folder of the active data and displayed in the Data Viewer.

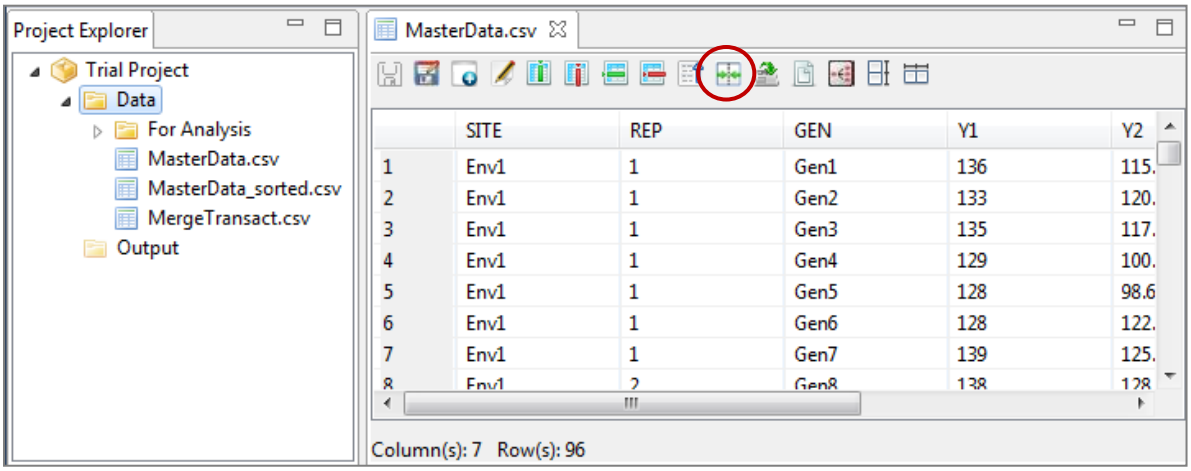


Merging datasets

This feature can be used to combine the active data file (known as the master data) with another data file (referred to as the transaction data) that contains the same cases or rows but different variables.

The steps to merge datasets are listed below:

- For the example, two datasets will be used. The first one is the *MasterData.csv* from the *Data* folder of the project named *Trial Project*. The second one is the file named *MergeTransact.csv* which should be imported from the project named *SampleProject*. Double-click the file *MasterData.csv* to open it and view it in the Data Viewer.
- Choose **Data | Merge Datasets…** or click on the Merge Datasets icon  in the Data Viewer toolbar.



- The **Merge Data** dialog box will appear.

- Specify the required fields and appropriate options.

  Transaction File Name

  Specify this file by selecting a file using the drop-down combo box or by locating it using the **Browse** button. The files included in the drop-down combo box are files inside the *Data* folder not in its sub-folder(s). Only *.txt* and *.csv* data file format can be selected as the transaction file.

Text File Delimeter

If the transaction file selected is a text file, the delimeter should be specified. Four delimeters are available, namely: comma, space, tab and semi-colon.

Observations to Include

This option pertains to how rows or cases will be included in the new merged data file. The options available are to include common observations (default value), all observations in the active data, all observations in the transaction data and all observations.

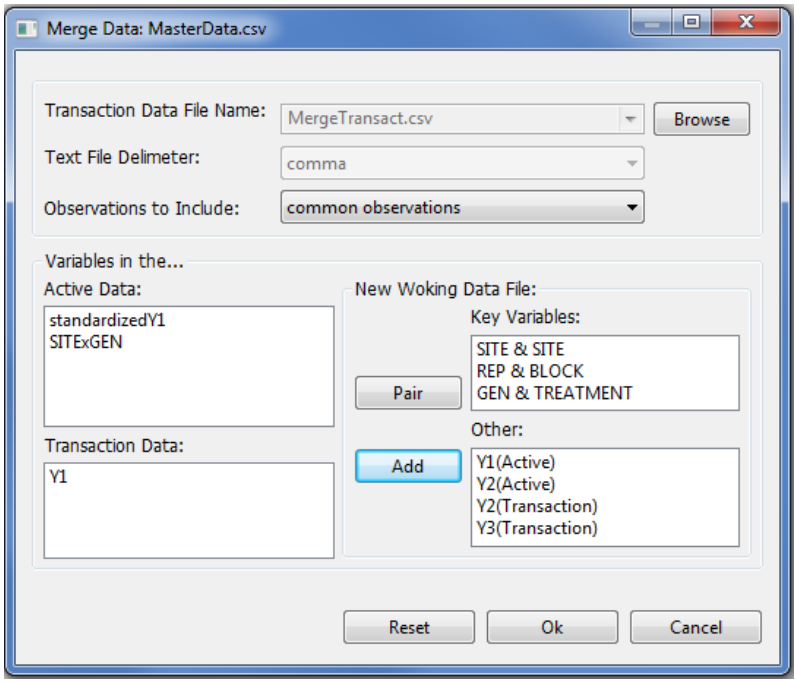
Key Variables

These are pair(s) of variables that are present in the active and transaction files and will be the basis for the merging. Variables can be paired by clicking one variable in the active data and clicking one variable in the transaction data then clicking the **Pair** button. If the paired variables have different names, the merged data file that will be created will use the column name of the active data.

Other

These are the other variables the user wants to include in the merged data file. If columns in this list have the same name, **.1** will be appended to the column name coming from the active data and **.2** to the one from the transaction data.

For the example, select *MergeTransact.csv* file as the transaction data file from the drop-down combo box. The completed dialog box should appear as illustrated below:

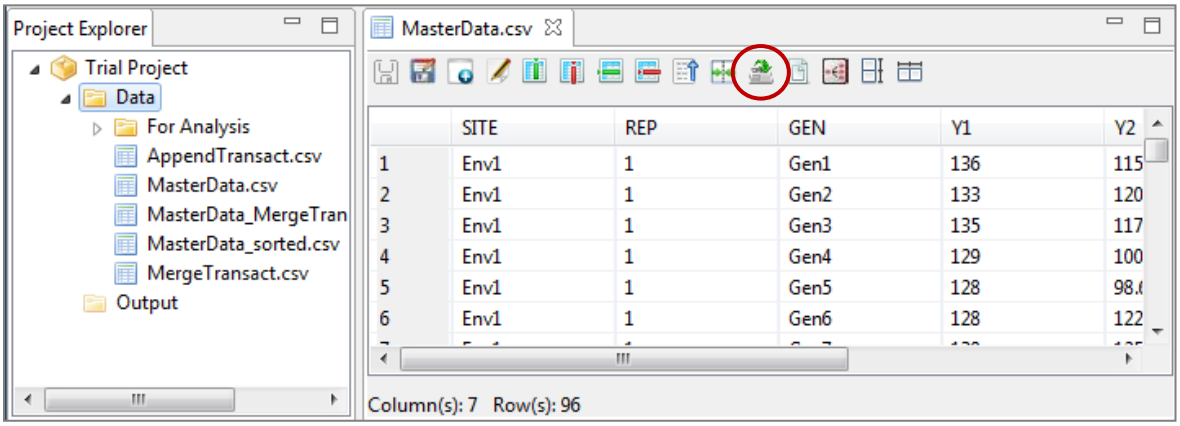


- Click **OK**. The merged data is saved in the parent folder of the active data and displayed in the Data Viewer tab. The default filename of the merged data set follows the format *<activeDataFilename>_<TransactionDataFilename>_merge.csv*.

Appending Datasets

This feature can be used to combine two data files with the same variables but different cases. The steps to merge datasets are listed below:

- For the example, two datasets will be used. The first one is the *MasterData.csv* from the *Data* folder of the project named *Trial Project*. The second one is the file named *AppendTransact.csv* which should be imported from the project named *SampleProject*. Double-click the file *MasterData.csv* to open it and view it in the Data Viewer.
- Choose **Data | Append Datasets…** or click on the Append Datasets icon  in the Data Viewer toolbar.



- The **Append Data** dialog box will appear.

- Specify the required fields and appropriate options.

  Transaction File Name

  Specify this file by selecting a file using the drop-down combo box or by locating it using the **Browse** button. The files included in the drop-down combo box are files inside the *Data* folder not in its sub-folder(s). Only *.txt* and *.csv* data file format can be selected as the transaction file.

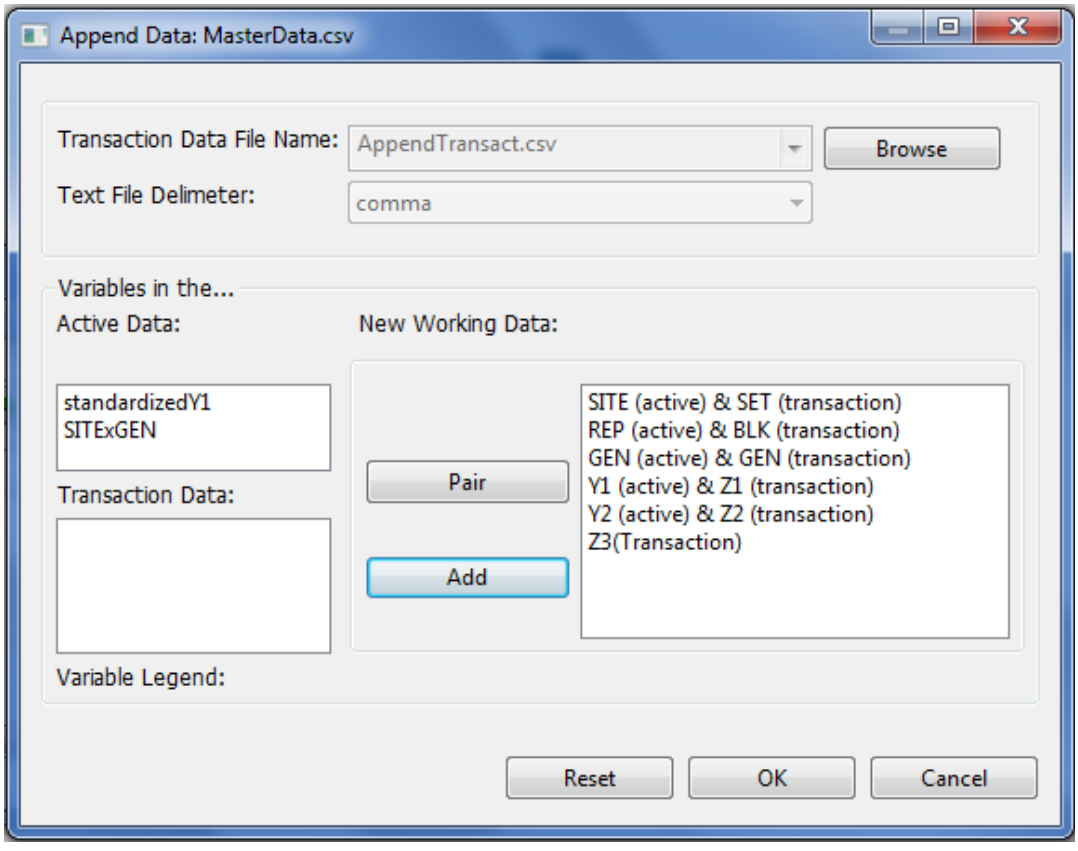
Text File Delimeter

If the transaction file selected is a text file, the delimeter should be specified. Four delimeters are available, namely: comma, space, tab and semi-colon

New Working Data

This is the list of variables or paired variables that will be included in the new data file. Variables can be paired by clicking one variable in the active data and clicking one variable in the transaction data then clicking the **Pair** button. Other variables can be added to this list by clicking the variable then clicking the **Add** button. For paired variables in this list, the appended data file that will be created will use the column name of the active data. For non-paired variables, **.1** will be appended to the column name coming from the active data and **.2** to the one from the transaction data.

For the example, select *AppendTransact.csv* file as the transaction data file from the drop-down combo box. The completed dialog box should appear as illustrated below:

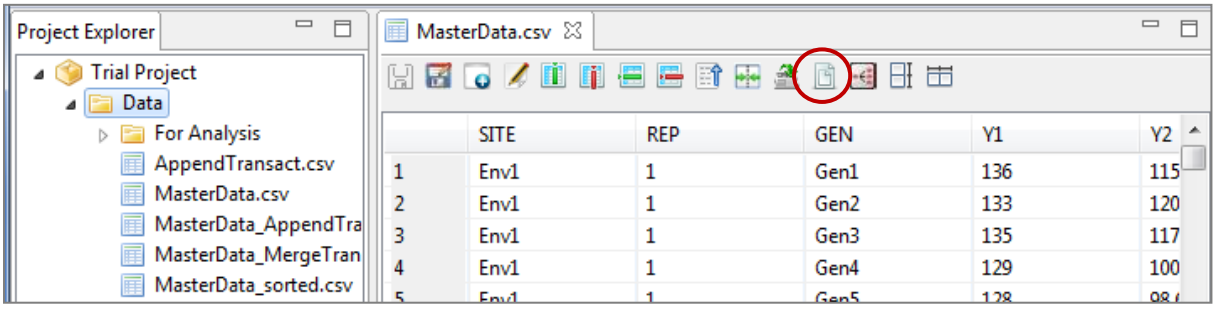


- Click **OK**. The generated file is saved in the parent folder of the active data and displayed in the Data Viewer tab. It will contain an additional column named *Source* which indicates where the data came from. The default filename of the appended data set follows the format <*activeDataFilename*>_<*TransactionDataFilename*>_*merge.csv*.

Creating Data Subset

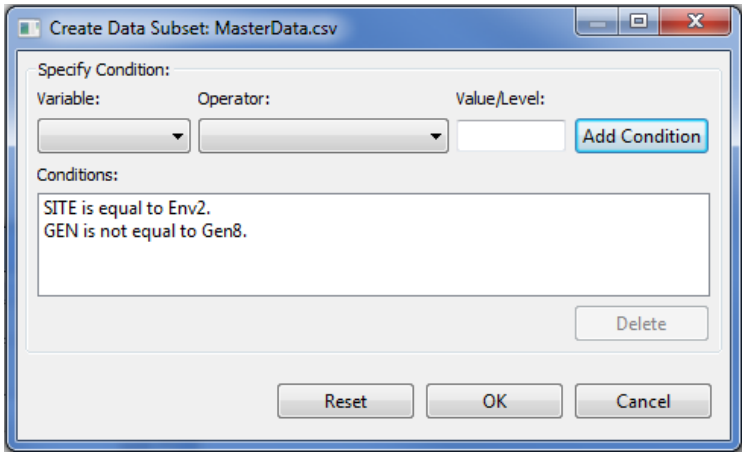
The steps in creating a subset of a data are listed below:

- On the Project Explorer, locate the *MasterData.csv* from the *Data* folder of the project named *Trial Project*. Double-click the file to view it in the Data Viewer.
- Choose **Data | Create Data Subset…** or click on the Create Data Subset icon in the Data Viewer toolbar.



- The **Create Data Subset** dialog box will appear.
- For the example, suppose the user wants to select only the observations from *Env2* except those belonging to *Gen8*. This can be done by selecting variable *Site*, operator *equals (==),* typing *Env2* under Value/Level and clicking the **Add Condition** button. Do the same for the condition "GEN not equal to Gen8".

  For the *example*, the completed dialog box should appear as illustrated below:



- Click **OK**. The new data is saved in the *Data* folder and displayed in the Data Viewer.

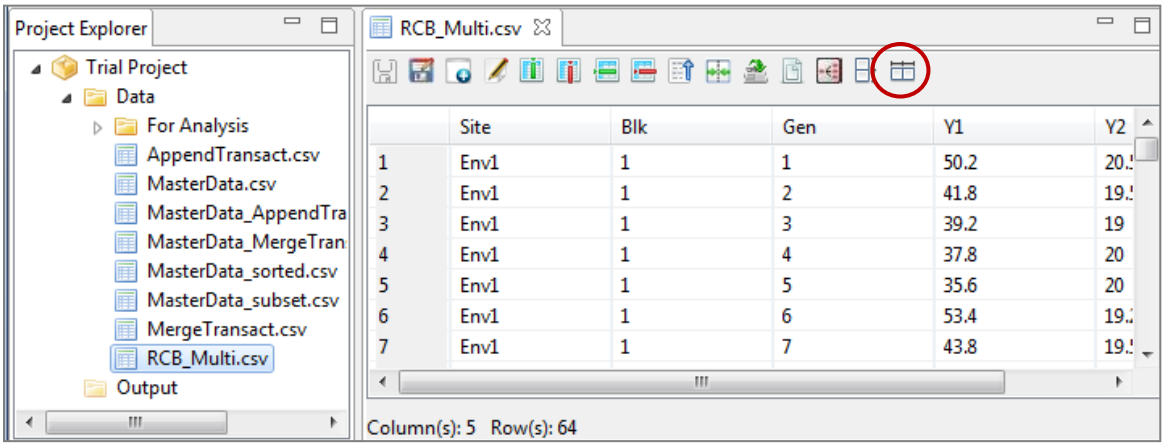
Reshaping Data From Long (Serial) to Wide (Parallel)

Reshaping data from long (serial) to wide (parallel) involves re-arranging a data file, such that, repeated measurements are in separate columns.

The steps for reshaping the data from long to wide format are listed below:

- Import *RCB_Multi* file from PBTools package. Double-click the file to view it in the Data Viewer.

- Click **Data | Reshape | Long (Serial) to Wide (Parallel) …** from the menu bar or click on the Reshape to Wide icon in the Data Viewer tool bar.



- The **Reshape Data (Long to Wide)** dialog box will appear.

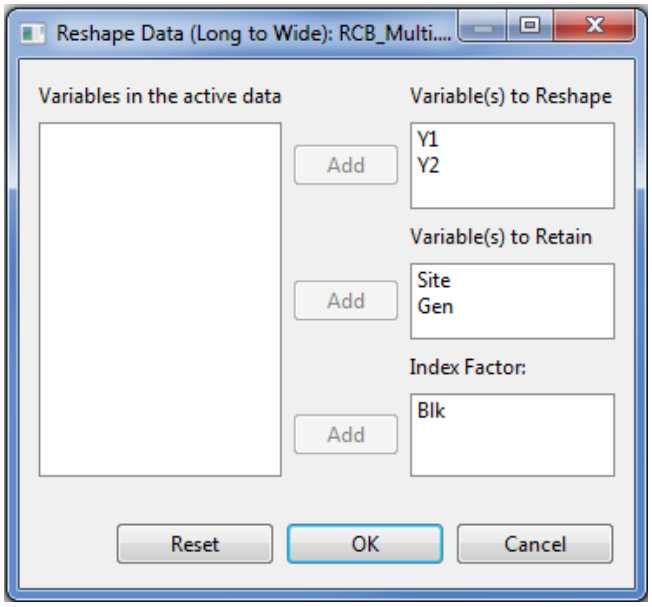
- Specify the required fields and appropriate options.

Variable(s) to Reshape
This is the list of variable(s) whose values are to be divided two or more columns and saved to different variables in terms of the levels of the index factor.

Variable(s) to Retain
This pertains to variable(s) to be retained to describe the individual cases or observations.

Index Factor
This pertains to variable(s) which will determine the groupings of the values of the variable(s) to be transposed.

For the example, the completed dialog box should appear as illustrated below:



- Click **OK**. The restructured data set is saved in the parent folder of the active data and displayed in the Data Viewer tab. The default filename of the restructured data set follows the format <*activeDataFilename*>*_ReshapeToWide.csv*.

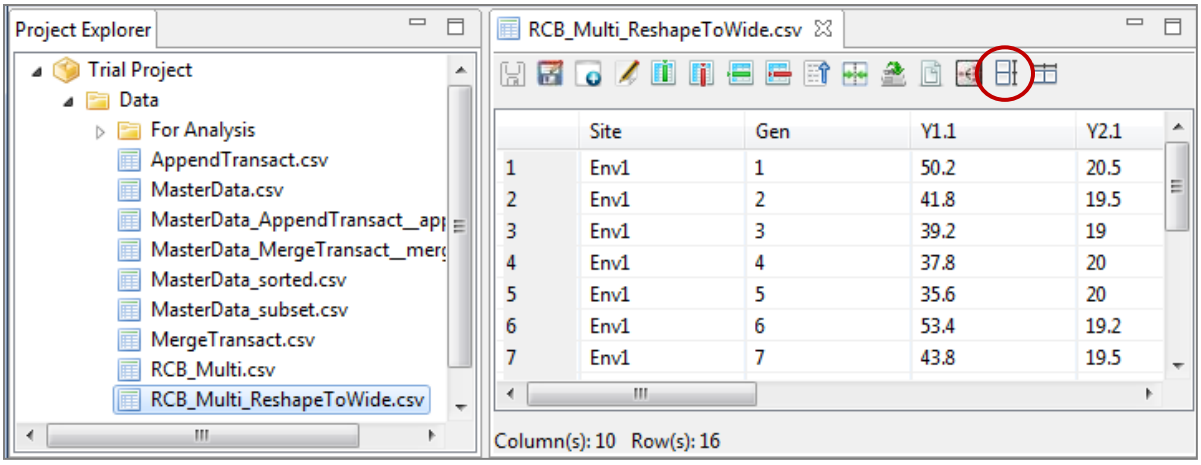
Reshaping Data From Wide (Parallel) to Long (Serial)

Manipulating data from wide (parallel) to long (serial) refers to re-arranging a multivariate into a univariate data. For instance, different columns representing measurements taken for a response variable over a period of time can be combined to form a single column, with the time variable used as an additional identifier variable.

The steps for reshaping the data from wide to long format are listed below:

- On the Project Explorer, locate the *RCB_Multi_ReshapeToWide.csv* file (the generated file from the previous section) from the *Data* folder of the project named *Trial Project*. Double-click the file to view it in the Data Viewer.

- Click **Data | Reshape | From Wide (Parallel) to Long (Serial) …** from the menu bar or click on the Reshape to Long icon  in the Data Viewer tool bar.



- The **Reshape Data (Wide to Long)** dialog box will appear.

- Specify the required fields and appropriate options.

Variables to Reshape

These are the variables that will be combined to form one variable. Click the variables in the list of variables in the active data while holding the Ctrl key then click the **Add** button. Moreover, the user should specify as the target variable the name of the variable to be created. The default target variable name is *targetvar1*.Then click the **Add Target Variable**. In specifying the name of the target variable, the following rules apply:

- The name must start with a letter while the remaining characters can be any letter, any digit, a period or underscore.
- The name must be different from the existing column names in the data.
- The name is case sensitive.

User can create at least one target variable. This target variable should be unique and the length of the *Reshape Variables* should be the equal for all target variables to be created.

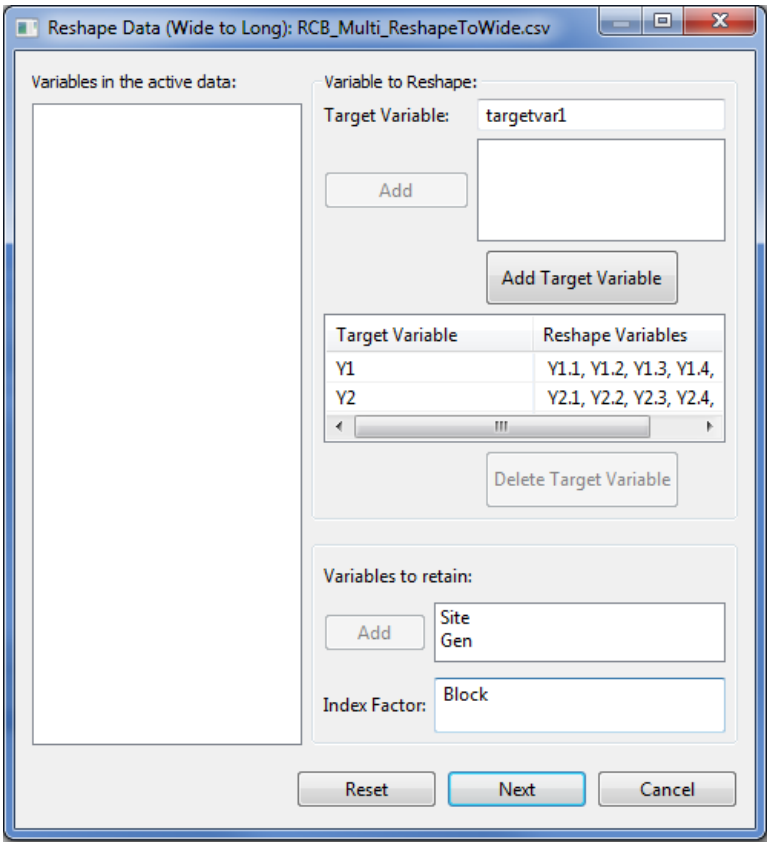
Variables to Retain

These are the variables that will be included in the restructured data set and pertains to variables that describe the individual cases.

Index Factor

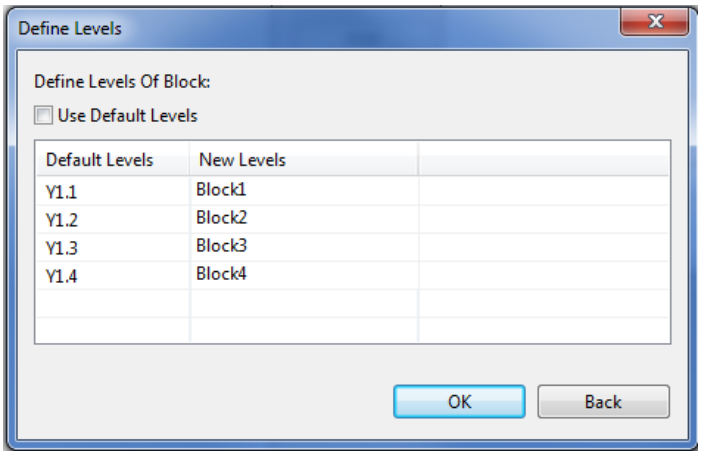
This is the name of the grouping variables that will be created.

For the *example*, the completed dialog box should appear as illustrated below:



- Click the **Next** button to proceed.

- The **Define Levels** dialog box will appear. To specify the levels of the Index Factor that will be created, the user can use the default values or type the desired levels in the column labeled 'New Levels'. For now, PBTools only accepts non-numeric values for the levels.



- Click the **OK** button. The restructured data set is saved in the parent folder of the active data and displayed in the Data Viewer tab. The default filename of the restructured data set follows the format <*activeDataFilename*>*_ReshapeToLong.csv*.

Aggregating Data

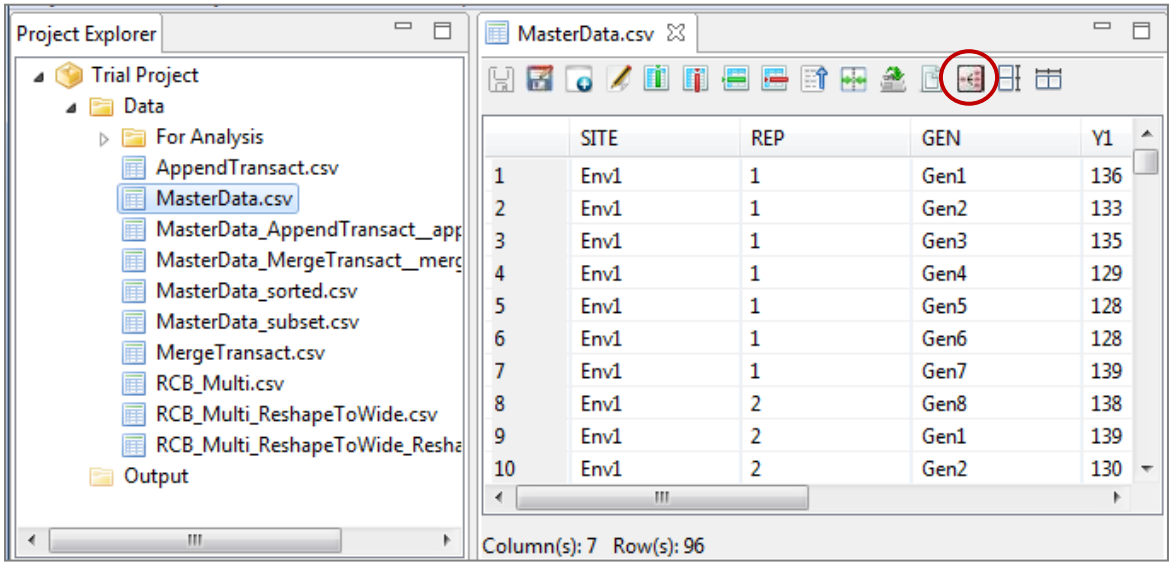
This feature aggregates group of observations into single observation and creates a new, aggregated data file or creates new columns in the active data file that contain the aggregated data. Observations are aggregated based on the value of one or more grouping variables.

The steps for aggregating the data are listed below:

- On the Project Explorer, locate the *MasterData.csv* from the *Data* folder of the project named *Trial Project*. Double-click the file to open it and view it in the Data Viewer.

  Opening the data for the first-time, *REP* field is regarded by PBTools as numerical variable. To change its type, go to **Data | Edit Variable Information** and change the *Variable Type* of *REP* from *Numeric* to *Factor*.

- Choose **Data | Aggregate…** or click on the Aggregate Data icon  in the Data Viewer tool bar.



- The **Aggregate Data** dialog box will appear.

- Specify the required fields and appropriate options.

Variable(s) to Aggregrate

This is the list of variables whose values will be summarized using the selected functions.

Grouping Factor(s)

Specify this if the user wants to summarize the values per level of a grouping factor.

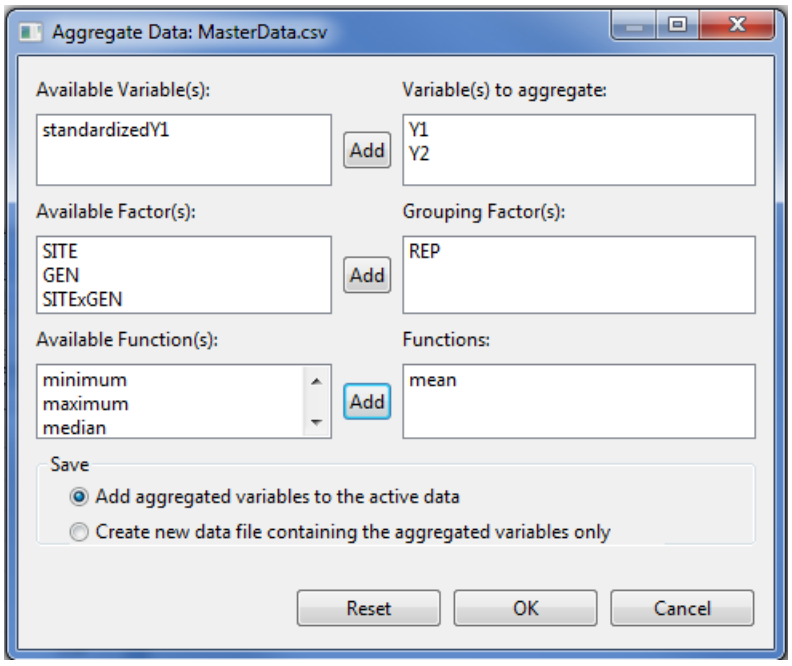
Functions

Available functions are minimum, maximum, mean, median, sum, variance and standard deviation.

Save

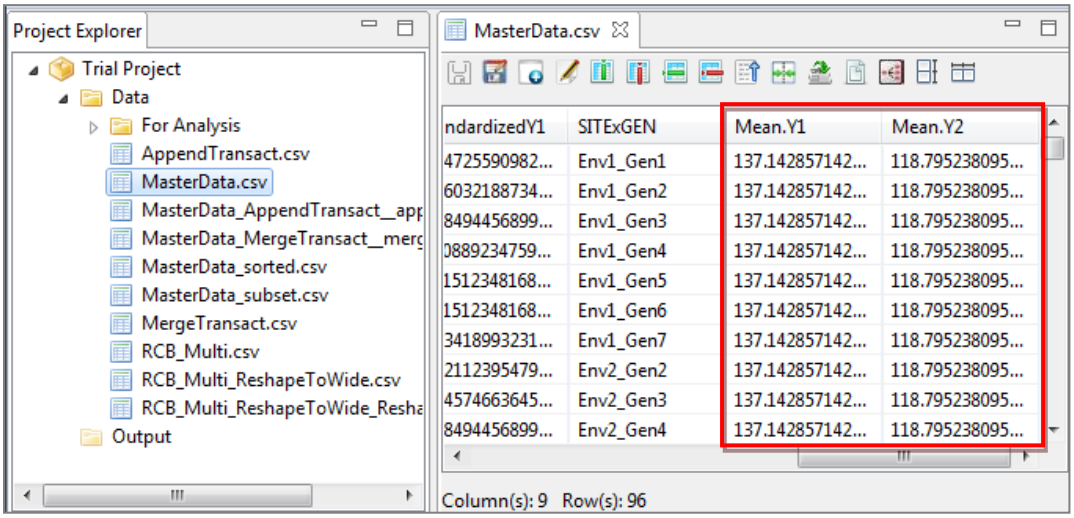
Specify how the aggregated data will be saved. User can either add aggregated variables to the active data (default option) or create a new data file containing the aggregated variables only. If user chooses the default option, the resulting data file is not aggregated. Each observations with the save value(s) of the grouping variable(s) will receive the same values of the new aggregated variables. If user chooses the option create a new data file containing the aggregated variables only, the new data files contains one observation for each category of the grouping variables.

For the *example*, suppose we want to compute the mean of the variables *Y1* and *Y2* for each level of *REP*. The completed dialog box should appear as illustrated below:

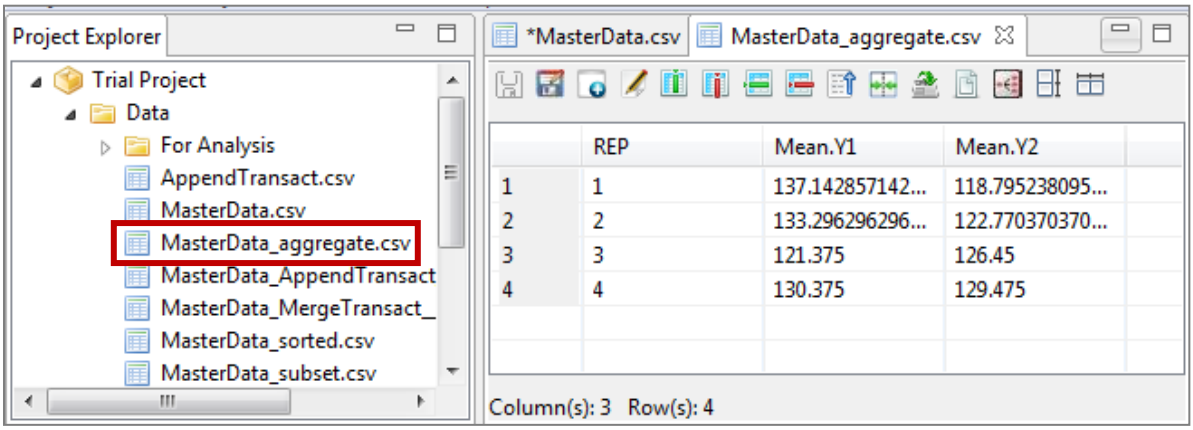


- Click the **OK** button.

If the user chooses the save option: *add aggregated variables to the active data* (default option), the active data is saved with the additional column(s) and displayed in the Data Viewer. The default column name(s) which contain the aggregate data follows the format *<Function>.< Variable to aggregate >*.



If the user chooses the save option: *create a new data file containing the aggregated variables only*, the aggregated data file is saved in the parent folder of the active data and displayed in the Data Viewer. The default column name(s) which contain the aggregated data follows the format *<Function>.<Variable name to aggregate>* and the default filename of the aggregated data set follows the format *<activeDataFilename>_aggregate.csv*.



5. Randomization for Some Experimental Designs

The **Randomization** menu allows user to generate randomization of treatments for single- and multi-factor designs. If the menu is used for the first-time in the active project, a *Randomization* folder will be created inside the *Output* folder. Sub-folders will be created inside the *Randomization* folder where the generated field book in *csv* format and a text file will be saved. The default sub-folder name follows the format <*design*>_<*time stamp*>.

Randomized Complete Block Design

The steps to generate randomization for Randomized Complete Block design are listed below:

- Click **Randomization | Randomized Complete Block Design …**. The **Randomization and Layout** dialog box will appear.

- Specify the required fields and appropriate options.

  Number of Factors

  Define the number of factors to be generated with default and minimum value equal to 1 and maximum value equal to 10. The number of factors specified will define the number of rows of the table inside the **Factor Definition** frame. To change the default value, user can either type a value inside the spin box or click the up-arrow key of the spin box to increase the value, down-arrow key, otherwise.

  Factor Definition Table

  Name

  This column contains the default name for each factor which can be changed. In specifying the factor name, the following rules apply:

  - it should not contain any space;
  - it must begin with a letter or a period (.);
  - its succeeding characters can be a combination of letters, numbers, period (.) and underscore (_); and
  - it must be different from other factor names already specified.

Factor ID

This column contains the default code for each factor. User can change the factor ID by typing the desired factor ID. Factor ID should consist of 1 to 4 letters only and should be unique.

Levels

The column contains the level for each factor. User can change the level of the factor by either typing a value inside the spin box or clicking the up-arrow (down-arrow) key of the spin box to increase (decrease) the value. The default and minimum value of the levels is equal to 2 while the maximum value of the levels is 500.

Number of Blocks

Specify in this field the number of blocks to be generated. The default and minimum value is equal to 2.

Number of Rows per Block

Specify in this field the number of rows per block. The default and minimum value is equal to 1.

Number of Field Rows

Specify in this field the number of rows in the field layout for each trial. The default and minimum value is equal to 1.

Number of Trial

Specify in this field the number of trials to be generated. The default and minimum value is equal to 1.

Field Book Filename

This field is required and it specifies the filename where the result of the randomization will be saved. The default filename is *fieldbookRCBD*.

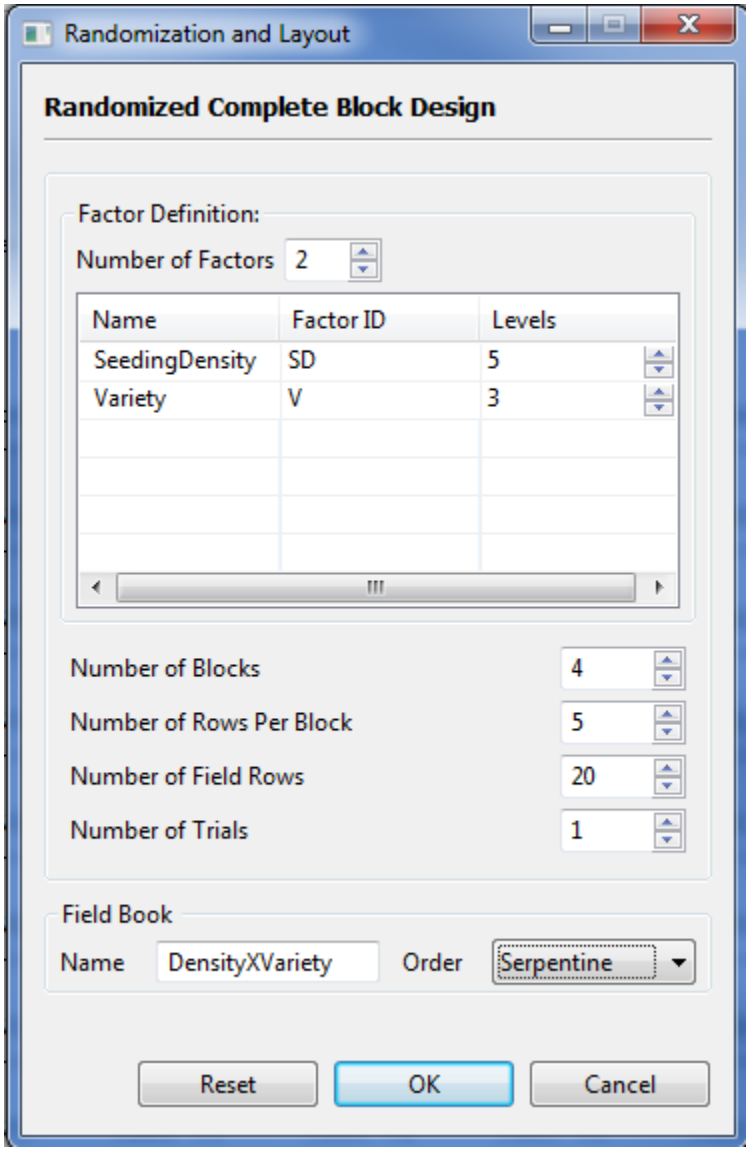
Field Book Order

Specify in this field how the plot number will be arranged. There are two options available: *Plot Order* (default value) and *Serpentine*.

For the example, suppose we want to generate a randomization and layout for an experiment whose aim is to compare the effects of five seeding densities on the grain yield of three rice varieties. The experiment will be conducted in randomized complete block design with four blocks in one trial. The planned field layout is shown below:

	Field Column 1	Field Column 2	Field Column 3	
Field Row 1				
Field Row 2				
Field Row 3				Block 1
Field Row 4				
Field Row 5				
Field Row 6				
Field Row 7				
Field Row 8				Block 2
Field Row 9				
Field Row 10				
Field Row 11				
Field Row 12				
Field Row 13				Block 3
Field Row 14				
Field Row 15				
Field Row 16				
Field Row 17				
Field Row 18				Block 4
Field Row 19				
Field Row 20				

The completed dialog box should appear as shown below:



- Click the **OK** button to generate the randomization. The **Randomization and Layout** dialog box will be minimized and PBTools will automatically display the *csv* data file and the *txt* file created in the Data Viewer tab.

Sample *txt* file displayed in the Data Viewer tab is shown below:

```
DESIGN PROPERTIES:
      Factorial Design
      Randomized Complete Block Design

DESIGN PARAMETERS:
      Number of Trials = 1
      Number of Blocks = 4
      Factor 1 = SeedingDensity
      Levels = SD1, SD2, SD3, SD4, SD5
      Factor 2 = Variety
      Levels = V1, V2, V3

      Number of Field Row = 20
      Number of Field Column = 3

Layout for Randomized Complete Block Design

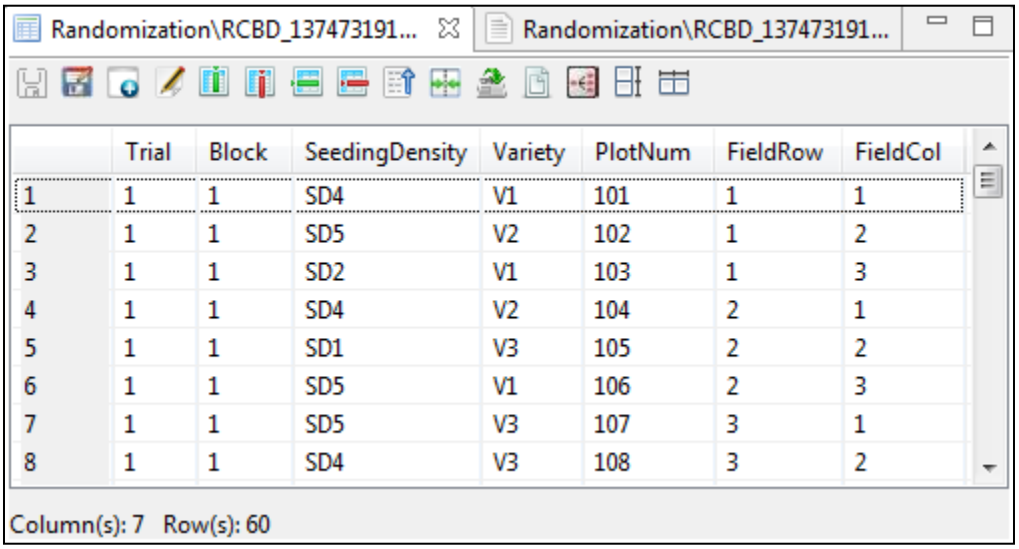

 Trial = 1

             FieldCol1   FieldCol2   FieldCol3
            +-----------+-----------+-----------+
FieldRow1   |        101|        102|        103|
            |SD4 V1     |SD5 V2     |SD2 V1     |
            +-----------+-----------+-----------+
FieldRow2   |        104|        105|        106|
            |SD4 V2     |SD1 V3     |SD5 V1     |
            +-----------+-----------+-----------+
FieldRow3   |        107|        108|        109|
            |SD5 V3     |SD4 V3     |SD1 V1     |
            +-----------+-----------+-----------+
FieldRow4   |        110|        111|        112|
            |SD1 V2     |SD2 V3     |SD3 V2     |
            +-----------+-----------+-----------+
FieldRow5   |        113|        114|        115|
            |SD3 V3     |SD2 V2     |SD3 V1     |
            +-----------+-----------+-----------+
FieldRow6   |        201|        202|        203|
            |SD1 V1     |SD4 V1     |SD5 V1     |
            +-----------+-----------+-----------+
FieldRow7   |        204|        205|        206|
            |SD1 V2     |SD4 V2     |SD3 V1     |
            +-----------+-----------+-----------+
FieldRow8   |        207|        208|        209|
            |SD1 V3     |SD3 V3     |SD2 V1     |
            +-----------+-----------+-----------+
FieldRow9   |        210|        211|        212|
            |SD2 V3     |SD3 V2     |SD2 V2     |
            +-----------+-----------+-----------+
FieldRow10  |        213|        214|        215|
            |SD5 V2     |SD4 V3     |SD5 V3     |
            +-----------+-----------+-----------+
FieldRow11  |        301|        302|        303|
            |SD5 V3     |SD4 V2     |SD2 V1     |
```

FieldRow12	304 SD5 V1	305 SD5 V2	306 SD2 V2
FieldRow13	307 SD3 V2	308 SD1 V1	309 SD1 V3
FieldRow14	310 SD3 V3	311 SD3 V1	312 SD4 V3
FieldRow15	313 SD1 V2	314 SD4 V1	315 SD2 V3
FieldRow16	401 SD4 V2	402 SD5 V3	403 SD2 V3
FieldRow17	404 SD3 V1	405 SD4 V1	406 SD1 V3
FieldRow18	407 SD5 V2	408 SD4 V3	409 SD2 V1
FieldRow19	410 SD3 V2	411 SD1 V1	412 SD5 V1
FieldRow20	413 SD3 V3	414 SD2 V2	415 SD1 V2

**Note: Cells contain plot numbers on top, treatments/entries below

Sample *csv* data file displayed in the Data Viewer tab is shown below:



	Trial	Block	SeedingDensity	Variety	PlotNum	FieldRow	FieldCol
1	1	1	SD4	V1	101	1	1
2	1	1	SD5	V2	102	1	2
3	1	1	SD2	V1	103	1	3
4	1	1	SD4	V2	104	2	1
5	1	1	SD1	V3	105	2	2
6	1	1	SD5	V1	106	2	3
7	1	1	SD5	V3	107	3	1
8	1	1	SD4	V3	108	3	2

Lattice Design

The steps to generate randomization for Lattice design are listed below:

- Click **Randomization | Lattice Design…** from the main window. The **Randomization and Layout** dialog box will appear.

- Specify the required fields and appropriate options.

  Number of Treatments
  Specify in this field the number of treatments (*trmt*) to be included in the randomization. The default and minimum value is 9 while the maximum value is 144. The entry should be a perfect square.

  Number of Replicates
  Specify in this field the number of replicates to be generated. The default value and minimum value is equal to 2. If the $\sqrt{trmt}$ is odd, the maximum number of replicates is $\sqrt{trmt} + 1$. If the $\sqrt{trmt}$ is equal to 4, the maximum number of replicates is 5. If the $\sqrt{trmt}$ is even and not equal to 4, the maximum number of replicates is 3.

  Number of Field Rows
  Specify in this field the total number of rows in the field layout for each trial. The default and minimum value is equal to 3.

  Number of Trial
  Specify in this field the number of trials to be generated. The default value and minimum value is equal to 1.

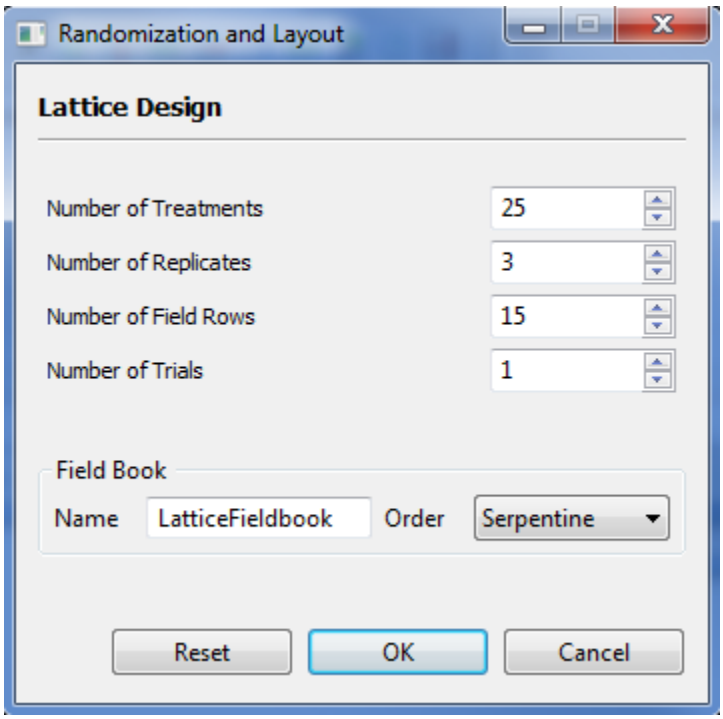
  Field Book Filename
  This field is required and it specifies the filename where the result of the randomization will be saved. The default filename is *fieldbookLattice*.

  Field Book Order
  Specify in this field how the plot number will be arranged. There are two options available: *Plot Order* (default value) and *Serpentine*.

  For this example, suppose we want to generate a randomization for an experiment that will be conducted using Lattice design with 25 treatments in three replicates. The planned field layout is shown below:

	Field Column 1	Field Column 2	Field Column 3	Field Column 4	Field Column 5	
Field Row 1						Rep
Field Row 2						
Field Row 3						
Field Row 4						
Field Row 5						
Field Row 6						Rep 2
Field Row 7						
Field Row 8						
Field Row 9						
Field Row 10						
Field Row 11						Rep 3
Field Row 12						
Field Row 13						
Field Row 14						
Field Row 15						

The completed dialog box should appear as shown below:



- Click the **OK** button to generate the randomization. The **Randomization and Layout** dialog box will be minimized and PBTools will automatically display the *csv* data file and the *txt* file created in the Data Viewer tab.

  Sample *txt* file displayed in the Data Viewer is shown below:

```
DESIGN PROPERTIES:
       Incomplete Block Design
       5 x 5 Partially Balanced Lattice Design

DESIGN PARAMETERS:
       Number of Trials = 1
       Number of Treatments = 25
       Number of Replicates = 3
       Plots per Block (Block Size) = 5
       Block per Replicate = 5
       Number of Field Row = 15
       Number of Field Column = 5


Layout for Lattice Design

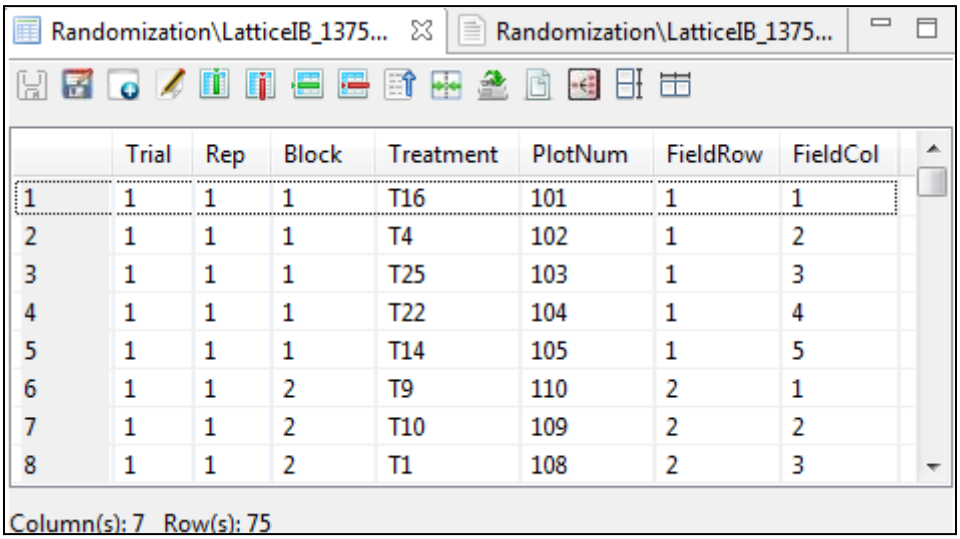

 Trial = 1

             FieldCol1   FieldCol2   FieldCol3   FieldCol4   FieldCol5
            +-----------+-----------+-----------+-----------+-----------+
FieldRow1   |        101|        102|        103|        104|        105|
            |T16        |T4         |T25        |T22        |T14        |
            +-----------+-----------+-----------+-----------+-----------+
FieldRow2   |        110|        109|        108|        107|        106|
            |T9         |T10        |T1         |T20        |T2         |
            +-----------+-----------+-----------+-----------+-----------+
FieldRow3   |        111|        112|        113|        114|        115|
            |T6         |T15        |T12        |T19        |T8         |
            +-----------+-----------+-----------+-----------+-----------+
FieldRow4   |        120|        119|        118|        117|        116|
            |T13        |T18        |T21        |T17        |T11        |
            +-----------+-----------+-----------+-----------+-----------+
FieldRow5   |        121|        122|        123|        124|        125|
            |T3         |T24        |T5         |T7         |T23        |
            +-----------+-----------+-----------+-----------+-----------+
FieldRow6   |        201|        202|        203|        204|        205|
            |T5         |T12        |T1         |T25        |T21        |
            +-----------+-----------+-----------+-----------+-----------+
FieldRow7   |        210|        209|        208|        207|        206|
            |T3         |T6         |T9         |T16        |T13        |
            +-----------+-----------+-----------+-----------+-----------+
FieldRow8   |        211|        212|        213|        214|        215|
            |T23        |T8         |T2         |T14        |T11        |
            +-----------+-----------+-----------+-----------+-----------+
FieldRow9   |        220|        219|        218|        217|        216|
            |T7         |T19        |T20        |T22        |T17        |
            +-----------+-----------+-----------+-----------+-----------+
FieldRow10  |        221|        222|        223|        224|        225|
            |T24        |T15        |T10        |T4         |T18        |
            +-----------+-----------+-----------+-----------+-----------+
FieldRow11  |        301|        302|        303|        304|        305|
            |T13        |T24        |T12        |T20        |T14        |
            +-----------+-----------+-----------+-----------+-----------+
```

```
FieldRow12  |        310|        309|        308|        307|        306|
            |T3         |T15        |T1         |T22        |T11        |
            +-----------+-----------+-----------+-----------+-----------+
FieldRow13  |        311|        312|        313|        314|        315|
            |T6         |T10        |T25        |T17        |T23        |
            +-----------+-----------+-----------+-----------+-----------+
FieldRow14  |        320|        319|        318|        317|        316|
            |T16        |T18        |T5         |T19        |T2         |
            +-----------+-----------+-----------+-----------+-----------+
FieldRow15  |        321|        322|        323|        324|        325|
            |T9         |T4         |T21        |T7         |T8         |
            +-----------+-----------+-----------+-----------+-----------+
```

**Note: Cells contain plot numbers on top, treatments/entries below

Sample *csv* data file displayed in the Data Viewer is shown below:



	Trial	Rep	Block	Treatment	PlotNum	FieldRow	FieldCol
1	1	1	1	T16	101	1	1
2	1	1	1	T4	102	1	2
3	1	1	1	T25	103	1	3
4	1	1	1	T22	104	1	4
5	1	1	1	T14	105	1	5
6	1	1	2	T9	110	2	1
7	1	1	2	T10	109	2	2
8	1	1	2	T1	108	2	3

Alpha Lattice Design

The steps to generate randomization for Alpha Lattice Design are listed below:

- Click **Randomization | Alpha Lattice Design…** from the main window. The **Randomization** dialog box will appear.

- Specify the required field and appropriate options.

  Number of Treatments
  Specify in this field the number of treatments to be included in the randomization. The default and minimum value is 9.

Number of Replicates
Specify in this field the number of replicates to be generated per trial. The default and minimum value is 2.

Number of Plots per Block
Specify in this field the number of plots per block or the block size. The default value is 3 and the minimum is 2.

Number of Blocks per Replicate
PBTools automatically computes this value.

Number of Rows in Each Block
Specify in this field the number of rows per block. The default value is 3, minimum is 1 and the maximum is equal to the block size. If Latinized alpha lattice design is chosen, this field is not needed.

Number of Rows in Each Replicate
Specify in this field the number of rows per replicate. The default value is 3, minimum is equal to the number of rows per block and the maximum is equal to the number of treatments. If Latinized alpha lattice design is chosen, this field is not needed.

Number of Field Rows
Specify in this field the total number of rows in the field layout for each trial. The default value is 3, minimum is equal to the number of rows per replicate and the maximum is equal to the product of the number of rows per replicate and the number of replicates.

Number of Trial
Specify in this field the number of trials to be generated. The default value and minimum value is equal to 1.

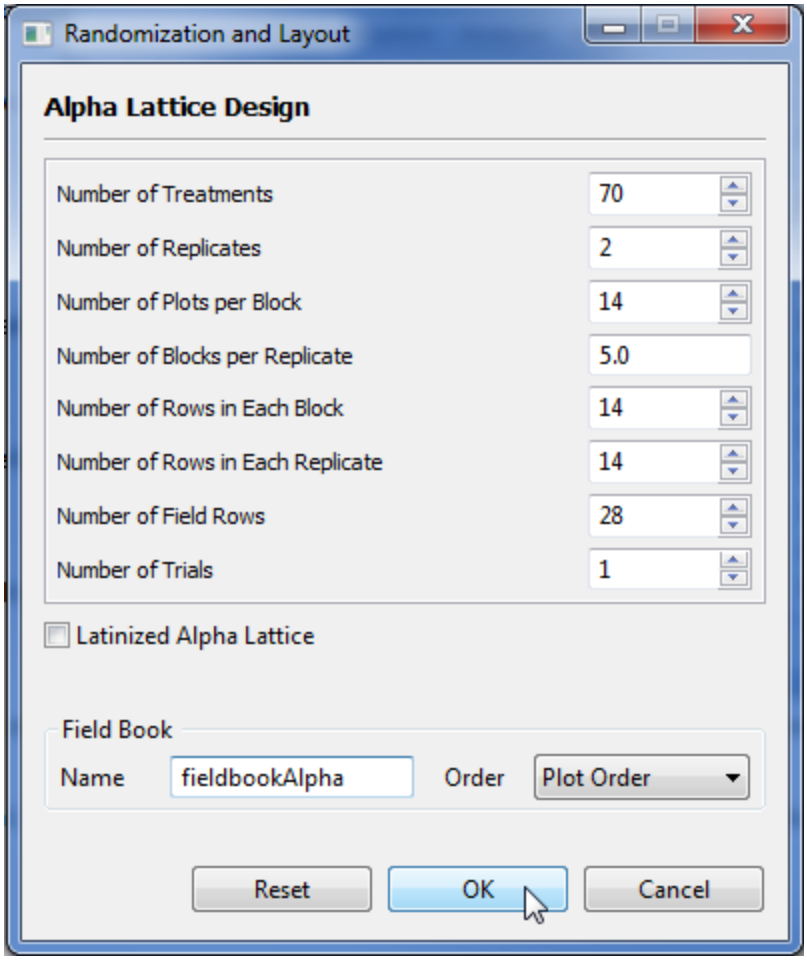
Field Book Filename
This field is required and it specifies the filename where the result of the randomization will be saved. The default filename is *fieldbookAlpha*.

Field Book Order
Specify in this field how the plot number will be arranged. There are two options available: *Plot Order* (default value) and *Serpentine*.

For this example, suppose we want to generate a randomization and layout for an experiment that will be conducted using Alpha Lattice design with 70 entries grouped into 14 with two replicates. The planned field layout is shown below:

	Block 1	Block 2	Block 3	Block 4	Block 5	
Field Row 1						Replicate 1
Field Row 2						
⋮	⋮	⋮	⋮	⋮	⋮	
Field Row 14						
Field Row 15						Replicate 2
Field Row 16						
⋮	⋮	⋮	⋮	⋮	⋮	
Field Row 28						

The completed dialog box should appear as illustrated below:



- Click the **OK** button to generate the randomization. The **Randomization and Layout** dialog box will be minimized and PBTools will automatically display the *csv* data file and the *txt* file created in the Data Viewer tab.

  Sample *txt* file displayed in the Data Viewer is shown below:

```
DESIGN PROPERTIES:
       Incomplete Block Design
       Alpha Lattice Design

DESIGN PARAMETERS:
       Number of Trials = 1
       Number of Treatments = 70
       Number of Replicates = 2
       Number of Plots per Block = 14
       Number of Blocks per Replicate = 5

       Number of Field Rows = 28
       Number of Field Columns = 5


Layout for Alpha Lattice Design:

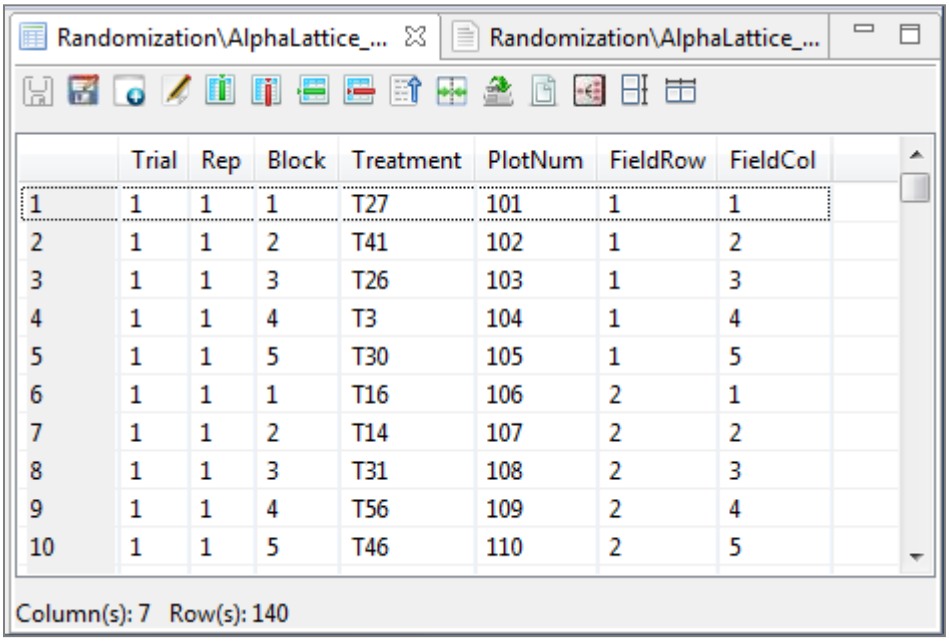

Trial = 1

             FieldCol1   FieldCol2   FieldCol3   FieldCol4   FieldCol5
            +-----------+-----------+-----------+-----------+-----------+
FieldRow1   |        101|        102|        103|        104|        105|
            |T27        |T41        |T26        |T3         |T30        |
            +-----------+-----------+-----------+-----------+-----------+
FieldRow2   |        106|        107|        108|        109|        110|
            |T16        |T14        |T31        |T56        |T46        |
            +-----------+-----------+-----------+-----------+-----------+
FieldRow3   |        111|        112|        113|        114|        115|
            |T15        |T23        |T45        |T13        |T28        |
            +-----------+-----------+-----------+-----------+-----------+
FieldRow4   |        116|        117|        118|        119|        120|
            |T34        |T54        |T39        |T62        |T70        |
            +-----------+-----------+-----------+-----------+-----------+
FieldRow5   |        121|        122|        123|        124|        125|
            |T29        |T60        |T7         |T40        |T36        |
            +-----------+-----------+-----------+-----------+-----------+
FieldRow6   |        126|        127|        128|        129|        130|
            |T22        |T6         |T69        |T32        |T55        |
            +-----------+-----------+-----------+-----------+-----------+
FieldRow7   |        131|        132|        133|        134|        135|
            |T21        |T43        |T47        |T51        |T25        |
            +-----------+-----------+-----------+-----------+-----------+
FieldRow8   |        136|        137|        138|        139|        140|
            |T44        |T5         |T66        |T24        |T17        |
            +-----------+-----------+-----------+-----------+-----------+
FieldRow9   |        141|        142|        143|        144|        145|
            |T18        |T58        |T64        |T53        |T65        |
            +-----------+-----------+-----------+-----------+-----------+
FieldRow10  |        146|        147|        148|        149|        150|
            |T50        |T48        |T35        |T8         |T4         |
            +-----------+-----------+-----------+-----------+-----------+
FieldRow11  |        151|        152|        153|        154|        155|
            |T38        |T1         |T2         |T63        |T10        |
```

FieldRow12	156 T59	157 T33	158 T42	159 T67	160 T37
FieldRow13	161 T57	162 T52	163 T49	164 T9	165 T12
FieldRow14	166 T11	167 T19	168 T20	169 T61	170 T68
FieldRow15	201 T68	202 T4	203 T67	204 T60	205 T49
FieldRow16	206 T53	207 T57	208 T28	209 T31	210 T19
FieldRow17	211 T48	212 T20	213 T62	214 T65	215 T63
FieldRow18	216 T39	217 T12	218 T14	219 T2	220 T59
FieldRow19	221 T33	222 T25	223 T38	224 T22	225 T9
FieldRow20	226 T61	227 T69	228 T24	229 T37	230 T1
FieldRow21	231 T17	232 T64	233 T21	234 T52	235 T32
FieldRow22	236 T47	237 T8	238 T55	239 T44	240 T56
FieldRow23	241 T35	242 T16	243 T13	244 T54	245 T26
FieldRow24	246 T10	247 T3	248 T58	249 T15	250 T42
FieldRow25	251 T23	252 T70	253 T29	254 T66	255 T18
FieldRow26	256 T5	257 T51	258 T11	259 T41	260 T7
FieldRow27	261 T46	262 T27	263 T6	264 T36	265 T34
FieldRow28	266 T40	267 T45	268 T30	269 T50	270 T43

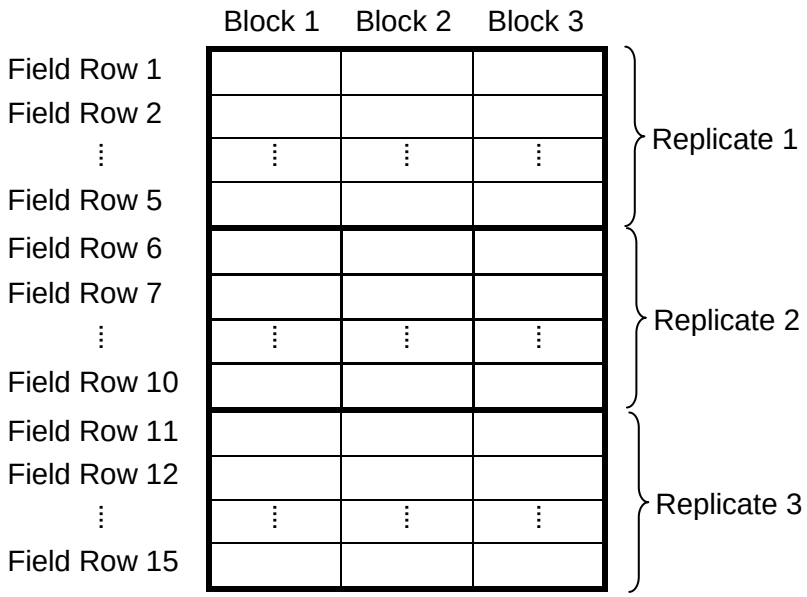
**Note: Cells contain plot numbers on top, treatments/entries below

Sample *csv* data file displayed in the Data Viewer is shown below:

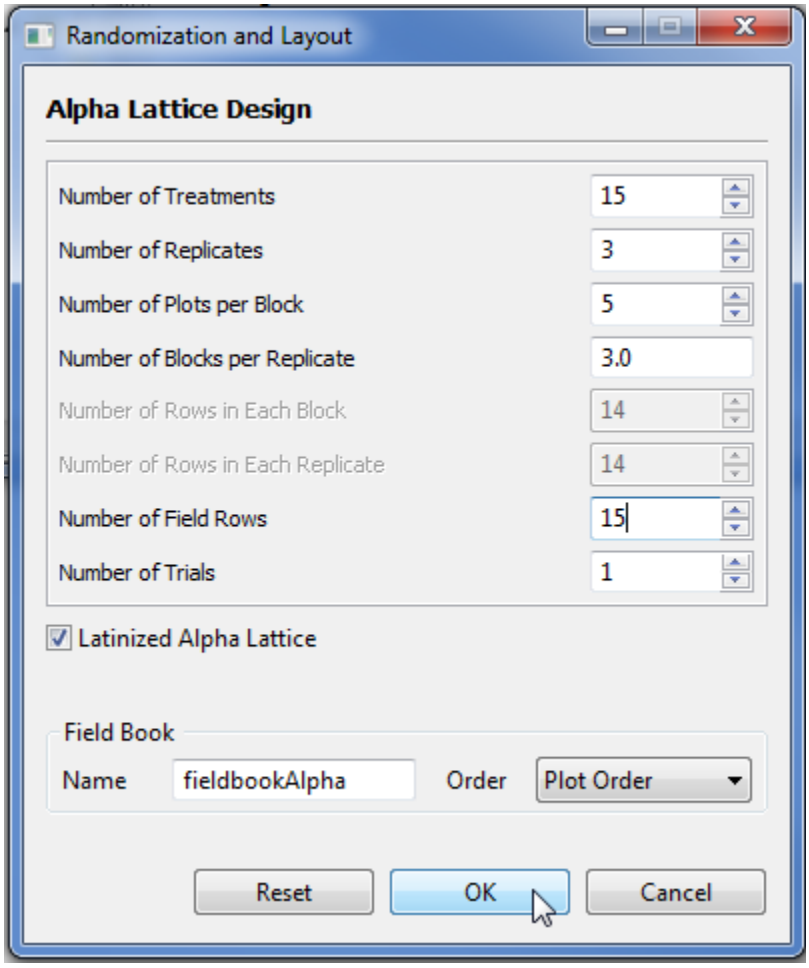


	Trial	Rep	Block	Treatment	PlotNum	FieldRow	FieldCol
1	1	1	1	T27	101	1	1
2	1	1	2	T41	102	1	2
3	1	1	3	T26	103	1	3
4	1	1	4	T3	104	1	4
5	1	1	5	T30	105	1	5
6	1	1	1	T16	106	2	1
7	1	1	2	T14	107	2	2
8	1	1	3	T31	108	2	3
9	1	1	4	T56	109	2	4
10	1	1	5	T46	110	2	5

Suppose we want to generate a randomization and layout for an experiment that will be conducted using Latinized Alpha Lattice design with 15 entries grouped into 5 with three replicates. The planned field layout is shown below:



The completed dialog box should appear as illustrated below:



- Click the **OK** button to generate the randomization. The **Randomization and Layout** dialog box will be minimized and PBTools will automatically display the *csv* data file and the *txt* file created in the Data Viewer tab.

Sample *txt* file displayed in the Data Viewer is shown below:

```
DESIGN PROPERTIES:
      Incomplete Block Design
      Latinized Alpha Lattice Design

DESIGN PARAMETERS:
      Number of Trials = 1
      Number of Treatments = 15
      Number of Replicates = 3
      Number of Plots per Block = 5
      Number of Blocks per Replicate = 3
      Number of Field Rows = 15
```

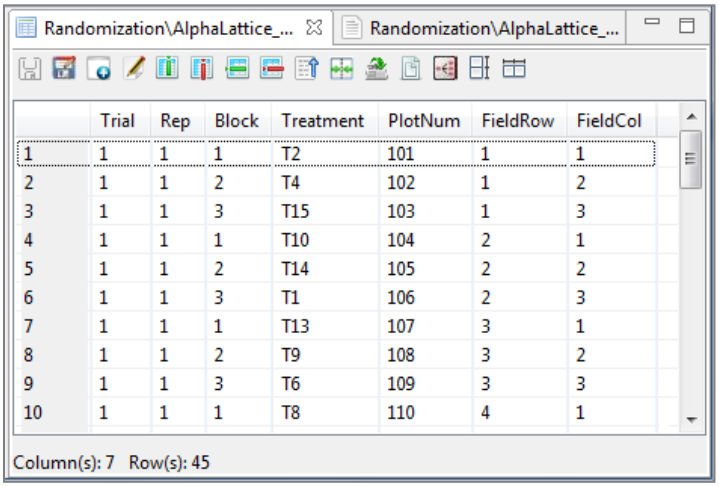
Layout for Latinized Alpha Lattice Design:

Trial = 1

	FieldCol1	FieldCol2	FieldCol3
FieldRow1	101 T2	102 T4	103 T15
FieldRow2	104 T10	105 T14	106 T1
FieldRow3	107 T13	108 T9	109 T6
FieldRow4	110 T8	111 T3	112 T7
FieldRow5	113 T11	114 T5	115 T12
FieldRow6	201 T15	202 T7	203 T9
FieldRow7	204 T6	205 T2	206 T5
FieldRow8	207 T3	208 T11	209 T10
FieldRow9	210 T4	211 T1	212 T8
FieldRow10	213 T12	214 T13	215 T14
FieldRow11	301 T9	302 T8	303 T2
FieldRow12	304 T1	305 T12	306 T3
FieldRow13	307 T5	308 T15	309 T13
FieldRow14	310 T7	311 T10	312 T4
FieldRow15	313 T14	314 T6	315 T11

**Note: Cells contain plot numbers on top, treatments/entries below

- Sample *csv* data file displayed in the Data Viewer is shown below:



	Trial	Rep	Block	Treatment	PlotNum	FieldRow	FieldCol
1	1	1	1	T2	101	1	1
2	1	1	2	T4	102	1	2
3	1	1	3	T15	103	1	3
4	1	1	1	T10	104	2	1
5	1	1	2	T14	105	2	2
6	1	1	3	T1	106	2	3
7	1	1	1	T13	107	3	1
8	1	1	2	T9	108	3	2
9	1	1	3	T6	109	3	3
10	1	1	1	T8	110	4	1

Augmented Design in Randomized Complete Block

The steps to generate randomization for augmented randomized complete block design are listed below:

- Click **Randomization | Augmented Randomized Complete Block Design…** from the main window. The **Randomization and Layout** dialog box will appear.

- Specify the required fields and appropriate options.

  Number of Replicated Treatments
  Specify in this field the number of replicated treatments to be included in the randomization. The default and minimum value is 2 while the maximum value is 500.

  Number of Blocks
  Specify in this field the number of blocks to be generated. The default and minimum value is 2.

  Number of Unreplicated Treatments
  Specify in this field the number of unreplicated treatments to be included in the randomization. The default and minimum value is 2 while the maximum value is 2000.

  Number of Field Rows
  Specify in this field the number of rows in the field layout. It should be greater than or equal to the number of blocks. The default and minimum value is equal to 2.

Number of Trials

Specify in this field the number of trials to be generated. The default and minimum value is 1.

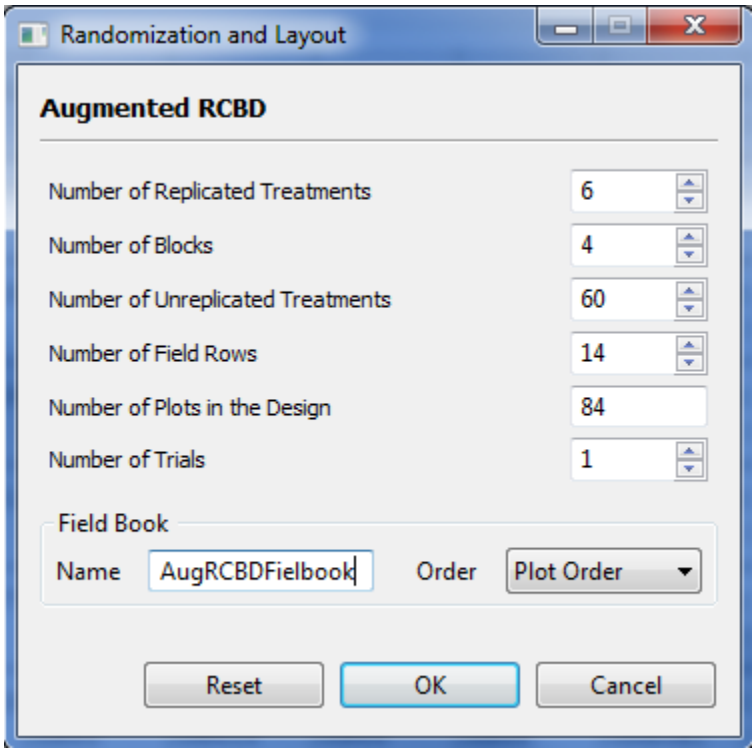
Field Book Filename

This field is required and it specifies the filename where the result of the randomization will be saved. The default filename is *fieldbookAugRCBD*.

Field Book Order

Specify in this field how the plot number will be arranged. There are two options available: *Plot Order* (default value) and *Serpentine*.

For this example, suppose we want to generate a randomization for an experiment which will be conducted using augmented randomized complete block design involving four blocks. The experiment will use six replicated treatments and 60 unreplicated treatments (test entries). The completed dialog box should appear as shown below:



- Click the **OK** button to generate the randomization. The **Randomization and Layout** dialog box will be minimized and PBTools will automatically display the *csv* data file and the *txt* file created in the Data Viewer tab.

Sample *txt* file displayed in the Data Viewer is shown below:

```
DESIGN PROPERTIES:
       Augmented Randomized Complete Block Design (Augmented RCBD)

DESIGN PARAMETERS:
       Number of Trials = 1
       Number of Replicated Treatments = 6
       Levels of Replicated Treatments = check1, check2, check3, ..., check6
       Number of Replicates = 4
       Number of Unreplicated Treatments = 60
       Levels of UnReplicated Treatments = new1, new2, new3, ..., new60
       Number of Field Row =  14

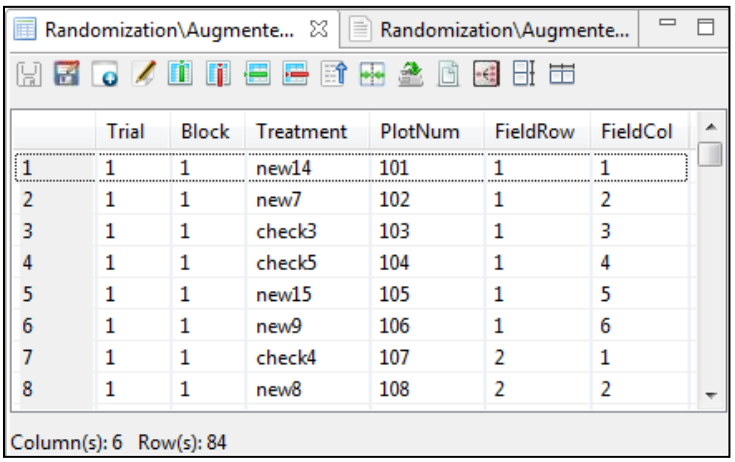

Layout for Augmented Randomized Complete Block Design

Trial = 1
```

FieldRow1	101	102	103	104	105	106
	new14	new7	check3	check5	new15	new9
FieldRow2	107	108	109	110	111	112
	check4	new8	new6	check2	new5	check6
FieldRow3	113	114	115	116	117	118
	new3	new17	new2	new16	new10	new13
FieldRow4	119	120	121	122	123	124
	new11	new18	check1	new4	new12	new1
FieldRow5	201	202	203	204	205	206
	new4	new14	new15	new2	new13	check3
FieldRow6	207	208	209	210	211	212
	new1	new7	new10	new8	check6	check5
FieldRow7	213	214	215	216	217	218
	new16	new3	new18	new9	new6	check1
FieldRow8	219	220	221	222	223	224
	check2	new12	new11	check4	new5	new17
FieldRow9	301	302	303	304	305	306
	new7	new9	new4	new1	new5	new8
FieldRow10	307	308	309	310	311	312
	check6	new11	new2	check5	new10	check1
FieldRow11	313	314	315	316	317	318
	new12	check4	check3	new3	check2	new6
FieldRow12	401	402	403	404	405	406
	new3	new8	check4	new7	check5	check6
FieldRow13	407	408	409	410	411	412
	new10	new2	new9	new11	new12	check1
FieldRow14	413	414	415	416	417	418
	new4	check3	check2	new1	new6	new5

```
 **Note: Cells contain plot numbers on top, treatments/entries below
```

Sample *csv* data file displayed in the Data Viewer is shown below:



	Trial	Block	Treatment	PlotNum	FieldRow	FieldCol
1	1	1	new14	101	1	1
2	1	1	new7	102	1	2
3	1	1	check3	103	1	3
4	1	1	check5	104	1	4
5	1	1	new15	105	1	5
6	1	1	new9	106	1	6
7	1	1	check4	107	2	1
8	1	1	new8	108	2	2

Augmented Design in Latin Square

The steps to generate randomization for augmented Latin square design are listed below:

- Click **Randomization | Augmented Latin Square Design…** from the main window. The **Randomization and Layout** dialog box will appear.

- Specify the required fields and appropriate options.

  Number of Replicated Treatments
  Specify in this field the number of replicated treatments to be included in the randomization. The default and minimum value is 2 while the maximum value is 11.

  Number of Unreplicated Treatments
  Specify in this field the number of unreplicated treatments to be included in the randomization. The default and minimum value is 2 while the maximum value is 2000.

  Number of Field Rows
  Specify in this field the number of rows in the field layout. It should be greater than or equal to the number of replicated treatments. The default and minimum value is equal to 2.

  Number of Trials
  Specify in this field the number of trials to be generated. The default and minimum value is 1.

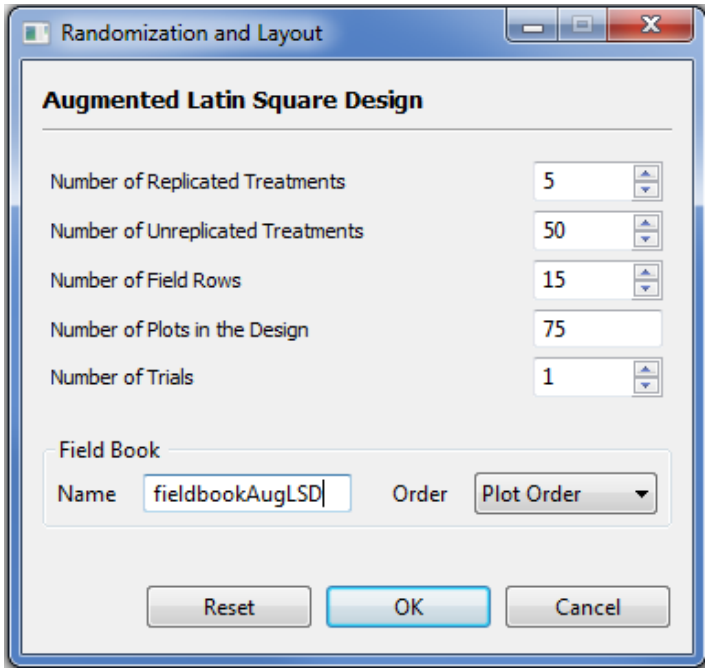
Field Book Filename

This field is required and it specifies the filename where the result of the randomization will be saved. The default filename is *fieldbookAugLSD*.

Field Book Order

Define how the plot number will be arranged. There are two options available: *Plot Order* (default value) and *Serpentine*.

For this example, suppose we want to generate a randomization for an experiment which will be conducted using augmented Latin square design. The experiment will use five replicated treatment and 50 unreplicated treatments (test entries). The completed dialog box should appear as illustrated below:



- Click the **OK** button to generate the randomization. The **Randomization and Layout** dialog box will be minimized and PBTools will automatically display the *csv* data file and *txt* file created in the Data Viewer tab.

Sample *txt* file displayed in the Data Viewer is shown below:

```
DESIGN PROPERTIES:
       Augmented Latin Square Design (Augmented LSD)

DESIGN PARAMETERS:
       Number of Trials = 1
       Number of Replicated Treatments = 5
       Levels of Replicated Treatments = check1, check2, check3, check4, check5
```

```
Number of Unreplicated Treatments = 50
Levels of UnReplicated Treatments = new1, new2, new3, ..., new50
Number of Field Row = 15
Number of Field Column = 5
```

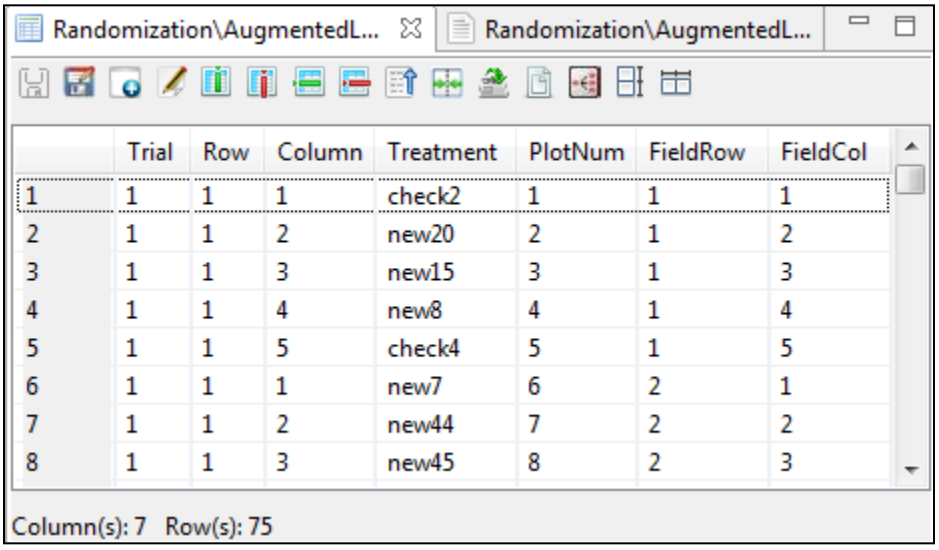
Layout for Augmented Latin Square Design

Trial = 1

	FieldCol1	FieldCol2	FieldCol3	FieldCol4	FieldCol5
FieldRow1	1 check2	2 new20	3 new15	4 new8	5 check4
FieldRow2	6 new7	7 new44	8 new45	9 check5	10 new46
FieldRow3	11 new5	12 check1	13 check3	14 new9	15 new35
FieldRow4	16 check3	17 new25	18 check4	19 new19	20 check5
FieldRow5	21 new6	22 check2	23 new23	24 new31	25 new32
FieldRow6	26 new43	27 new22	28 new1	29 check1	30 new11
FieldRow7	31 new37	32 check3	33 check5	34 new42	35 check1
FieldRow8	36 new50	37 new26	38 new34	39 check2	40 new13
FieldRow9	41 check4	42 new16	43 new14	44 new48	45 new49
FieldRow10	46 new30	47 new21	48 check1	49 new27	50 new29
FieldRow11	51 new3	52 new39	53 new24	54 new33	55 check2
FieldRow12	56 check5	57 check4	58 new10	59 check3	60 new17
FieldRow13	61 new47	62 new41	63 new4	64 check4	65 new2
FieldRow14	66 check1	67 new18	68 new28	69 new38	70 check3
FieldRow15	71 new12	72 check5	73 check2	74 new40	75 new36

**Note: Cells contain plot numbers on top, treatments/entries below

Sample *csv* data file displayed in the Data Viewer is shown below:



	Trial	Row	Column	Treatment	PlotNum	FieldRow	FieldCol
1	1	1	1	check2	1	1	1
2	1	1	2	new20	2	1	2
3	1	1	3	new15	3	1	3
4	1	1	4	new8	4	1	4
5	1	1	5	check4	5	1	5
6	1	1	1	new7	6	2	1
7	1	1	2	new44	7	2	2
8	1	1	3	new45	8	2	3

Row-Column Design

The steps to generate randomization for row-column design are listed below:

- Click **Randomization | Row-Column Design…** from the main window. The **Randomization and Layout** dialog box will appear.

- Specify the required fields and appropriate options.

  Number of Treatments
  Specify in this field the number of treatments to be included in the randomization. The default and minimum value is 9.

  Number of Replicates
  Specify in this field the number of replicates to be generated. The default and minimum value is 2.

  Number of Rows in Each Replicate
  Specify in this field the number of rows per replicate. The default value is 3 and the minimum is 2.

  Number of Field Rows
  Specify in this field the number of rows in the field layout. It should be greater than or equal to the number of rows in each replicate. The default value is 3. If the user chooses

Latinized Row-Column design, this should be equal to the number of rows in each replicate or equal to the number of treatments or equal to the product of the number of replicates and the number of rows in each replicate.

Number of Trials
Specify in this field the number of trials to be generated. The default and minimum value is 1.

Latinized Row-Column Option
Check this option if the user wants Latinized Row-Column design.

Field Book Filename
This field is required and it specifies the filename where the result of the randomization will be saved. The default filename is *fieldbookRowCol*.

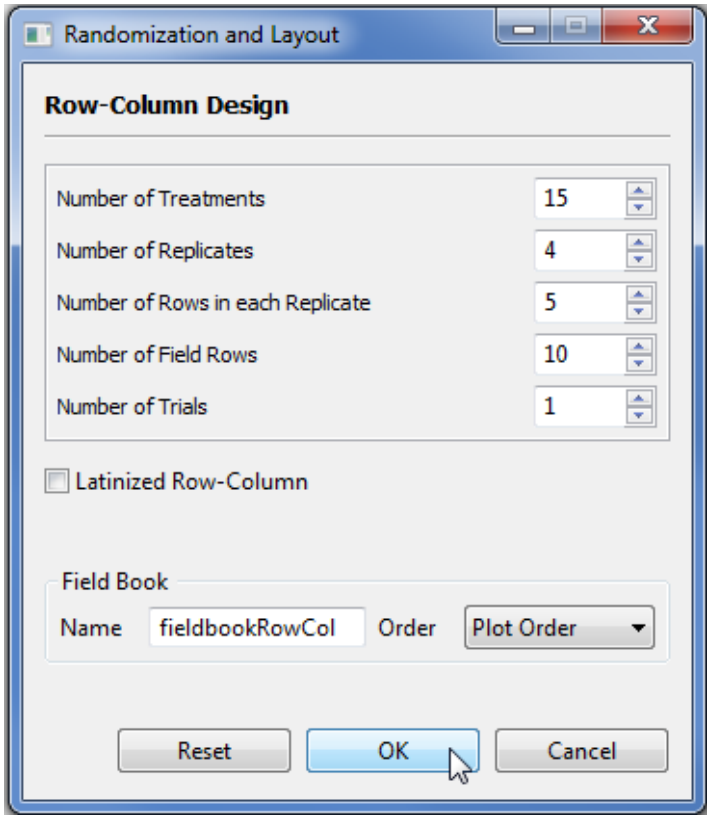
Field Book Order
Specify in this field how the plot number will be arranged. There are two options available: *Plot Order* (default value) and *Serpentine*.

For this example, suppose we want to generate a randomization for an experiment which will be conducted using row-column design with 15 treatments in 5 rows, 3 columns and four replicates. The planned field layout is shown below:

	Column 1	Column 2	Column 3	Column 4	Column 5	Column 6
Field Row 1						
Field Row 2						
Field Row 3						
Field Row 4						
Field Row 5						
Field Row 6						
Field Row 7						
Field Row 8						
Field Row 9						
Field Row 10						

NOTE: ▭ - represents one replicate

The completed dialog box should appear as illustrated below:



- Click the **OK** button to generate the randomization. The **Randomization and Layout** dialog box will be minimized and PBTools will automatically display the *csv* data file and *txt* file created in the Data Viewer tab.

  Sample *txt* file displayed in the Data Viewer is shown below:

```
DESIGN PROPERTIES:
       Incomplete Block Design
       Row-Column Design

DESIGN PARAMETERS:
       Number of Trials = 1
       Number of Treatments = 15
       Number of Replicates = 4
       Number of Rows per Replicate = 5
       Number of Columns per Replicate = 3
       Number of Field Rows = 10
       Number of Field Columns = 6
```

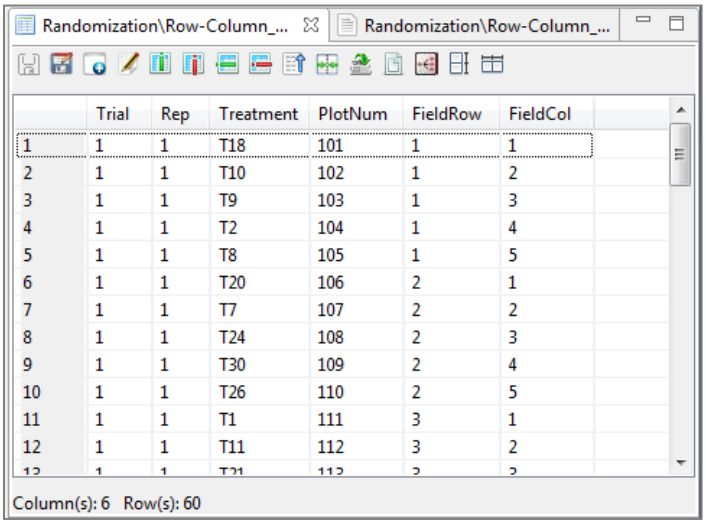
Layout for Row-Column Design:

Trial = 1

	FieldCol1	FieldCol2	FieldCol3	FieldCol4	FieldCol5	FieldCol6
FieldRow1	101 T11	102 T3	103 T9	301 T12	302 T14	303 T10
FieldRow2	104 T14	105 T12	106 T6	304 T9	305 T8	306 T15
FieldRow3	107 T15	108 T2	109 T5	307 T4	308 T6	309 T11
FieldRow4	110 T1	111 T10	112 T4	310 T13	311 T7	312 T2
FieldRow5	113 T13	114 T8	115 T7	313 T5	314 T3	315 T1
FieldRow6	201 T8	202 T4	203 T11	401 T7	402 T1	403 T5
FieldRow7	204 T10	205 T7	206 T12	404 T13	405 T11	406 T8
FieldRow8	207 T5	208 T15	209 T13	407 T6	408 T10	409 T12
FieldRow9	210 T3	211 T6	212 T2	410 T14	411 T9	412 T15
FieldRow10	213 T9	214 T1	215 T14	413 T4	414 T2	415 T3

**Note: Cells contain plot numbers on top, treatments/entries below

Sample *csv* data file displayed in the Data Viewer is shown below:

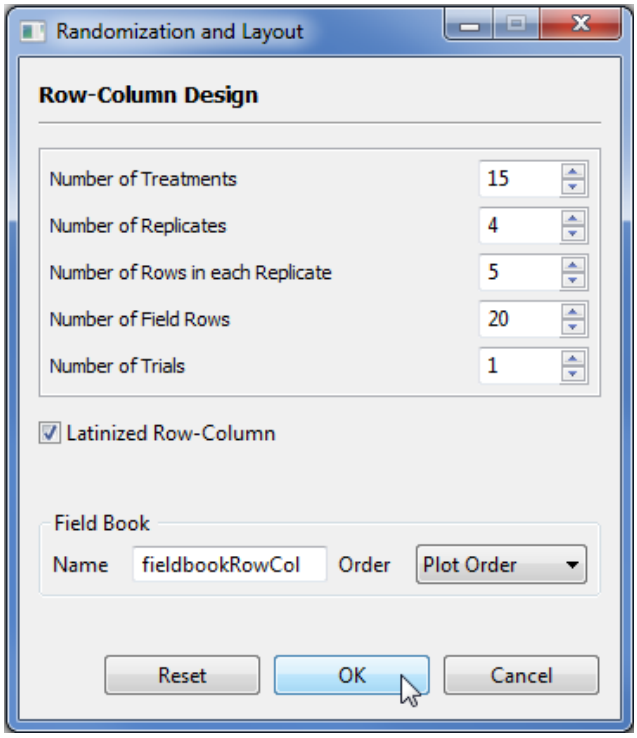


Suppose we want to generate a randomization for the same experiment (15 treatments in 5 rows, 3 columns and four replicates) but this time in Latinized row-column design. The planned field layout is shown below:

	Column 1	Column 2	Column 3
Field Row 1			
Field Row 2			
Field Row 3			
Field Row 4			
Field Row 5			
Field Row 6			
Field Row 7			
Field Row 8			
Field Row 9			
Field Row 10			
Field Row 11			
Field Row 12			
Field Row 13			
Field Row 14			
Field Row 15			
Field Row 16			
Field Row 17			
Field Row 18			
Field Row 19			
Field Row 20			

NOTE: ☐ - represents one replicate

- The completed dialog box should appear as illustrated below:



- Click the **OK** button to generate the randomization. The **Randomization and Layout** dialog box will be minimized and PBTools will automatically display the *csv* data file and *txt* file created in the Data Viewer tab.

  Sample *txt* file displayed in the Data Viewer is shown below:

```
DESIGN PROPERTIES:
       Incomplete Block Design
       Latinized Row-Column Design

DESIGN PARAMETERS:
       Number of Trials = 1
       Number of Treatments = 15
       Number of Replicates = 4
       Number of Rows per Replicate = 5
       Number of Field Rows = 20


Layout for Latinized Row-Column Design:

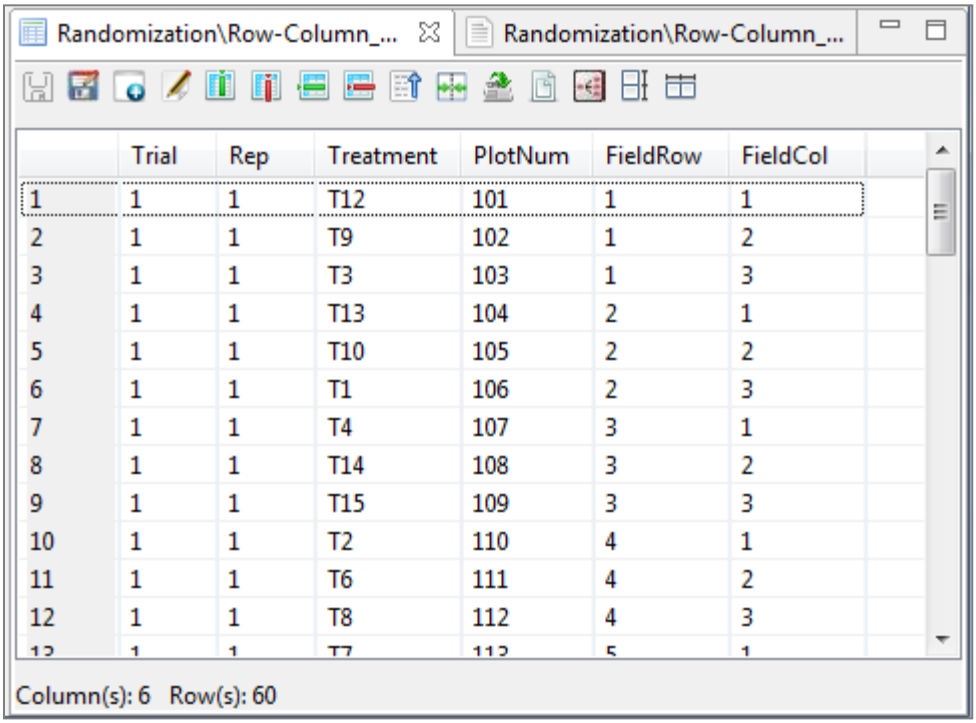

Trial = 1

             FieldCol1   FieldCol2   FieldCol3
            +-----------+-----------+-----------+
FieldRow1   |        101|        102|        103|
            |T12        |T9         |T3         |
            +-----------+-----------+-----------+
```

FieldRow2	104	105	106
	T13	T10	T1
FieldRow3	107	108	109
	T4	T14	T15
FieldRow4	110	111	112
	T2	T6	T8
FieldRow5	113	114	115
	T7	T11	T5
FieldRow6	201	202	203
	T6	T5	T7
FieldRow7	204	205	206
	T15	T2	T9
FieldRow8	207	208	209
	T4	T3	T10
FieldRow9	210	211	212
	T1	T14	T12
FieldRow10	213	214	215
	T11	T13	T8
FieldRow11	301	302	303
	T14	T8	T11
FieldRow12	304	305	306
	T5	T1	T13
FieldRow13	307	308	309
	T9	T4	T6
FieldRow14	310	311	312
	T3	T12	T15
FieldRow15	313	314	315
	T7	T10	T2
FieldRow16	401	402	403
	T8	T7	T12
FieldRow17	404	405	406
	T10	T15	T5
FieldRow18	407	408	409
	T2	T13	T14
FieldRow19	410	411	412
	T11	T9	T4
FieldRow20	413	414	415
	T3	T1	T6

**Note: Cells contain plot numbers on top, treatments/entries below

Sample *csv* data file displayed in the Data Viewer is shown below:



	Trial	Rep	Treatment	PlotNum	FieldRow	FieldCol
1	1	1	T12	101	1	1
2	1	1	T9	102	1	2
3	1	1	T3	103	1	3
4	1	1	T13	104	2	1
5	1	1	T10	105	2	2
6	1	1	T1	106	2	3
7	1	1	T4	107	3	1
8	1	1	T14	108	3	2
9	1	1	T15	109	3	3
10	1	1	T2	110	4	1
11	1	1	T6	111	4	2
12	1	1	T8	112	4	3
13	1	1	T7	113	5	1

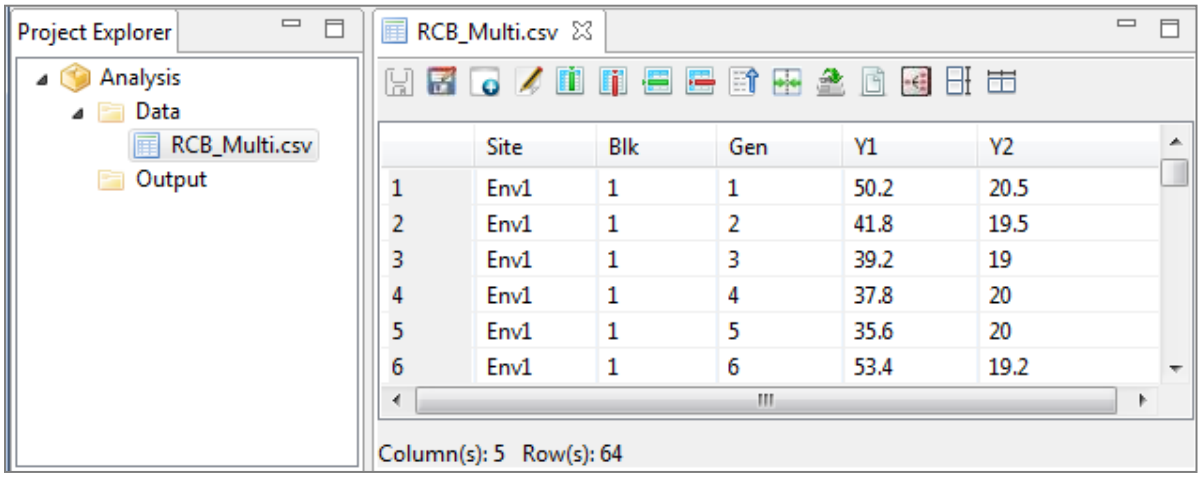
6. Phenotypic Analysis

Single environment analysis

The steps to perform Single-environment Analysis are listed below:

- Create a new PBTools project named *Analysis*.

- Import *RCB_Multi* file from PBTools package. Double-click the file to view it in the Data Viewer. The file contains data for an experiment conducted using Randomized Complete Block (RCB) design for two response variables, *Y1* and *Y2*. The environment variable *Site* has two levels (*Env1* and *Env2*), the blocking variable *Blk* has four levels (1, 2, 3, and 4) and the genotype variable *Gen* has eight levels (1, 2, …, 8).



- Choose **Analysis | Single-environment Analysis…** from the menu bar.

- *Blk* and *Gen* fields are already defined in PBTools package as factors. In some cases wherein the data is not imported from package and is opened for the first time, columns with numeric values will be classified by PBTools as numeric variables. To set these as factors, choose these variables and click the **Set to Factor** button.

- Specify the required fields and appropriate options for the analysis.

Model Specifications Tab

Type of Design

There are seven available experimental designs, namely: Randomized Complete Block Design (RCB), Augmented RCB, Augmented Latin Square, Alpha-Lattice and Row-Column, Latinized Alpha-Lattice and Latinized Row-Column. For the example, select RCB.

Response Variables(s)

This is a required field. This list may contain one or more numeric variables. The analysis will be done per response variable.

Environment

If this field is specified, the analysis will be done per environment level. Otherwise, the data will be treated as if it came from one environment.

Genotype

This is a required field.

'Genotype as Fixed' Option

Select this if genotype is considered as a fixed factor.

'Genotype as Random' Option

Select this if genotype is considered as a random factor.

Block

This field is required if the design is RCB, Augmented RCB, Alpha-Lattice or Latinized Alpha-Lattice.

Replicate

This field is required if the design is Augmented Latin Square, Alpha-Lattice, Latinized Alpha-Lattice, Row-Column or Latinized Row-Column.

Row

This field is required if the design is Augmented Latin Square, Row-Column or Latinized Row-Column.

Column

This field is required if the design is Augmented Latin Square, Row-Column or Latinized Row-Column.

Options Tab

Perform pairwise mean comparisons: Compare with control(s)
If this is selected, the user should specify the control level(s) to be compared with the rest of the genotype levels. This can be done by selecting the level(s) of genotype that are considered as control(s) then click the **Add** button.

Perform pairwise mean comparisons: Perform all comparisons
This option is not recommended when the number of genotype levels is very large. For now, PBTools only executes this option if the number of genotype levels is at most fifteen.

Exclude controls in the estimation of genotypic variance
If this is selected and the design is RCB, Alpha-Lattice or Row-Column, the user should specify the control level(s) to exclude. This can be done by selecting the level(s) of genotype that are considered as control(s) then click the **Add** button.

If this is selected and the design is Augmented RCB or Augmented Latin Square, PBTools will automatically identify the replicated genotypes and set these as controls.

Estimate genotypic and phenotypic correlations
This option is enabled if two or more response variables are specified in the Model Specifications tab.

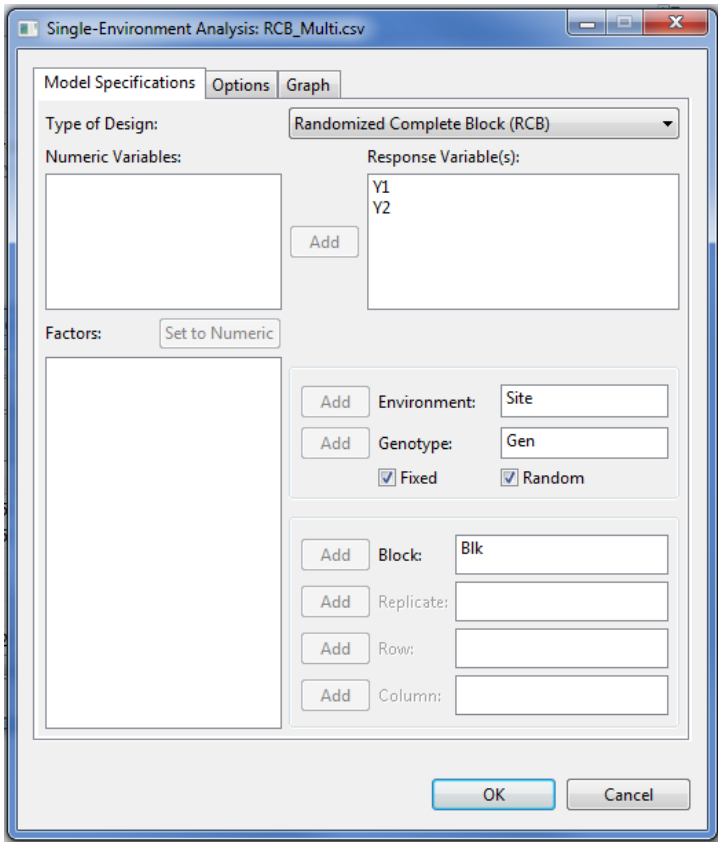
Display Descriptive Statistics
If selected, a summary table with number of missing observations, mean and standard deviation will be displayed.

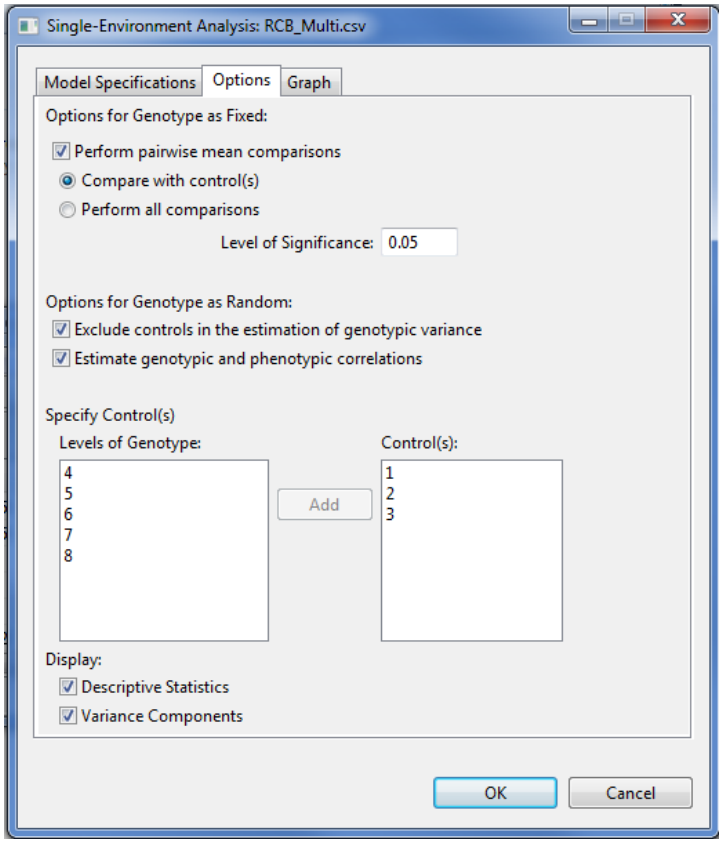
Display Variance Components
If selected, a table with the variance estimates of the some model components will be displayed.

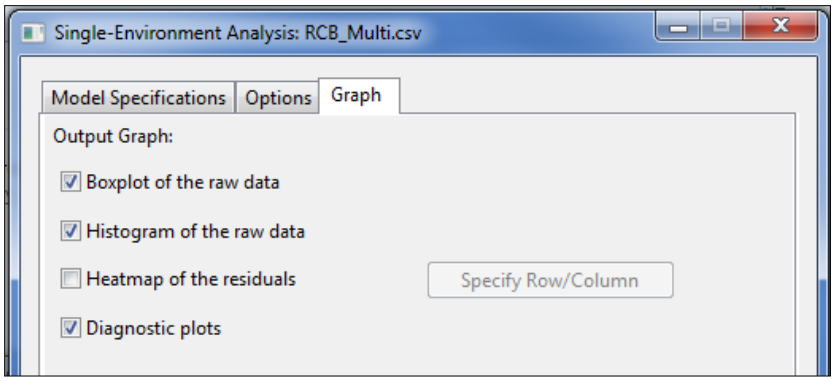
Graph Tab

Graphs like boxplot, histogram, heatmap and diagnostic plots can be generated as part of the output of the analysis. If heatmap option is selected, field row and field column variables should be specified. These columns must be numeric and each observation should have a corresponding unique row-column label.

For the *example*, the completed dialog box should appear as illustrated below:


Single-Environment Analysis: RCB_Multi.csv
Model Specifications
Options
Graph
Type of Design:
Randomized Complete Block (RCB)
Numeric Variables:
Response Variable(s):
Y1
Y2
Add
Factors:
Set to Numeric
Add
Environment:
Site
Add
Genotype:
Gen
Fixed
Random
Add
Block:
Blk
Add
Replicate:
Add
Row:
Add
Column:
OK
Cancel


Single-Environment Analysis: RCB_Multi.csv
Model Specifications
Options
Graph
Options for Genotype as Fixed:
Perform pairwise mean comparisons
Compare with control(s)
Perform all comparisons
Level of Significance:
0.05
Options for Genotype as Random:
Exclude controls in the estimation of genotypic variance
Estimate genotypic and phenotypic correlations
Specify Control(s)
Levels of Genotype:
4
5
6
7
8
Add
Control(s):
1
2
3
Display:
Descriptive Statistics
Variance Components
OK
Cancel



- Click **OK**. Processing the analysis may take a while.

- Sample output of the analysis (only results from *response variable* = *Y1* and *Site* = *Env1*) is shown below:

```
DATA FILE: E:/PBTools/Projects/Analysis/Data/RCB_Multi.csv

SINGLE-ENVIRONMENT ANALYSIS

DESIGN: Randomized Complete Block (RCB)

==============================
GENOTYPE AS: Fixed
==============================

------------------------------
RESPONSE VARIABLE: Y1
------------------------------

DESCRIPTIVE STATISTICS:

  Variable Site N_NonMissObs     Mean   StdDev
1       Y1 Env1           32 42.49063 6.739166
2       Y1 Env2           32 42.99063 5.741213

------------------------------
ANALYSIS FOR: Site  =  Env1
------------------------------

DATA SUMMARY:

Number of observations read:  32
Number of observations used:  32

 Factors   Number of Levels   Levels
 Gen       8                  1  2  3 ... 8
 Blk       4                  1  2  3  4
```

```
VARIANCE COMPONENTS TABLE:

    Groups  Variance Std.Dev.
1      Blk  2.857857 1.690520
2 Residual 13.563289 3.682837


TESTING FOR THE SIGNIFICANCE OF GENOTYPIC EFFECT:

Analysis of Variance Table with Satterthwaite Denominator Df
    Df Sum Sq  Mean Sq F value   Denom Pr(>F)
Gen  7 1013.8 144.8285  10.678 20.9994 0.0000


GENOTYPE LSMEANS AND STANDARD ERRORS:

  Gen LSMean StdErrMean
1   1 41.900   2.026107
2   2 43.800   2.026107
3   3 37.300   2.026107
4   4 41.150   2.026107
5   5 32.500   2.026107
6   6 52.625   2.026107
7   7 43.900   2.026107
8   8 46.750   2.026107


STANDARD ERROR OF THE DIFFERENCE (SED):

           Estimate
 Minimum   2.6042
 Average   2.6042
 Maximum   2.6042


SIGNIFICANT PAIRWISE COMPARISONS (IF ANY):
Compared with control(s)

  Trmt[i] Trmt[j] Difference      Lower     Upper
1       5       1     -9.400 -16.206616 -2.593384
2       6       1     10.725   3.918384 17.531616
3       5       2    -11.300 -18.104529 -4.495471
4       6       2      8.825   2.020471 15.629529
5       6       3     15.325   8.520974 22.129026
6       8       3      9.450   2.645974 16.254026
```

```
==============================
GENOTYPE AS: Random
==============================

------------------------------
RESPONSE VARIABLE: Y1
------------------------------

DESCRIPTIVE STATISTICS:

  Variable Site N_NonMissObs     Mean   StdDev
1       Y1 Env1           32 42.49063 6.739166
2       Y1 Env2           32 42.99063 5.741213

------------------------------
ANALYSIS FOR: Site  =  Env1
------------------------------

DATA SUMMARY:
Number of observations read:  32
Number of observations used:  32

 Factors   Number of Levels   Levels
 Gen       8                 1  2  3 ... 8
 Blk       4                 1  2  3  4


VARIANCE COMPONENTS TABLE:

      Groups  Variance Std.Dev.
1 Test:Check 51.720292 7.191682
2        Blk  2.857874 1.690525
3   Residual 13.563282 3.682836


TESTING FOR THE SIGNIFICANCE OF GENOTYPIC EFFECT USING -2 LOGLIKELIHOOD RATIO
TEST:

Formula for Model1:  Y1 ~ 1 + Check + (1|Blk) + (1|Test:Check)
Formula for Model2:  Y1 ~ 1 + Check + (1|Blk)

            AIC      BIC    logLik   Chisq Df Pr(>Chisq)
Model2 205.5525 214.3469 -96.77626
Model1 187.7383 197.9985 -86.86916 19.8142  1     0.0000


TESTING FOR THE SIGNIFICANCE OF CHECK EFFECT:

Analysis of Variance Table with Satterthwaite Denominator Df
      Df Sum Sq Mean Sq F value  Denom Pr(>F)
Check  3 8.1229  2.7076  0.1996 3.9998 0.8917
```

```
PREDICTED MEANS:

  Gen    Means
1   4 41.28751
2   5 33.16972
3   6 52.05649
4   7 43.86831
5   8 46.54296


CHECK/CONTROL LSMEANS:

  Gen LSMean StdErrMean
1   1   41.9   7.471644
2   2   43.8   7.471644
3   3   37.3   7.471644


HERITABILITY:

0.94

==============================

GENOTYPIC CORRELATIONS:

Site =  Env1

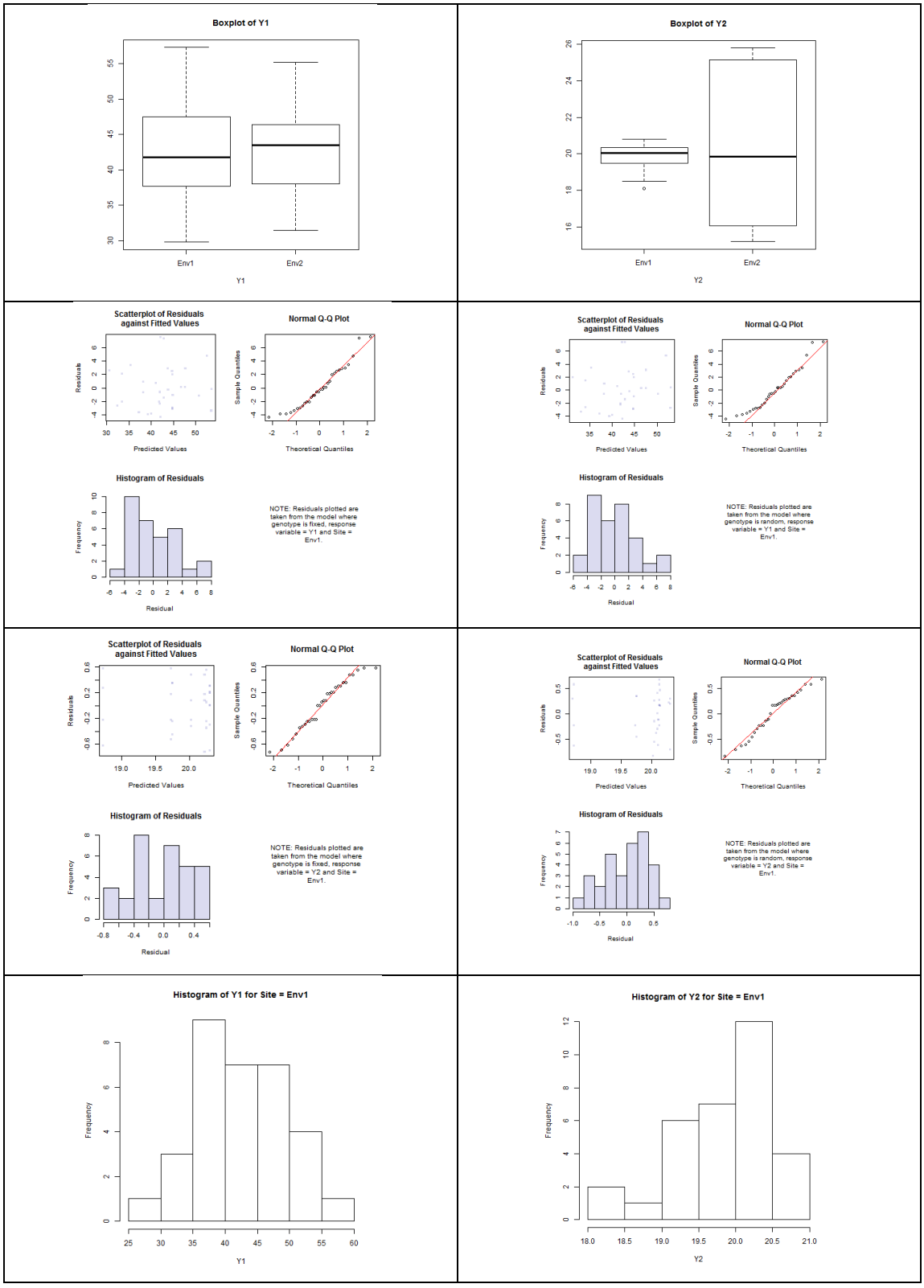
   Y1      Y2
Y1         -1.0000
Y2 -1.0000


PHENOTYPIC CORRELATIONS:

Site =  Env1

   Y1      Y2
Y1         -0.8154
Y2 -0.8154
```

- Generated graphs can be viewed by clicking the Graph Tab of the displayed analysis folder. Sample generated graphs are shown below:



Multi-environment analysis

One-Stage Analysis

The steps to perform Multi-environment Analysis using the raw data are listed below:

- Import *RCB_ME.csv* from the project named *SampleProject*. Double-click the file to view it in the Data Viewer. The file contains data for an experiment conducted using Randomized Complete Block (RCB) design for response variable *Yield*. The environment variable *Env* has eleven levels (*E1*, *E2, …, E11*), the blocking variable *Block* has three levels (1, 2 and 3) and the genotype variable *Genotype* has fifteen levels (GEN1, GEN2, …, GEN15).

- Choose **Analysis | Multi-environment Analysis | One-Stage Analysis…** from the menu bar.

- Opening the data for the first time, *Block* is regarded by R as a numerical variable. To set this as factor, choose this variable and click on the **Set to Factor** button.

- Specify the required fields and appropriate options for the analysis.

  Model Specifications Tab

  Type of Design
  There are three available experimental designs, namely: Randomized Complete Block Design (RCB), Alpha-Lattice, Row-Column, Latinized Alpha-Lattice and Latinized Row-Column. For the example, select RCB.

  Response Variables(s)
  This is a required field. This list may contain one or more numeric variables. The analysis will be done per response variable.

  Environment
  This is a required field.

  Genotype
  This is a required field.

  'Genotype as Fixed' Option
  Select this if genotype is considered as a fixed factor.

'Genotype as Random' Option
Select this if genotype is considered as a random factor.

Block
This field is required if the design is RCB, Augmented RCB or Alpha-Lattice.

Replicate
This field is required if the design is Augmented Latin Square, Alpha-Lattice or Row-Column.

Row
This field is required if the design is Augmented Latin Square or Row-Column.

Column
This field is required if the design is Augmented Latin Square or Row-Column.

Options Tab

Perform pairwise mean comparisons: Compare with control(s)
If this is selected, the user should specify the control level(s) to be compared with the rest of the genotype levels. This can be done by selecting the level(s) of genotype that are considered as control(s) then click the **Add** button.

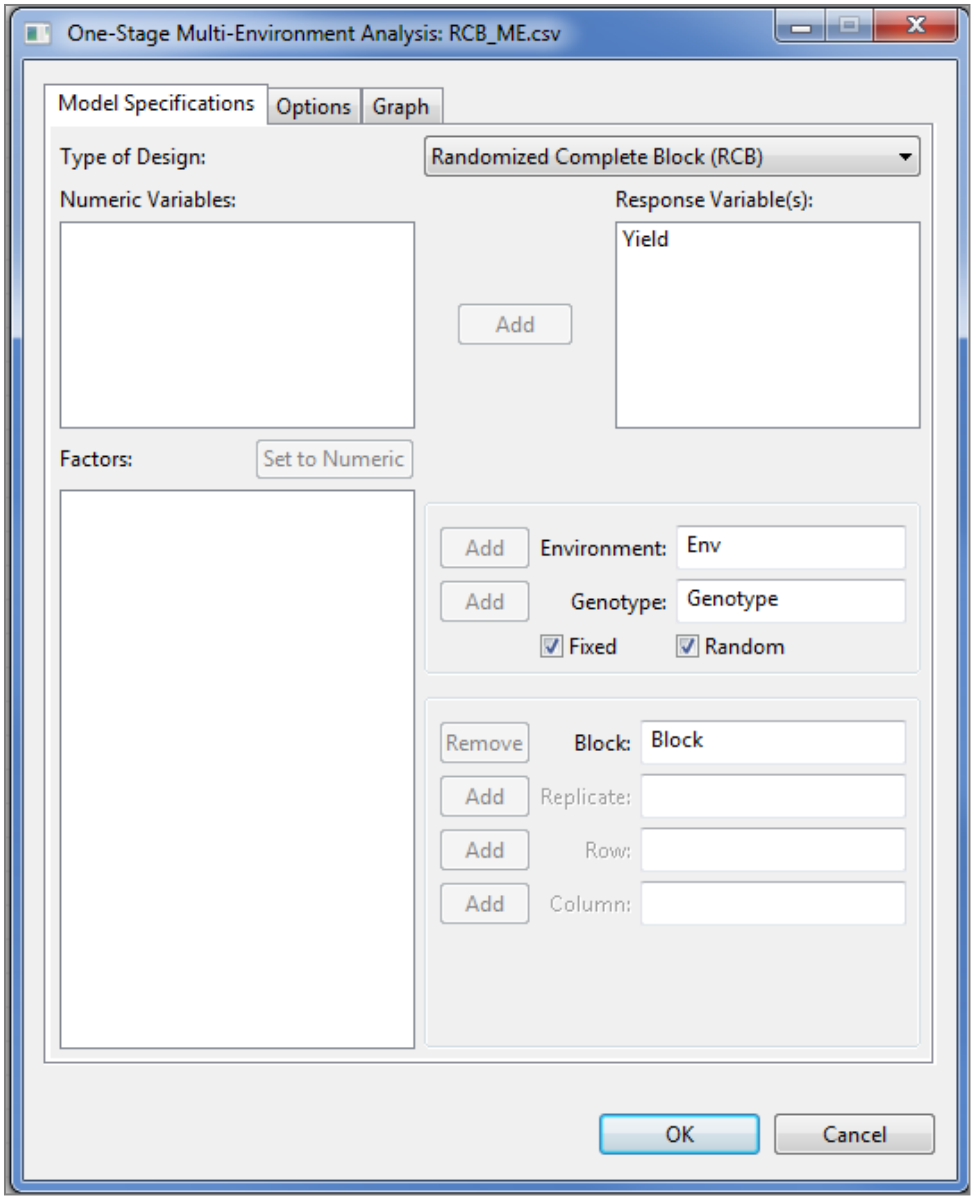
Perform pairwise mean comparisons: Perform all comparisons
This option is not recommended when the number of genotype levels is very large. For now, PBTools only executes this option if the number of genotype levels is at most fifteen.

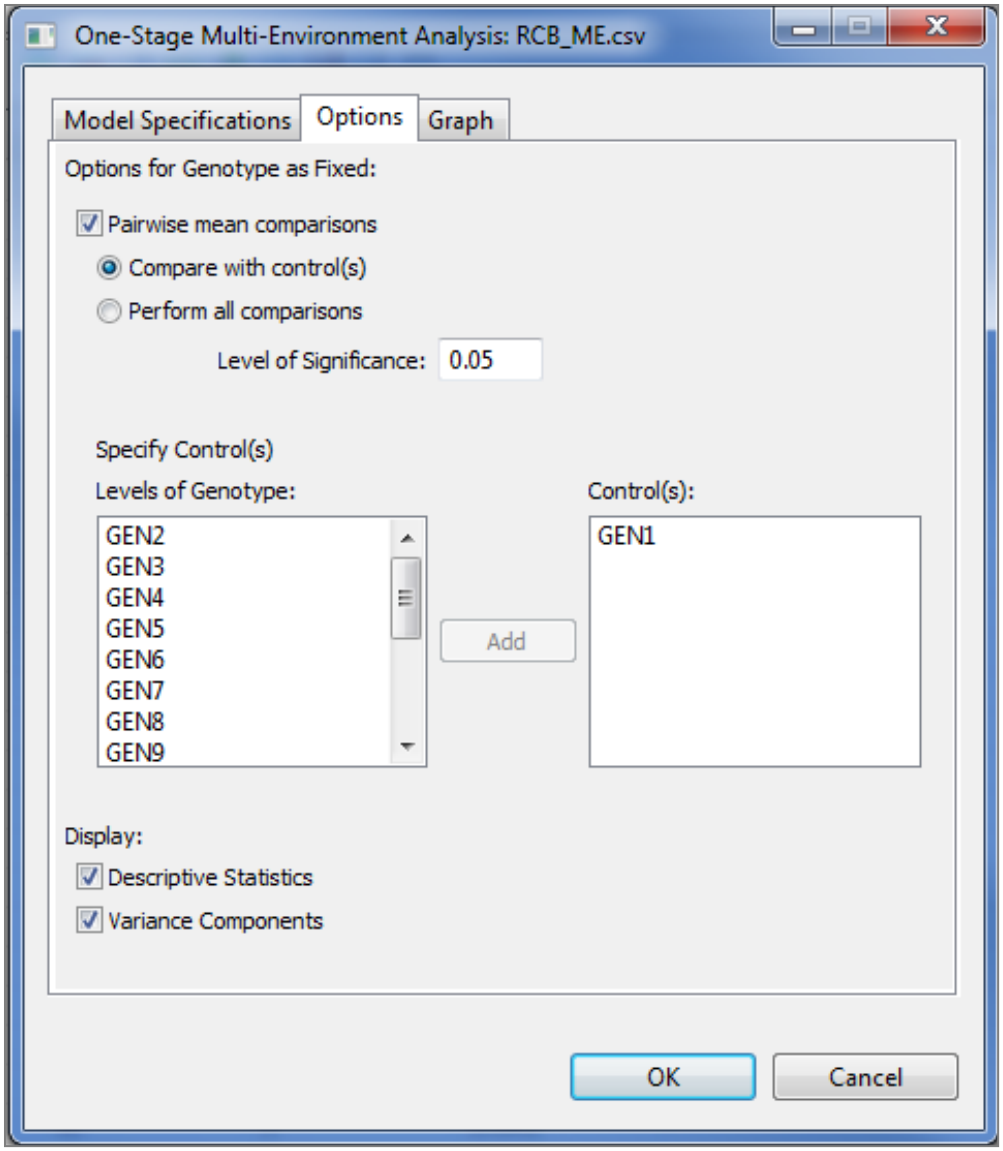
Display
The user has the options to display descriptive statistics and variance components.

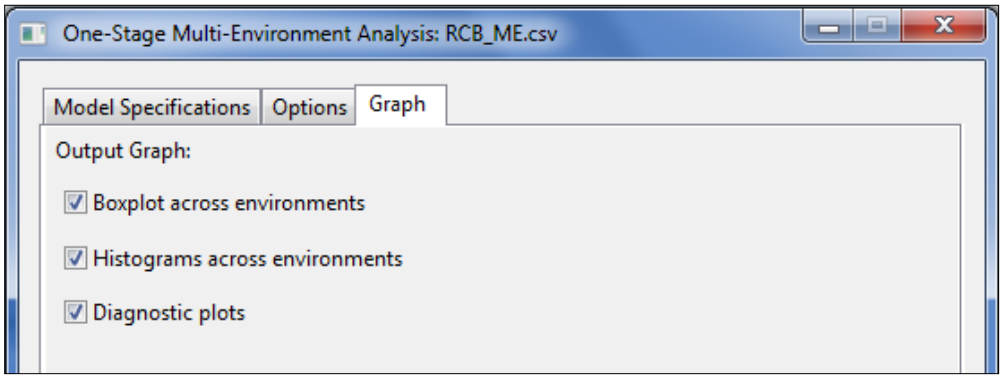
Graph Tab

Graphs like boxplot, histogram and diagnostic plots can be generated as part of the output of the analysis.

For the *example*, the completed dialog box should appear as illustrated below:


One-Stage Multi-Environment Analysis: RCB_ME.csv
Model Specifications
Options
Graph
Type of Design:
Randomized Complete Block (RCB)
Numeric Variables:
Response Variable(s):
Yield
Add
Factors:
Set to Numeric
Add
Environment:
Env
Add
Genotype:
Genotype
Fixed
Random
Remove
Block:
Block
Add
Replicate:
Add
Row:
Add
Column:
OK
Cancel


One-Stage Multi-Environment Analysis: RCB_ME.csv
Model Specifications
Options
Graph
Options for Genotype as Fixed:
Pairwise mean comparisons
Compare with control(s)
Perform all comparisons
Level of Significance:
0.05
Specify Control(s)
Levels of Genotype:
GEN2
GEN3
GEN4
GEN5
GEN6
GEN7
GEN8
GEN9
Add
Control(s):
GEN1
Display:
Descriptive Statistics
Variance Components
OK
Cancel


One-Stage Multi-Environment Analysis: RCB_ME.csv
Model Specifications
Options
Graph
Output Graph:
Boxplot across environments
Histograms across environments
Diagnostic plots

- Click **OK**.

- Sample output of the analysis is shown below:

```
DATA FILE: E:/PBTools/Projects/Analysis/Data/RCB_ME.csv

MULTI-ENVIRONMENT ANALYSIS (ONE-STAGE)

DESIGN: Randomized Complete Block (RCB)

==============================
GENOTYPE AS: Fixed
==============================

------------------------------
RESPONSE VARIABLE: Yield
------------------------------

DATA SUMMARY:

Number of observations read:  495
Number of observations used:  495

 Factors   Number of Levels   Levels
 Env       11                E1  E10  E11 ... E9
 Genotype  15                GEN1  GEN10  GEN11 ... GEN9
 Block     3                 1  2  3


DESCRIPTIVE STATISTICS:

  Variable N_NonMissObs     Mean   StdDev
1    Yield          495 1718.596 1117.791


VARIANCE COMPONENTS TABLE:

        Groups   Variance  Std.Dev.
1 Genotype:Env 664747.696 815.32061
2    Block:Env   6497.161  80.60497
3          Env 530979.863 728.68365
4     Residual  94709.288 307.74874


TESTING FOR THE SIGNIFICANCE OF GENOTYPIC EFFECT:

Analysis of Variance Table with Satterthwaite Denominator Df
         Df  Sum Sq Mean Sq F value    Denom Pr(>F)
Genotype 14 1386210   99015  1.0455 139.9947 0.4127
```

TESTING FOR THE SIGNIFICANCE OF ENVIRONMENT EFFECT USING -2 LOGLIKELIHOOD RATIO TEST:

Formula for Model1: Yield ~ 1 + Genotype + (1|Env) + (1|Block:Env) + (1|Genotype:Env)
Formula for Model2: Yield ~ 1 + Genotype + (1|Block:Env) + (1|Genotype:Env)

	AIC	BIC	logLik	Chisq	Df	Pr(>Chisq)
Model2	7513.743	7589.425	-3738.871			
Model1	7457.593	7537.480	-3709.797	58.1496	1	0.0000

TESTING FOR THE SIGNIFICANCE OF GENOTYPE X ENVIRONMENT EFFECT USING -2 LOGLIKELIHOOD RATIO TEST:

Formula for Model1: Yield ~ 1 + Genotype + (1|Env) + (1|Block:Env) + (1|Genotype:Env)
Formula for Model2: Yield ~ 1 + Genotype + (1|Env) + (1|Block:Env)

	AIC	BIC	logLik	Chisq	Df	Pr(>Chisq)
Model2	7942.529	8018.211	-3953.264			
Model1	7457.593	7537.480	-3709.797	486.9356	1	0.0000

GENOTYPE X ENVIRONMENT MEANS:

	Genotype	E1	E5	E4	E11	…	E8
1	GEN1	2958.164	352.6978	1294.798	2072.8784	…	299.9992
2	GEN10	2199.114	1012.1162	1856.555	2351.7368	…	717.7737
3	GEN11	2777.579	1125.8198	1359.307	3161.8980	…	302.4992
4	GEN12	2663.602	954.9771	1964.795	3042.9571	…	358.8341
5	GEN13	2726.812	1655.7099	2150.647	3539.0740	…	581.7587
6	GEN14	2397.222	1256.6844	1706.243	6681.1756	…	694.8774
7	GEN15	1955.552	1785.8101	2090.610	1542.3468	…	309.5644
8	GEN2	2981.912	1391.8241	1311.927	1558.7381	…	314.3661
9	GEN3	2099.569	280.0127	1609.770	519.7356	…	534.5689
10	GEN4	2325.432	426.4795	2225.294	1778.8608	…	698.1242
11	GEN5	1764.656	484.0382	1470.849	2145.3480	…	313.7380
12	GEN6	1962.796	368.6027	2626.068	1677.6422	…	722.2527
13	GEN7	2731.895	496.6144	1934.025	2424.5930	…	521.3835
14	GEN8	2247.593	858.3659	2006.610	2280.9466	…	244.3071
15	GEN9	2192.044	542.0348	1291.421	1071.0839	…	274.4865

GENOTYPE LSMEANS AND STANDARD ERRORS:

	Genotype	LSMean	StdErrMean
1	GEN1	1500.965	334.2864
2	GEN10	1804.918	334.2864
3	GEN11	1902.555	334.2864
4	GEN12	2023.497	334.2864

```
5     GEN13 1953.552   334.2864
6     GEN14 2163.924   334.2864
7     GEN15 1829.575   334.2864
8      GEN2 1826.130   334.2864
9      GEN3 1615.296   334.2864
10     GEN4 1557.518   334.2864
11     GEN5 1246.419   334.2864
12     GEN6 1606.112   334.2864
13     GEN7 1523.806   334.2864
14     GEN8 1871.903   334.2864
15     GEN9 1352.776   334.2864


STANDARD ERROR OF THE DIFFERENCE (SED):

           Estimate
 Minimum   355.8134
 Average   355.8134
 Maximum   355.8134


SIGNIFICANT PAIRWISE COMPARISONS (IF ANY):
Compared with control(s)

(No significant pairwise comparisons.)



==============================
GENOTYPE AS: Random
==============================

------------------------------
RESPONSE VARIABLE: Yield
------------------------------

DATA SUMMARY:

Number of observations read:  495
Number of observations used:  495

 Factors   Number of Levels   Levels
 Env       11                E1  E10  E11 ... E9
 Genotype  15                GEN1  GEN10  GEN11 ... GEN9
 Block     3                 1  2  3


DESCRIPTIVE STATISTICS:

  Variable N_NonMissObs     Mean   StdDev
1    Yield          495 1718.596 1117.791
```

```
VARIANCE COMPONENTS TABLE:

        Groups   Variance  Std.Dev.
1 Genotype:Env 664747.522 815.32050
2    Block:Env   6497.219  80.60533
3     Genotype   2876.978  53.63747
4          Env 530984.967 728.68715
5     Residual  94709.311 307.74878


TESTING FOR THE SIGNIFICANCE OF GENOTYPIC EFFECT USING -2 LOGLIKELIHOOD RATIO
TEST:

Formula for Model1:   Yield ~ 1 + (1|Genotype) + (1|Env) + (1|Block:Env) +
(1|Genotype:Env)
Formula for Model2:  Yield ~ 1 + (1|Env) + (1|Block:Env) + (1|Genotype:Env)

            AIC      BIC    logLik  Chisq Df Pr(>Chisq)
Model2 7627.446 7648.469 -3808.723
Model1 7629.433 7654.661 -3808.717 0.0127  1     0.9102


TESTING FOR THE SIGNIFICANCE OF ENVIRONMENT EFFECT USING -2 LOGLIKELIHOOD RATIO
TEST:

Formula for Model1:   Yield ~ 1 + (1|Genotype) + (1|Env) + (1|Block:Env) +
(1|Genotype:Env)
Formula for Model2:     Yield  ~  1  +  (1|Genotype)  +  (1|Block:Env)  +
(1|Genotype:Env)

            AIC      BIC    logLik   Chisq Df Pr(>Chisq)
Model2 7686.930 7707.953 -3838.465
Model1 7629.433 7654.661 -3808.717 59.4969  1     0.0000


TESTING  FOR  THE  SIGNIFICANCE  OF  GENOTYPE  X  ENVIRONMENT  EFFECT  USING  -2
LOGLIKELIHOOD RATIO TEST:

Formula for Model1:   Yield ~ 1 + (1|Genotype) + (1|Env) + (1|Block:Env) +
(1|Genotype:Env)
Formula for Model2:  Yield ~ 1 + (1|Genotype) + (1|Env) + (1|Block:Env)

            AIC      BIC    logLik    Chisq Df Pr(>Chisq)
Model2 8114.369 8135.392 -4052.184
Model1 7629.433 7654.661 -3808.717 486.9356  1     0.0000
```

GENOTYPE X ENVIRONMENT MEANS:

	Genotype	E1	E5	E4	E11	…	E8
1	GEN1	2967.628	362.1033	1304.239	2082.3420	…	309.3893
2	GEN10	2195.396	1008.3402	1852.814	2348.0189	…	713.9822
3	GEN11	2769.627	1117.8096	1351.332	3153.9459	…	294.4734
4	GEN12	2650.405	941.7219	1951.575	3029.7600	…	345.5635
5	GEN13	2716.648	1645.4881	2140.460	3528.9102	…	571.5213
6	GEN14	2377.935	1237.3393	1686.933	6661.8885	…	675.5168
7	GEN15	1950.765	1780.9648	2085.800	1537.5596	…	304.7036
8	GEN2	2977.274	1387.1282	1307.267	1554.1003	…	309.6547
9	GEN3	2104.075	284.4601	1614.252	524.2411	…	539.0008
10	GEN4	2332.444	433.4326	2232.282	1785.8719	…	705.0617
11	GEN5	1785.159	504.4827	1491.329	2165.8505	…	334.1670
12	GEN6	1967.700	373.4483	2630.949	1682.5460	…	727.0829
13	GEN7	2740.368	505.0294	1942.476	2433.0661	…	529.7830
14	GEN8	2240.970	851.6849	1999.964	2274.3237	…	237.6107
15	GEN9	2207.934	557.8670	1307.289	1086.9741	…	290.3031

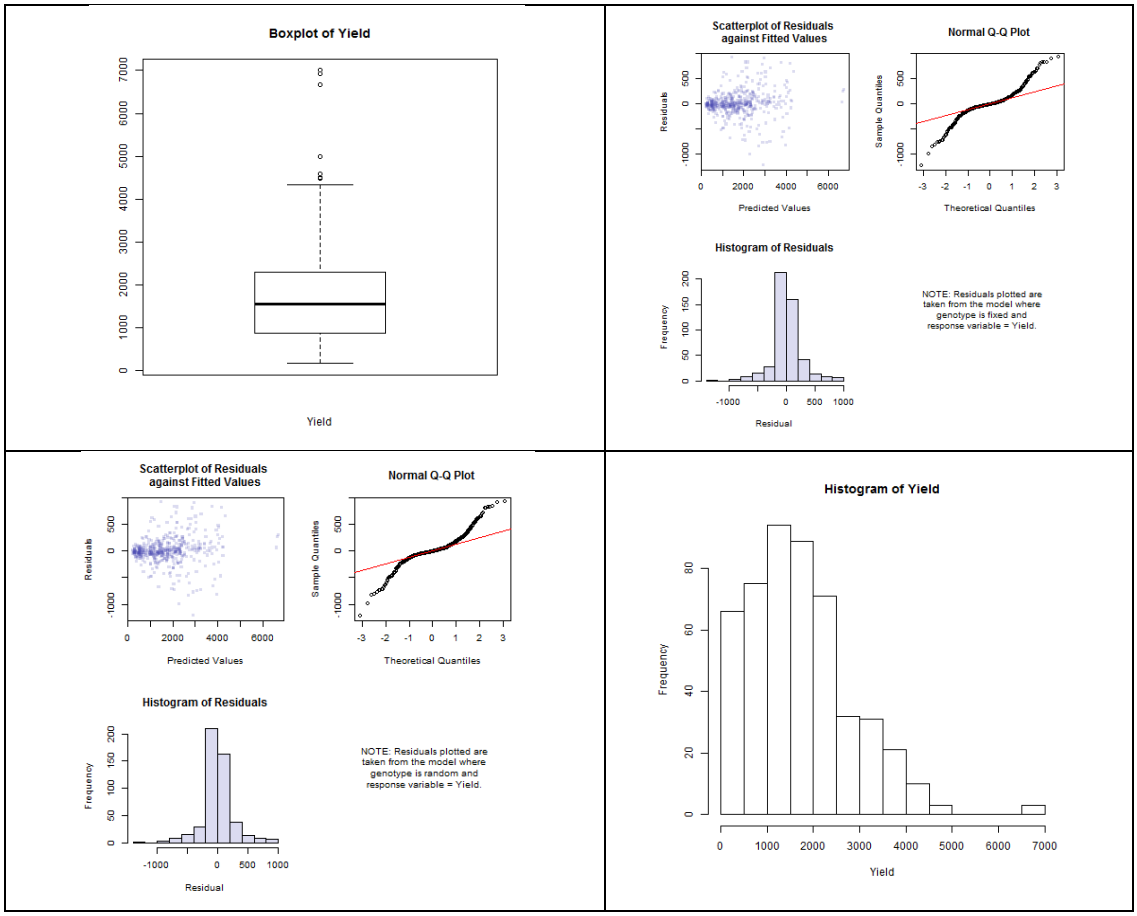
PREDICTED GENOTYPE MEANS:

	Genotype	Mean
1	GEN1	1709.135
2	GEN10	1722.349
3	GEN11	1726.594
4	GEN12	1731.851
5	GEN13	1728.811
6	GEN14	1737.956
7	GEN15	1723.421
8	GEN2	1723.271
9	GEN3	1714.106
10	GEN4	1711.594
11	GEN5	1698.069
12	GEN6	1713.706
13	GEN7	1710.128
14	GEN8	1725.261
15	GEN9	1702.693

HERITABILITY:

0.05

- Generated graphs can be viewed by clicking the Graph Tab of the displayed analysis folder. Sample generated graphs are shown below:



Two-Stage Analysis

The steps to perform Multi-environment Analysis using the summary of data from single-environment analysis as input are listed below:

- Import *Two_stage* file from PBTools package. Double-click the file to view it in the Data Viewer. The file contains four columns (variables) for yield: *Yield_Mean*, *Yield_StdErrMean*, *Yield_sigma2*, *Yield_No.rep* and the *Env* and *Genotype* factors, with 10 and 15 levels, respectively.

- Choose **Analysis | Multi-environment Analysis | Two-Stage Analysis…** from the menu bar.

- Specify the required fields and appropriate options for the analysis.

Model Specifications Tab

Weight Option
The user has the option to either apply no weights or use the reciprocal of the variance of the mean as weights.

Response Variables
This is a required field which contains the column name corresponding to the mean of a trait.

Standard Error
If the selected weight option is the reciprocal of the variance of the mean (*1/(sem^2)*), this field should be specified to proceed to the entire process of the analysis.

Residual Variance
This field should be specified to proceed to the entire process of the analysis.

Number of Replicates
This field should be specified to proceed to the entire process of the analysis.

For each trait, the user has to specify the response variable, its standard error (if required), residual variance and number of replicates. To indicate that this set of variables corresponds to one trait, click the **Add to Table** button. The variables corresponding to one trait will be displayed in a row of the summary table. Do the same for the rest of the traits.

Environment
This is a required field.

Genotype
This is a required field.

'Genotype as Fixed' Option
Select this if genotype is considered as a fixed factor

'Genotype as Random' Option
Select this if genotype is considered as a random factor

If genotype is considered as a fixed factor and PBTools detects that in the summary table a row has missing standard error or residual variance or number of replicates, a dialog box will appear prompting the user that only AMMI and stability analysis can be done.

Options Tab

Perform pairwise mean comparisons: Compare with control(s)

If this is selected, the user should specify the control level(s) to be compared with the rest of the genotype levels. This can be done by selecting the level(s) of genotype that are considered as control(s) then click the **Add** button.

Perform pairwise mean comparisons: Perform all comparisons

This option is not recommended when the number of genotype levels is very large. For now, PBTools only executes this option if the number of genotype levels is at most fifteen.

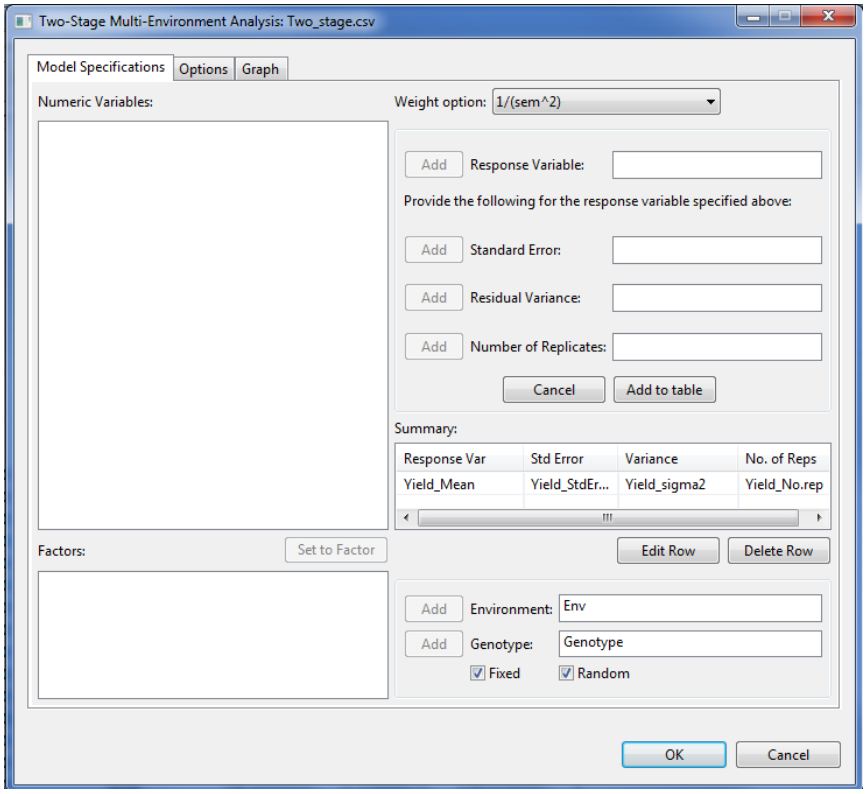
Display

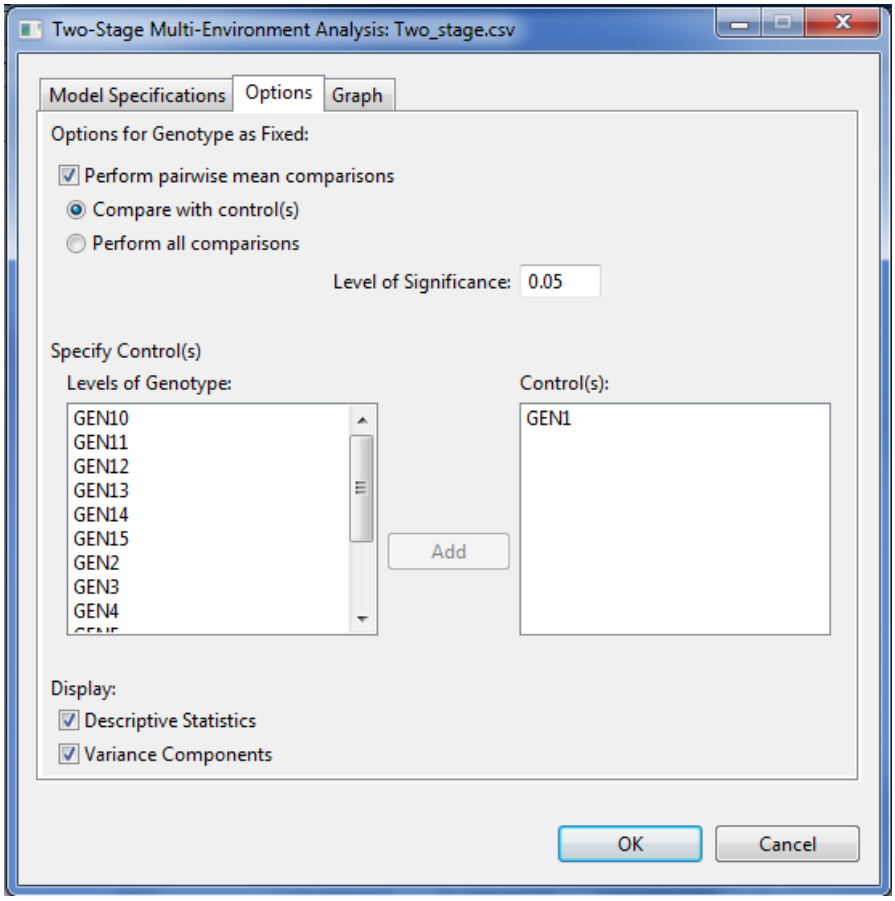
The user has the options to display descriptive statistics and variance.

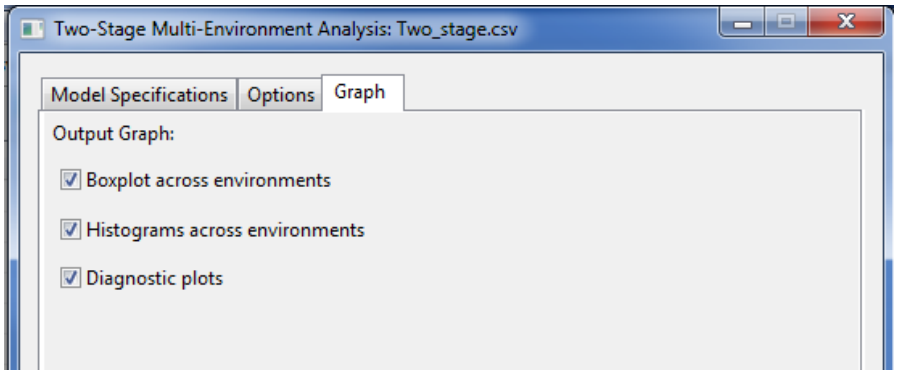
Graph Tab

Graphs like boxplot, histogram and diagnostic plots can be generated as part of the output of the analysis.

For the *example*, the completed dialog box should appear as illustrated below:




Two-Stage Multi-Environment Analysis: Two_stage.csv
Model Specifications
Options
Graph
Options for Genotype as Fixed:
Perform pairwise mean comparisons
Compare with control(s)
Perform all comparisons
Level of Significance: 0.05
Specify Control(s)
Levels of Genotype:
GEN10
GEN11
GEN12
GEN13
GEN14
GEN15
GEN2
GEN3
GEN4
Add
Control(s):
GEN1
Display:
Descriptive Statistics
Variance Components
OK
Cancel


Two-Stage Multi-Environment Analysis: Two_stage.csv
Model Specifications
Options
Graph
Output Graph:
Boxplot across environments
Histograms across environments
Diagnostic plots

- Click **OK**.

- Sample output of the analysis is shown below:

```
DATA FILE: E:/PBTools/Projects/Analysis/Data/Two_stage.csv

MULTI-ENVIRONMENT ANALYSIS (TWO-STAGE)

WEIGHT OPTION: 1/(sem^2)

==============================
GENOTYPE AS: Fixed
==============================

------------------------------
RESPONSE VARIABLE: Yield_Mean
------------------------------

DATA SUMMARY:
Number of observations read:  150
Number of observations used:  150

 Factors   Number of Levels   Levels
 Genotype  15                GEN1  GEN10  GEN11 ... GEN9
 Env       10                E1  E10  E2 ... E9


DESCRIPTIVE STATISTICS:
    Variable N_NonMissObs     Mean   StdDev
1 Yield_Mean          150 1629.058 1054.553


VARIANCE COMPONENTS TABLE:
    Groups     Variance   Std.Dev.
1      Env 1536420379.9 39197.1986
2 Residual     404328.3   635.8681


TESTING FOR THE SIGNIFICANCE OF GENOTYPIC EFFECT:
Analysis of Variance Table with Satterthwaite Denominator Df
         Df   Sum Sq  Mean Sq F value Denom Pr(>F)
Genotype 14 10504543 750324.5  1.1602   126 0.3140


GENOTYPE LSMEANS AND STANDARD ERRORS:
   Genotype    LSMean StdErrMean
1      GEN1  909.7275   18597.93
2     GEN10 2388.7966   18597.93
3     GEN11 1521.2375   18597.93
4     GEN12 2108.5082   18597.93
5     GEN13 2095.4355   18597.93
6     GEN14 1777.4761   18597.93
```

```
7     GEN15 1979.3168   18597.93
8      GEN2 1552.8571   18597.93
9      GEN3 1073.8184   18597.93
10     GEN4 1592.7507   18597.93
11     GEN5  918.8396   18597.93
12     GEN6 1179.8754   18597.93
13     GEN7 1118.9574   18597.93
14     GEN8 1572.1282   18597.93
15     GEN9  817.5437   18597.93


STANDARD ERROR OF THE DIFFERENCE (SED):
           Estimate
 Minimum   16828.4000
 Average   16828.4000
 Maximum   16828.4000


SIGNIFICANT PAIRWISE COMPARISONS (IF ANY):
Compared with control(s)

(No significant pairwise comparisons.)



==============================
GENOTYPE AS: Random
==============================

------------------------------
RESPONSE VARIABLE: Yield_Mean
------------------------------

DATA SUMMARY:

Number of observations read:  150
Number of observations used:  150

 Factors   Number of Levels   Levels
 Genotype  15                GEN1  GEN10  GEN11 ... GEN9
 Env       10                E1  E10  E2 ... E9


DESCRIPTIVE STATISTICS:

    Variable N_NonMissObs     Mean   StdDev
1 Yield_Mean          150 1629.058 1054.553
```

```
VARIANCE COMPONENTS TABLE:

    Groups     Variance   Std.Dev.
1 Genotype  713658226.5 26714.3824
2      Env 1536420294.8 39197.1975
3 Residual     404328.3   635.8681


TESTING FOR THE SIGNIFICANCE OF GENOTYPIC EFFECT USING -2 LOGLIKELIHOOD RATIO
TEST:

Formula for Model1:  Yield_Mean ~ 1 + (1|Genotype) + (1|Env)
Formula for Model2:  Yield_Mean ~ 1 + (1|Env)

           AIC      BIC    logLik   Chisq Df Pr(>Chisq)
Model2 1206.641 1215.673 -600.3206
Model1 1176.498 1188.540 -584.2490 32.1432  1     0.0000


TESTING FOR THE SIGNIFICANCE OF ENVIRONMENT EFFECT USING -2 LOGLIKELIHOOD RATIO
TEST:

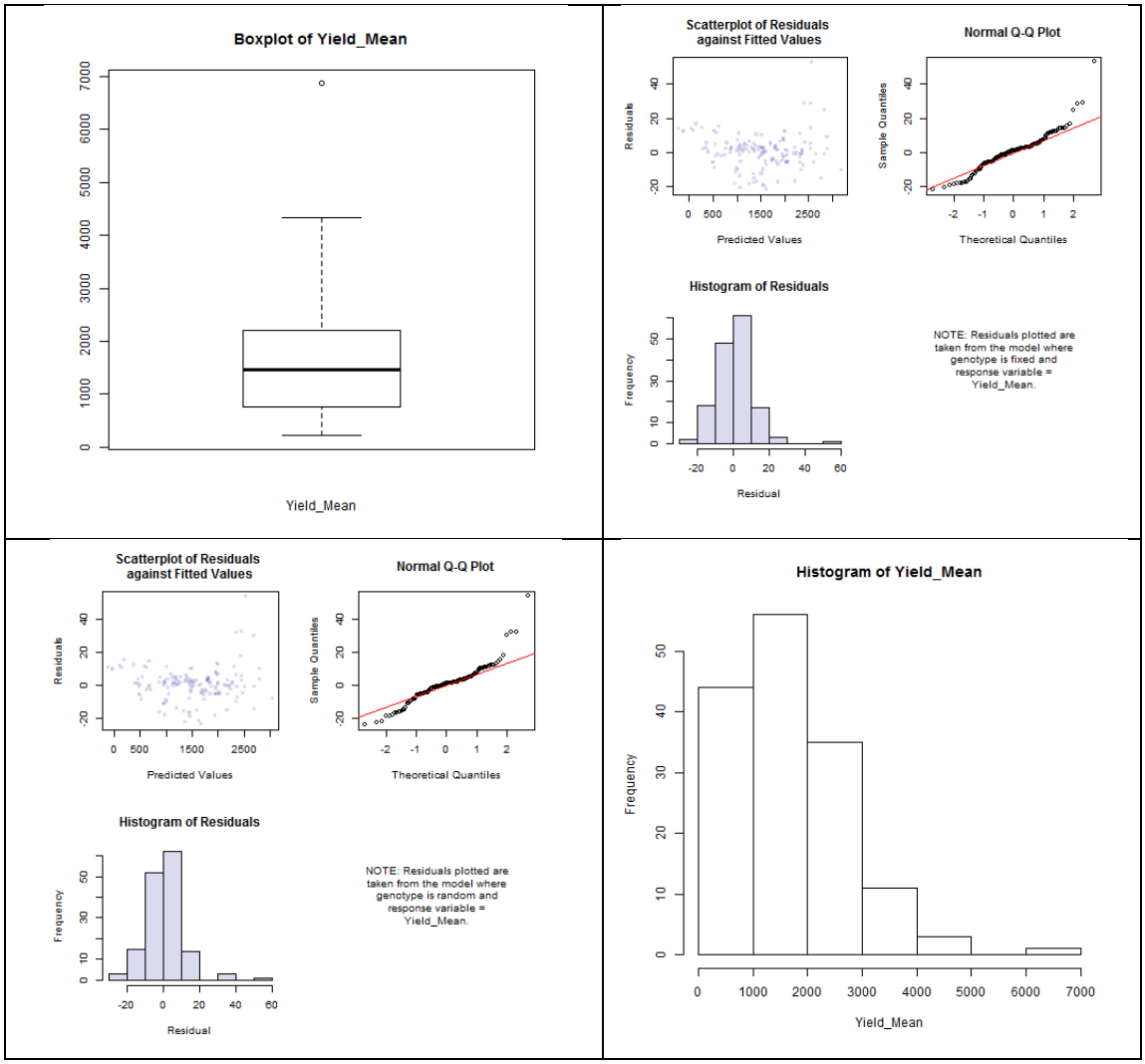
Formula for Model1:  Yield_Mean ~ 1 + (1|Genotype) + (1|Env)
Formula for Model2:  Yield_Mean ~ 1 + (1|Genotype)

           AIC     BIC    logLik   Chisq Df Pr(>Chisq)
Model2 1271.648 1280.68 -632.8242
Model1 1176.498 1188.54 -584.2490 97.1504  1     0.0000


PREDICTED GENOTYPE MEANS:

   Genotype      Mean
1      GEN1 1008.6335
2     GEN10 2242.8254
3     GEN11 1518.9009
4     GEN12 2008.9421
5     GEN13 1998.0337
6     GEN14 1732.7162
7     GEN15 1901.1398
8      GEN2 1545.2855
9      GEN3 1145.5572
10     GEN4 1578.5742
11     GEN5 1016.2370
12     GEN6 1234.0552
13     GEN7 1183.2229
14     GEN8 1561.3660
15     GEN9  931.7118
```

- Generated graphs can be viewed by clicking the Graph Tab of the displayed analysis folder. Sample generated graphs are shown below:



G×E Analysis

Stability Models

The steps to perform stability analysis using the summary of data from single-environment analysis as input are listed below:

- Import *Two_stage* file from PBTools package. Double-click the file to view it in the Data Viewer. The file contains four columns (variables) for yield: *Yield_Mean*, *Yield_StdErrMean*, *Yield_sigma2*, *Yield_No.rep* and the *Env* and *Genotype* factors, with 10 and 15 levels, respectively.

- Choose **Analysis | Multi-environment Analysis | G×E Analysis | Stability Models…** from the menu bar.

- Specify the required fields and appropriate options for the analysis.

  Model Specifications Tab

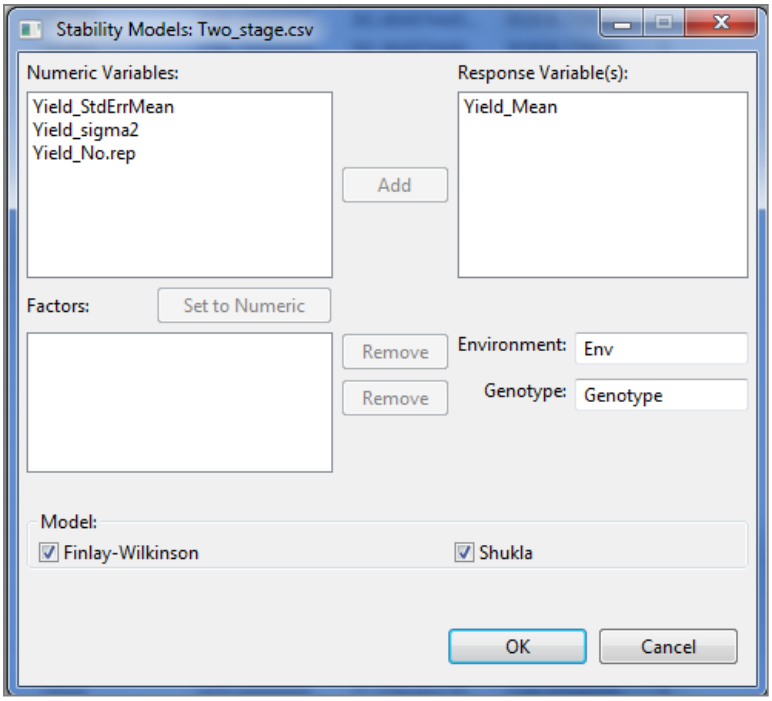
  Response Variables
  This is a required field which contains the column name corresponding to the mean of a trait.

  Environment
  This is a required field. PBTools requires that the number of environment levels is at least five to perform the analysis.

  Genotype
  This is a required field.

  Model
  The user has to specify the stability model to be used. The available models are Finlay-Wilkinson and Shukla.

  For the *example*, the completed dialog box should appear as illustrated below:



- Click **OK**.

- Sample output of the analysis is shown below:

```
STABILITY ANALYSIS

------------------------------
RESPONSE VARIABLE:  Yield_Mean
------------------------------

MODEL: FINLAY-WILKINSON

          Slope        SE   t.value          Prob      MSReg     MSDev
GEN1  0.9959142 0.2131061 4.6733257 0.0015956208  4873724.0  223156.1
GEN10 0.8484889 0.2784564 3.0471163 0.0158905007  3537605.8  381005.6
GEN11 1.6656718 0.2693899 6.1831271 0.0002642465 13633140.2  356598.5
GEN12 1.6542959 0.2703369 6.1193868 0.0002833593 13447559.3  359110.1
GEN13 1.2874242 0.2567362 5.0145802 0.0010338390  8144424.4  323885.3
GEN14 1.6300465 0.6835077 2.3848253 0.0442114578 13056206.8 2295640.6
GEN15 1.1784947 0.4083819 2.8857661 0.0203308103  6824522.9  819502.1
GEN2  1.2331710 0.3151059 3.9135133 0.0044589293  7472461.1  487899.1
GEN3  0.5118599 0.3610126 1.4178452 0.1939889247  1287417.1  640415.2
GEN4  0.2911745 0.3470663 0.8389593 0.4258490364   416604.4  591891.4
GEN5  0.6689718 0.1451654 4.6083411 0.0017364237  2199038.0  103548.4
GEN6  0.5243238 0.2717067 1.9297422 0.0897563679  1350878.4  362758.6
GEN7  0.7307594 0.2697616 2.7089079 0.0267031998  2624012.4  357583.4
GEN8  0.9286961 0.4162939 2.2308664 0.0562209397  4238033.0  851563.8
GEN9  0.8507071 0.2896187 2.9373351 0.0187860797  3556126.5  412164.1
```

```
MODEL: SHUKLA

          lower      est.    upper
GEN1  199.20782  382.9876  736.314
GEN10 159.53690  694.5456 3023.712
GEN11 225.37755  955.7149 4052.715
GEN12 228.97306  972.1788 4127.698
GEN13 172.46276  741.8346 3190.941
GEN14 368.07146 1564.2532 6647.861
GEN15 243.28824 1055.8627 4582.408
GEN2  197.85128  844.6951 3606.293
GEN3  178.11266  793.6803 3536.685
GEN4  159.50509  731.1260 3351.274
GEN5   41.00058  245.7681 1473.198
GEN6  112.10707  527.8490 2485.344
GEN7   91.33933  435.5675 2077.080
GEN8  195.09817  879.5450 3965.180
GEN9  144.81779  638.4840 2814.998
```

Multiplicative Models

The steps to perform AMMI and GGE analyses using the summary of data from single-environment analysis as input are listed below:

- Import *Two_stage* file from PBTools package. Double-click the file to view it in the Data Viewer. The file contains four columns (variables) for yield: *Yield_Mean*, *Yield_StdErrMean*, *Yield_sigma2*, *Yield_No.rep* and the *Env* and *Genotype* factors, with 10 and 15 levels, respectively.

- Choose **Analysis | Multi-environment Analysis | G×E Analysis | Multiplicative Models…** from the menu bar.

- Specify the required fields and appropriate options for the analysis.

  Model Specifications Tab

  Response Variables
  This is a required field which contains the column name corresponding to the mean of a trait.

  Residual Variance
  If a column corresponding to residual variances is available, specify it in this field.

Number of Replicates
If a column corresponding to number of replicates is available, specify it in this field.

To indicate that a set of response variable, residual variance and number of replicates corresponds to one trait, click the **Add to Table** button. The variables corresponding to one trait will be displayed in a row of the summary table. Do the same for the rest of the traits.

Environment
This is a required field.

Genotype
This is a required field.

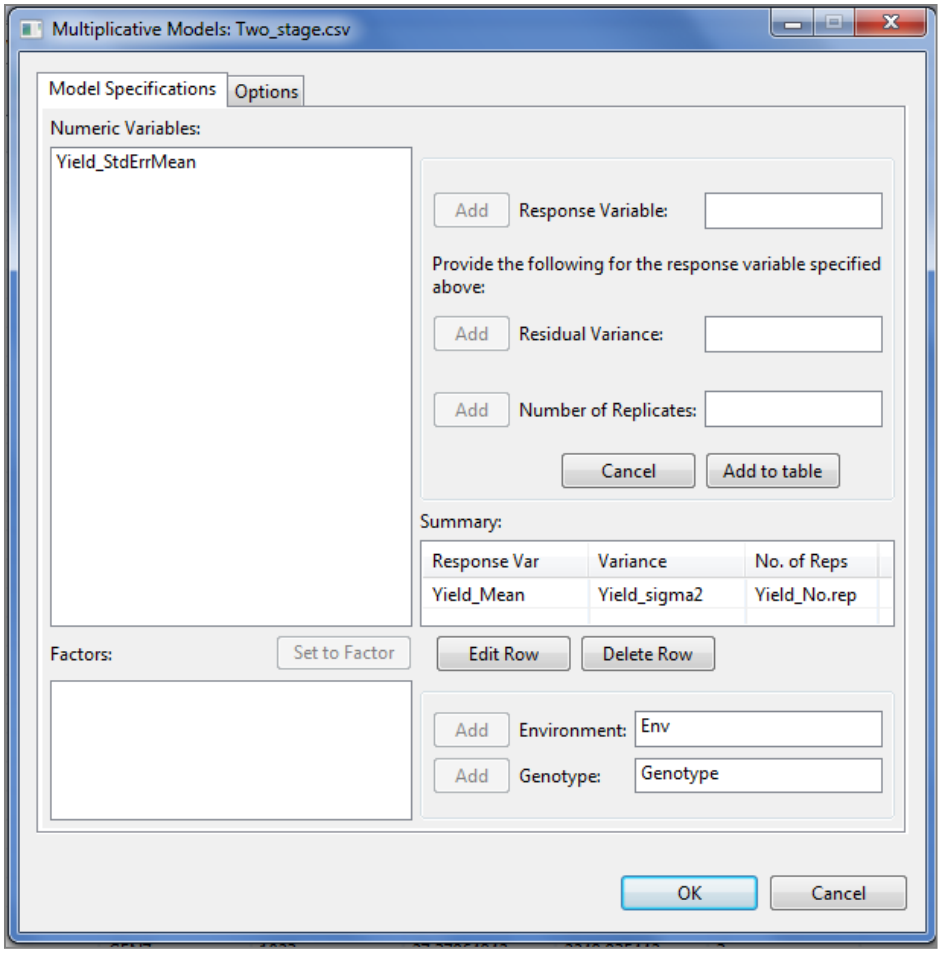
If PBTools detects that in the summary table a row has missing residual variance or number of replicates, a dialog box will appear prompting the user to specify the numerical value of MSE and number of replicates.

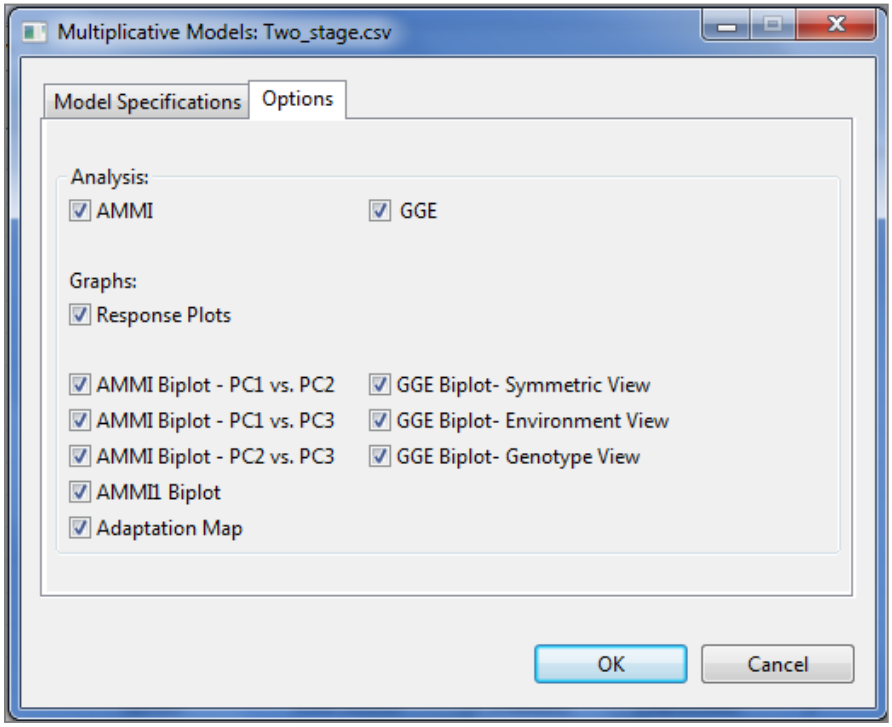
Options Tab

The user has to specify the multiplicative model to be used. The available models are AMMI and GGE. If AMMI analysis is selected, the user can request for graphs such as response plot, AMMI biplots (PC1 vs. PC2, PC1 vs. PC3, PC2 vs. PC3), AMMI1 biplot and adaptation map. If GGE analysis is selected, PBTools can present the GGE biplot in three views, namely, symmetric, environment and genotype.

Note: If the levels of the environment and genotype factors are coded using more than four characters, PBTools will automatically recode the levels of environment and genotype to E1, E2, etc. and G1, G2, etc, respectively. If recoding of the levels was done, the coding used will be reflected in the text output and all the requested graphs.

For the *example*, the completed dialog box should appear as illustrated below:


Multiplicative Models: Two_stage.csv
Model Specifications
Options
Numeric Variables:
Yield_StdErrMean
Add
Response Variable:
Provide the following for the response variable specified above:
Add
Residual Variance:
Add
Number of Replicates:
Cancel
Add to table
Summary:
Response Var
Variance
No. of Reps
Yield_Mean
Yield_sigma2
Yield_No.rep
Factors:
Set to Factor
Edit Row
Delete Row
Add
Environment:
Env
Add
Genotype:
Genotype
OK
Cancel


Multiplicative Models: Two_stage.csv
Model Specifications
Options
Analysis:
AMMI
GGE
Graphs:
Response Plots
AMMI Biplot - PC1 vs. PC2
AMMI Biplot - PC1 vs. PC3
AMMI Biplot - PC2 vs. PC3
AMMI1 Biplot
Adaptation Map
GGE Biplot- Symmetric View
GGE Biplot- Environment View
GGE Biplot- Genotype View
OK
Cancel

- Click **OK**.

- Sample output of the analysis is shown below:

```
MULTIPLICATIVE MODELS

------------------------------
RESPONSE VARIABLE:  Yield_Mean
------------------------------

AMMI ANALYSIS:

      percent acum Df      Sum.Sq    Mean.Sq F.value   Pr.F
PC1      45.7 45.7 22 111669055.7 5075866.17   51.22 0.0000
PC2      28.5 74.2 20  69670255.6 3483512.78   35.15 0.0000
PC3      11.8 86.0 18  28835650.3 1601980.57   16.16 0.0000
PC4       8.0 94.0 16  19596895.3 1224805.96   12.36 0.0000
PC5       2.8 96.8 14   6918224.5  494158.89    4.99 0.0000
PC6       1.4 98.2 12   3458554.3  288212.85    2.91 0.0008
PC7       1.1 99.3 10   2773777.9  277377.79    2.80 0.0025
PC8       0.5 99.8  8   1267436.4  158429.55    1.60 0.1245
PC9       0.1 99.9  6    275967.4   45994.57    0.46 0.8376
PC10      0.0 99.9  4         0.0       0.00    0.00 1.0000


GGE ANALYSIS:

      percent  acum Df       Sum.Sq    Mean.Sq F.value   Pr.F
PC1      46.2  46.2 22 127504474.60 5795657.94   58.48 0.0000
PC2      25.4  71.6 20  70051541.68 3502577.08   35.34 0.0000
PC3      11.3  82.9 18  31082789.22 1726821.62   17.42 0.0000
PC4      10.3  93.2 16  28316136.22 1769758.51   17.86 0.0000
PC5       2.6  95.8 14   7262841.82  518774.42    5.23 0.0000
PC6       1.6  97.4 12   4374564.38  364547.03    3.68 0.0000
PC7       1.2  98.6 10   3295838.89  329583.89    3.33 0.0004
PC8       1.0  99.6  8   2765865.54  345733.19    3.49 0.0007
PC9       0.5 100.1  6   1261950.34  210325.06    2.12 0.0512
PC10      0.0 100.1  4     63443.89   15860.97    0.16 0.9583


------------------------------

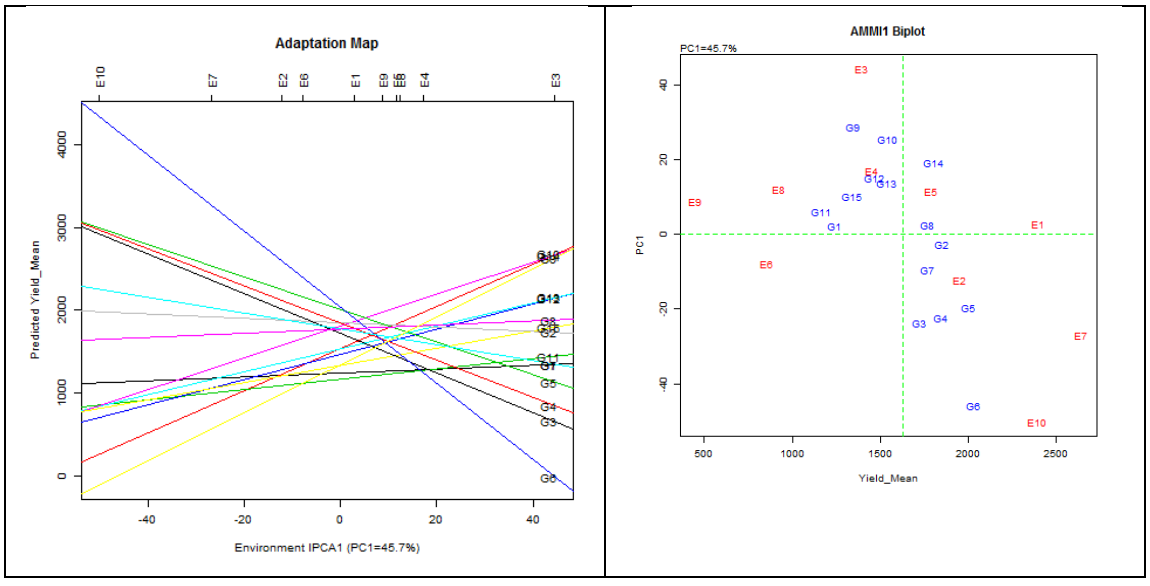
CODES USED IN GRAPHS:

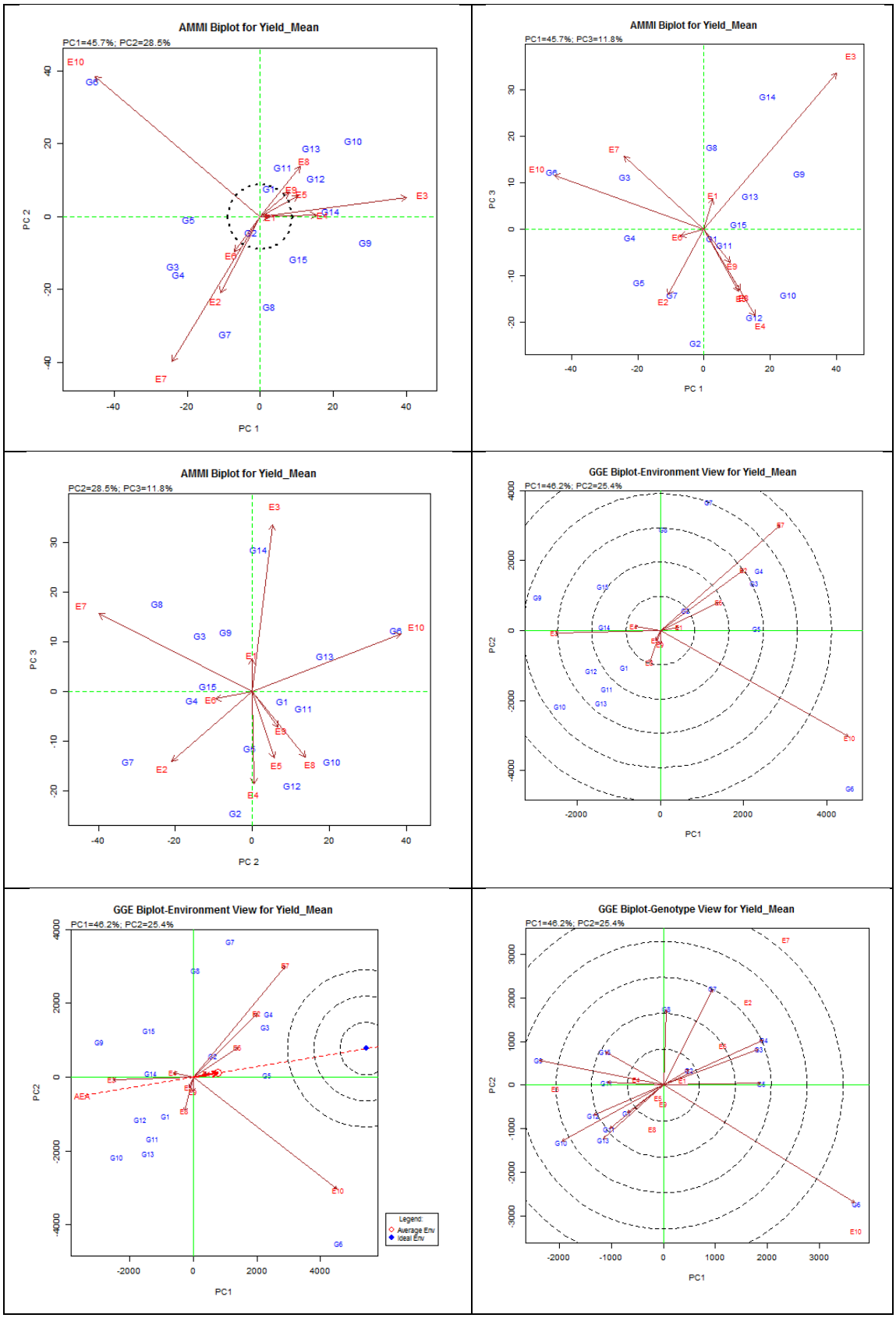
   Genotype Code
1      GEN1   G1
2     GEN10   G2
3     GEN11   G3
4     GEN12   G4
5     GEN13   G5
```

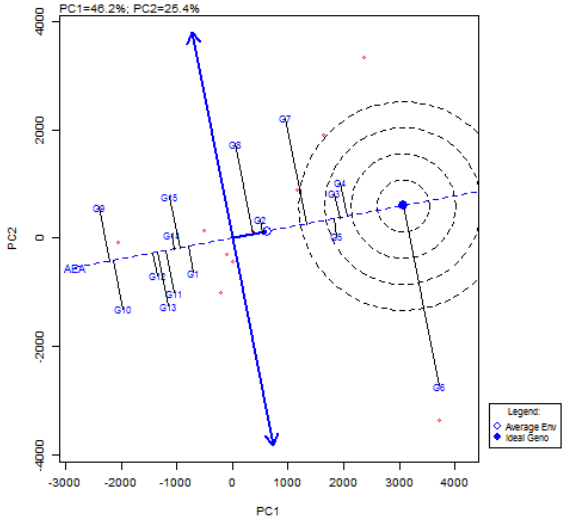
```
6      GEN14    G6
7      GEN15    G7
8       GEN2    G8
9       GEN3    G9
10      GEN4   G10
11      GEN5   G11
12      GEN6   G12
13      GEN7   G13
14      GEN8   G14
15      GEN9   G15

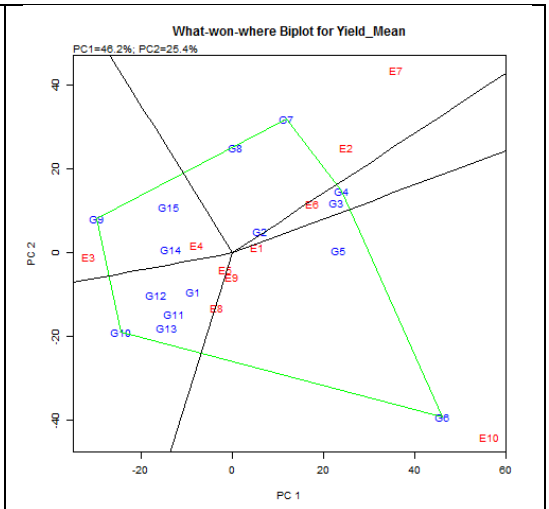
   Environment Code
1           E1    E1
2          E10    E2
3           E2    E3
4           E3    E4
5           E4    E5
6           E5    E6
7           E6    E7
8           E7    E8
9           E8    E9
10          E9   E10
```

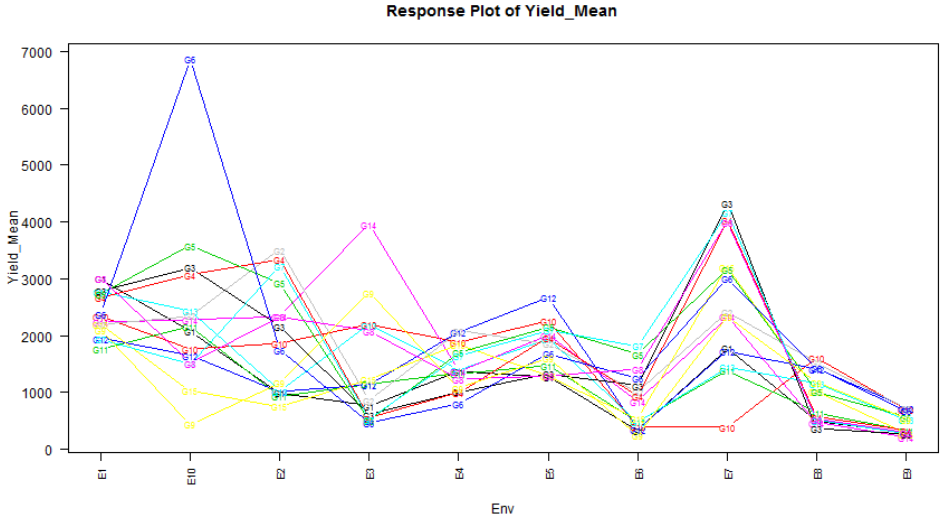
- Generated graphs can be viewed by clicking the Graph Tab of the displayed analysis folder. Sample generated graphs are shown below:

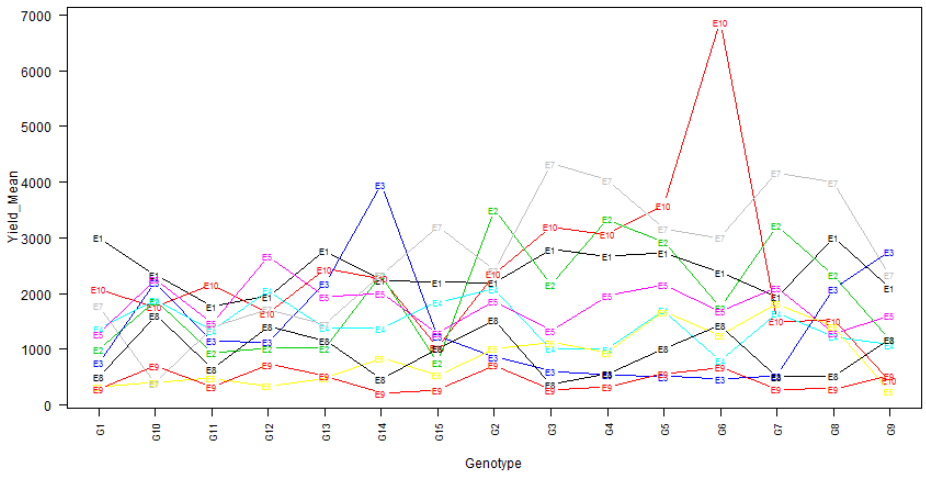



AMMI Biplot for Yield_Mean
PC1=45.7%; PC2=28.5%
PC 1
PC 2
AMMI Biplot for Yield_Mean
PC1=45.7%; PC3=11.8%
PC 1
PC 3
AMMI Biplot for Yield_Mean
PC2=28.5%; PC3=11.8%
PC 2
PC 3
GGE Biplot-Environment View for Yield_Mean
PC1=46.2%; PC2=25.4%
PC1
PC2
GGE Biplot-Environment View for Yield_Mean
PC1=46.2%; PC2=25.4%
AEA
Legend:
Average Env
Ideal Env
PC1
PC2
GGE Biplot-Genotype View for Yield_Mean
PC1=46.2%; PC2=25.4%
PC1
PC2


GGE Biplot-Genotype View for Yield_Mean
PC1=46.2%; PC2=25.4%
PC2
PC1
ABA
Legend:
Average Env
Ideal Geno


What-won-where Biplot for Yield_Mean
PC1=46.2%; PC2=25.4%
PC 2
PC 1


Response Plot of Yield_Mean
Yield_Mean
Env


Response Plot of Yield_Mean
Yield_Mean
Genotype

7. QTL Analysis

In this module, the QTL analysis is done per environment level only.

Predicted Means as Input

The steps to perform QTL Analysis using predicted means as input are listed below:

- Open the project named *SampleProject*. Double-click the *QTL_pred_height.csv* file to view it in the Data Viewer.

- Choose **Analysis | QTL Analysis**… from the menu bar.

- Specify the required fields and appropriate options for the analysis.

Data Input Tab

Phenotypic Data Used

The *Estimated Means* button is selected if the values in the phenotypic data displayed in the Data Viewer are estimated means. If the values in the phenotypic data are raw data, select the *Raw* button.

Genotypic data file

Specify this file by selecting from the project *Data* folder using the drop-down combo box or by locating it using the **Browse** button. For the example, select *QTL_geno.txt* file from the drop-down combo box.

Genetic map file

Specify this file by selecting from the project *Data* folder using the drop-down combo box or by locating it using the **Browse** button. For the example, select *QTL_map.txt* file from the drop-down combo box.

Model Specifications Tab

- Opening the data for the first time, *env* field is regarded by R as numerical variable. To set this as factors, choose this variable and click on the **Set to Factor** button.

Response Variable(s)

This is a required field. This list may contain one or more numeric variables. The analysis will be done per response variable.

Genotype

This is a required field.

Environment

Specifying the environment factor is optional.

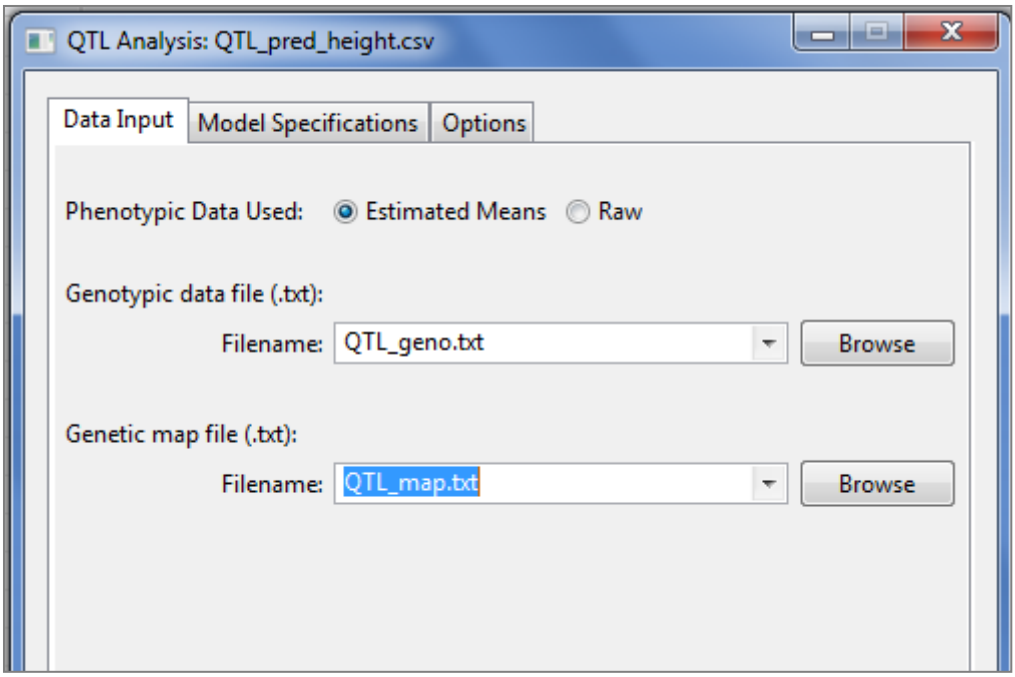
Environment Level(s)

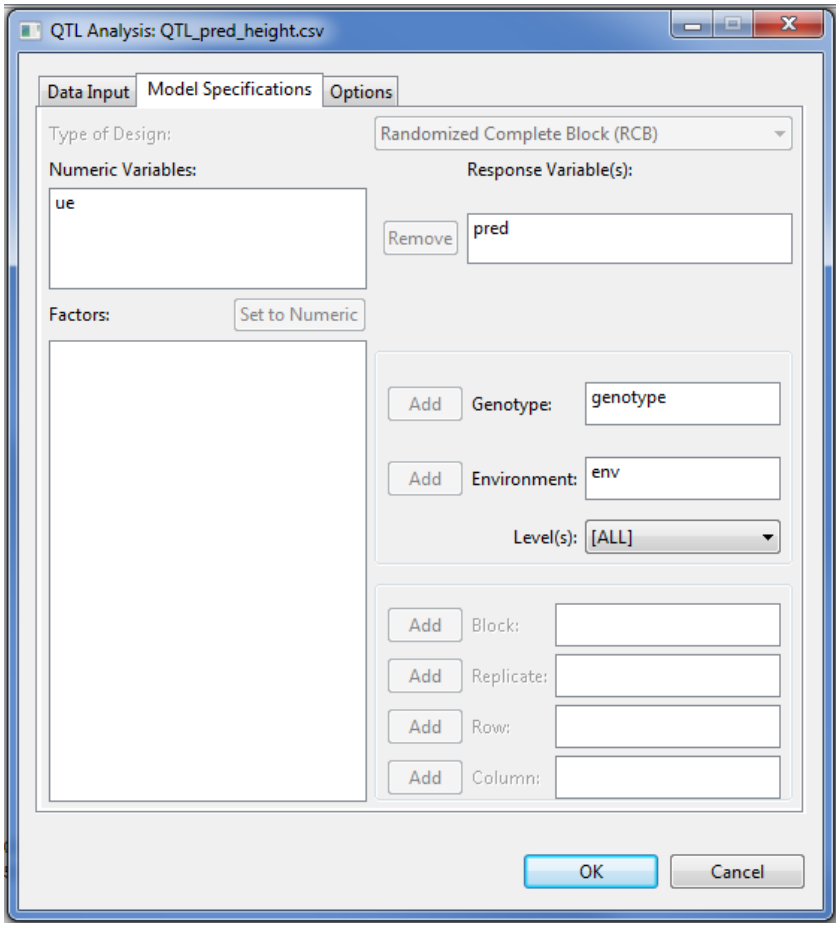
If the environment factor is specified, the user has the option to perform the analysis for each of the environment levels by selecting "[ALL]". To perform the analysis on a specific environment level, select environment level of interest in the combo box.

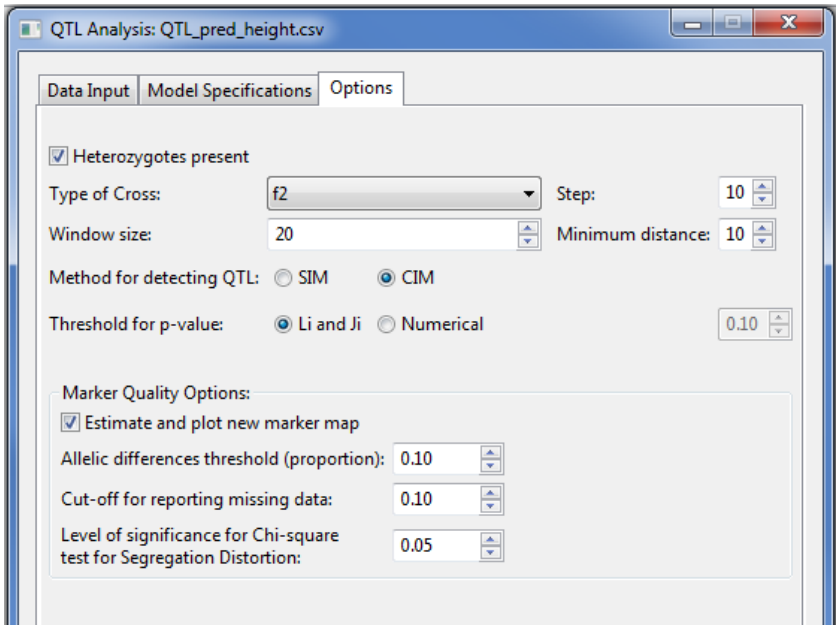
Options Tab

- Examine and change, if desired, the default options set.

For the *example*, the completed dialog box should appear as illustrated below:




QTL Analysis: QTL_pred_height.csv
Data Input
Model Specifications
Options
Type of Design:
Randomized Complete Block (RCB)
Numeric Variables:
ue
Response Variable(s):
Remove
pred
Factors:
Set to Numeric
Add
Genotype:
genotype
Add
Environment:
env
Level(s):
[ALL]
Add
Block:
Add
Replicate:
Add
Row:
Add
Column:
OK
Cancel


QTL Analysis: QTL_pred_height.csv
Data Input
Model Specifications
Options
Heterozygotes present
Type of Cross:
f2
Step:
10
Window size:
20
Minimum distance:
10
Method for detecting QTL:
SIM
CIM
Threshold for p-value:
Li and Ji
Numerical
0.10
Marker Quality Options:
Estimate and plot new marker map
Allelic differences threshold (proportion):
0.10
Cut-off for reporting missing data:
0.10
Level of significance for Chi-square test for Segregation Distortion:
0.05

- Click **OK**.

- Sample output of the analysis is shown below:

```
DATA FILE: E:/PBTools/Projects/SampleProject/Data/QTL_pred_height.csv

QTL ANALYSIS

Method: CIM

TRAIT: pred

ENVIRONMENT: 1
```

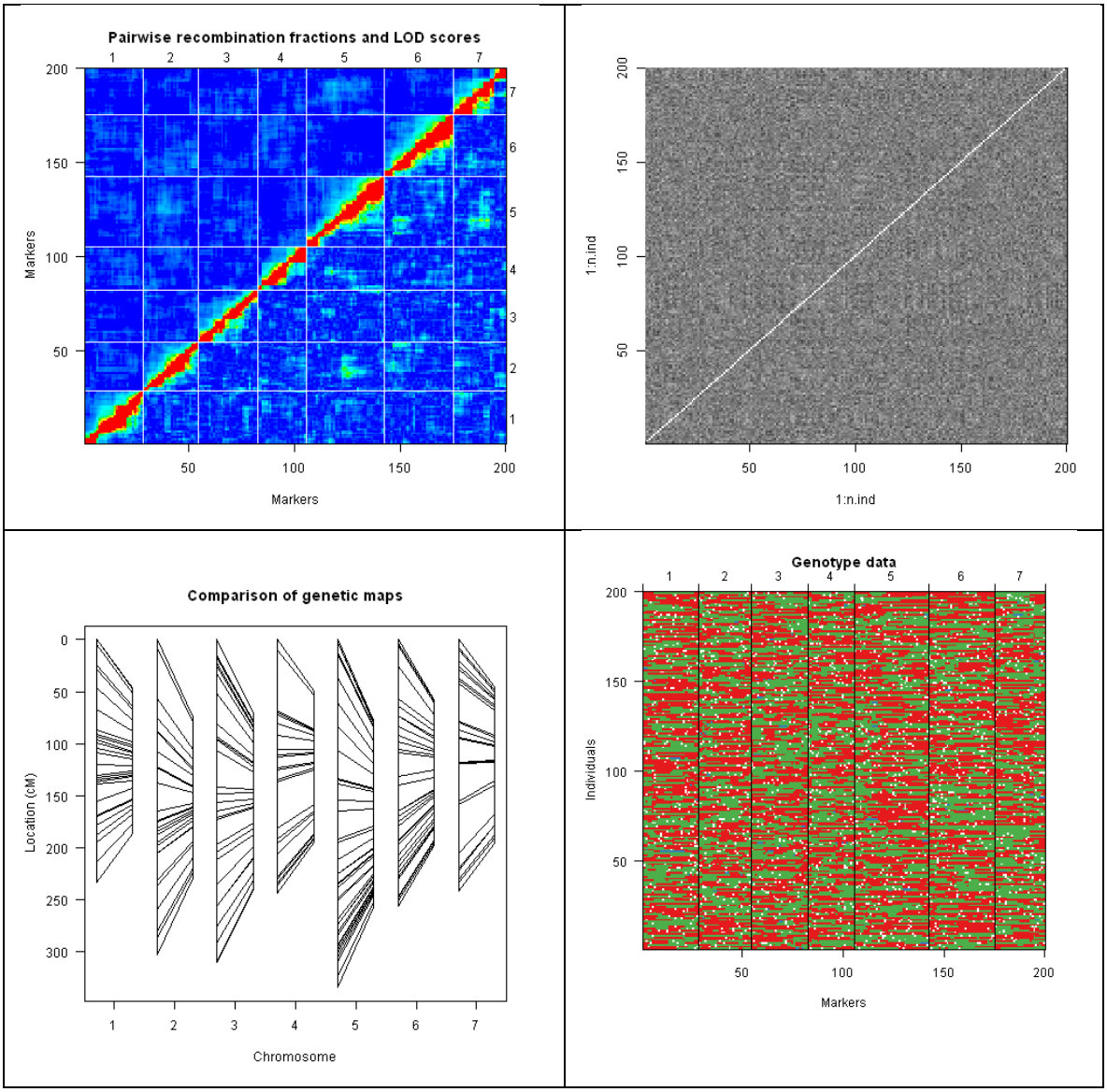
LOD of All Markers (partial results are shown below)

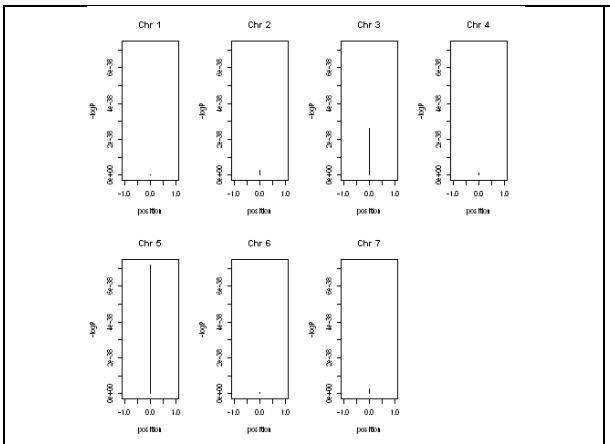
```
QTL RESULT (ALL):
      marker Chr Pos          LOD
1     M_0001   1   0  1.174800566
2     M_0002   1   1  1.203821631
3     M_0006   1   5  1.668347468
4    1_loc10   1  10  1.188279464
5     M_0018   1  17  0.402935738
6    1_loc20   1  20  0.225377662
7     M_0022   1  21  0.167893203
8    1_loc30   1  30  0.027011398
9     M_0032   1  31  0.010962483
10   1_loc40   1  40  0.354145428
11    M_0042   1  41  7.571604389
12   1_loc50   1  50 10.147830671
13    M_0053   1  52  9.607553808
14    M_0056   1  55  7.779334005
15    M_0058   1  57  7.203216436
16   1_loc60   1  60  0.115636404
17    M_0062   1  61  0.121921684
18    M_0063   1  62  0.204758238
19    M_0066   1  65  0.205360985
20    M_0069   1  68  0.638024571
21   1_loc70   1  70  0.520645664
22    M_0076   1  75  0.153868813
23    M_0081   1  80  0.245741558
24    M_0083   1  82  1.527724377
25    M_0085   1  84  1.944514227
26    M_0086   1  85  2.453003945
27    M_0090   1  89  2.255778133
28   1_loc90   1  90  2.461440368
29    M_0097   1  96  2.747008898
30  1_loc100   1 100  0.592397083
31    M_0108   1 107  0.296877445
32    M_0109   1 108  0.227384432
```

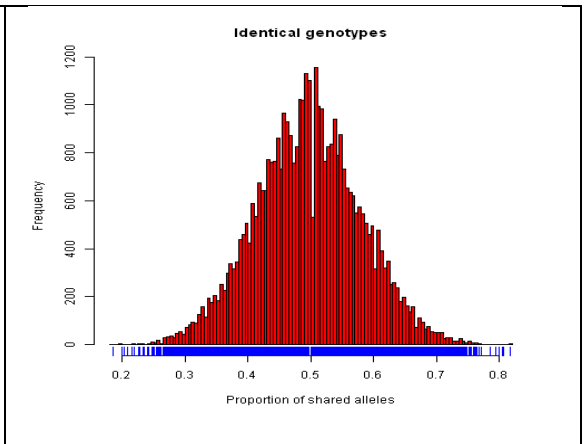
Characteristics of the Selected Markers

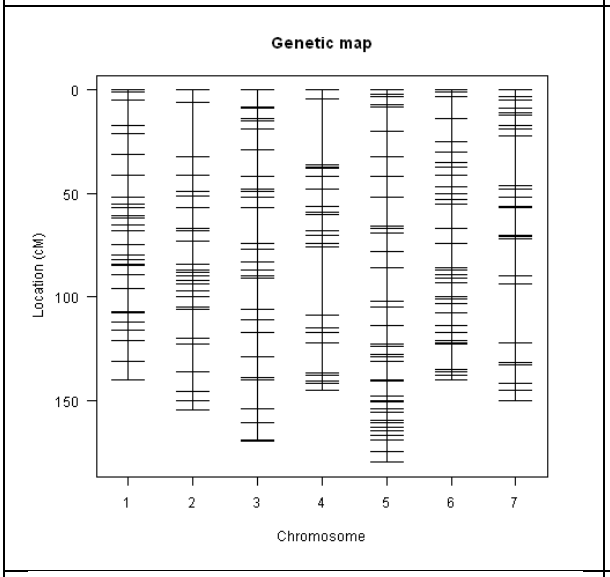
```
QTL RESULT (SELECTED):
   marker Chr Pos       LOD     m.eff        Rsq
1 1_loc50   1  50 10.147831  8.550234 0.01500826
2 2_loc10   2  10  7.869992 -7.398572 0.05434154
3  M_0340   3  42  5.963655 -4.560166 0.06830667
4  M_0543   4  74  3.793779  5.271883 0.09855792
5  M_0617   5   2  3.673243 -2.450220 0.15389838
6  M_0989   7  52 11.673238  7.733852 0.17982716
```

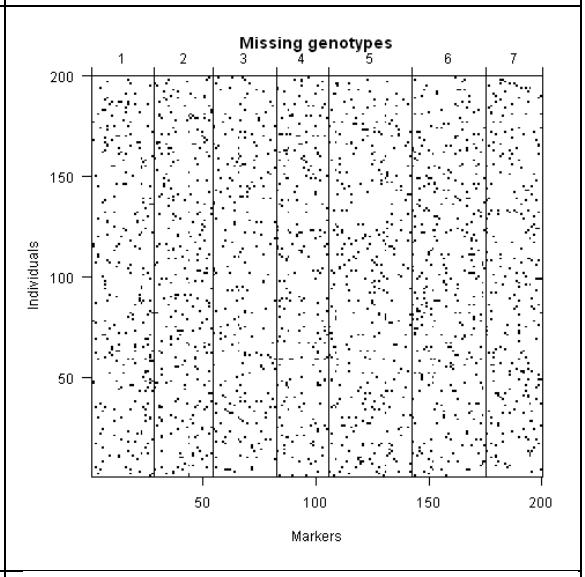
- Generated graphs can be viewed by clicking the Graph Tab of the displayed analysis folder. Sample generated graphs are shown below:

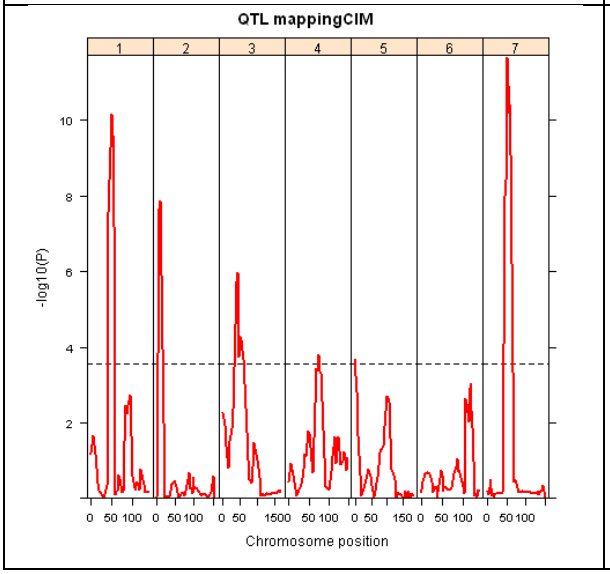


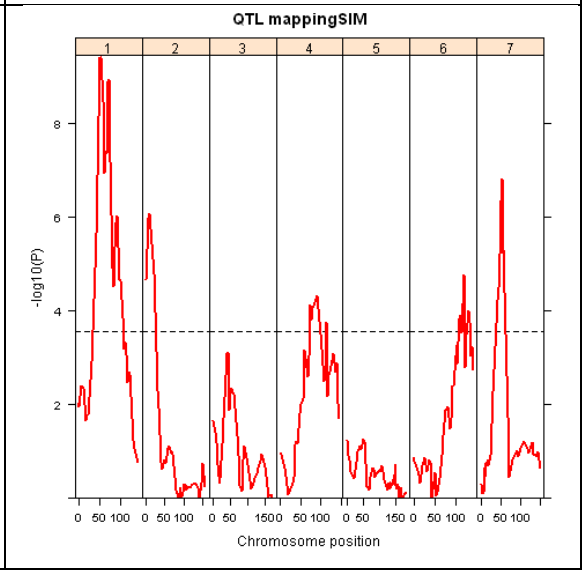

Chr 1
Chr 2
Chr 3
Chr 4
Chr 5
Chr 6
Chr 7


Identical genotypes
Frequency
Proportion of shared alleles


Genetic map
Location (cM)
Chromosome


Missing genotypes
Individuals
Markers


QTL mappingCIM
-log10(P)
Chromosome position


QTL mappingSIM
-log10(P)
Chromosome position

Raw Data as Input

The steps to perform QTL Analysis using raw data as input are listed below:

- Open the project named *SampleProject*. Double-click the *QTL_pheno.csv* file to view it in the Data Viewer.

- Choose **Analysis | QTL Analysis…** from the menu bar.

- Specify the required fields and appropriate options for the analysis.

  Data Input Tab

  Phenotypic Data Used

  The *Predicted* button is selected if the values in the phenotypic data displayed in the Data Viewer are predicted means. If the values in the phenotypic data are raw data, select the *Raw* button.

  Genotypic data file

  Specify this file by selecting from the project *Data* folder using the drop-down combo box or by locating it using the **Browse** button. For the example, select *QTL_geno.txt* file from the drop-down combo box.

  Genetic map file

  Specify this file by selecting from the project *Data* folder using the drop-down combo box or by locating it using the **Browse** button. For the example, select *QTL_map.txt* file from the drop-down combo box.

  Model Specifications Tab

  - Opening the data for the first time, *ENV* and *REP* fields are regarded by R as numerical variables. To set these as factors, choose these variable and click on the **Set to Factor** button.

  Type of Design

  There are five available experimental designs, namely: Randomized Complete Block Design (RCB), Augmented RCB, Augmented Latin Square, Alpha-Lattice and Row-Column. For the example, select RCB.

Response Variable(s)
This is a required field. This list may contain one or more numeric variables. The analysis will be done per response variable.

Genotype
This is a required field.

Environment
Specifying the environment factor is optional.

Environment Level(s)
If the environment factor is specified, the user has the option to perform the analysis for each of the environment levels by selecting "[ALL]". To perform the analysis on a specific environment level, select environment level of interest in the combo box.

Block
This field is required if the design is RCB, Augmented RCB or Alpha-Lattice.

Replicate
This field is required if the design is Augmented Latin Square, Alpha-Lattice or Row-Column.

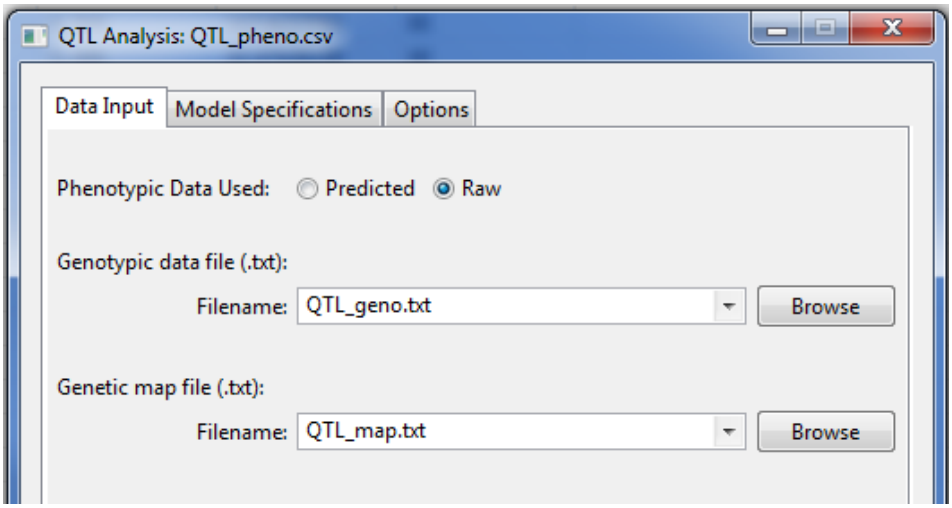
Row
This field is required if the design is Augmented Latin Square or Row-Column.

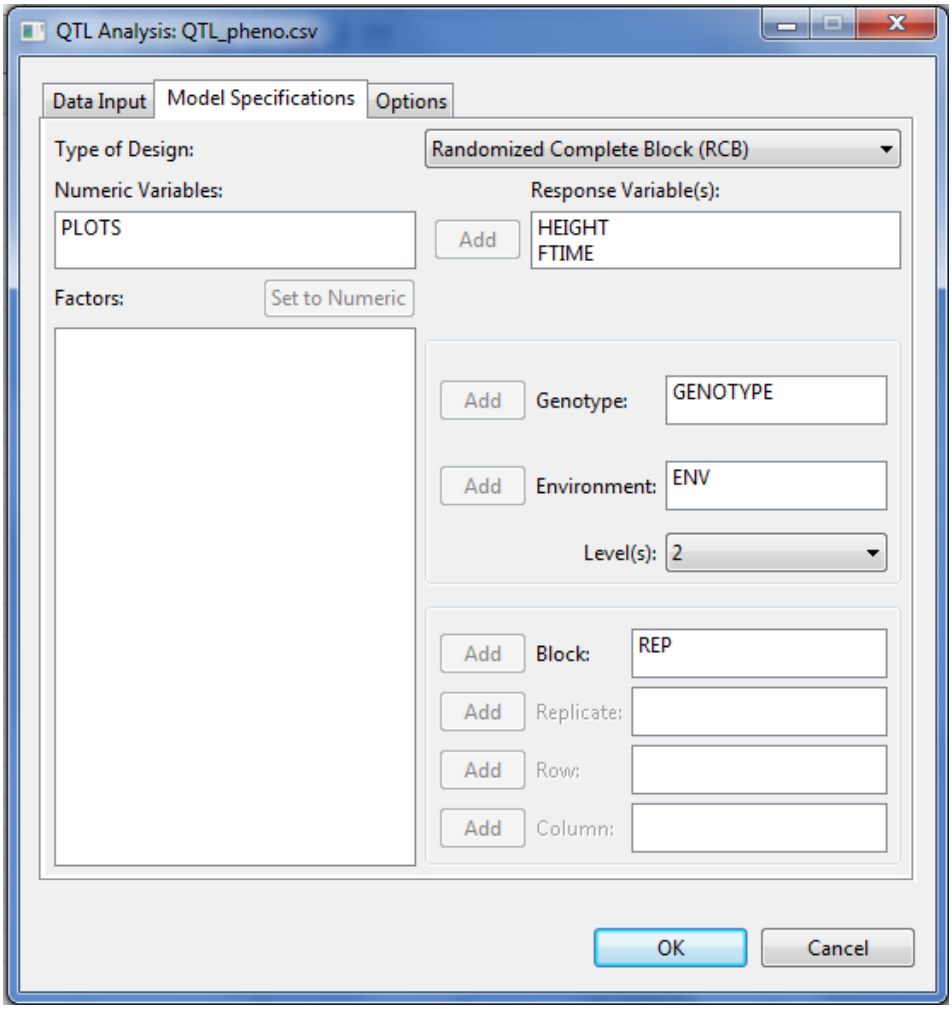
Column
This field is required if the design is Augmented Latin Square or Row-Column.

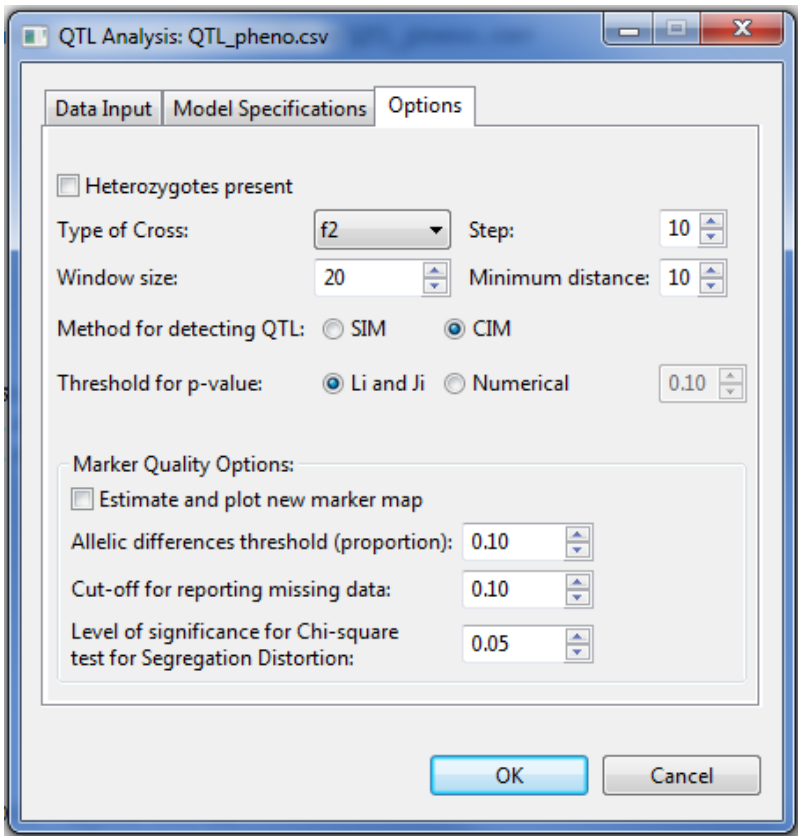
Options Tab

- Examine and change, if desired, the default options set.

For the *example*, the completed dialog box should appear as illustrated below:


QTL Analysis: QTL_pheno.csv
Data Input
Model Specifications
Options
Phenotypic Data Used:
Predicted
Raw
Genotypic data file (.txt):
Filename:
QTL_geno.txt
Browse
Genetic map file (.txt):
Filename:
QTL_map.txt
Browse


QTL Analysis: QTL_pheno.csv
Data Input
Model Specifications
Options
Type of Design:
Randomized Complete Block (RCB)
Numeric Variables:
PLOTS
Add
Response Variable(s):
HEIGHT
FTIME
Factors:
Set to Numeric
Add
Genotype:
GENOTYPE
Add
Environment:
ENV
Level(s):
2
Add
Block:
REP
Add
Replicate:
Add
Row:
Add
Column:
OK
Cancel



- Click **OK**.

- Sample output of the analysis (only results from *response variable* = *HEIGHT*) is shown below:

```
DATA FILE: E:/PBTools/Projects/SampleProject/Data/QTL_pheno.csv

SINGLE-ENVIRONMENT ANALYSIS

DESIGN: Randomized Complete Block (RCB)

------------------------------
RESPONSE VARIABLE: HEIGHT
------------------------------

Descriptive Statistics:

  Variable ENV N_NonMissObs     Mean   StdDev
1   HEIGHT   2          606 86.69455 31.62887
```

```
------------------------------
ANALYSIS FOR: ENV  =  2
------------------------------

Trial Summary:

Number of observations read:  606
Number of observations used:  606

 Factors   Number of Levels   Levels
 GENOTYPE  202               G_001  G_002  G_003 ... P2
 REP       3                 1  2  3
```

Estimates of the Variance Components

```
Variance Components Table:

    Groups Variance Std.Dev.
1      REP 611.3356 24.72520
2 Residual 302.7346 17.39927
```

Test for Significance of Genotypic Effect

```
Testing for the Significance of Genotypic Effect:

Models:
model2: HEIGHT ~ 1 + (1|REP)
model1: HEIGHT ~ 1 + GENOTYPE + (1|REP)

        Df    AIC    BIC  logLik  Chisq Chi Df Pr(>Chisq)
model2   3 5610.1 5623.3 -2802.1
model1 204 5362.1 6261.1 -2477.0 650.03    201  < 2.2e-16

          Df   Sum Sq  Mean Sq  F value p-value
GENOTYPE 201 235948.7 1173.874 3.877569  0.0000
```

Genotype Means and Standard Errors

```
Genotype means and standard errors:

    GENOTYPE      Mean StdErrMean
1      G_001 104.09469   17.45067
2      G_002  67.11523   17.45067
3      G_003 129.68455   17.45067
4      G_004 114.59435   17.45067
5      G_005 108.58281   17.45067
.
.
.
202       P2  75.78512   17.45067
```

```
QTL ANALYSIS

Method: CIM

TRAIT: HEIGHT

ENVIRONMENT: 2
```

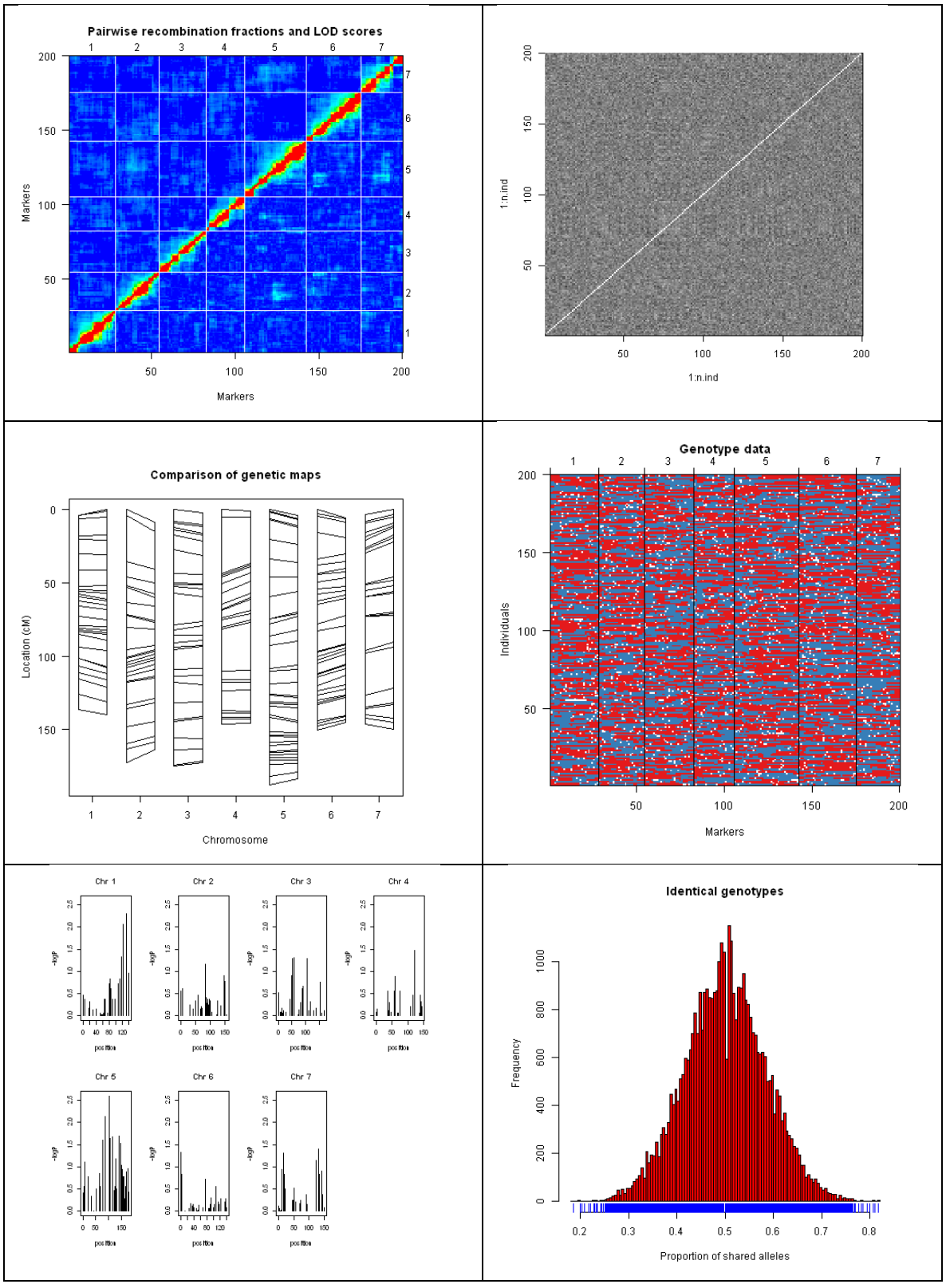
LOD of All Markers (partial results are shown below)

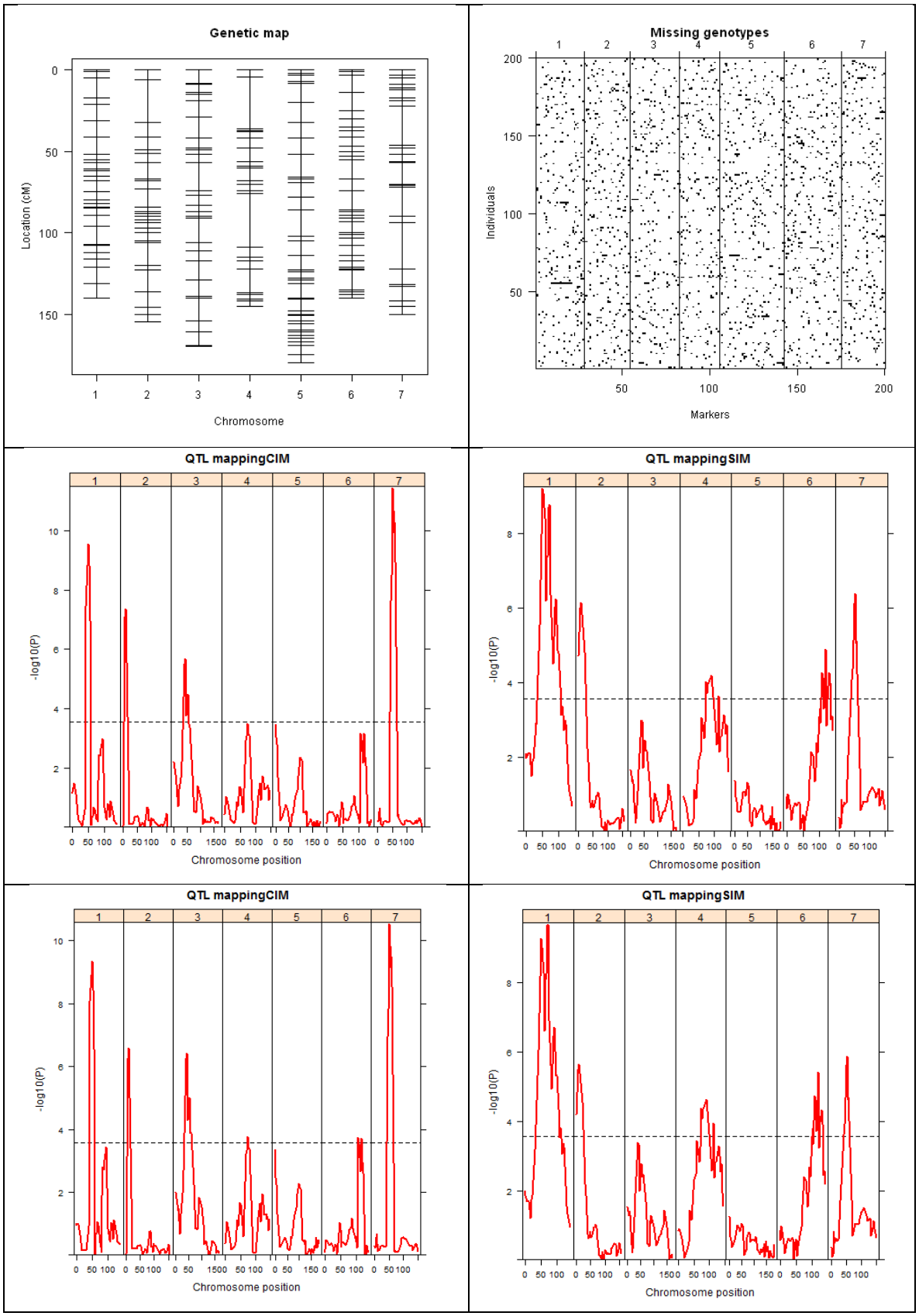
```
QTL RESULT (ALL):
      marker Chr Pos          LOD
1     M_0001   1   0 9.661112e-01
2     M_0002   1   1 1.016853e+00
3     M_0006   1   5 9.769430e-01
4    1_loc10   1  10 6.512969e-01
5     M_0018   1  17 1.608477e-01
.
.
.
284   M_1087   7 150 1.331439e-01
```

Characteristics of the Selected Markers

```
QTL RESULT (SELECTED):
   marker Chr Pos       LOD     m.eff        Rsq
1 1_loc50   1  50  9.350702  8.328936 0.03822631
2 2_loc10   2  10  6.579351 -6.526687 0.05935303
3  M_0340   3  42  6.414814 -4.899655 0.07238491
4  M_0543   4  74  3.766788  5.009470 0.08592566
5  M_0904   6 108  3.753643  3.984999 0.13875675
6  M_0989   7  52 10.544196  7.416265 0.17716468
```

- Generated graphs can be viewed by clicking the Graph Tab of the displayed analysis folder. Sample generated graphs are shown below:


Pairwise recombination fractions and LOD scores
Markers
Markers
1:n.ind
1:n.ind
Comparison of genetic maps
Location (cM)
Chromosome
Genotype data
Individuals
Markers
Chr 1
Chr 2
Chr 3
Chr 4
Chr 5
Chr 6
Chr 7
Identical genotypes
Frequency
Proportion of shared alleles


Genetic map
Location (cM)
Chromosome
Missing genotypes
Individuals
Markers
QTL mappingCIM
-log10(P)
Chromosome position
QTL mappingSIM
-log10(P)
Chromosome position
QTL mappingCIM
-log10(P)
Chromosome position
QTL mappingSIM
-log10(P)
Chromosome position

8. Selection Index

The steps to perform Selection Index Analysis are listed below:

- Open the project named *SampleProject*. Double-click the *SI_traits.csv* file to view it in the Data Viewer. The file contains the *REP*, *Block*, and *ENTRY* columns and several traits (*MFL1*, *FFL1*, *EHT1*, *PHT1*, *GY1*, …).

- Choose **Analysis** | **Selection Index…** from the menu bar.

- Specify the required fields and appropriate options for the analysis.

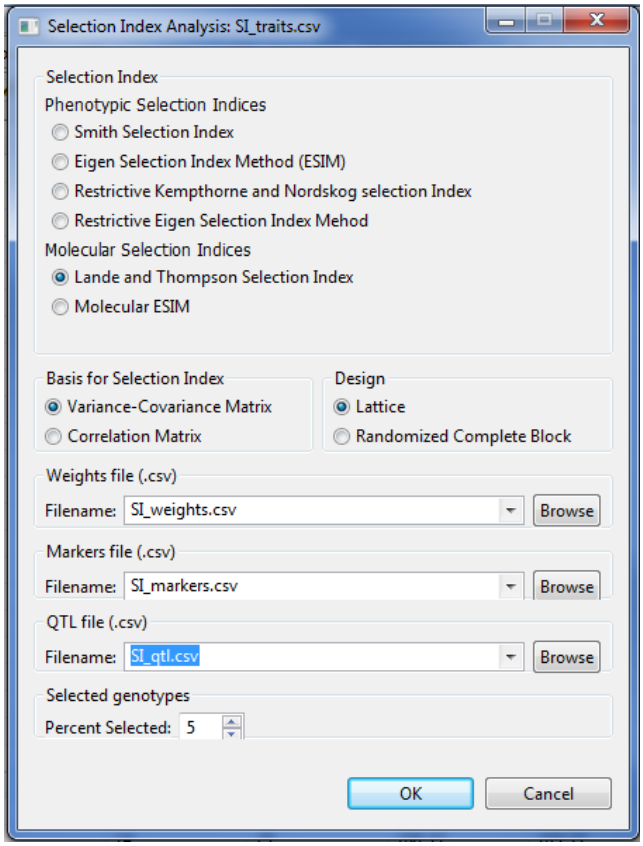
  Selection Index
  Choose one selection index from among the four under *Phenotypic selection indices* and two under *Molecular selection indices*. Note that the molecular selection indices require a markers and QTL file to be specified, in addition to the weights file. For the example, click on the Lande and Thompson Selection Index and leave Variance-Covariance Matrix as basis for selection index and Lattice as design.

  Weights file
  Specify this file by selecting from the project *Data* folder using the drop-down combo box or by locating it using the **Browse** button. For the example, select *SI_weights.csv* file from the drop-down combo box.

  Markers file and QTL file
  If needed, specify these by selecting them from the project *Data* folder using the respective drop-down combo box or by locating them using the **Browse** buttons. For the example, select *SI_markers.csv* and *SI_qtl.csv* files from the drop-down combo box.

  Percent Selected Genotypes
  If desired, change the value of the percentage using the spin box. The default value is 5%.

  For the *example*, the completed dialog box should appear as illustrated below:



- Click **OK.**

- Sample output of the analysis is shown below:

```
DATA FILE: E:/PBTools/Projects/SampleProject/Data/SI_traits.csv

Lande and Thompson Selection Index

DESIGN: Lattice
```

Genetic and Phenotypic Correlation Matrices

```
GENETIC CORRELATION MATRIX

        MFL1   FFL1   EHT1   PHT1    GY1 MFL.2   FFL2   EHT2   PHT2    GY2
MFL1    1.00   0.88   0.19 -0.33 -0.62   0.96   0.88   0.23 -0.25 -0.36
FFL1    0.88   1.00   0.18 -0.20 -0.77   0.83   0.68   0.15 -0.28 -0.41
EHT1    0.19   0.18   1.00   0.75 -0.03   0.18   0.04   1.09   0.96   0.13
PHT1   -0.33 -0.20   0.75   1.00   0.32 -0.23 -0.28   0.75   1.09   0.28
GY1    -0.62 -0.77 -0.03   0.32   1.00 -0.59 -0.70   0.09   0.40   0.98
MFL.2   0.96   0.83   0.18 -0.23 -0.59   1.00   0.91   0.31 -0.23 -0.52
FFL2    0.88   0.68   0.04 -0.28 -0.70   0.91   1.00   0.27 -0.23 -0.61
EHT2    0.23   0.15   1.09   0.75   0.09   0.31   0.27   1.00   0.77 -0.08
```

```
PHT2  -0.25 -0.28  0.96  1.09  0.40 -0.23 -0.23  0.77  1.00  0.20
GY2   -0.36 -0.41  0.13  0.28  0.98 -0.52 -0.61 -0.08  0.20  1.00
```

```
PHENOTYPIC CORRELATION MATRIX

       MFL1  FFL1 EHT1  PHT1   GY1 MFL.2  FFL2 EHT2  PHT2   GY2
MFL1   1.00  0.71 0.01 -0.37 -0.47  0.63  0.56 0.10 -0.17 -0.29
FFL1   0.71  1.00 0.04 -0.26 -0.48  0.54  0.51 0.03 -0.20 -0.31
EHT1   0.01  0.04 1.00  0.80  0.07  0.13  0.06 0.71  0.51  0.06
PHT1  -0.37 -0.26 0.80  1.00  0.32 -0.14 -0.15 0.49  0.53  0.17
GY1   -0.47 -0.48 0.07  0.32  1.00 -0.35 -0.39 0.04  0.15  0.44
MFL.2  0.63  0.54 0.13 -0.14 -0.35  1.00  0.72 0.05 -0.31 -0.40
FFL2   0.56  0.51 0.06 -0.15 -0.39  0.72  1.00 0.05 -0.21 -0.45
EHT2   0.10  0.03 0.71  0.49  0.04  0.05  0.05 1.00  0.81  0.15
PHT2  -0.17 -0.20 0.51  0.53  0.15 -0.31 -0.21 0.81  1.00  0.34
GY2   -0.29 -0.31 0.06  0.17  0.44 -0.40 -0.45 0.15  0.34  1.00
```

Molecular Covariance Matrix

```
MOLECULAR COVARIANCE MATRIX

       MFL1  FFL1  EHT1  PHT1   GY1 MFL.2  FFL2  EHT2  PHT2   GY2
MFL1   1.00  0.33  0.69  0.63  0.11  0.43 -0.23  0.48  0.79  0.37
FFL1   0.33  1.00  0.46  0.37  0.02  0.24  0.22  0.16  0.23 -0.12
EHT1   0.69  0.46  1.00  0.48  0.23  0.51 -0.41  0.58  0.41  0.13
PHT1   0.63  0.37  0.48  1.00 -0.21  0.47  0.25  0.48  0.50 -0.10
GY1    0.11  0.02  0.23 -0.21  1.00 -0.04 -0.34 -0.01  0.27 -0.18
MFL.2  0.43  0.24  0.51  0.47 -0.04  1.00  0.15  0.73  0.31  0.04
FFL2  -0.23  0.22 -0.41  0.25 -0.34  0.15  1.00 -0.14 -0.04 -0.16
EHT2   0.48  0.16  0.58  0.48 -0.01  0.73 -0.14  1.00  0.32  0.01
PHT2   0.79  0.23  0.41  0.50  0.27  0.31 -0.04  0.32  1.00 -0.02
GY2    0.37 -0.12  0.13 -0.10 -0.18  0.04 -0.16  0.01 -0.02  1.00
```

Covariance, Variances and Correlation of Selection Index and Breeding Value

```
COVARIANCE BETWEEN SELECTION INDEX AND BREEDING VALUE:  3.205508

VARIANCE OF THE SELECTION INDEX:  1.50582

VARIANCE OF THE BREEDING VALUE:  6.145354

CORRELATION BETWEEN SELECTION INDEX AND BREEDING VALUE:  0.9999
```

Characteristics of the Selected Individuals

```
VALUES OF THE TRAITS, SELECTION INDEX, MEANS, GAINS FOR THE 5% SELECTED
INDIVIDUALS

                                MFL1   FFL1   EHT1   PHT1    GY1  MFL.2   FFL2
```

Entry 240	102.01	102.12	80.53	140.12	0.00	100.70	100.74
Entry 2	104.88	104.42	100.22	148.82	28.72	99.60	100.36
Entry 21	101.23	100.75	93.45	163.42	34.45	100.23	98.34
Entry 132	104.69	104.71	75.75	126.00	44.00	97.88	100.35
Entry 167	106.75	105.49	81.67	132.64	0.83	106.21	108.25
Entry 179	104.39	101.50	76.00	126.03	23.78	100.74	99.34
Entry 220	105.34	104.37	78.75	137.92	97.16	99.84	98.31
Entry 174	103.13	102.54	96.84	146.33	47.44	96.59	99.25
Entry 165	104.61	103.66	90.75	150.25	79.00	99.32	103.80
Entry 161	103.12	99.05	85.75	148.50	82.50	101.46	103.00
Entry 87	101.53	100.88	103.66	156.84	68.33	102.34	99.20
Mean of Selected Individuals	103.79	102.68	87.58	143.35	46.02	100.44	100.99
Mean of all Individuals	102.01	102.12	80.53	140.12	75.99	100.70	100.74
Selection Differential	1.78	0.56	7.04	3.23	-29.97	-0.26	0.26
Expected Genetic Gain for 5%	1.57	1.64	1.52	1.19	-1.13	1.95	1.37

	EHT2	PHT2	GY2	LT index
Entry 240	78.47	133.44	0.00	2.88
Entry 2	106.75	161.75	140.00	2.59
Entry 21	89.69	156.81	1.50	2.34
Entry 132	76.09	127.28	33.39	2.24
Entry 167	75.00	113.64	0.00	2.14
Entry 179	74.31	124.14	9.24	2.14
Entry 220	87.00	143.50	129.50	1.94
Entry 174	94.93	149.59	18.61	1.83
Entry 165	96.25	160.25	41.00	1.80
Entry 161	76.50	125.69	12.75	1.74
Entry 87	91.95	140.56	35.00	1.70
Mean of Selected Individuals	86.08	139.69	38.27	NA
Mean of all Individuals	78.47	133.44	51.70	NA
Selection Differential	7.61	6.26	-13.43	NA
Expected Genetic Gain for 5%	1.73	1.08	-1.02	NA

Values of the Traits and Selection Index for All Individuals (*partial results are shown below*)

VALUES OF THE TRAITS AND THE SELECTION INDEX FOR ALL INDIVIDUALS

	MFL1	FFL1	EHT1	PHT1	GY1	MFL.2	FFL2	EHT2	PHT2	GY2
Entry 1	102.21	100.25	71.45	123.75	42.45	99.29	98.95	68.51	117.47	16.87
Entry 2	104.88	104.42	100.22	148.82	28.72	99.60	100.36	106.75	161.75	140.00
Entry 3	98.97	100.44	80.17	154.36	77.78	97.93	96.82	66.75	133.75	116.00
Entry 4	102.73	103.32	78.16	137.57	488.89	100.89	100.92	85.76	145.25	52.78
Entry 5	94.09	96.60	80.94	152.67	225.25	95.00	96.10	72.94	130.42	63.89
Entry 6	99.90	101.92	94.04	153.39	121.00	100.75	100.09	93.06	149.92	66.00

.
.
.

9. Mating Designs

North Carolina I (NCI)

In this design, n_1 male parents are chosen and each male is crossed to a different set of n_2 females. This design follows the hierarchical or nested design in Statistics. The progenies of all the crosses are tested for the traits of interest using the chosen experimental design.

The steps to perform such analysis are listed below:

- Import *NCI_ME* file from package. Double-click the file to view it in the Data Viewer. The example file contains data for an experiment conducted for crosses derived from a NCI mating design with four inbred males and 11 inbred females per male using Randomized Complete Block (RCB) field design with two blocks (*1* and *2*) in two environments (*A* and *B*) and one trait, *Y*.

- Choose **Analysis | Mating Designs | North Carolina I…** from the menu bar.

- Opening the data for the first time, *Block*, *Male* and *Female* fields are regarded by R as numerical variables. To set these as factors, choose these variables and click on the **Set to Factor** button.

- Specify the required fields and appropriate options for the analysis.

  Type of Design
  There are four available experimental designs, namely: Completely Randomized Design (CRD), Randomized Complete Block Design (RCB), Alpha-Lattice and Row-Column Design.

  'Inbred' Option
  This is selected if the parental lines are inbreds.

  'Crossed' Option
  This is selected if the parental lines are from random-mating population (outcrossing species).

Data Input
The input data for the analysis may be plot-means or observations of individual plants (*not yet implemented*).

Response Variable(s)
This is a required field. This list may contain one or more numeric variables.

'Perform Analysis Per Site' Option
If this option is selected and the Environment factor is specified, the analysis will be performed for each level of the Environment factor.

If this option is selected but the Environment factor is **not** specified, PBTools will perform the analysis as if the entire data came from only one site.

'Perform Multiple Environment Analysis' Option
If this option is selected, the *Environment* factor must be specified.

Environment
This is a required field if the 'Perform Multiple Environment Analysis' option is selected. Moreover, the number of its levels must be at least two.

Female
This is a required field.

Male
This is a required field.

Block
This is a required field if the selected experimental design is Randomized Complete Block or Alpha-Lattice.

Replicate
This is a required field if the selected experimental design is Completely Randomized Design, Alpha-Lattice or Row-Column.

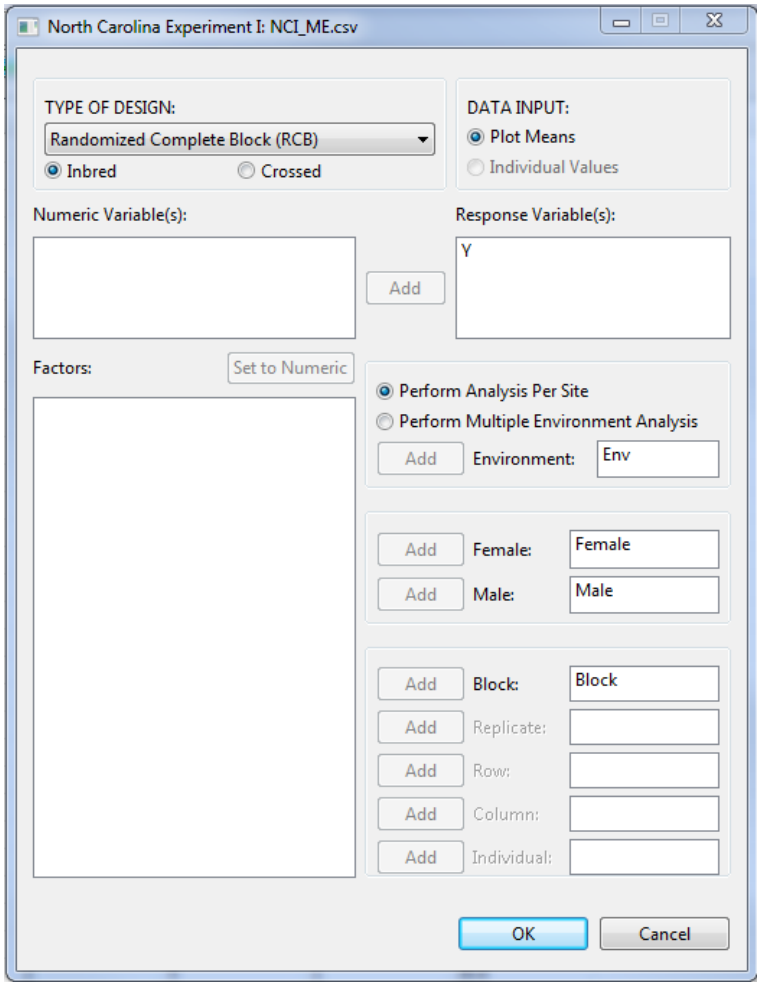
Row
This is a required field if the selected experimental design is Row-Column.

Column
This is a required field if the selected experimental design is Row-Column.

Individual

This is a required field if the selected data input is Individual Values.

For the example, the completed dialog box should appear as illustrated below:



- Click **OK**.

- Sample output of the analysis (only results from *Env* = *A*) is shown below:

```
DATA FILE: E:/PBTools/Projects/SampleProject/Data/NCI_ME.csv

DESIGN: NORTH CAROLINA EXPERIMENT I IN RCB (INBRED)

-----------------------------
RESPONSE VARIABLE: Y
-----------------------------
```

```
------------------------------
ANALYSIS FOR: Env = A
------------------------------

DATA SUMMARY:

 Factors   No of Levels   Levels
 Env       1             A
 Male      4             1  2  3  4
 Female    11            1  10  11 ... 9
 Block     2             1  2

Number of observations read: 88
Number of observations used: 88
```

Analysis of Variance Table

```
ANOVA TABLE FOR THE EXPERIMENT:

            Df    Sum Sq  Mean Sq F value Pr(>F)
Block        1   15.9801  15.9801    2.17 0.1483
Male         3  479.9476 159.9825   21.70 0.0000
Male:Female 40 1161.6870  29.0422    3.94 0.0000
Residuals   43  317.0749   7.3738
-------
REMARK: Raw dataset is balanced.
```

Estimated Variance Components of Random Effects

```
LINEAR MIXED MODEL FIT BY RESTRICTED MAXIMUM LIKELIHOOD:

 Formula:  Y ~ 1 + (1|Block) + (1|Male/Female)

      AIC      BIC    logLik deviance  REMLdev
 500.0305 512.4172 -245.0152 492.5127 490.0305

 Random Effects:
 Groups      Variance Std. Deviation
 Female:Male  10.8342         3.2915
 Male          5.9518         2.4396
 Block         0.1956         0.4423
 Residual      7.3738         2.7155
```

Genetic Variance Components

```
ESTIMATES OF GENETIC VARIANCE COMPONENTS:

                  Estimate
VA              11.903600
VD               4.882400
h2-narrow sense  0.492703
H2-broad sense   0.694791
Dominance Ratio  0.905717
```

North Carolina II (NCII)

In this design, n_1 male parents and n_2 female parents are chosen. Each male is crossed to every female. This design follows the factorial design in Statistics. The progenies of all the crosses are tested for the traits of interest using the chosen experimental design.

The steps to perform such analysis are listed below:

- Import *NCII_ME* file from the package. Double-click the file to view it in the Data Viewer. The example file contains data for an experiment conducted for crosses derived from a NCII mating design with eight inbred males and eight inbred females using Randomized Complete Block (RCB) field design with three blocks (*1, 2* and *3*) in two environments (*A* and *B*) and one trait, *Y*.

- Choose **Analysis | Mating Designs | North Carolina II…** from the menu bar.

- Opening the data for the first time, *Block*, *Male* and *Female* fields in the data file are regarded by R as numerical variables. To set these as factors, choose these variables and click on the **Set to Factor** button.

- Specify the required fields and appropriate options for the analysis.

  Type of Design
  There are four available experimental designs, namely: Completely Randomized Design (CRD), Randomized Complete Block Design (RCB), Alpha-Lattice and Row-Column Design.

  'Inbred' Option
  This is selected if the parental lines are inbreds.

'Crossed' Option
This is selected if the parental lines are from random-mating population (outcrossing species).

Data Input
The input data for the analysis may be plot-means or observations of individual plants (*not yet implemented*).

Response Variable(s)
This is a required field. This list may contain one or more numeric variables.

'Perform Analysis Per Site' Option
If this option is selected and the Environment factor is specified, the analysis will be performed for each level of the Environment factor.

If this option is selected but the Environment factor is **not** specified, PBTools will perform the analysis as if the entire data came from only one site.

'Perform Multiple Environment Analysis' Option
If this option is selected, the *Environment* factor must be specified.

Environment
This is a required field if the 'Perform Multiple Environment Analysis' option is selected. Moreover, the number of its levels must be at least two.

Female
This is a required field.

Male
This is a required field.

Block
This is a required field if the selected experimental design is Randomized Complete Block or Alpha-Lattice.

Replicate
This is a required field if the selected experimental design is Completely Randomized Design, Alpha-Lattice or Row-Column.

Row
This is a required field if the selected experimental design is Row-Column.

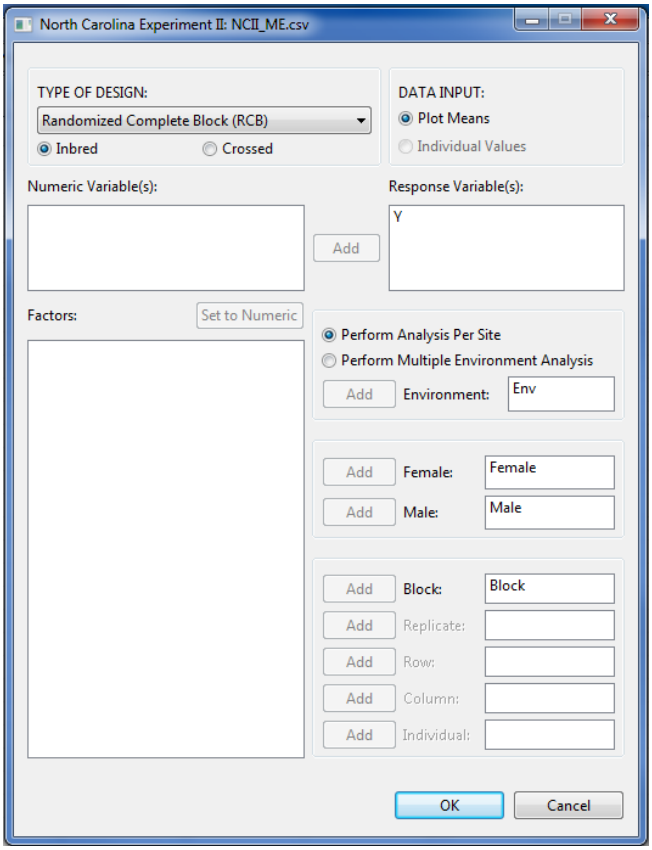
Column

This is a required field if the selected experimental design is Row-Column.

Individual

This is a required field if the selected data input is Individual Values.

For the example, the completed dialog box should appear as illustrated below:



- Click **OK**.

- Sample output of the analysis (only results from Env = A) is shown below:

```
DATA FILE: E:/PBTools/Projects/SampleProject/Data/NCII_ME.csv

DESIGN: NORTH CAROLINA EXPERIMENT II IN RCB (INBRED)

-----------------------------
RESPONSE VARIABLE: Y
-----------------------------
-----------------------------
ANALYSIS FOR: Env = A
-----------------------------
```

```
DATA SUMMARY:

 Factors   No of Levels   Levels
 Env       1              A
 Female    8              1  2  3 ... 8
 Male      8              10  11  12 ... 9
 Block     3              1  2  3

Number of observations read: 192
Number of observations used: 192
```

Analysis of Variance Table

```
ANOVA TABLE FOR THE EXPERIMENT:

             Df    Sum Sq  Mean Sq F value Pr(>F)
Block         2   22.3531  11.1766    1.14 0.3220
Male          7 1112.6490 158.9498   16.26 0.0000
Female        7  735.2669 105.0381   10.75 0.0000
Male:Female  49  823.1469  16.7989    1.72 0.0086
Residuals   126 1231.4730   9.7736
-------
REMARK: Raw dataset is balanced.
```

Estimated Variance Components of Random Effects

```
LINEAR MIXED MODEL FIT BY RESTRICTED MAXIMUM LIKELIHOOD:

 Formula:  Y ~ 1 + (1|Block) + (1|Male) + (1|Female) + (1|Male:Female)

      AIC     BIC    logLik deviance  REMLdev
 1057.665 1077.21 -522.8323 1047.758 1045.665

 Random Effects:
 Groups      Variance Std. Deviation
 Male:Female   2.3418         1.5303
 Female        3.6767         1.9175
 Male          5.9229         2.4337
 Block         0.0219         0.1481
 Residual      9.7736         3.1263
```

Genetic Variance Components

```
ESTIMATES OF GENETIC VARIANCE COMPONENTS:
                                            Estimate
 VA                                         9.599670
 VD                                         2.341800
 Narrow sense heritability(plot-mean based) 0.442075
 Broad sense heritability(plot-mean based)  0.549917
 Dominance Ratio                            0.698492
```

- Suppose the user opted to perform Multiple Environment Analysis, sample output of the analysis is shown below:

```
DATA FILE: E:/PBTools/Projects/SampleProject/Data/NCII_ME.csv

MULTIPLE ENVIRONMENT ANALYSIS

DESIGN: NORTH CAROLINA EXPERIMENT II IN RCB (INBRED)

-----------------------------
RESPONSE VARIABLE: Y
-----------------------------

DATA SUMMARY:
 Factors   No of Levels   Levels
 Env       2              A  B
 Female    8              1  2  3 ... 8
 Male      8              10  11  12 ... 9
 Block     3              1  2  3

 Number of observations read: 384
 Number of observations used: 384
```

ANOVA Table

```
ANOVA TABLE:
                 Df    Sum Sq  Mean Sq F value Pr(>F)
Env               1    8.7665   8.7665    0.38 0.5724
Env:Block         4   93.0194  23.2549    2.21 0.0682
Male              7  840.6569 120.0938   11.43 0.0000
Female            7  821.7501 117.3929   11.17 0.0000
Male:Female      49 1233.6320  25.1762    2.40 0.0000
Env:Male          7  493.0000  70.4286    6.70 0.0000
Env:Female        7  215.9051  30.8436    2.93 0.0057
Env:Male:Female  49  974.2153  19.8819    1.89 0.0009
Residuals       252 2648.7840  10.5110
-------
REMARK: Raw dataset is balanced.
```

Estimated Variance Components of Random Effects

```
LINEAR MIXED MODEL FIT BY RESTRICTED MAXIMUM LIKELIHOOD:

 Formula:    Y  ~  1  +  (1|Env)  +  (1|Env:Block)  +  (1|Male)  +  (1|Female)  +
(1|Male:Female) + (1|Env:Male) + (1|Env:Female) + (1|Env:Male:Female)

      AIC      BIC   logLik deviance  REMLdev
 2147.079 2186.586 -1063.54 2128.376 2127.079
```

```
Random Effects:
Groups          Variance Std. Deviation
Env:Male:Female   3.1506         1.7750
Male:Female       0.8681         0.9317
Env:Female        0.3985         0.6313
Env:Male          1.8441         1.3580
Female            1.7219         1.3122
Male              1.0554         1.0273
Env:Block         0.1792         0.4233
Env               0.0000         0.0000
Residual         10.5157         3.2428
```

Genetic Variance Components

```
ESTIMATES OF GENETIC VARIANCE COMPONENTS:

                Estimate
VA              2.777330
VAxE            2.242660
VD              0.868100
VDxE            3.150620
h2-narrow sense 0.142031
H2-broad sense  0.186425
Dominance Ratio 0.790653
```

North Carolina III (NCIII)

This design involves taking a number of F_2 individuals from a cross between two inbred lines (P_1 and P_2) and crossing each back to these same two lines. The two parental inbreds thus act as testers against which the F_2 are assessed.

The steps to perform such analysis using are listed below:

- Import *NCIII_ME* file from the package. Double-click the file to view it in the Data Viewer. The example file contains data for an experiment conducted for crosses derived from a NCIII mating design with 16 F_2 lines labeled as *Male* and two testers labeled as *Female* using Randomized Complete Block (RCB) field design with two blocks (*1* and *2*) in two environments (*A* and *B*) and one trait, *Y*.

- Choose **Analysis | Mating Designs | North Carolina III…** from the menu bar.

- Opening the data for the first time, *Block*, *Male* and *Female* fields are regarded by R as numerical variables. To set these as factors, choose these variables and click on the **Set to Factor** button.

- Specify the required fields and appropriate options for the analysis.

Type of Design
There are four available experimental designs, namely: Completely Randomized Design (CRD), Randomized Complete Block Design (RCB), Alpha-Lattice and Row-Column Design.

Data Input
The input data for the analysis may be plot-means or observations of individual plants (*not yet implemented*).

Response Variable(s)
This is a required field. This list may contain one or more numeric variables.

'Perform Analysis Per Site' Option
If this option is selected and the Environment factor is specified, the analysis will be performed for each level of the Environment factor.

If this option is selected but the Environment factor is **not** specified, PBTools will perform the analysis as if the entire data came from only one site.

'Perform Multiple Environment Analysis' Option
If this option is selected, the *Environment* factor must be specified.

Environment
This is a required field if the 'Perform Multiple Environment Analysis' option is selected. Moreover, the number of its levels must be at least two.

Tester
This is a required field.

F2
This is a required field.

Block
This is a required field if the selected experimental design is Randomized Complete Block or Alpha-Lattice.

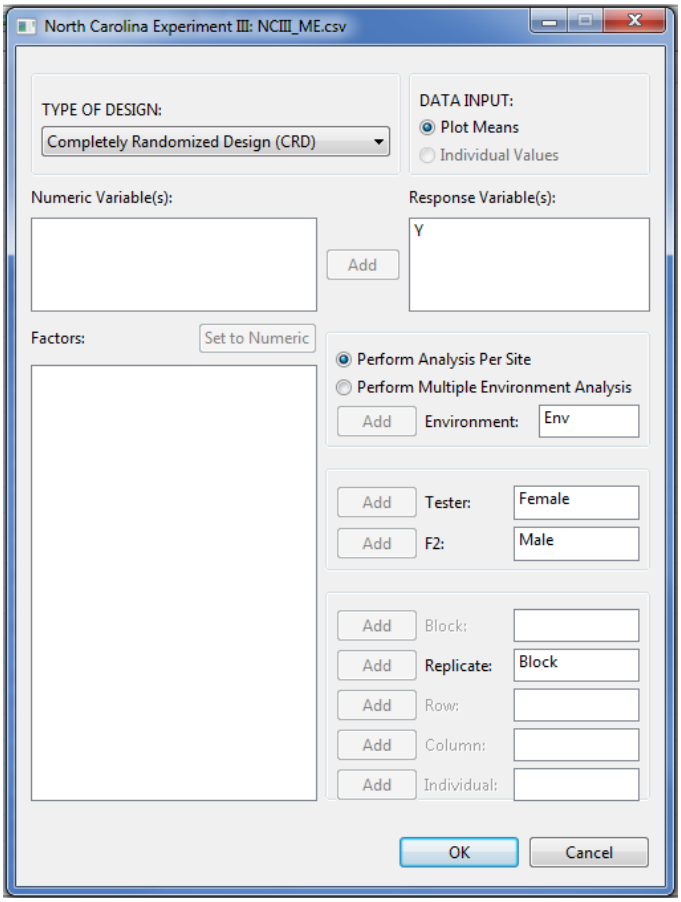
Replicate
This is a required field if the selected experimental design is Completely Randomized Design, Alpha-Lattice or Row-Column.

Row
This is a required field if the selected experimental design is Row-Column.

Column
This is a required field if the selected experimental design is Row-Column.

Individual
This is a required field if the selected data input is Individual Values.

For the example, the completed dialog box should appear as illustrated below:



- Click **OK**.

- Sample output of the analysis (only results from *Env* = *A*) is shown below:

```
DATA FILE: E:/PBTools/Projects/SampleProject/Data/NCIII_ME.csv

DESIGN: NORTH CAROLINA EXPERIMENT III IN RCB
```

```
 ----------------------------
 RESPONSE VARIABLE: Y
 ----------------------------
 ----------------------------
 ANALYSIS FOR: Env = A
 ----------------------------

 DATA SUMMARY:
  Factors   No of Levels   Levels
  Env       1              A
  Male      16             1  10  11 ... 9
  Female    2              1  2
  Block     2              1  2
Number of observations read: 64
Number of observations used: 64
```

Analysis of Variance Table

```
ANOVA TABLE FOR THE EXPERIMENT:

            Df Sum Sq Mean Sq F value Pr(>F)
Block        1 3.8025  3.8025   13.13 0.0010
Female       1 0.9025  0.9025    3.12 0.0874
Male        15 6.8294  0.4553    1.57 0.1400
Female:Male 15 5.1925  0.3462    1.20 0.3255
Residuals   31 8.9775  0.2896
-------
REMARK: Raw dataset is balanced.
```

Estimated Variance Components of Random Effects

```
LINEAR MIXED MODEL FIT BY RESTRICTED MAXIMUM LIKELIHOOD:

 Formula:  Y ~ 1 + Female + (1|Block) + (1|Male) + (1|Female:Male)

      AIC      BIC    logLik deviance  REMLdev
 130.0835 143.0368 -59.04175 115.1131 118.0835

 Random Effects:
 Groups      Variance Std. Deviation
 Female:Male   0.0283         0.1682
 Male          0.0273         0.1652
 Block         0.1098         0.3313
 Residual      0.2896         0.5381
```

Genetic Variance Components

```
ESTIMATES OF GENETIC VARIANCE COMPONENTS:

                Estimate
```

```
VA              0.109124
VD              0.056570
h2-narrow sense 0.239680
H2-broad sense  0.363930
h2-family       0.158537
Dominance Ratio 1.018235
```

Triple Test Cross

This design involves taking a number of F_2 individuals from a cross between two inbred lines (P_1 and P_2) and crossing each back to these same two parental lines and F_1. The two parental inbreds and F_1 thus act as testers against which the F_2 are assessed.

The steps to perform such analysis are listed below:

- Import *TTC_ME* file from package. Double-click the file to view it in the Data Viewer. The example file contains data for an experiment conducted for crosses derived from a Triple Test Cross design with 10 F_2 lines labeled as *Male* and three testers labeled as *Female* using Randomized Complete Block (RCB) field design with three blocks (*1, 2* and *3*) in two environments (*A* and *B*) and one trait, *Y*.

- Choose **Analysis | Mating Designs | Triple Test Cross | Triple Test Cross (no parents)…** from the menu bar.

- Opening the data for the first time, *Block*, *Male* fields are regarded by R as numerical variables, they need to be changed as factors. Choose these variables and click on the **Set to Factor** button.

- Specify the required fields and appropriate options for the analysis.

  Type of Design
  There are four available experimental designs, namely: Completely Randomized Design (CRD, Randomized Complete Block Design (RCB), Alpha-Lattice and Row-Column Design.

  Data Input
  The input data for the analysis may be plot-means or observations of individual plants (*not yet implemented*).

  Response Variable(s)
  This is a required field. This list may contain one or more numeric variables.

'Perform Analysis Per Site' Option

If this option is selected and the Environment factor is specified, the analysis will be performed for each level of the Environment factor.

If this option is selected but the Environment factor is **not** specified, PBTools will perform the analysis as if the entire data came from only one site.

'Perform Multiple Environment Analysis' Option

If this option is selected, the *Environment* factor must be specified.

Environment

This is a required field if the 'Perform Multiple Environment Analysis' option is selected. Moreover, the number of its levels must be at least two.

Tester

This is a required field.

Levels of Tester

Once the Tester factor is specified, its levels appear on the 'Levels of Tester' text box. The user should specify which among these levels correspond to P_1, P_2 and F_1.

P1

This is a required field. This corresponds to the level of the Tester factor that represents P1.

P2

This is a required field. This corresponds to the level of the Tester factor that represents P2.

F1

This is a required field. This corresponds to the level of the Tester factor that represents F1.

F2

This is a required field.

Block

This is a required field if the selected experimental design is Randomized Complete Block or Alpha-Lattice.

Replicate

This is a required field if the selected experimental design is Completely Randomized Design, Alpha-Lattice or Row-Column.

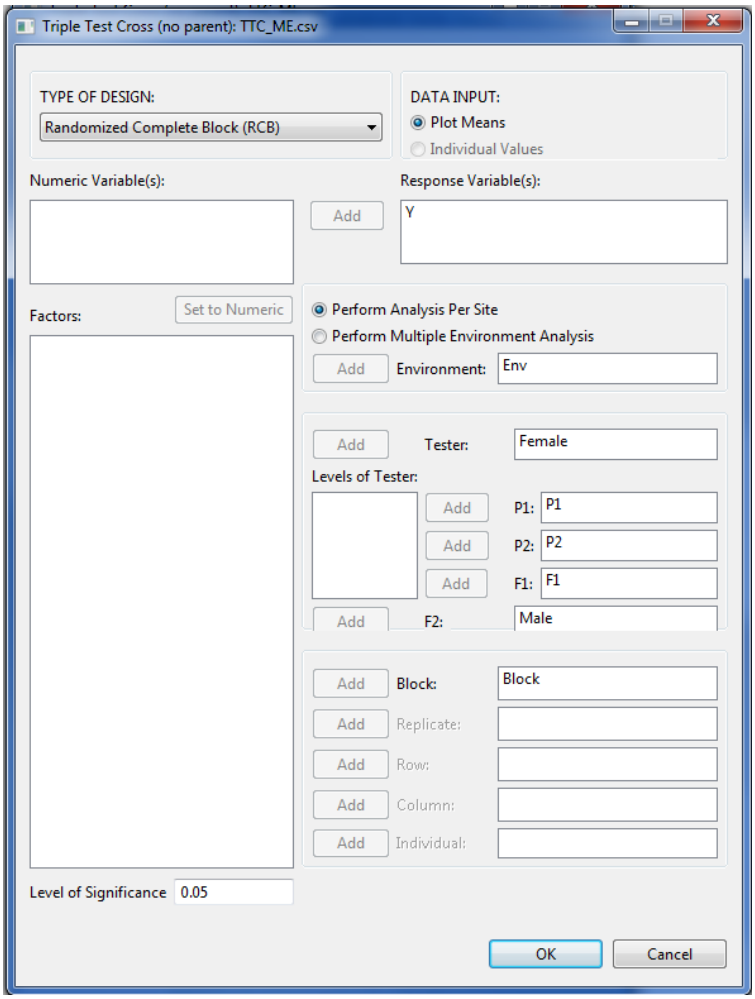
Row
This is a required field if the selected experimental design is Row-Column.

Column
This is a required field if the selected experimental design is Row-Column.

Individual
This is a required field if the selected data input is Individual Values.

Level of Significance
Its default value is 0.05. The user can change this value by specifying a numeric value from zero to 1.

For the example, the completed dialog box should appear as illustrated below:



- Click **OK**.

- Sample output of the analysis (only results from *Env* = *A*) is shown below:

Data Summary

```
DATA FILE: E:/PBTools/Projects/SampleProject/Data/TTC_ME.csv

DESIGN: TRIPLE TEST CROSS (NO PARENTS) IN RCB

------------------------------
RESPONSE VARIABLE: Y
------------------------------
------------------------------
ANALYSIS FOR: Env  =  A
------------------------------
DATA SUMMARY:

 Factors   No of Levels   Levels
 Env       1              A
 Block     3              1  2  3
 Female    3              F1  P1  P2
 Male      10             1  10  2 ... 9

Number of observations read:  90
Number of observations used:  90
```

Analysis of Variance Table

```
ANOVA TABLE FOR THE EXPERIMENT:

            Df Sum Sq Mean Sq F value Pr(>F)
Block        2 1.8842  0.9421   12.05 0.0000
Female       2 1.0709  0.5354    6.85 0.0021
Male         9 9.4432  1.0492   13.42 0.0000
Female:Male 18 1.9891  0.1105    1.41 0.1604
Residuals   58 4.5358  0.0782
-------
REMARK: Raw dataset is balanced.
```

Testing for the Differences Among Crosses

```
TESTING FOR THE SIGNIFICANCE OF CROSS EFFECT: (Crosses = Female:Male)

Model: Y ~ Crosses + (1|Block)

Analysis of Variance Table with Satterthwaite Denominator Df
        Df  Sum Sq Mean Sq F value   Denom Pr(>F)
Crosses 29 12.5032  0.4311  5.5132 57.9793 0.0000
```

Testing for Epistasis

```
ANOVA FOR TESTING EPISTASIS:

            Df Sum Sq Mean Sq F value Pr(>F)
AxA          1 0.0569  0.0569    0.78 0.3891
AxD and DxD  9 0.8142  0.0905    1.24 0.3328
Total       10 0.8711  0.0871    1.19 0.3572
Residuals   18 1.3149  0.0730
-------
REMARK: Raw dataset is balanced.


ANOVA TABLE:

     Df Sum Sq Mean Sq F value Pr(>F)
s     9 9.4432  1.0492   16.25 0.0000
e(s) 18 1.1624  0.0646
d     9 0.8227  0.0914    1.21 0.3476
e(d) 18 1.3553  0.0753
-------
REMARK: Raw dataset is balanced.
```

Genetic Variance Components and Heritability

```
ESTIMATES OF GENETIC VARIANCE COMPONENTS:

                           Estimate       SE
VA                         0.437630 0.039622
VD                         0.005370 0.000232
VE                         0.064580
VP                         0.507580
Broad Sense Heritability  0.872768 0.004519
Narrow Sense Heritability 0.862188 0.005221
Dominance Ratio           0.156662
```

Diallel Analysis (Griffing Method)

Method 1

In this design, all parents are crossed to make hybrids in all possible combinations. Parents and reciprocal crosses are involved along with F_1.

The steps to perform such analysis are listed below:

- Import *Diallel_M1* file from package. Double-click the file to view it in the Data Viewer. The example file contains data for an experiment conducted for crosses derived from full diallel design with seven parents using Randomized Complete Block (RCB) design with three blocks (*1, 2* and *3*) in two environments (*Normal* and *Saline*) and three traits, *Plant_height, Grain_yield* and *Th_grain_weight*.

- Choose **Analysis | Mating Designs | Diallel Analysis | Griffing Method I…** from the menu bar.

- Opening the data for the first time, *Block*, *P1* and *P2* fields are regarded by R as numerical variables. To set these as factors, choose these variables and click on the **Set to Factor** button.

- Specify the required fields and appropriate options for the analysis.

  Type of Design
  There are four available experimental designs, namely: Completely Randomized Design (CRD), Randomized Complete Block Design (RCB), Alpha-Lattice and Row-Column Design.

  'Cross' Option
  This is selected if the parental lines are from random-mating population (outcrossing species).

  'Self' Option
  This is selected if the parental lines are inbreds.

  Data Input
  The input data for the analysis may be plot-means or observations of individual plants (*not yet implemented*).

  Response Variable(s)
  This is a required field. This list may contain one or more numeric variables.

  'Perform Analysis Per Site' Option
  If this option is selected and the Environment factor is specified, the analysis will be performed for each level of the Environment factor.

  If this option is selected but the Environment factor is **not** specified, PBTools will perform the analysis as if the entire data came from only one site.

'Perform Multiple Environment Analysis' Option
If this option is selected, the *Environment* factor must be specified.

Environment
This is a required field if the 'Perform Multiple Environment Analysis' option is selected. Moreover, the number of its levels must be at least two.

P1
This is a required field.
P2
This is a required field.

Block
This is a required field if the selected experimental design is Randomized Complete Block or Alpha-Lattice.

Replicate
This is a required field if the selected experimental design is Completely Randomized Design, Alpha-Lattice or Row-Column.

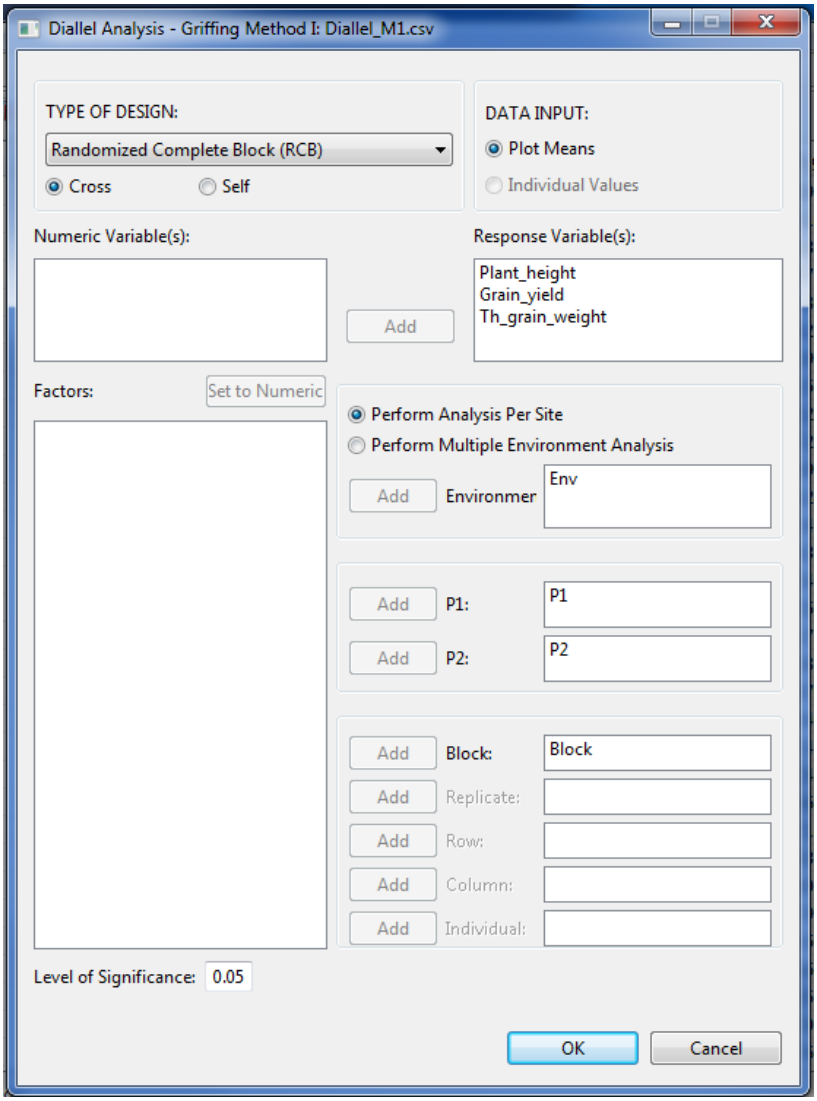
Row
This is a required field if the selected experimental design is Row-Column.

Column
This is a required field if the selected experimental design is Row-Column.

Individual
This is a required field if the selected data input is Individual Values.

Level of Significance
Its default value is 0.05. The user can change this value by specifying a numeric value from zero to 1.

For the example, the completed dialog box should appear as illustrated below:



- Click **OK**.

- Sample output of the analysis (only results from *response variable* = *Plant_height* and *Env* = *Normal*) is shown below:

```
DATA FILE: E:/PBTools/Projects/SampleProject/Data/Diallel_M1.csv

DIALLEL ANALYSIS: GRIFFING METHOD I IN RCB (CROSS)

------------------------------
RESPONSE VARIABLE: Plant_height
------------------------------
------------------------------
ANALYSIS FOR: Env  =  Normal
------------------------------
```

```
 DATA SUMMARY:

 Factors   No of Levels   Levels
 Env       1             Normal
 P1        7             1  2  3  4  5  6  7
 P2        7             1  2  3  4  5  6  7
 Block     3             1  2  3

Number of observations read:  147
Number of observations used:  147
```

Analysis of Variance Table

```
ANOVA TABLE FOR THE EXPERIMENT:

          Df     Sum Sq   Mean Sq F value Pr(>F)
Block      2  3026.5310 1513.2660   19.74 0.0000
P1         6 32496.7700 5416.1280   70.67 0.0000
P2         6 21855.2900 3642.5480   47.53 0.0000
P1:P2     36 22865.0000  635.1388    8.29 0.0000
Residuals 96  7357.5760   76.6414
-------
REMARK: Raw dataset is balanced.
```

Testing for the Differences among Crosses

```
TESTING FOR THE SIGNIFICANCE OF CROSS EFFECT: (Crosses = P1:P2)

Model: Plant_height ~ Crosses + (1|Block)

Analysis of Variance Table with Satterthwaite Denominator Df
        Df   Sum Sq  Mean Sq F value   Denom Pr(>F)
Crosses 48 77217.05 1608.689 20.9898 95.9989 0.0000

MATRIX OF MEANS:

     P2=1     P2=2     P2=3     P2=4     P2=5     P2=6     P2=7
P1=1 142.9000 148.3333 163.9000 152.9000 142.3667 160.6667 191.3333
P1=2 144.7667 129.6667 142.9000 143.9000 131.5667 143.5333 186.6667
P1=3 167.5667 149.0333 131.5667 163.5667 136.9000 166.4333 189.7667
P1=4 154.1000 156.1000 147.7667 159.3333 149.8000 164.8000 200.2333
P1=5 142.1000 133.8000 138.9000 149.6667 122.4333 138.9000 175.7667
P1=6 173.1000 146.2000 167.7667 184.3333 156.7667 146.1333 195.5667
P1=7 195.7667 194.5667 193.8000 213.4667 169.1000 218.0333 157.5333
-------
REMARK: Raw dataset is balanced.
```

Combining Ability Analysis

```
ANALYSIS OF VARIANCE:

           Df     Sum Sq   Mean Sq F value Pr(>F)
GCA         6 17675.6700 2945.9450    9.05 0.0000
SCA        21  6983.5130  332.5482   13.02 0.0000
Reciprocal 21  1079.8370   51.4208    2.01 0.0118
Error      96  2452.5120   25.5470
-------
NOTE: MS* = 325.408666   Error used for GCA MS with df = 1920.5

GENERAL COMBINING ABILITY EFFECTS (diagonal), SPECIFIC COMBINING
ABILITY EFFECTS (above diagonal) AND RECIPROCAL EFFECTS (below diagonal):

     P2=1     P2=2     P2=3     P2=4     P2=5     P2=6     P2=7
P1=1  -1.9718  -0.0997  11.1741  -8.7639   0.5980   3.9718   7.0122
P1=2   1.7833 -12.1146   1.5503  -2.1211   1.1908  -7.9020  14.2218
P1=3  -1.8333  -3.0667  -4.2051  -4.3639  -1.5020   6.4218   7.4789
P1=4  -0.6000  -6.1000   7.9000   3.4997   2.6265   6.1837  14.8408
P1=5   0.1333  -1.1167  -1.0000   0.0667 -17.1289   0.0789   1.0527
P1=6  -6.2167  -1.3333  -0.6667  -9.7667  -8.9333   4.1473  14.1432
P1=7  -2.2167  -3.9500  -2.0167  -6.6167   3.3333 -11.2333  27.7735

ESTIMATES OF VARIANCE COMPONENTS:

           Estimate Std. Error
GCA        187.1811   121.6995
SCA        174.9193    58.5110
Reciprocal  12.9369     8.1458
Error       25.5470     3.6874

ESTIMATES OF GENETIC VARIANCE COMPONENTS:

                  Estimate
VA              748.724542
VD              699.677221
h2-narrow sense   0.507972
H2-broad sense    0.982668
Dominance Ratio   1.367108

TABLE OF STANDARD ERRORS AND LSDs:

        Std. Error     LSD
Gi          1.2506
Sii         4.3323
Sij         3.1057
Rij         3.5740
Gi-Gj       1.9104  3.7921
Sii-Sjj     6.0412 11.9916
```

```
Sii-Sij     5.8882 11.6880
Sii-Sjk     5.2318 10.3851
Sij-Sik     4.6795  9.2887
Sij-Skl     4.2718  8.4794
Rij-Rkl     5.0544 10.0329
```

Method 2

If no reciprocal differences exist, there is no need to make reciprocal crosses. The parents and one set of F_1's can be analyzed.

The steps to perform such analysis are listed below:

- Import *Diallel_M2* file from package. Double-click the file to view it in the Data Viewer. The example file contains data for an experiment conducted for crosses derived from partial diallel design (parents and one set of F_1's) with seven parents using Randomized Complete Block (RCB) design with three blocks (*1, 2* and *3*) in two environments (*Normal* and *Saline*) and three traits, *Plant_height, Grain_yield* and *Th_grain_weight*.

- Choose **Analysis | Mating Designs | Diallel Analysis | Griffing Method II…** from menu bar.

- Opening the data for the first time, *Block*, *P1* and *P2* fields are regarded by R as numerical variables. To set these as factors, choose these variables and click on the **Set to Factor** button.

- Specify the required fields and appropriate options for the analysis.

  Type of Design
  There are four available experimental designs, namely: Completely Randomized Design (CRD), Randomized Complete Block Design (RCB), Alpha-Lattice and Row-Column Design.

  'Cross' Option
  This is selected if the parental lines are from random-mating population (outcrossing species).

  'Self' Option
  This is selected if the parental lines are inbreds.

Data Input
The input data for the analysis may be plot-means or observations of individual plants (*not yet implemented*).

Response Variable(s)
This is a required field. This list may contain one or more numeric variables.

'Perform Analysis Per Site' Option
If this option is selected and the Environment factor is specified, the analysis will be performed for each level of the Environment factor.

If this option is selected but the Environment factor is **not** specified, PBTools will perform the analysis as if the entire data came from only one site.

'Perform Multiple Environment Analysis' Option
If this option is selected, the *Environment* factor must be specified.

Environment
This is a required field if the 'Perform Multiple Environment Analysis' option is selected. Moreover, the number of its levels must be at least two.

P1
This is a required field.

P2
This is a required field.

Block
This is a required field if the selected experimental design is Randomized Complete Block or Alpha-Lattice.

Replicate
This is a required field if the selected experimental design is Completely Randomized Design, Alpha-Lattice or Row-Column.

Row
This is a required field if the selected experimental design is Row-Column.

Column
This is a required field if the selected experimental design is Row-Column.

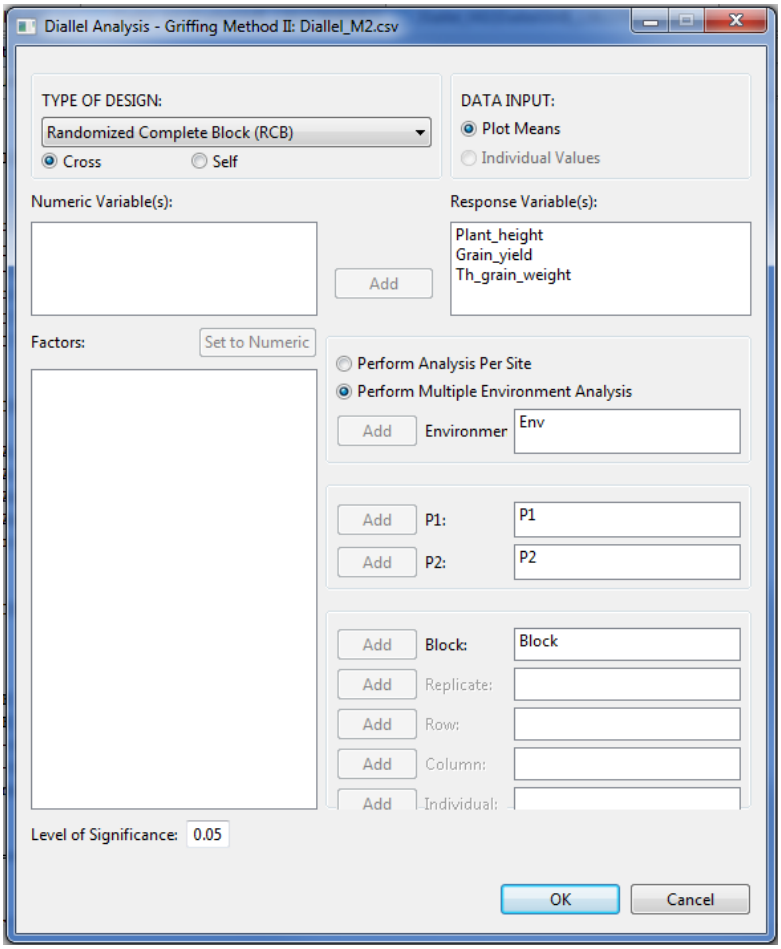
Individual

This is a required field if the selected data input is Individual Values.

Level of Significance

Its default value is 0.05. The user can change this value by specifying a numeric value from zero to 1.

For the example, the completed dialog box should appear as illustrated below:



- Click **OK**.

- Sample output of the analysis (only results from *response variable* = *Plant_height* and *Env* = *Normal*) is shown below:

```
DATA FILE: E:/PBTools/Projects/SampleProject/Data/Diallel_M2.csv

DIALLEL ANALYSIS: GRIFFING METHOD II IN RCB (CROSS)
```

```
------------------------------
RESPONSE VARIABLE: Plant_height
------------------------------
------------------------------
ANALYSIS FOR: Env  =  Normal
------------------------------
DATA SUMMARY:

 Factors   No of Levels   Levels
 P1        7              1  2  3  4  5  6  7
 P2        7              1  2  3  4  5  6  7
 Block     3              1  2  3
 Env       1              Normal

Number of observations read:  84
Number of observations used:  84


ANOVA TABLE FOR THE EXPERIMENT:

          Df     Sum Sq   Mean Sq F value Pr(>F)
Block      2   852.4474  426.2237    9.65 0.0003
P1:P2     27 36854.8400 1364.9940   30.92 0.0000
Residuals 54  2383.9790   44.1478
-------
REMARK: Raw dataset is balanced.
```

Testing for the Differences among Crosses

```
TESTING FOR THE SIGNIFICANCE OF CROSS EFFECT: (Crosses = P1:P2)

Model: Plant_height ~ Crosses + (1|Block)

Analysis of Variance Table with Satterthwaite Denominator Df
        Df   Sum Sq  Mean Sq F value   Denom Pr(>F)
Crosses 27 36854.84 1364.994 30.9188 53.9992 0.0000

MATRIX OF MEANS:

  1        2        3        4        5        6        7
1 142.9000 148.3333 163.9000 152.9000 142.3667 160.6667 191.3333
2          129.6667 142.9000 143.9000 131.5667 143.5333 186.6667
3                   131.5667 163.5667 136.9000 166.4333 189.7667
4                            159.3333 149.8000 164.8000 200.2333
5                                     122.4333 138.9000 175.7667
6                                              146.1333 195.5667
7                                                       157.5333
-------
REMARK: Raw dataset is balanced.
```

Combining Ability Analysis

ANALYSIS OF VARIANCE:

```
      Df    Sum Sq   Mean Sq F value Pr(>F)
GCA    6 7784.3680 1297.3950    6.05 0.0008
SCA   21 4500.5790  214.3133   14.56 0.0000
Error 54  794.6640   14.7160
```

GENERAL COMBINING ABILITY EFFECTS (diagonal), SPECIFIC COMBINING ABILITY EFFECTS (above diagonal) AND RECIPROCAL EFFECTS (below diagonal):

```
  1        2        3        4        5        6        7
1  -0.6608   3.1454  10.8935  -7.5806   1.1861   3.7083  13.0157
2          -10.5571  -0.2102  -6.6843   0.2824  -3.5287  18.2454
3                    -2.7386   5.1639  -2.2028  11.5528  13.5269
4                              4.7354   3.2231   2.4454  16.5194
5                                      -14.5646  -4.1546  11.3528
6                                                1.2132  15.3750
7                                                        22.5725
```

ESTIMATES OF VARIANCE COMPONENTS:

```
      Estimate Std. Error
GCA   120.3424   217.0737
SCA   199.5973    66.1426
Error  14.7160     2.8321
```

ESTIMATES OF GENETIC VARIANCE COMPONENTS:

```
                  Estimate
VA              481.369471
VD              798.389175
h2-narrow sense   0.371865
H2-broad sense    0.988632
Dominance Ratio   1.821307
```

TABLE OF STANDARD ERRORS AND LSDs:

```
        Std. Error     LSD
Gi          1.1839
Sii         2.9299
Sij         3.4430
Gi-Gj       1.8084  3.6256
Sii-Sjj     4.0437  8.1070
Sij-Sik     5.1149 10.2547
Sij-Skl     4.7845  9.5924
```

- Suppose the user opted to perform Multiple Environment Analysis, sample output of the analysis (only results from *response variable = Plant_height*) is shown below:

```
DATA FILE: E:/PBTools/Projects/SampleProject/Data/Diallel_M2.csv

MULTIPLE ENVIRONMENT ANALYSIS

DIALLEL ANALYSIS: GRIFFING METHOD II IN RCB (CROSS)

------------------------------
RESPONSE VARIABLE: Plant_height
------------------------------

DATA SUMMARY:
 Factors   No of Levels   Levels
 P1        7              1  2  3  4  5  6  7
 P2        7              1  2  3  4  5  6  7
 Env       2              Normal  Saline
 Block     3              1  2  3

 Number of observations read: 168
 Number of observations used: 168
```

Testing the Significance of Genotypic Differences

```
ANOVA TABLE: (Crosses = P1:P2)

               Df     Sum Sq    Mean Sq F value Pr(>F)
Env             1 17111.4300 17111.4300   28.18 0.0061
Block(Env)      4  2429.2920   607.3231   14.44 0.0000
Crosses        27 80041.5800  2964.5030   31.42 0.0000
Crosses x Env  27  2547.4580    94.3503    2.24 0.0018
Residuals     108  4543.8540    42.0727
-------
REMARK: Raw dataset is balanced.
```

Combining Ability Analysis

```
ANOVA TABLE:

           Df     Sum Sq   Mean Sq F value Pr(>F)
GCA         6 54367.6700 9061.2780   51.14 0.0001
SCA        21 25673.9100 1222.5670   17.30 0.0000
GCAxE       6  1063.1420  177.1904    4.21 0.0008
SCAxE      21  1484.3160   70.6817    1.68 0.0451
Residuals 108  4543.8540   42.0727
-------
REMARK: Raw dataset is balanced.
```

MATRIX OF MEANS:

	1	2	3	4	5	6	7
1	131.0000	133.8333	153.5500	140.7333	128.9000	154.4333	182.2833
2		116.8333	130.5500	133.8500	122.7167	129.0500	175.9000
3			122.0000	154.5000	131.2167	153.2167	177.7833
4				137.3333	138.5667	155.7833	195.2333
5					111.8833	132.5667	166.3333
6						143.0167	188.6167
7							155.1000

REMARK: Raw dataset is balanced.

GENERAL COMBINING ABILITY EFFECTS, SPECIFIC COMBINING ABILITY EFFECTS (above diagonal):

	1	2	3	4	5	6	7
1		1.4884	11.7329	-6.4745	-1.7227	6.5181	12.5181
2			-0.5819	-2.6727	2.7792	-8.1801	16.8199
3				8.5051	1.8069	6.5144	9.2310
4					3.7662	3.6903	21.2903
5						-2.9412	8.9755
6							13.9662
7							
GCA	-1.6418	-12.3270	-2.8548	2.5360	-14.0492	3.2434	25.0934

TABLE OF STANDARD ERRORS AND LSDs:

	Std. Error	LSD
Gi	0.8172	
Sii	2.3767	
Sij	2.0225	
Gi-Gj	1.2483	2.4743
Sii-Sjj	2.7913	5.5328
Sij-Sik	3.5307	6.9985
Sij-Skl	3.3027	6.5465

ESTIMATES OF VARIANCE COMPONENTS:

	Estimate
GCA	143.1889
SCA	191.9809
GCAxE	3.9448
SCAxE	9.5363
Residuals	42.0727

```
ESTIMATES OF GENETIC VARIANCE COMPONENTS:

                   Estimate
VA               572.755708
VAxE              15.779068
VD               767.923774
VDxE              38.145306
h2-narrow sense    0.398667
H2-broad sense     0.933181
Dominance Ratio    1.637530
```

Method 3

In this design, both F1's and reciprocal crosses are the only ones being considered.

The steps to perform such analysis are listed below:

- Import *Diallel_M3* file from package. Double-click the file to view it in the Data Viewer. The example file contains data for an experiment conducted for crosses derived from partial diallel design (F1's and reciprocal crosses) with seven parents using Randomized Complete Block (RCB) design with three blocks (*1, 2* and *3*) in two environments (*Normal* and *Saline*) and three traits, *Plant_height, Grain_yield* and *Th_grain_weight*.

- Choose **Analysis | Mating Designs | Diallel Analysis | Griffing Method III…** from the menu bar.

- Opening the data for the first time, *Block*, *P1* and *P2* fields are regarded by R as numerical variables, they need to be changed as factors. Choose these variables and click on the **Set to Factor** button.

- Specify the required fields and appropriate options for the analysis.

Type of Design

There are four available experimental designs, namely: Completely Randomized Design (CRD), Randomized Complete Block Design (RCB), Alpha-Lattice and Row-Column Design.

'Cross' Option

This is selected if the parental lines are from random-mating population (outcrossing species).

'Self' Option
This is selected if the parental lines are inbreds.

Data Input
The input data for the analysis may be plot-means or observations of individual plants (*not yet implemented*).

Response Variable(s)
This is a required field. This list may contain one or more numeric variables.

'Perform Analysis Per Site' Option
If this option is selected and the Environment factor is specified, the analysis will be performed for each level of the Environment factor.

If this option is selected but the Environment factor is **not** specified, PBTools will perform the analysis as if the entire data came from only one site.

'Perform Multiple Environment Analysis' Option
If this option is selected, the *Environment* factor must be specified.

Environment
This is a required field if the 'Perform Multiple Environment Analysis' option is selected. Moreover, the number of its levels must be at least two.

P1
This is a required field.

P2
This is a required field.

Block
This is a required field if the selected experimental design is Randomized Complete Block or Alpha-Lattice.

Replicate
This is a required field if the selected experimental design is Completely Randomized Design, Alpha-Lattice or Row-Column.

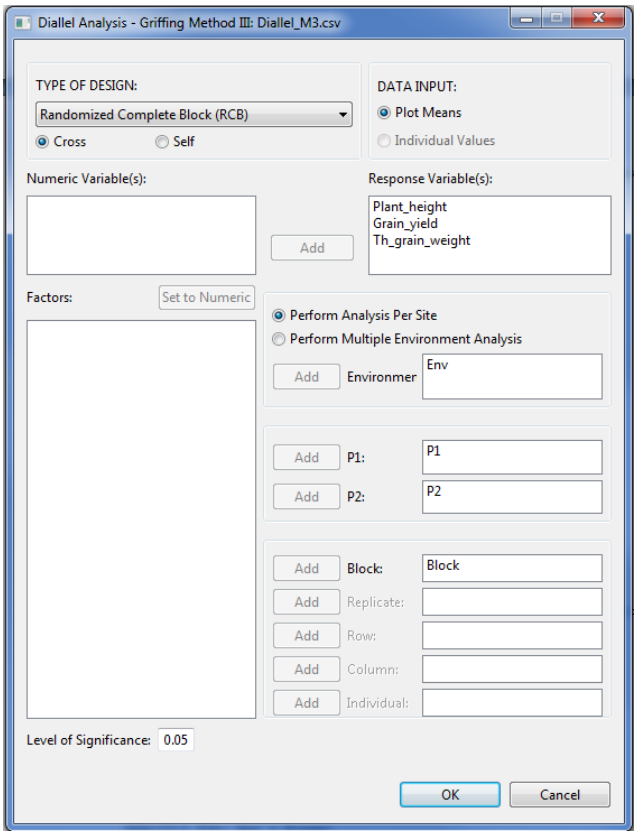
Row
This is a required field if the selected experimental design is Row-Column.

Column
This is a required field if the selected experimental design is Row-Column.

Individual
This is a required field if the selected data input is Individual Values.

Level of Significance
Its default value is 0.05. The user can change this value by specifying a numeric value from zero to 1.

For the example, the completed dialog box should appear as illustrated below:



- Click **OK.**

- Sample output of the analysis (only results from *response variable* = *Plant_height* and *Env* = *Normal*) is shown below:

```
DATA FILE: E:/PBTools/Projects/SampleProject/Data/Diallel_M3.csv

DIALLEL ANALYSIS: GRIFFING METHOD III IN RCB (CROSS)
```

```
------------------------------
RESPONSE VARIABLE: Plant_height
------------------------------
------------------------------
ANALYSIS FOR: Env  =  Normal
------------------------------

DATA SUMMARY:

 Factors   No of Levels   Levels
 Env       1              Normal
 P1        7              1  2  3  4  5  6  7
 P2        7              1  2  3  4  5  6  7
 Block     3              1  2  3

Number of observations read:  126
Number of observations used:  126


ANOVA TABLE FOR THE EXPERIMENT:

          Df     Sum Sq   Mean Sq F value Pr(>F)
Block      2  3024.4780 1512.2390   18.44 0.0000
P1:P2     41 64423.4000 1571.3030   19.16 0.0000
Residuals 82  6723.2290   81.9906
-------
REMARK: Raw dataset is balanced.
```

Testing for the Differences among Crosses

```
TESTING FOR THE SIGNIFICANCE OF CROSS EFFECT: (Crosses = P1:P2)

Model: Plant_height ~ Crosses + (1|Block)

Analysis of Variance Table with Satterthwaite Denominator Df
        Df  Sum Sq  Mean Sq F value   Denom Pr(>F)
Crosses 41 64423.4 1571.303 19.1644 81.9991 0.0000

MATRIX OF MEANS:

     P2=1     P2=2     P2=3     P2=4     P2=5     P2=6     P2=7
P1=1          148.3333 163.9000 152.9000 142.3667 160.6667 191.3333
P1=2 144.7667          142.9000 143.9000 131.5667 143.5333 186.6667
P1=3 167.5667 149.0333          163.5667 136.9000 166.4333 189.7667
P1=4 154.1000 156.1000 147.7667          149.8000 164.8000 200.2333
P1=5 142.1000 133.8000 138.9000 149.6667          138.9000 175.7667
P1=6 173.1000 146.2000 167.7667 184.3333 156.7667          195.5667
P1=7 195.7667 194.5667 193.8000 213.4667 169.1000 218.0333
-------
REMARK: Raw dataset is balanced.
```

Combining Ability Analysis

ANALYSIS OF VARIANCE:

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
GCA	6	19425.2300	3237.5380	46.76	0.0000
SCA	14	969.4004	69.2429	2.53	0.0047
Reciprocal	21	1079.8370	51.4208	1.88	0.0231
Error	82	2241.0870	27.3303		

GENERAL COMBINING ABILITY EFFECTS (diagonal), SPECIFIC COMBINING ABILITY EFFECTS (above diagonal) AND RECIPROCAL EFFECTS (below diagonal):

	P2=1	P2=2	P2=3	P2=4	P2=5	P2=6	P2=7
P1=1	-3.0671	0.2733	8.7633	-8.7033	1.5300	1.1333	-2.9967
P1=2	1.7833	-14.6205	0.5500	-0.6500	3.5333	-9.3300	5.6233
P1=3	-1.8333	-3.0667	-3.9271	-5.6767	-1.9433	2.2100	-3.9033
P1=4	-0.6000	-6.1000	7.9000	1.3062	4.6567	4.4433	5.9300
P1=5	0.1333	-1.1167	-1.0000	0.0667	-20.1938	-0.7900	-6.9867
P1=6	-6.2167	-1.3333	-0.6667	-9.7667	-8.9333	4.8529	2.3333
P1=7	-2.2167	-3.9500	-2.0167	-6.6167	3.3333	-11.2333	35.6495

ESTIMATES OF VARIANCE COMPONENTS:

	Estimate	Std. Error
GCA	316.8295	186.9377
SCA	20.9563	13.2586
Reciprocal	12.0452	8.2164
Error	27.3303	4.2683

ESTIMATES OF GENETIC VARIANCE COMPONENTS:

	Estimate
VA	1267.318169
VD	83.825111
h2-narrow sense	0.919363
H2-broad sense	0.980173
Dominance Ratio	0.363713

TABLE OF STANDARD ERRORS AND LSDs:

	Std. Error	LSD
Gi	1.5306	
Sii	3.0183	
Rij	3.6966	
Gi-Gj	2.3380	4.6510
Sij-Sik	4.6759	9.3019
Sij-Skl	4.0495	8.0557

Method 4

In this design, only one set of F1's are considered.

The steps to perform such analysis are listed below:

- Import *Diallel_M4* file from package. Double-click the file to view it in the Data Viewer. The example file contains data for an experiment conducted for crosses derived from partial diallel design (one set of F_1's) with seven parents using Randomized Complete Block (RCB) design with three blocks (*1, 2* and *3*) in two environments (*Normal* and *Saline*) and three traits, *Plant_height, Grain_yield* and *Th_grain_weight*.

- Choose **Analysis | Mating Designs | Diallel Analysis | Griffing Method IV…** from the menu bar.

- Opening the data for the first time, *Block*, *P1* and *P2* fields are regarded by R as numerical variables, they need to be changed as factors. Choose these variables and click on the **Set to Factor** button.

- Specify the required fields and appropriate options for the analysis.

  Type of Design
  There are four available experimental designs, namely: Completely Randomized Design (CRD), Randomized Complete Block Design (RCB), Alpha-Lattice and Row-Column Design.

  'Cross' Option
  This is selected if the parental lines are from random-mating population (outcrossing species).

  'Self' Option
  This is selected if the parental lines are inbreds.

  Data Input
  The input data for the analysis may be plot-means or observations of individual plants (*not yet implemented*).

  Response Variable(s)
  This is a required field. This list may contain one or more numeric variables.

'Perform Analysis Per Site' Option
If this option is selected and the Environment factor is specified, the analysis will be performed for each level of the Environment factor.

If this option is selected but the Environment factor is **not** specified, PBTools will perform the analysis as if the entire data came from only one site.

'Perform Multiple Environment Analysis' Option
If this option is selected, the *Environment* factor must be specified.

Environment
This is a required field if the 'Perform Multiple Environment Analysis' option is selected. Moreover, the number of its levels must be at least two.

P1
This is a required field.
P2
This is a required field.

Block
This is a required field if the selected experimental design is Randomized Complete Block or Alpha-Lattice.

Replicate
This is a required field if the selected experimental design is Completely Randomized Design, Alpha-Lattice or Row-Column.

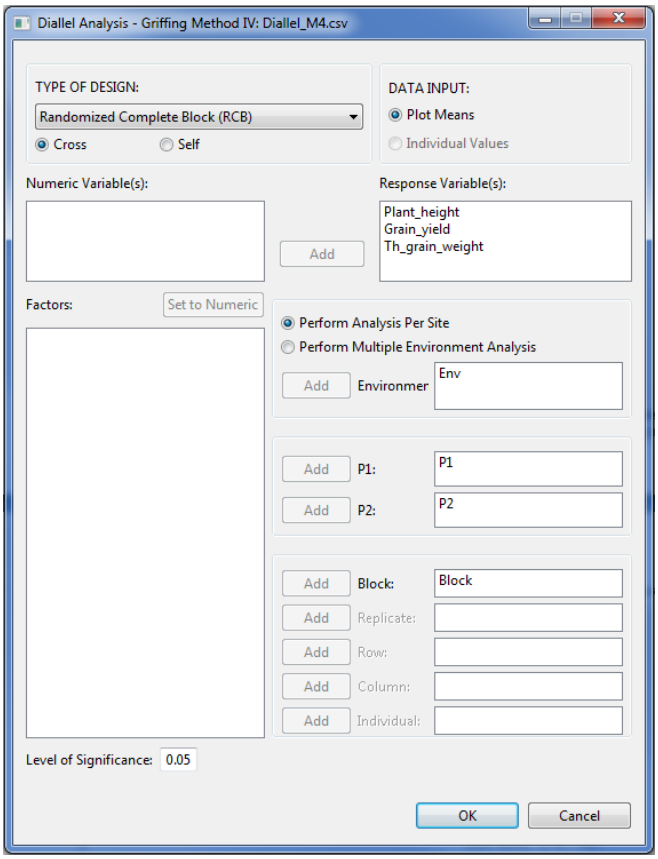
Row
This is a required field if the selected experimental design is Row-Column.

Column
This is a required field if the selected experimental design is Row-Column.

Individual
This is a required field if the selected data input is Individual Values.

Level of Significance
Its default value is 0.05. The user can change this value by specifying a numeric value from zero to 1.

For the example, the completed dialog box should appear as illustrated below:



- Click **OK**.

- Sample output of the analysis (only results from *response variable* = *Plant_height* and *Env* = *Normal*) is shown below:

```
DATA FILE: E:/PBTools/Projects/SampleProject/Data/Diallel_M4.csv

DIALLEL ANALYSIS: GRIFFING METHOD IV IN RCB (CROSS)

------------------------------
RESPONSE VARIABLE: Plant_height
------------------------------
------------------------------
ANALYSIS FOR: Env  =  Normal
------------------------------

DATA SUMMARY:

 Factors   No of Levels   Levels
 Env       1              Normal
 P1        6              1  2  3  4  5  6
 P2        6              2  3  4  5  6  7
 Block     3              1  2  3
```

```
Number of observations read:  63
Number of observations used:  63


ANOVA TABLE FOR THE EXPERIMENT:

          Df     Sum Sq   Mean Sq F value Pr(>F)
Block      2   869.0552  434.5276   10.04 0.0003
P1:P2     20 26919.8900 1345.9950   31.10 0.0000
Residuals 40  1730.9710   43.2743
-------
REMARK: Raw dataset is balanced.
```

Testing for the Differences Among Crosses

```
TESTING FOR THE SIGNIFICANCE OF CROSS EFFECT: (Crosses = P1:P2)

Model: Plant_height ~ Crosses + (1|Block)

Analysis of Variance Table with Satterthwaite Denominator Df
        Df   Sum Sq  Mean Sq F value   Denom Pr(>F)
Crosses 20 26919.89 1345.995 31.1038 39.9994 0.0000


MATRIX OF MEANS:

  1        2        3        4        5        6        7
1          148.3333 163.9000 152.9000 142.3667 160.6667 191.3333
2                   142.9000 143.9000 131.5667 143.5333 186.6667
3                            163.5667 136.9000 166.4333 189.7667
4                                     149.8000 164.8000 200.2333
5                                              138.9000 175.7667
6                                                       195.5667
7
-------
REMARK: Raw dataset is balanced.
```

Combining Ability Analysis

```
ANALYSIS OF VARIANCE:

      Df    Sum Sq   Mean Sq F value Pr(>F)
GCA    6 8630.0020 1438.3340   58.66 0.0000
SCA   14  343.2947   24.5210    1.70 0.0945
Error 40  576.9867   14.4247
```

GENERAL COMBINING ABILITY EFFECTS (diagonal), SPECIFIC COMBINING ABILITY EFFECTS (above diagonal) AND RECIPROCAL EFFECTS (below diagonal):

	1	2	3	4	5	6	7
1	-1.8029	3.0400	5.2933	-8.0533	1.3933	0.7733	-2.4467
2		-14.3229	-3.1867	-4.5333	3.1133	-3.8400	5.4067
3			-1.0095	1.8200	-4.8667	5.7467	-4.8067
4				1.3371	5.6867	1.7667	3.3133
5					-18.6429	-4.1533	-1.1733
6						0.2771	-0.2933
7							34.1638

ESTIMATES OF VARIANCE COMPONENTS:

	Estimate	Std. Error
GCA	282.7625	166.0948
SCA	10.0964	9.8133
Error	14.4247	0.8493

ESTIMATES OF GENETIC VARIANCE COMPONENTS:

	Estimate
VA	1131.050116
VD	40.385524
h2-narrow sense	0.953780
H2-broad sense	0.987836
Dominance Ratio	0.267231

TABLE OF STANDARD ERRORS AND LSDs:

	Std. Error	LSD
Gi	1.5725	
Sii	3.1010	
Gi-Gj	2.4021	4.8547
Sij-Sik	4.8041	9.7095
Sij-Skl	4.1605	8.4086

10. Generation Mean Analysis

In this analysis, *PBTools* uses the weighted regression approach. Two full models are fitted to the data. The first one is "mean = 0 + m + a + d + aa + ad" and the other one is "mean = 0 + m + a + d + aa + dd". For each model, backward regression procedure is used to obtain the best model. Due to the number of parameters to be estimated in each full model, *PBTools* requires that there should be at least six generations available in the data set.

Raw Data as Input

The steps to perform such analysis are listed below:

- Import *GenMean(raw)* file from package. Double-click the file to view it in the Data Viewer.

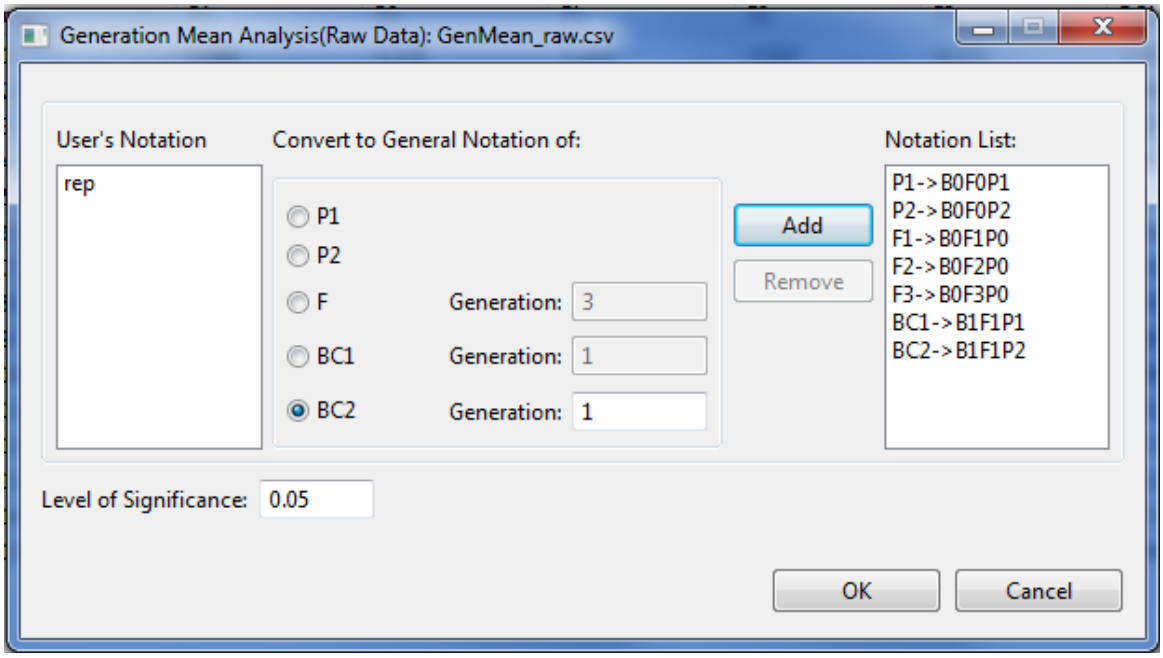
- Choose **Analysis | Generation Mean Analysis | Raw data as input…** from the menu bar.

- For each column that represents a generation, convert it to the general notation that PBTools recognizes by clicking one variable in the user's notation box, selecting the appropriate radio button that corresponds to its generation classification, specifying the generation number if necessary and clicking the **Add** button. A text indicating the conversion will be displayed in the Notation List.

  For the example, do the following:
  - select P1 column in the user's notation box, then classify it as P1 (the default selection) and click **Add** button
  - select P2 column in the user's notation box, then classify it as P2 and click **Add** button
  - select F1 column in the user's notation box, then classify it as F, specify the generation number as 1 and click **Add** button
  - select F2 column in the user's notation box, then classify it as F, specify the generation number as 2 and click **Add** button
  - select F3 column in the user's notation box, then classify it as F, specify the generation number as 3 and click **Add** button
  - select BC1 column in the user's notation box, then classify it as BC1, specify the generation number as 1 and click **Add** button

- select BC2 column in the user's notation box, then classify it as BC2, specify the generation number as 1 and click **Add** button

For the *example*, the completed dialog box should appear as illustrated below:



- Click **OK**.

- Sample output of the analysis is shown below:

```
GENERATION MEAN ANALYSIS


DATA SUMMARY:
 Generation     Mean    Std Dev    No of Obsn
         P1   3.7030     0.2106             9
         P2   4.6010     0.0956             9
         F1   3.2317     0.1697            15
         F2   3.5159     0.1263            15
         F3   4.1406     0.1262            15
        BC1   3.7756     0.1502            15
        BC2   4.4830     0.3794            15


-----------------------------------------------------
FULL MODEL: mean ~ 0 + m + a + d + aa + ad + dd
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MODEL 1 :

Regression Model:  mean ~ 0 + m + a + d + aa + ad + dd

Regression Coefficients:
   Estimate Std. Error t value Pr(>|t|)
m    4.3303     0.9640  4.4923   0.1394
a   -0.4130     0.4293 -0.9618   0.5124
d   -1.1949     3.5035 -0.3410   0.7908
aa  -0.1235     1.0144 -0.1218   0.9228
ad   0.6042     2.2432  0.2693   0.8325
dd   0.0726     2.7793  0.0261   0.9834

 Residual Standard Error: 11.1775
 Multiple R-square:        0.9982
 Adjusted R-square:        0.9872
 F-value:                 91.0825
 p-value:                  0.0800

Observed and Predicted Values of Generation Means:
  Generation Observed Predicted
1         P1 3.703000  3.793826
2         P2 4.601000  4.619727
3         F1 3.231667  3.208086
4         F2 3.515867  3.751047
5         F3 4.140600  4.036147
6        BC1 3.775600  3.664735
7        BC2 4.483000  3.775585

Goodness-of-fit Test
Chi-square value:  124.9361
p-value:           5.255899e-29

------------------------------------------------
MODEL 2 :

Regression Model:  mean ~ 0 + m + a + d + aa + ad

Regression Coefficients:
   Estimate Std. Error t value Pr(>|t|)
m    4.3084     0.3351 12.8574   0.0060
a   -0.4138     0.3028 -1.3665   0.3051
d   -1.1058     0.5687 -1.9443   0.1913
aa  -0.1029     0.4499 -0.2287   0.8403
ad   0.5778     1.4172  0.4077   0.7230

 Residual Standard Error:   7.9064
 Multiple R-square:         0.9982
 Adjusted R-square:         0.9936
 F-value:                 218.4485
 p-value:                   0.0046
```

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Observed and Predicted Values of Generation Means:
  Generation Observed Predicted
1         P1 3.703000  3.791690
2         P2 4.601000  4.619287
3         F1 3.231667  3.202629
4         F2 3.515867  3.755507
5         F3 4.140600  4.031947
6        BC1 3.775600  3.667342
7        BC2 4.483000  3.792224

Goodness-of-fit Test
Chi-square value:  125.0215
p-value:           7.111018e-28

------------------------------------------------
MODEL 3 :

Regression Model:  mean ~ 0 + m + a + d + ad

Regression Coefficients:
   Estimate Std. Error t value Pr(>|t|)
m    4.2513     0.1850 22.9814   0.0002
a   -0.3836     0.2255 -1.7015   0.1874
d   -1.0222     0.3604 -2.8360   0.0659
ad   0.4871     1.1253  0.4328   0.6944

 Residual Standard Error:   6.5394
 Multiple R-square:         0.9981
 Adjusted R-square:         0.9956
 F-value:                 399.1306
 p-value:                   0.0002

Observed and Predicted Values of Generation Means:
  Generation Observed Predicted
1         P1 3.703000  3.867672
2         P2 4.601000  4.634953
3         F1 3.231667  3.229141
4         F2 3.515867  3.740227
5         F3 4.140600  3.995770
6        BC1 3.775600  3.670172
7        BC2 4.483000  3.810281

Goodness-of-fit Test
Chi-square value:  128.2918
p-value:           1.26231e-27
------------------------------------------------

MODEL 4 :

Regression Model:  mean ~ 0 + m + a + d
```

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Regression Coefficients:
  Estimate Std. Error t value Pr(>|t|)
m   4.2772     0.1563 27.3709   0.0000
a  -0.3389     0.1789 -1.8946   0.1311
d  -1.0359     0.3205 -3.2322   0.0319

 Residual Standard Error:  5.8375
 Multiple R-square:         0.998
 Adjusted R-square:        0.9965
 F-value:                 667.778
 p-value:                  0.0000

Observed and Predicted Values of Generation Means:
  Generation Observed Predicted
1         P1 3.703000  3.938297
2         P2 4.601000  4.616091
3         F1 3.231667  3.241292
4         F2 3.515867  3.759243
5         F3 4.140600  4.018219
6        BC1 3.775600  3.589795
7        BC2 4.483000  3.928692

Goodness-of-fit Test
Chi-square value:  136.304
p-value:           1.74492e-28

------------------------------------------------

MODEL 5 :

Regression Model:  mean ~ 0 + m + d

Regression Coefficients:
  Estimate Std. Error t value Pr(>|t|)
m   4.4151     0.1704 25.9159   0.0000
d  -1.3037     0.3544 -3.6786   0.0143

 Residual Standard Error:   7.1919
 Multiple R-square:         0.9962
 Adjusted R-square:         0.9947
 F-value:                 658.7258
 p-value:                   0.0000

Observed and Predicted Values of Generation Means:
  Generation Observed Predicted
1         P1 3.703000  4.415106
2         P2 4.601000  4.415106
3         F1 3.231667  3.111449
4         F2 3.515867  3.763277
5         F3 4.140600  4.089191
6        BC1 3.775600  3.763277
```

```
7        BC2 4.483000  3.763277

Goodness-of-fit Test
Chi-square value:  258.6178
p-value:           7.774878e-54
```

Summary Statistics as Input

The steps to perform such analysis using *PBTools* are listed below:

- Import *GenMean(summaryStats)* file from package. Double-click the file to view it in the Data Viewer.

- Choose **Analysis | Generation Mean Analysis | Summary statistics as input…** from the menu bar.

- Specify the required fields and appropriate options for the analysis.

  No Computed Weights
  This is selected if in the data set there is no column that corresponds to weights, hence, PBTools will automatically compute the weights needed for the analysis.

  With Computed Weights
  This is selected if in the data set there is a column that corresponds to weights.

  Mean
  This is the column that corresponds to the generation mean.

  Weights
  This is required if 'With Computed Weights' option is selected.

  Standard Deviation
  This is required if 'No Computed Weights' option is selected.

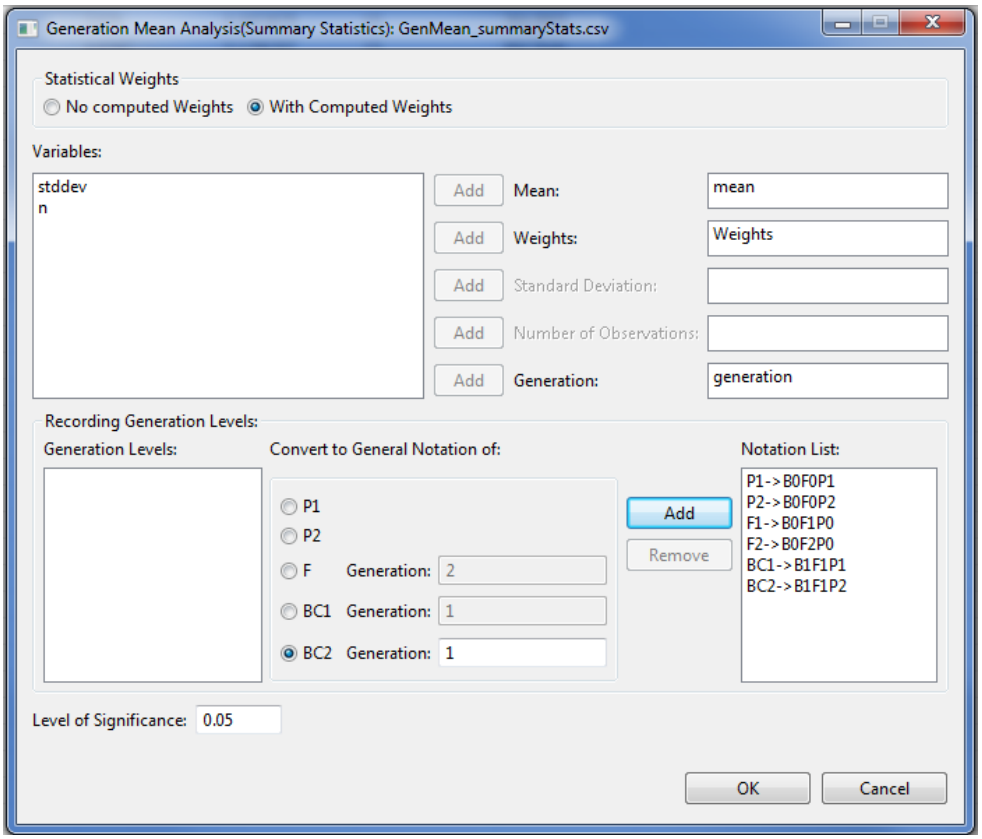
  Number of Observations
  This is required if 'No Computed Weights' option is selected.

  Generation
  This is the column that corresponds to the generation classification.

- For each generation level, convert it to the general notation that PBTools recognizes by clicking one level in the user's notation box, selecting the appropriate radio button that corresponds to its generation classification, specifying the generation number if necessary and clicking the **Add** button. A text indicating the conversion will be displayed in the Notation List.

  For the example, do the following:
  - select P1 in the user's notation box, then classify it as P1 (the default selection) and click **Add** button
  - select P2 in the user's notation box, then classify it as P2 and click **Add** button
  - select F1 in the user's notation box, then classify it as F, specify the generation number as 1 and click **Add** button
  - select F2 in the user's notation box, then classify it as F, specify the generation number as 2 and click **Add** button
  - select BC1 column in the user's notation box, then classify it as BC1, specify the generation number as 1 and click **Add** button
  - select BC2 column in the user's notation box, then classify it as BC2, specify the generation number as 1 and click **Add** button

  For the *example*, the completed dialog box should appear as illustrated below:



- Click **OK**.

- Sample output of the analysis is shown below:

```
GENERATION MEAN ANALYSIS

------------------------------------------------
SET A FULL MODEL: mean ~ 0 + m + a + d + aa + ad
------------------------------------------------

SET A MODEL 1 :

Regression Model:  mean ~ 0 + m + a + d + aa + ad

Regression Coefficients:
   Estimate Std. Error t value Pr(>|t|)
m    3.8952     0.7203  5.4076   0.1164
a   -0.4640     0.2816 -1.6481   0.3472
d   -0.5853     1.0100 -0.5795   0.6656
aa   0.2978     0.7773  0.3831   0.7671
ad   0.8323     1.5665  0.5313   0.6891

 Residual Standard Error:  9.2487
 Multiple R-square:        0.9984
 Adjusted R-square:        0.9903
 F-value:                123.755
 p-value:                  0.0681

Observed and Predicted Values of Generation Means:
  Generation Observed Predicted
1         P1 3.676000  3.728934
2         P2 4.634200  4.657033
3         F1 3.231667  3.309876
4         F2 3.515867  3.602534
5        BC1 3.775600  3.653035
6        BC2 4.483000  3.700929

Goodness-of-fit Test
Chi-square value:  85.53851
p-value:           2.272328e-20

------------------------------------------------
SET A MODEL 2 :

Regression Model:  mean ~ 0 + m + a + d + ad

Regression Coefficients:
   Estimate Std. Error t value Pr(>|t|)
m    4.1527     0.1960 21.1824   0.0022
a   -0.4800     0.2109 -2.2767   0.1505
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d   -0.9260     0.3628 -2.5525   0.1253
ad   0.8271     1.1861  0.6973   0.5578

 Residual Standard Error:   7.0034
 Multiple R-square:         0.9981
 Adjusted R-square:         0.9944
 F-value:                 269.7224
 p-value:                   0.0037

Observed and Predicted Values of Generation Means:
  Generation Observed Predicted
1         P1 3.676000  3.672681
2         P2 4.634200  4.632768
3         F1 3.231667  3.226763
4         F2 3.515867  3.689744
5        BC1 3.775600  3.656493
6        BC2 4.483000  3.722995

Goodness-of-fit Test
Chi-square value:  98.09447
p-value:           5.000995e-22

------------------------------------------------
SET A MODEL 3 :

Regression Model:  mean ~ 0 + m + a + d

Regression Coefficients:
  Estimate Std. Error t value Pr(>|t|)
m   4.1910     0.1713 24.4589   0.0001
a  -0.4197     0.1750 -2.3977   0.0961
d  -0.9292     0.3302 -2.8140   0.0671

 Residual Standard Error:   6.3755
 Multiple R-square:         0.9977
 Adjusted R-square:         0.9954
 F-value:                 433.7527
 p-value:                   0.0002

Observed and Predicted Values of Generation Means:
  Generation Observed Predicted
1         P1 3.676000  3.771248
2         P2 4.634200  4.610674
3         F1 3.231667  3.261740
4         F2 3.515867  3.726351
5        BC1 3.775600  3.516494
6        BC2 4.483000  3.936207

Goodness-of-fit Test
Chi-square value:  121.9421
p-value:           2.945444e-26
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------------------------------------------------
SET A MODEL 4 :

Regression Model:  mean ~ 0 + m + d

Regression Coefficients:
  Estimate Std. Error t value Pr(>|t|)
m   4.3104     0.2425 17.7778   0.0001
d  -1.1646     0.4663 -2.4975   0.0669

 Residual Standard Error:   9.4291
 Multiple R-square:         0.9933
 Adjusted R-square:         0.9899
 F-value:                 296.1461
 p-value:                   0.0000

Observed and Predicted Values of Generation Means:
  Generation Observed Predicted
1         P1 3.676000  4.310434
2         P2 4.634200  4.310434
3         F1 3.231667  3.145865
4         F2 3.515867  3.728150
5        BC1 3.775600  3.728150
6        BC2 4.483000  3.728150

Goodness-of-fit Test
Chi-square value:  355.6286
p-value:           1.068149e-75

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SET A MODEL 5 :
Regression Model:  mean ~ 0 + m

Regression Coefficients:
  Estimate Std. Error t value Pr(>|t|)
m   3.8537     0.2278 16.9184        0

 Residual Standard Error:  13.4921
 Multiple R-square:         0.9828
 Adjusted R-square:         0.9794
 F-value:                 286.2308
 p-value:                   0.0000

Observed and Predicted Values of Generation Means:
  Generation Observed Predicted
1         P1 3.676000  3.853678
2         P2 4.634200  3.853678
3         F1 3.231667  3.853678
4         F2 3.515867  3.853678
```

```
5        BC1 3.775600  3.853678
6        BC2 4.483000  3.853678

Goodness-of-fit Test
Chi-square value:  910.1826
p-value:           1.664558e-194

------------------------------------------------

------------------------------------------------
SET B FULL MODEL: mean ~ 0 + m + a + d + aa + dd
------------------------------------------------

SET B MODEL 1 :

Regression Model:  mean ~ 0 + m + a + d + aa + dd

Regression Coefficients:
   Estimate Std. Error t value Pr(>|t|)
m    2.0015     0.4243  4.7171   0.1330
a   -0.4967     0.0610 -8.1481   0.0777
d    4.8273     1.1225  4.3004   0.1455
aa   2.1466     0.4144  5.1796   0.1214
dd  -3.5971     0.7296 -4.9306   0.1274

 Residual Standard Error:    2.0817
 Multiple R-square:          0.9999
 Adjusted R-square:          0.9995
 F-value:                 2446.4808
 p-value:                    0.0153

Observed and Predicted Values of Generation Means:
  Generation Observed Predicted
1         P1 3.676000  3.651347
2         P2 4.634200  4.644834
3         F1 3.231667  3.231667
4         F2 3.515867  3.515867
5        BC1 3.775600  3.804141
6        BC2 4.483000  4.300884

Goodness-of-fit Test
Chi-square value:  4.333597
p-value:           0.03736719

------------------------------------------------
SET B MODEL 2 :

Regression Model:  mean ~ 0 + m + a + aa + dd

Regression Coefficients:
   Estimate Std. Error t value Pr(>|t|)
```

```
m    3.7883     0.2687 14.0978   0.0050
a   -0.4175     0.1814 -2.3012   0.1480
aa   0.4303     0.3487  1.2341   0.3425
dd  -0.5247     0.4612 -1.1377   0.3732

 Residual Standard Error:   6.4991
 Multiple R-square:         0.9984
 Adjusted R-square:         0.9952
 F-value:                 313.2789
 p-value:                   0.0032

Observed and Predicted Values of Generation Means:
  Generation Observed Predicted
1         P1 3.676000  3.801075
2         P2 4.634200  4.636054
3         F1 3.231667  3.263525
4         F2 3.515867  3.657078
5        BC1 3.775600  3.555909
6        BC2 4.483000  3.973399

Goodness-of-fit Test
Chi-square value:  84.47771
p-value:           4.527921e-19

------------------------------------------------
SET B MODEL 3 :

Regression Model:  mean ~ 0 + m + a + aa

Regression Coefficients:
   Estimate Std. Error t value Pr(>|t|)
m    3.5361     0.1593 22.2024   0.0002
a   -0.4025     0.1896 -2.1225   0.1239
aa   0.6958     0.2715  2.5628   0.0830

 Residual Standard Error:   6.8106
 Multiple R-square:         0.9974
 Adjusted R-square:         0.9948
 F-value:                 379.9835
 p-value:                   0.0002

Observed and Predicted Values of Generation Means:
  Generation Observed Predicted
1         P1 3.676000  3.829457
2         P2 4.634200  4.634389
3         F1 3.231667  3.536143
4         F2 3.515867  3.536143
5        BC1 3.775600  3.508855
6        BC2 4.483000  3.911321
```

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Goodness-of-fit Test
Chi-square value:  139.1521
p-value:           5.758125e-30

------------------------------------------------
SET B MODEL 4 :

Regression Model:  mean ~ 0 + m + aa

Regression Coefficients:
   Estimate Std. Error t value Pr(>|t|)
m    3.4809     0.2152 16.1730   0.0001
aa   0.8898     0.3502  2.5413   0.0639

 Residual Standard Error:    9.329
 Multiple R-square:         0.9934
 Adjusted R-square:         0.9902
 F-value:                 302.5775
 p-value:                   0.0000

Observed and Predicted Values of Generation Means:
  Generation Observed Predicted
1         P1 3.676000  4.370741
2         P2 4.634200  4.370741
3         F1 3.231667  3.480892
4         F2 3.515867  3.480892
5        BC1 3.775600  3.703354
6        BC2 4.483000  3.703354

Goodness-of-fit Test
Chi-square value:  348.1192
p-value:           4.467431e-74

------------------------------------------------
SET B MODEL 5 :

Regression Model:  mean ~ 0 + m

Regression Coefficients:
  Estimate Std. Error t value Pr(>|t|)
m   3.8537     0.2278 16.9184        0

 Residual Standard Error:  13.4921
 Multiple R-square:         0.9828
 Adjusted R-square:         0.9794
 F-value:                 286.2308
 p-value:                   0.0000

Observed and Predicted Values of Generation Means:
  Generation Observed Predicted
1         P1 3.676000  3.853678
```

```
2          P2 4.634200  3.853678
3          F1 3.231667  3.853678
4          F2 3.515867  3.853678
5         BC1 3.775600  3.853678
6         BC2 4.483000  3.853678

Goodness-of-fit Test
Chi-square value:  910.1826
p-value:           1.664558e-194

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